C-1

GENERAL INFORMATION

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1.1 INTRODUCTION

GENERAL INFORMATION

The Contractor shall provide all services, materials, supplies, facilities, supervision, labor, and equipment as specified herein and in accordance with the terms, conditions, and specifications of this Contract. The Contractor shall perform facilities engineering functions at Fort Benning, Georgia, and at satellite facilities.

1.1.1 BROAD WORK DESCRIPTION

1.1.1.1 Work Responsibility

Contractor work and responsibility shall include all Contractor planning, programming, administration, management, and execution necessary to provide the specified services. The Contractor shall perform work in accordance with this Contract and all applicable Federal, State, and local laws, regulations, and directives to include, but not limited to, applicable publications in Section C-6. The Contractor shall ensure that all work meets critical reliability rates or tolerances specified in the Contract Specifications, the Performance Requirements Summary (PRS) (see Technical Exhibit 1-001), or in applicable referenced documents. The Contractor shall perform all related Contractor administrative services required to perform work such as materiel requisitioning, procurement of environmental clearance, quality control (QC), financial control, and correspondence. The Contractor shall also maintain accurate and complete records, files, and libraries of documents to include Federal, State, and local regulations, codes, laws, technical manuals, and manufacturer's instructions and recommendations, which are necessary and related to the functions being performed. The Contractor shall compile historical data, prepare required reports, and submit information as specified by Contract Deliverable List (CD) presented in this Contract. In addition, there are numerous formal and informal requirements throughout the PWS for the contractor to “report” data to the KO. It is expected that the contractor shall comply with these requirements also.

1.1.1.2 Functional Areas

Work to be performed by the Contractor is categorized under Functional Areas. Functional Areas covered under this Contract include:

- Buildings and Structures Maintenance
- Dining Facility Equipment Maintenance
- Utility Systems Operation and Maintenance
- HVAC Systems Operation and Maintenance
- Grounds Maintenance
- Surfaced Area Maintenance
- Material Recycle Facility Operation and Maintenance
- Self-Help Services
- Remote Camp Operations, Maintenance, and Support

- Range Maintenance
- Cemetery Services
- Chiller Maintenance and Repair

1.1.2 INSTALLATION MISSION

The mission of Fort Benning, a Training and Doctrine Command (TRADOC) installation, is to provide the world's best Infantry soldiers and trained units; act as a power projection platform capable of deploying soldiers and units anywhere in the world on short notice; and to be the Army's premier installation and home for soldiers, families, civilian employees, and military services.

1.1.3 INSTALLATION CHARACTERISTICS

1.1.3.1 Background

Fort Benning, the "Home of the Infantry," was named for Henry Lewis Benning a Confederate Major General and an accomplished Georgian lawyer. The famed United States Army Infantry School was established in 1918 and over the years emerged as the most influential infantry center in the modern world. From 1918 to the present the mission of Fort Benning as well as the school has been to produce the world's finest combat infantrymen. After establishment in 1918, Fort Benning saw the addition of the 11th Air Assault Division, Airborne School, Ranger School, 29th Infantry Regiment, 36th Engineer Group, and battle labs to make it a truly combined arms center. Combined with the leadership training courses, there isn't a General or Private that has not benefited from the experience and teaching at the Home of the Infantry.

Fort Benning has several primary activities. It is the home of the US Army Infantry Center and School (USAIC&S), Infantry Training Brigade, 30th Adjutant General Battalion (Reception), Ranger Training Brigade, 75th Ranger Regiment, 3/75th Ranger Battalion, 11th Infantry Regiment, Officers Candidate School (OCS), Basic Combat Training Brigade (BCTB), 29th Infantry Regiment, US Army Physical Fitness School, Lawson Army Airfield (LAAF), US Army Marksmanship Unit, US Army Research Institute, Western Hemisphere Institute for Security Cooperation (WHISC) , CONUS Replacement Center (CRC), The Henry Caro Noncommissioned Officers Academy and US Air Force Units.

USAIC&S is the home of the Infantry. It was created to supervise the training of Infantry personnel and troop units. The primary mission of USAIC&S is to provide training and professional development for military personnel of the Active Army and Reserve Components, members of other branches of the Armed Forces, civilians, and personnel of allied countries. Along with its training missions, USAIC&S serves as the focal point for the development of the Infantry missions, organizations, and support requirements. The training battalions and curriculum are supported by the Ranger Training Brigade, Drill Sergeant School, 11th Infantry Regiment, US Air force units, and 29th Infantry Regiment, The 75th Ranger Regiment.

The US Army Garrison manages the Installation functions and services that keep the US Army Infantry Center and Fort Benning community operating so that organizations and activities may concentrate on their primary missions. The garrison staff office is committed to providing quality of life support to Fort Benning personnel, their families, and to retired service members and their family members who live in the surrounding area.

Fort Benning's primary mission is to provide training and professional development for military personnel of the active Army and Reserve Components and members of the other Armed Forces. A mission of equal importance has been Force Projection. Since the commencement of Desert Shield and Desert Storm in 1990, Fort Benning has been a major Force Projection platform for mobilization, deployment, and redeployment of forces. Desert Shield alone required Fort Benning to deploy 79 units with over 17,490 troops, 501 vehicles, and 687 air pallets with a total cargo weight of 25,526.9 tons by air on 265 aircraft out of LAAF. Twenty-nine units and 2,104 pieces of equipment were conveyed to port. In addition, 4,750 troops were bused to Fort Stewart for transport to the Middle East. Desert Shield required 24 hour/7 days a week Emergency Operations Center (EOC) operations by logistical support activities for almost one year. Since Desert Shield/Desert Storm, Fort Benning has been a Force Projection platform for 7 additional real world contingencies, deploying over 20,000 personnel on over 107 aircraft from LAAF. Over 2,000 pieces of equipment deployed by air, rail, or long haul. During this same period, Fort Benning supported 59 major Force Projection JCS/JRT Readiness training exercises deploying troops and equipment throughout CONUS and OCONUS. Fort Benning continues to support Force Projection with troops deploying and redeploying to Bosnia, Haiti, Kuwait, EOC 24-hour operation, and Port of Jacksonville PSA operations.

1.1.3.2 Location

Fort Benning is located in the lower Piedmont Region of Central Georgia and Alabama, six miles southwest of Columbus, Georgia, and supports satellite locations in Dahlonega, GA and Eglin Air Force Base, FL. The requirements of this contract only apply to Fort Benning and the satellite location (Camp Merrill) at Dahlonega. The moderate climate and various terrains are well suited for Infantry training and support missions. Fort Benning has excellent accessibility, with the terminus of I-185 and LAAF.

1.1.3.3 Physical Description

The Installation consists of approximately 184,089 acres of river valley terraces and rolling terrain. There are 2,851 active buildings, 18,935 enlisted spaces, 108 bachelor officer quarters, and 26 senior enlisted quarters located on Fort Benning. Currently, there are 4,110 family housing units (3,468 designated for enlisted personnel and 642 designated for officers). The 115 guest house rooms on the Installation provide temporary lodging for married soldiers and for the guests of military personnel stationed at Fort Benning. A summary of the facilities and land that make up the Fort Benning installation is presented in Technical Exhibit 1-002. Also see Technical Exhibit C-5.2-004 for the National Register Eligibility Inventory Table; Assessment of National Register Eligibility for Fort Benning Historic Resources Constructed Prior to the Year 1953 as published in the 1997 Fort Benning Resource Survey Update. A series of maps can be found in the Technical Library, highlighting different aspects of the Installation. A complete listing of all the buildings that the Contractor shall be responsible for maintaining is provided in the Real Property Inventory in Technical Exhibit 1-003.

1.1.3.4 Population and Demographics

The population of Fort Benning is approximately 21,000 military personnel and 7,000 civilian personnel. More than 25,000 students graduate from training, education, and professional development courses at the Installation each year. The Installation supports more than 21,000 family members and 29,000 retirees, plus survivors and family members living in the area. Fort Benning services a total average daily population of 103,000.

Permanent party man-strengths remain relatively consistent throughout the year at approximately 3711 for TRADOC personnel, 829 for Tenant personnel, and 9388 for Tactical Troop (TOE) personnel. However, the Training mission varies rather significantly throughout the year. The cyclic load associated with the training mission weekly average strength is presented in Technical Exhibit 1-006.

Technical Exhibit 1-006 also presents the Basic Combat Training Brigade (BTCB) training strength workload that the contractor shall support. The BCTB training strength represent workload that is presented in the technical exhibits of this document. BCTB training has started and contract proposals shall include estimated requirements for contractor support of this requirement based on Technical Exhibits 1-006 and 1-008 and other impact information and data provided in this document. BTCB training has resulted in an increase of 350 to the permanent party man-strength for TRADOC personnel.

1.1.3.5 Climate and Terrain

Average monthly temperature ranges from 33.9°F to 89.0°F, with an overall average temperature of 59.7°F. The extreme temperatures range from a low of 4°F in the winter to an average high of 102°F in the summer. The average annual precipitation is
45.3 inches. The average annual snowfall is 0.0 inches.

1.1.4 LAYOUT OF SECTION C

1.1.4.1 Section C Contents

Section C of this Contract is structured as follows:

- C-1: General Information
- C-2: Definitions
- C-3: Government-Furnished Property and Services
- C-4: Contractor-Furnished Property
- C-5: Technical Description of Work
- C-6: Publications and Forms
- C-7: Technical Exhibits

Paragraphs in Section C-1 all begin with the number “1", paragraphs in Section C-2 all begin with the number “2", and the pattern continues for the other sections.

1.1.4.2 Functional Areas

Functional Areas are found in Section C-5, Technical Description of Work. The Functional Areas covered by this Contract are numbered as follows:

- Buildings and Structures Maintenance, C-5.2
- Dining Facility Equipment Maintenance, C-5.3
- Utility Systems Operation and Maintenance, C-5.4
- HVAC Systems Operation and Maintenance, C-5.5
- Grounds Maintenance, C-5.6
- Surfaced Area Maintenance, C-5.7

- Material Recycle Facility Operation and Maintenance, C-5.8
- Self-Help Services, C-5.9
- Remote Camp Operations, Maintenance, and Support, C-5.10
- Range Maintenance, C-5.11
- Cemetery Services, C-5.12
- Chiller Maintenance and Repair, C-5-13

1.1.4.3 Workload Data

Historical workload data is presented in Section 5.1 and applicable functional areas. This data has been summarized for this Contract. The workload information displayed in the text and in technical exhibits is based on annual data, where available, or extrapolated to represent estimated annual workload, where less than one year's data was available. When possible, the data has been updated to show current levels of work for the various functional areas. This workload is provided to assist offerors in proposal preparation and shall not be a limiting factor on the Contractor's obligation to perform all services described in this Contract to the required level of effort.

1.1.4.4 Other Information

1.1.4.4.1 Technical Exhibits. Technical Exhibits are used to provide supplementary information and can be in the form of tables, graphs, or maps. Technical Exhibits may be referenced from any part of the Contract. Technical Exhibits for Section C have a 3- digit number which links them to a designated Contract Section; e.g., Technical Exhibit 5.2-002 is the second technical exhibit referenced from Functional Area 5.2.

1.1.4.4.2 Pagination. Pagination for all parts of Section C begins with the letter "C". Examples are: C-1-12 or C-5-4 (at the Section C level) or C-5.2-6 or C-5.4-10 (at the Functional Area level). Technical Exhibits will have a page number that looks like C- 7.5.2-002-2 (which would designate the second page of Technical Exhibit 5.2-002 referenced from Functional Area 5.2).

1.2 MANAGEMENT AND ADMINISTRATION

1.2.1 WORK BY THE GOVERNMENT

The Government reserves the right to utilize Government forces or other contractors to accomplish the same type or similar work as solicited for in this Contract.

1.2.2 PHASE-IN AND PHASE-OUT PERIODS

The Contractor shall develop comprehensive procedures for phasing in Contractor performance to the level prescribed and within the time allowed under the terms of this Contract. The Contractor shall submit a completed Phase-In Plan within 15 calendar days of Contract award date (CD 102R001). The Contractor shall also establish and implement plans for an orderly phase-out of the contracted operations at the termination o f this Contract. This phase-in plan shall be approved by the KO prior to implementation. The Contractor's phase-out procedures shall not disrupt or adversely impact the day-to- day conduct of Government business. These plans shall be submitted by the Contractor for approval by the Government, and shall be in accordance with the Government's transition plan. The Contractor shall provide the KO copies of changes and revisions for review and approval no later than 30 calendar days prior to implementation.

1.2.2.1 Phase-In Period

The phase-in period is part of the base year of the Contract and it starts immediately on the Contract start date. During the phase-in period, the Contractor shall prepare to assume full responsibility for all areas of operation in accordance with the terms and conditions of this Contract.

The Contractor shall take all actions necessary for a smooth transition of the contracted operations. This period will be 60 calendar days in duration. The Government will make all facilities and equipment accessible to the Contractor for a maximum of 60 calendar days prior to the Contract start date. During the last 30 calendar days of this period (i.e., during the 30 calendar days immediately prior to the Contract start date), the Contractor's management personnel will be permitted to observe any Facilities engineering operations at Fort Benning, as approved by the KO. During the phase-in period, the Contractor shall at a minimum:

- Establish the Project Management Office.
- Recruit and hire necessary personnel.
- Obtain all required certifications and clearances, including personnel security clearances (this can take longer than 30 to 60 calendar days).
- Participate in joint inventories and sign for Government-furnished Property (GFP) or real property.
- Develop and submit any required deliverables.
- Attend post-award meetings as required.
- Receive natural resources briefings.
- Accomplish any necessary training of its personnel to support the functions listed in Section C-5.

1.2.2.2 Phase-Out Period

Sixty calendar days prior to the completion of this Contract (to include option periods), an observation period shall occur, at which time management personnel of the incoming workforce may observe operations and performance methods of the incumbent Contractor. This will allow for orderly turnover of facilities, equipment, and records and will help to ensure continuity of service. The Contractor shall not defer any requirements for the purpose of avoiding responsibility or of transferring such responsibility to the succeeding Contractor. The Contractor shall fully cooperate with the succeeding Contractor and the Government so as not to interfere with their work or duties.

1.2.2.2.1 Phase-Out Plan. The Contractor shall develop a Phase-Out Plan (CD 102R002) to effect a smooth and orderly transfer of Contract responsibility to a successor. The Contractor shall submit the completed Phase-Out Plan to the KO no later than 60 calendar days prior to Contract completion date. The plan shall fully describe how the Contractor shall, at a minimum, approach the following issues: employee notification; retention of key personnel; turn-over of work-in-progress, inventories, and Government property; removal of Contractor property; data and information transfer; and any other actions required to ensure continuity of operations. The Contractor's Phase-Out Plan shall, at a minimum, require an inventory by the incumbent and the Government before conduct of a joint inventory between the incumbent and the successor.

The Plan shall, at a minimum, also include: reconciliation of all property accounts, requisitions, and work-in-progress; turn-in of excess property; clean-up of Contractor work areas; provision for training of the successor's personnel on Government-furnished automated information systems (AIS) used in performance of this Contract, specialized equipment, utilities systems, and ongoing work that the successor would be required to complete; and, security debriefings in accordance with AR 380-5 for incumbent personnel holding security clearances.

1.2.3 CONTRACTOR ADMINISTRATION

1.2.3.1 Project Management

Contractor shall perform continual Project Management and shall provide a Project Manager or his designated representative. The Project Manager shall conduct overall management coordination and shall be the central point of contact with the Government for performance of all work under the Contract. A Contractor employee shall be designated to act for the Project Manager when work is being performed outside of duty hours, or during the Project Manager's absence. The Contractor shall provide verbal, 48 hour advance notice of such absences to the KO. The Project Manager, and any individuals designated to act in that capacity, shall have full authority to contractually bind the Contractor for prompt action on matters pertaining to execution of the Contract. A general description of how this will be accomplished shall be provided in a Management Approach Report as required in CD 102R008.

1.2.3.2 Responsiveness

The Project Manager or alternate shall return all calls from the KO within ten minutes during normal Contractor personnel working hours unless otherwise specified herein.

During non-working hours, the Project Manager or alternate shall return all calls from the KO within twenty minutes unless otherwise specified herein.

1.2.3.3 Adherence to Garrison Regulations

The Contractor and Contractor personnel shall abide by the current USAIC Regulation 210-5, Garrison Regulations, in effect during performance of this Contract. A copy of this regulation is available for review in the Directorate of Contracting (DOC) office in Building 6.

1.2.3.4 Meetings, Conferences, and Briefings

The Contractor shall attend, participate in, and furnish input to scheduled and unscheduled meetings, conferences, and briefings that relate to the contracted functions and services as required by the Government to provide effective communication and impart necessary information. These meetings include, but are not limited to, the Performance Evaluation meetings in Paragraph 1.3.1.2 below. The Contractor shall participate in meetings with Installation customers, Federal, State, and local agencies and their representatives, and other contractors as required by the Government. These meetings include both on-site and off-site meetings, and the frequency may be weekly, monthly, or as otherwise required.

An average of 5 to 10 hours per week is estimated for briefings, meetings, and conferences; however, this requirement may vary based upon Contractor performance. Meetings may start or end outside of normal working hours. The Contractor shall also attend pre and post construction meetings with other contractors and assist them with inspections.

1.2.3.4.1 Attendees. The Project Manager or designated representative shall be required to attend meetings as required by the Government. Meeting attendees shall at times include Contractor managerial, supervisory, or other personnel knowledgeable of the subject matter, e.g., maintenance, supply, equipment readiness, services, transportation, and costing.

1.2.3.4.2 Reporting Requirement. When the Contractor is present at meetings, conferences, or trips, a report shall be furnished to the KO within two working days a f t e r event completion for approval (CD102R003). The report shall include identifying information, general observations and conclusions or recommended actions, and any additional information, such as handouts.

1.2.3.5 Installation Closures

When an unforeseen Installation closure or curtailment of activities occurs on a regularly scheduled day of work, the Government will have the following options:

- reschedule the work to be performed the following workday
- reschedule the work on any day acceptable to the Government
- have the Contractor continue the work as scheduled

1.2.3.5.1 Notification of Closure. Announcements of Installation closures will be made in the following manner: during normal duty hours, notification will be given through normal chain of command; during non-duty hours, notification will be made through local radio and television channels. Installation closings shall in no way interfere with the Contractor operation or maintenance of the critical systems listed in Paragraph 5.1.4.2.

1.2.3.6 Federal Holidays

All Government offices will be closed, except for minimum essential personnel required for in-house operations, during Federal holidays. Except as otherwise specified, the Contractor shall not schedule routine work on Federal holidays. When a scheduled service is required less than three times per week and the schedule for that work falls on a Federal holiday, the Contractor shall accomplish the work on the workday following or preceding the holiday. The ten Federal holidays per year are as follows:

New Years Day	first day of January (or observed)
Martin Luther King, Jr. Birthday	third Monday of January
Presidents Day	third Monday of February
Memorial Day	last Monday of May
Independence Day	fourth day of July (or observed)
Labor Day	first Monday of September
Columbus Day	second Monday of October
Veterans Day	11th day of November (or observed)
Thanksgiving Day	fourth Thursday of November
Christmas Day	25th day of December (or observed)

When such holidays fall on a Saturday, the preceding Friday will be considered a holiday. When such holidays fall on a Sunday, the succeeding Monday will be considered a holiday.

1.2.3.7 Training Holidays

TRADOC Installation Commands have historically granted military personnel (Enlisted and Officers) training holidays (for example, Christmas Exodus) in addition to the ten Federal holidays listed above. Training holidays are not considered a day of excused absence for the Government civilian work force or Contractor employees. If a training holiday affects scheduled Contract work or access to facilities, the work in affected facilities may be rescheduled to compensate for the training holiday.

However, announced training holidays or exodus of military personnel may provide the Contractor with an opportunity for unrestricted access to facilities/areas or uninterrupted performance of work, i.e. training facility/area maintenance and supply inventories. The Contractor shall take full advantage of such opportunities by planning maintenance that is better accomplished when personnel are not using facilities during those times.

1.2.3.8 Hours of Operation

1.2.3.8.1 Regular Duty Hours. The Contractor's regular duty hours shall be 0800 to 1630 Monday through Friday, excluding Federal Holidays, except as stated otherwise within Section C-5. The Contractor shall ensure that service to customers is not interrupted during break and lunch periods. In these and other such cases individual employees shall be required to perform shift work which shall represent regular duty hours for those employees. Variances in operating hours, other than those specified in Section C-5, must be approved by the KO.

1.2.3.8.2 Contact Personnel. The Contractor shall provide fully qualified contact personnel on a 24-hour basis to perform services and satisfy emergency requirements as discussed in Section C-5.

1.2.4 INTERFACES

1.2.4.1 Restricted Areas

Compliance by the Contractor shall be mandatory for security access requirements to restricted areas that include, but are not limited to, the communication center, ammunition storage area, computer centers, 75th Ranger Regiment, and research and development areas. Work in restricted areas shall be cleared by the Government's Security Officer for that area. A list of Government security points of contact shall be furnished to the Contractor prior to Contract start date. See TE 1-0011.

1.2.4.2 Administrative Office Areas

Contractor operations shall not unduly interfere with Government work in any area where any service or maintenance work is being performed by the Contractor. The Contractor shall schedule operations so as to minimize interference with Government work.

1.2.4.3 Emergency Operations Center

The Contractor shall coordinate with the EOC in preparation for and during contingency missions, inclement weather, storms, and other emergency conditions. Contingency missions include, but are not limited to, contingency plans, Emergency Deployment Readiness Exercises (EDREs), deployments, redeployments, mobilization, demobilization, Joint Readiness Training Center (JRTC) exercises, sea EDREs, and CRC/Individual Deployment Site (IDS) operations. During emergency operations, the Contractor shall have a representative on site in the EOC to provide required services and coordination in accordance with the Emergency, Mobilization, Contingency, and Disaster plans and procedures defined in Paragraph 1.2.7.

1.2.4.4 Other Contractors

Other Government contractors will be performing required services in areas associated with the requirements of this Contract. Some contractors will be providing services associated with, and in support of, work identified in this Contract. The Government will facilitate initial contact between Contractors performing other contracts and this Contract.

1.2.4.5 Disputes With Customers or Other Government Contractors

The Contractor shall verbally notify the KO of unresolved disputes in receiving support from or providing support to customers or other contractors within two hours from the time the dispute occurs. The Contractor shall provide written responses to customer complaints and submit them to the KO within five working days. Unresolved disputes must be settled within five working days. The Contractor shall also answer and resolve complaints or problems that are called in via 5-BOSS, the Installation customer hotline, within five working days.

1.2.4.6 Other Government Agencies

The Contractor shall be required to provide input to other Government agencies to support mission changes and facility MCA, plans and studies, footprint studies and actions, and Maintenance and Repair (M&R) funding support projects. Approximately 296 such actions occur each year within the facilities engineering and logistics areas that shall require Contractor input.

1.2.5 DATA AND INFORMATION

The Contractor shall respond to requests for information, including programmed (scheduled) and un-programmed requests. All responses shall meet the criteria defined below.

1.2.5.1 Data Criteria

The Contractor shall submit programmed and un-programmed information, subject to Government review for adequacy, utilizing the following criteria:

- Complete: To include all information
- Accurate: Factual and correctly tabulated data
- Preparation: In accordance with applicable publication, CD, or other specified format
- Name and signature of Project Manager

- Timely: Provided within the specified time frames
- Distribution: Provided to the specified distributees

1.2.5.2 Programmed Requirements

1.2.5.2.1 The Contractor shall furnish all recurring Contract data and information in digitally form as specified in the Contract Data Requirements Lists (CDs) in Technical Exhibit 1-004 and in the formats shown in TE 1-004a.

1.2.5.2.2 Cost and Performance Report. The Contractor shall submit to the KO monthly Cost and Performance Reports no later than the fifteenth calendar day of the following month (CD 102R004). This report shall contain budgeted hours and costs, actual hours and cost, explanation of variances, material cost, and estimates to complete.

1.2.5.3 Unprogrammed Requirements

Upon notification, the Contractor shall provide management and technical information including, but not limited to:

- Technical evaluation of suggestions
- Input for staff studies
- Fact sheets
- Audits
- Congressional Inquiries
- One-time reports
- Materiel, equipment, facilities, and other listings
- Materiel, equipment, facilities, and other property inventories, such as equipment density listings
- Equipment maintenance records
- Group tours to special interest groups (fielding teams, MWO teams, CVE teams, etc.)
- Recommendations for amending, revising, or originating Government regulations or policies within the scope of this Contract
- Information requested by the KO and other Contractor personnel performing official duties, to include monitoring Contract compliance
- Responses to the KO and other Contractor personnel conducting information and communication systems site surveys, information systems fielding, and communication engineering and construction as required by the Government
- Environmental Reporting Requirements.
- Cost estimates and other inputs.

1.2.5.4 Access to Data and Information

The Contractor shall ensure that all Contractor generated technical records, reports, files, and other documentation are made available to the KO, Facilities engineering, and other authorized Government representatives during the performance of this Contract. The Contractor shall obtain KO approval before releasing any information that has been stored, generated, or archived related to this Contract to the Contractor's corporate or other off-site offices, to other Government activities or agencies, to other contractors, or to private parties.

1.2.6 PERSONNEL

The Contractor shall provide qualified supervisory, technical, administrative, and clerical personnel to accomplish all work and services required by this Contract within specified time frames. This provision shall apply notwithstanding past historical records, estimates of personnel needed, or any minimum levels established elsewhere in this Contract. Contractor employees shall be trained, qualified, certified, or licensed as required in this Contract prior to starting work.

The Government will not provide initial skills training to Contractor employees. The Contractor shall maintain records of training qualifications, certifications, and licenses and keep them on hand for review by Government personnel upon request. The Contractor shall maintain the workforce in such a manner as to ensure that the employees remain fully qualified to perform the work assigned. The Contractor shall submit an organization chart with the technical proposal and identify personnel proposed to perform the Contract requirements.
Personnel performing under this Contract shall be employees of the Contractor and not the Government.

Common Access Card: The Contractor shall participate in the Contractor verification System. The Common Access Card (CAC) is a personalized smart card embedded with an integrated circuit chip for storing and processing date. Functional Description is to Logical access to DoD computer systems/networks. The COR is the Trusted Agent (TA) that initiate and approve applications for their assigned contractors for CAC issuance. Contractors access the CVS website (https://www.dmdc.osd.mil/appj/cvs/index.jsp) to complete and submit CAC applications for the TA's approval. The Contractor shall submit the "Common Access Card Re-verification Report", (CD102R010), every six (6) months.

1.2.6.1 Key Personnel

The Contractor may submit resumes for key personnel in the management and staffing section of the technical proposal. Key personnel include, but are not limited to, the following: Project Manager; a designated deputy Project Manager; a designated manager responsible for Environmental Coordinator; and those individuals responsible for managing the work in each functional area of the Contract. Any changes to the working status of these key personnel shall be provided to the KO for approval within five working days prior to the change.

1.2.6.2 Definition of Contractor Employees

For the purpose of this Contract, the term "Contractor employee(s)" applies to all Contractor personnel and Subcontractor employees performing work on this Contract and being paid either directly or indirectly by the Contractor.

1.2.6.3 Subcontractors

The Contractor shall provide a list of subcontractors used in the performance of this Contract to the KO. The list shall include names, addresses, and telephone numbers, and shall be submitted with the technical proposal. Any changes to this list must be submitted to the KO for approval.

1.2.6.4 Employment Limitations

1.2.6.4.1 Language Requirement. The Contractor shall employ only persons able to communicate and understand English for those positions interacting with Government and other personnel in the performance of this Contract, or where English is used or essential to provide the product, record, data, information, or service.

1.2.6.4.2 Mathematics Requirement. All Contractor personnel shall be capable of performing basic mathematics for the function(s) and task(s) which they are assigned.

1.2.6.4.3 Physical Fitness. The Contractor shall ensure that all Contractor personnel meet the physical requirements to perform the function(s) and task(s) to which they are assigned. This may include performing work in conditions of extreme heat and cold, high humidity, blowing sand and dust, and working in confined spaces and at heights.

1.2.6.4.4 Employment of Military or Government Personnel. Employment of off-duty military personnel or Government civilian personnel is generally permissible provided such employment does not contravene the policies set forth in DOD Joint Ethics Regulation 5500.7-R as determined by local SJA.

1.2.6.5 Identification of Employees

The Contractor shall provide each employee an identification (ID) card on Contract start date or on the employment start date. The ID card format shall be approved by the KO and include, at a minimum, the badge number, date issued and expiration date (not to exceed the end of the final option year), employee's name, Contractor's name, functional area of assignment, and color photograph.

1.2.6.5.1 Display of ID Card. Contractor personnel shall wear the ID card at all times when performing work under this Contract. Unless otherwise specified in the Contract, each Contractor employee shall wear the ID card in a conspicuous place on the front of exterior clothing and above the waist except when safety or health reasons prohibit. Vehicle mechanics may wear identifying badges inside the upper shirt pocket when performing maintenance or repair of equipment.

1.2.6.5.2 Contractor Accountability. Upon termination of this Contract or of employment of any individual, the Contractor shall collect and dispose of all Contractor- furnished identification cards. If any are not collected, the Contractor shall notify the KO within 24 hours.

1.2.6.6 Employee Uniforms

If the Contractor elects to provide uniforms, at the Contractor's expense, the selected uniforms shall not be similar to military or Installation guard uniforms. Employee uniforms shall consist of, at a minimum, shirts which identify company and functional area for which the employee works.

1.2.6.7 Conduct of Personnel

The Contractor shall be responsible for the performance and conduct of Contractor and Subcontractor employees at all times. Personnel employed by the Contractor in the performance of this Contract, or any representative of the Contractor entering the Installation, shall abide by the security regulations listed in the Contract and shall be subject to such checks by the Government as deemed necessary. The Contractor shall not employ for performance under this Contract any person whose employment would result in a conflict of interest with the Government's standards of conduct.

1.2.6.7.1 Personnel Removal. Government rules, regulations, laws, directives, and requirements which are issued during the Contract term relating to law and order, Installation administration, and security on the Installation shall be applicable to all Contractor employees or representatives who enter the Installation. Violation of such rules, regulations, laws, directives, or requirements shall be grounds for removal (permanently or temporarily as the Government determines) from the work site. Removal of employees does not relieve the Contractor from the responsibility for the work defined in this Contract.

a. Removal by Installation Commander. The Installation Commander may, at his discretion, bar an individual from the Installation under the authority of 18 USC 1382 (1972), and Paragraph B-15, Appendix B of AR 380-49 for conduct determined contrary to good order, discipline, or Installation security.

b. Removal by KO. The KO may require the Contractor to remove any employee, working under this Contract, for reason of misconduct or security. Contractor employees shall be subject to dismissal from the premises upon determination by the KO that such action is necessary in the interest of the Government.

c. Removal by Military Police. Contractor employees may be denied entry to or removed from the Installation by Military Police if it is determined that such entry may be contrary to good order, discipline, or the security of the Installation.

1.2.6.7.2 Appearance. Contractor employees shall be well groomed, neat in appearance, and dressed in accordance with Installation regulations.

1.2.6.7.3 Personnel Courtesy. Contractor employees shall not exhibit rude behavior toward civilian, military, or other contractor personnel.

1.2.6.7.4 Alcohol or Drug Use. The use of alcoholic beverages or illegal drugs by Contractor personnel is forbidden during performance of this Contract or while physically on the Government Installation. The Contractor shall not allow any employee who has possession of or who is under the influence of alcohol or controlled substances to perform work.

1.2.6.7.5 Loitering. Contractor employees shall not loiter in any working or patron area. Upon completion of their assigned duties, employees shall depart the facility. The Contractor shall allow only authorized personnel to be present in Contractor's work areas.

1.2.6.7.6 Fraud, Waste, and Abuse. In accordance with AR 11-2, Internal Management Controls, the Contractor shall be responsible for maintaining proper conduct and good discipline within Contractor occupied work areas. Contractor personnel shall be encouraged to be alert to and report suspected situations of fraud, waste, and abuse, or other intentionally dishonest conduct against the Government observed during or in the performance of this Contract.

1.2.6.8 Search and Seizure

All Contractor personnel and property may be subject to search and seizure upon entering, while on, and upon leaving Installation confines in accordance with paragraph 2-2, AR 190-22.

1.2.6.9 Personnel Training

1.2.6.9.1 Training and Recruitment Plan. The Contractor shall develop and implement plans that incorporate the Contractor's policy concerning the training of incumbent and replacement personnel, to include recruitment procedures. See paragraph 1.2.6.10.

1.2.6.10 Orientation Training. The Contractor shall conduct an orientation training class for all Contractor employees within five working days of initial employment or re- assignment to another functional area under the Contract. This training shall be relative to the respective functional areas and Contractor and Government offices with which the employees will interact. The Contractor shall maintain records of training accomplished. The training shall include, but is not limited to, the following topics:

- General Orientation. General orientation regarding Contract requirements and the role of the respective functional areas to the overall Contract
- Functional Area. The respective functional area's internal and external (customer) standing operating procedures and general work requirements
- Physical Configuration. Familiarization with the physical configuration of the Installation, to include at a minimum, the location and layout of functional area facilities, master publications library, storage areas, and shop areas
- Safety. The Contractor's safety plan, the Installation safety program (AR 385-10), functional area specific safety training, and Hazard Communication training pursuant to 29 CFR 1910.1200
- Fire Safety. Fire safety, including the Installation fire prevention program (DA Pam 420-2)
    - Environmental Requirements. Environmental considerations, to include functionally specific Hazardous Materials/Waste Handling training pursuant to 49 CFR (parts 171-177), 29 CFR, and 40 CFR.
    - Natural Resources Requirements. Natural resources considerations in accordance with the Endangered Species Act and the Integrated Natural Resources Management Plan.
    - Emergency Procedures. Emergency notification and response procedures
    - Conduct and standards, including standards of customer relations and dress
    - Applicable Installation regulations and directives
    - Security requirements and procedures

1.2.6.10 Formal Training during Contract Period. During the period of the Contract, some employees may require formal school training in order to update skills, licenses and certificates. Travel and per diem expenses shall be handled in compliance with Section H.

1.2.6.11 Training Requirements

The Contractor shall provide training on various Facilities engineering and Logistics functions as described herein or as requested by the KO. The Government will provide the Contractor any available training course outlines, presentation material, lecture material, or other training material. The Contractor shall provide instructors knowledgeable in the subject matter presented.

1.2.6.11.1 Facilities engineering Training. The Contractor shall provide Facilities engineering training as identified in functional area descriptions in Section C-5. Examples of training courses that will be required of the Contractor can be found in Technical Exhibit 1-005.

1.2.6.12 Specific Personnel Requirements

All Contractor personnel shall be fully qualified to perform the work to which they are assigned under this Contract.

1.2.6.12.1 Computer. The Contractor shall provide competent individuals to manage electronic management systems as specified in this Contract. The Contractor shall provide personnel certified at the level of systems administrator for those STAMISs for which the Contractor is required to perform systems administration, including facilities engineering specific systems, the Hazardous Substance Management System and any local unique systems provided to the Contractor. The Contractor shall provide at least one individual certified in Windows NT LAN administration (or having equivalent experience) to administer the Fort Benning DPW LANs.

1.2.6.12.2 Equipment Operators. All Contractor personnel operating mobile equipment and vehicles shall possess a valid state operator's license, to include Commercial Driver's Licenses (CDLs) with special permits to include air brakes and hazardous endorsements, , if required for the equipment and vehicles operated. Contractor motor vehicle drivers and equipment operators shall be appropriately qualified, certified, and licensed.

1.2.6.12.3 Boiler Maintenance. Operators and mechanics performing operations and foreman and supervisors of repair, maintenance, modification, and installation of heated pressure vessels. These individuals shall be certified Boiler Operators and Technicians capable of maintaining hot water boilers up to a capacity of 30,000,000 BTU, high pressure steam boilers up to a capacity of 56,000,000 BTU, and low pressure steam boilers up to a capacity of 2,000,000 BTU. Operators of boilers over 15 psi shall have a Fourth Class Power Engineer Certification by the National Institute for the Uniform Licensing of Power Engineers (NIULPE).

1.2.6.12.4 Backflow Prevention. All Contractor personnel performing any work involving backflow prevention shall be appropriately certified

1.2.6.12.5 Gas Leak Response. All Contractor personnel who will be responding to gas leaks shall be appropriately certified.

1.2.6.12.6 Fire Sprinkler System Maintenance. All Contractor personnel who will be inspecting and performing maintenance on fire sprinkler systems shall be appropriately certified.

1.2.6.12.7 Asbestos Removal and Disposal. All Contractor personnel who will be handling the removal and disposal of asbestos shall be appropriately certified.

1.2.6.12.8 Lead Paint Removal and Disposal. All Contractor personnel who will perform lead based paint activities shall be appropriately certified.

1.2.6.12.9 Cathodic Protection Systems Qualifications. All Contractor personnel who will be working with cathodic protection systems shall be appropriately certified.

1.2.6.12.10 Operation and Maintenance of Potable Water Systems. All Contractor personnel who will be operating the potable water treatment plant at Camp Merrill shall be appropriately certified by the Georgia Department of Professional and Occupational Regulation, State of Georgia. All personnel who maintain the potable water distribution systems shall have a maintenance certification or work under the direct supervision of a person who has a maintenance certification from the State of Georgia. All personnel performing bacterial laboratory analysis of the potable water shall have a laboratory certification from the State of Georgia

1.2.6.12.11 High Voltage Line Maintenance. All Contractor personnel who will be working on high voltage energized lines shall be at the journeyman level or be in an approved apprenticeship program working under the direct line of sight of a certified technician.

1.2.6.12.12 Locksmith Services. The Contractor shall employ bondable and security- cleared personnel to provide locksmith services.

1.2.6.12.13 Refrigeration, HVAC, and Boiler Work. Refrigeration; Heating, Ventilation, and Air Conditioning (HVAC); and Boiler personnel shall be certified in the handling of Chlorofluorocarbon (CFC) and halon gases in accordance with EPA approved Universal CFC handling certification to perform any work involving these gases.

1.2.6.12.14 Confined Space Entry. The Contractor shall be trained in Confined Space Entry in accordance with OSHA regulations when performing work in confined spaces. The Contractor shall maintain a log that shall record the following: confined space to be entered, location/building, purpose of entry, authorized

duration of permit (date and time), type of permit space hazards, preparation for entry- checks, equipment required for entry and work, list of authorized entrants, and testing record (recording the acceptable conditions to actual readings).

1.2.6.12.15 Wastewater Treatment. The Contractor shall be required to employ individuals who operate the wastewater treatment plant at Camp Merrill with a Class 3 certification in wastewater treatment with the Department of Professional and Occupational Regulations, State of Georgia.

1.2.6.12.16 Paragraph deleted

1.2.6.12.17 International Standards Organization (ISO) Container Inspector. The Contractor shall provide at least one certified inspector to inspect and certify containers leased for transportation use in support of deployments.

1.2.6.12.18 Communications Security (Classified) Equipment (COMSEC) Custodian. The Contractor shall provide a COMSEC custodian who is a graduate of the Army Standardized COMSEC Custodian Course.

1.2.6.12.19 Hazardous Material (HAZMAT) Transportation. Contractor personnel responsible for transporting HAZMAT shall be certified for the transportation of HAZMAT IAW 49 CFR, Part 106-178.

1.2.6.12.20 deleted

1.2.6.12.21Environmental POCs: Contractor personnel responsible for the handling of hazardous materials and hazardous waste shall designated appropriate environmental positions to ensure compliance with Ft. Benning, State and Federal requirements.
Guidance provided in Activity Specific Plan Section 5. To include, but not limited to:

Senior Environmental Compliance Officer (SECO)

Environmental Compliance Officer (ECO)

Environmental Coordinator (EC)

CAP/SAP hazardous Waste

Manager

Storm Water Activity Coordinator

1.2.6.13 Supervision of Work

The Project Manager or designated alternate shall be responsible for ensuring that required supervision of Contractor personnel is present during both regular and other than regular duty hours.

1.2.6.14 CONTRACTOR MANPOWER REPORTING APPLICATION (CMRA): The contractor shall report ALL contractor labor hours (including subcontractor labor hours) required for performance of services provided under this contract for the [NAMED COMPONENT] via a secure data collection site. The contractor is required to completely fill in all required data fields using the following web address: http://www.ecmra.mil/, and then click on "Department of the Army CMRA" or the icon of the DoD organization that is receiving or benefitting from the contracted services.

Reporting inputs will be for the labor executed during the period of performance during each Government fiscal year (FY), which runs October 1 through September 30. While inputs may be reported any time during the FY, all data shall be reported no later than October 31 of each calendar year, beginning with 2013. Contractors may direct questions to the help desk by clicking on "Send an email" which is located under the Help Resources ribbon on the right side of the login page of the applicable Service/ Component's CMR website.

1.2.7 CONTINGENCIES

The Government establishes contingency plans and contingency training missions to support mobilization and national emergencies, and to augment local governments in the event of natural disasters. The Government must be able to react to such events without undue delay. These sudden or unusual events could result in a great impact upon Contractor performance and Contract requirements. The Contractor shall establish, maintain, and implement, as necessary, plans and procedures to meet the Government's requirements for contingencies as specified below and in Functional Area 5.16, Military Operations, Logistical Planning, Support, and Execution. In the event that unusual conditions develop, the Contractor shall continue, and expand if necessary, Contract performance. Installation mobilization and disaster contingency plans are maintained and will be made available to the Contractor upon Contract award. These are located in the Office of the Chief, Plans and Operations.

1.2.7.1 Emergency/Disaster/Severe Weather Plan

The Government will provide an Emergency/Disaster/Severe Weather plan to the Contractor, and the Contractor shall adhere to and maintain the Government plans. The Contractor shall recommend revisions as necessary to these plans and shall submit a copy of the revisions to the KO for approval not later than 30 calendar days prior to the proposed effective date of the revised plan (CD102R005).

1.2.7.2 Strike Contingency Plan

The Contractor shall plan for labor strikes which impact upon either the Government's or Contractor's ability to perform. The Contractor shall submit a Strike Contingency Plan (CD 102R006) within 150 calendar days after Contract award date in accordance with FAR Clause 52.222.1, Notice of Labor Disputes The Contractor shall update this plan as changes occur and shall submit a copy of the proposed plan to the KO for approval not later than 30 calendar days prior to the proposed effective date of the updated plan. This plan shall contain the following information:

1.2.7.2.1 Provision of Essential Services. The plan shall include a description of the procedures which will be used to provide essential services, such as emergency service order response, and to operate essential functions such as heating (in season), refrigeration, and essential cooling, and to maintain water distribution systems, wastewater collection systems, water plant maintenance, power distribution systems, and electrical substations. Identification, by position, of the personnel who shall be responsible for performing the emergency services shall be included. These essential services shall be provided continuously without interruption pending any recruitment actions.

1.2.7.2.2 Personnel Recruitment. The plan shall include a description of how and where qualified personnel will be acquired, a description of the recruiting procedures to be used, and the time frames that may be needed to secure such personnel in the event of a work stoppage.

1.2.7.2.3 Collective Bargaining Experience. The plan shall include a description of the Contractor's experiences in the collective bargaining arena.

1.2.8 ORGANIZATIONAL CHART

The Contractor shall submit an Organizational Chart (CD 102R007) to the KO within 150 calendar days following Contract award date, showing the Contractor's organizational responsibilities, key personnel as defined in paragraph 1.2.6.1, and home and work phone numbers. The Contractor shall maintain and provide a current Organizational Chart to the KO upon request.

1.2.9 ADMINISTRATIVE REQUIREMENTS

The Contractor shall prepare all correspondence and maintain all functional files, blank forms, and the technical and administrative publication libraries required to accomplish the functions and tasks included in this Contract. The Contractor shall annotate the date received on all incoming documents and correspondence received. The Contractor shall conduct reports control and records administration programs in accordance with AR 335-15.

1.2.9.1 Release of Information

The Contractor shall not release any news (including photographs and films, public announcements, or denial or confirmation of same) or Installation related information of any subject matter within this Contract or any phase of any program herein to the media or any other unauthorized users without the prior written approval of the KO.

1.2.9.2 Files

Government-furnished classified and unclassified files in existence at the Contract start date, and those generated under this Contract, shall be maintained and retired in accordance with AR 25-400-2, TRADOC Supplements, and any Fort Benning Supplements thereto and other applicable Army regulations and directives listed in

Section C-6. The Contractor shall provide security of classified documents in accordance with AR 380-5 and DOD 5200.1-R, National Industrial Security Program Operating Manual. All such records and files shall be made available for review by any agency or individual authorized access by the KO. All functional files maintained by the Contractor under the provisions of this Contract are the property of the Government and shall be returned to the Government upon completion or termination of this Contract. The Contractor shall provide a List of Selective File Numbers, in compliance with Paragraph 5-13 of AR 25-400-2.

1.2.9.3 Regulations, Manuals, and Technical Documents

The Contractor shall become knowledgeable with and comply with all Government regulations as posted, or as required. Regulations, manuals, and technical documents applicable to this Contract are listed in Section C-6. This list is not meant to be all inclusive. All documents that are not unique to the installation are available at various WEB sites for your review and use. Some suggested sites, but not the only sites, are set forth in Section C-6.

1.2.9.3.1 Commercial Publications. The Contractor shall procure the commercial publications, including manufacturer's manuals, necessary to perform under the terms of this Contract when approved by the KO. The Government makes no representation that the Technical Library contains all applicable and up-to-date manuals and publications. Only those documents unique to the installation will be available in the Technical Library. Manuals or publications purchased under this clause are property of the Government and shall be turned over to the Government at Contract completion or termination.

1.2.9.3.2 Installation Regulations and Policies. The Contractor shall comply with all Installation regulations and policies specified herein and in Section C-6, as otherwise applicable.

1.2.9.3.3 Updates. The Contractor shall ensure timely distribution of updates and other publication requirements. Publications shall be kept current and maintained in accordance with DA Pam 25-40 and applicable Army regulations and other directives listed in Section C-6.

1.2.9.3.4 Use of Terms. Publications may contain references to divisions, such as "Supply Division," "Maintenance Division," "Transportation Division," or "Services Division," or to branches within those divisions. Such terms shall be interpreted to mean "the Contractor" as used herein, unless the cited function is specifically excluded herein.

1.2.9.4 Technical Reference Library

The Government will initially establish a Technical Reference Library for installation unique documents only to facilitate the bidding process of this Contract. All documents other than installation unique items are available at various sites on the WWW for review. Some suggested sites, but not the only sites, are set forth in Section C-6. For help in obtaining installation unique documents, offerors must contact the DPW project manager is responsible for unique documents. Request will be addressed only during normal operating hours of 0800- 1630 EST. Offerors shall expect a minimum of 3 workdays for the request to be answered.

Only technical document requests will be addressed. All other questions shall be directed by the offerors to the contracting officer. Following Contract start date, the Contractor shall maintain the Technical Reference Libraries, including all publications, data, exhibits, and other information provided therein. The Technical Reference Libraries shall be updated to ensure that all information therein is correct. The Technical Reference Libraries shall be considered Government property.

1.3 PERFORMANCE

1.3.1 QUALITY CONTROL

The Contractor shall be responsible for the quality of products and services provided under the terms of this Contract, to include those provided by subcontractors. To ensure that the requirements of this Contract are met, the Contractor shall implement an effective, economical Quality Control Program. Contractor shall comply with applicable Army Regulations and Supplements relative to the standards of operations necessary to maintain systems integrity and quality of service.

1.3.1.1 Quality Control Plan

1.3.1.1.1 The contractor shall develop a Quality Control Plan (CD 103R003) that implements a Quality Control Program on performance start date that assures the requirements of this contract are met. The contractor shall submit any changes in the Quality Control Plan to the Contracting Officer for approval 30 days prior to implementation.

1.3.1.1.2 An inspection system covering all services stated in this statement of work, appendices, and technical exhibits. It must specify the areas to be inspected both on a scheduled and unscheduled basis, how often inspections will be accomplished, the title of the individual(s) who will perform the inspection, and approximate time to complete the inspection. The Contractor shall submit the monthly inspection schedule to the COR by the last workday of the preceding month.

1.3.1.1.3 The methods for identifying and preventing deficiencies in the quality of service performed before the level of performance becomes unsatisfactory.

1.3.1.1.4 A file of all inspections conducted by Contractor and the corrective action taken. This documentation shall be made available to the Government at all times during the term of the contract.

1.3.1.1.5 Include a customer complaint feedback system for correction of complaints. All Customer Complaint Records shall be answered, all necessary corrective actions taken, and the completed forms returned to the COR within one workday.

1.3.1.2 Performance Evaluation Meetings

The Project Manager shall meet periodically with the KO and the Government COR to review Contract performance. Meetings shall include review and analyses of key process indicators, analyses of process deficiencies, and problem resolution. At these meetings, the KO and the Contractor will discuss Contractor's performance as viewed by the Government and problems, if any, being experienced.

1.3.1.2.1 Meeting Frequency. These meetings will be held weekly during the first 60 calendar days of the Contract period, and as needed, but not less than once a month, thereafter until it is determined by the KO to hold the meetings Quarterly. A meeting shall be held upon notification by the KO when a Contract Discrepancy Report (CDR) is issued.

1.3.1.2.2 Meeting Minutes. An electronic recording of these meetings may be kept by the KO. The Contractor shall prepare and maintain written minutes of these that will be signed by the KO, the COR, the Project Manager, or their designated representatives, as appropriate.

1.3.1.2.3 Management Review Meeting. On a quarterly basis, the Contractor shall participate in a Management Review Meeting to discuss major issues concerning the performance of the Contract. The Contractor shall provide an agenda for the Management Review Meeting five working days prior to the meeting.

1.3.1.3 Government Quality Assurance

The Government will inspect for compliance with Contract terms throughout the Contract period. Evaluation will be based on the Contractor's compliance with the requirements set forth in the Performance Requirement Summary (PRS), Technical Exhibit 1-001. The Government will monitor the Contractor's performance under this Contract by performing checks as contained in Quality Assurance Surveillance Plans (QASP) and as outlined in Clause 52.246-5 of the Federal Acquisition Regulation, "Inspection of Services - Cost Reimbursement" and the Award Fee Plan. Typical procedures include random sampling, planned sampling, scheduled inspections, observations, and validated customer comments.

1.3.1.3.1 Inspection Reports. Surveillance inspections will be recorded by the Government. The Government will provide copies of surveillance inspections to the Contractor on a monthly basis. Those inspection reports indicating deficient performance shall be signed by the Contractor indicating receipt of a copy of the report. When the Contractor's performance is unsatisfactory, a Contract Discrepancy Report (CDR) will be issued. The Contractor shall reply in writing within two working days after the date of the CDR, giving the reasons for the unsatisfactory performance, corrective action taken, and procedures to preclude recurrence (CD 103R001).

1.3.2 PERFORMANCE METRIC DEVELOPMENT, ANALYSIS, AND REPORTING

The Contractor shall maintain a set of metrics (METL Status) currently performed throughout the organization. In addition, the Contractor shall develop supplemental metrics or performance measures to evaluate additional factors not covered in the current performance standards that can be applied to data collected regarding the performance of work completed on the Contract. The additional set of metrics shall be submitted to the KO for review within 30 calendar days prior to Contract start data and then updated as necessary with concurrence of the KO (CKRL 103R002). All metrics shall be able to quantify the meeting of customer expectations and display added values to the Government. The Contractor shall analyze and report the metrics against baselines and benchmarks periodically. The metrics shall have the ability to assist the Contractor and Government to develop programs for continuous improvement and lift cycle cost reductions.

1.3.2.1 Metric Development

The Contractor shall review the Contract operation and develop a set of metrics to analyze the performance and work accomplishments. The metrics shall be trendable, capable of being visually displayed, and understandable by personnel outside of the respective functional area being measured. The metrics shall also encourage continual positive communication between the Government and the Contractor.

1.3.2.1.1 Included Attributes. Metric attributes shall include, but not be limited to, customer orientation, goal or objective linkage, process/action orientation, derivation from data that is readily collectable, trendable, and repeatable, distinguishes desirable from undesirable results, and is simple.

1.3.2.1.2 Areas Covered. Metrics shall be developed for performance of this Contract in addition to overall Contract operation and financial execution. The metrics shall include evaluation of work performance, quality, responsiveness, and long term effectiveness of the work performed.

1.3.2.2 Metric Analysis

The Contractor shall analyze the metrics on a schedule as proposed by the Contractor, to include, at a minimum, quarterly performance reviews and annual comprehensive program reviews.

1.3.2.2.1 Quarterly Analysis. The quarterly metric analysis shall include trends evaluated on resource expenditure and mission accomplishments. The Contractor shall identify deviations, reasons for the deviation, and make recommendations for adjustment if necessary.

1.3.2.2.2 Annual Analysis. The Contractor's annual analysis shall include revalidation of specific metrics or development of modified metrics or baselines, in addition to the required quarterly analysis.

1.3.2.3 Metric Reporting

The Contractor shall develop a program of performance reviews to report the status of the Contract using the metrics developed. The Contractor shall use metrics in the quarterly evaluation of the Contract in each of the Functional Areas and for overall Contract management.

1.4 COMPLIANCE

The Contractor and Contractor employees shall abide by Federal, State, and local laws, as well as DoD, Army, and Installation regulations as prescribed in, but not limited to, applicable directives listed in Section C-6 while engaged in the performance of all operations associated with this Contract.

1.4.1 PERMITS

The Contractor shall, without additional effort by the Government, obtain all certifications, licenses, and permits required for performance of work and for complying with all applicable Federal, State, and local laws and regulations and keep them on hand for review by Government personnel upon request. Evidence of such documents shall be maintained on file.

1.4.2 INSPECTION BY GOVERNMENT AGENCIES

The Contractor shall provide access to Government-owned, Contractor-operated (GOCO) facilities, Contractor-owned, Contractor-operated (COCO), and Government-furnished and Government-owned real or personal property ,EPMB inspectors, State regulators and cooperate with visiting Government personnel conducting official inspection visits and surveys at the Installation. The Contractor shall provide access to areas of functional responsibility under this Contract when approved by the KO.

1.4.2.1 KO Notification of Inspection

1.4.2.1.1 Planned Visits. The Contractor shall notify the KO of planned visits, investigation, or corrective actions required by Federal, State, local or any non- Installation Safety agencies for review and approval.

1.4.2.1.2 Unplanned Visits. The Contractor shall notify the KO by phone within ten minutes of unannounced arrival of any agents of any regulatory agency at Government facilities operated by the Contractor. The KO will issue instructions as to how to proceed in cooperating with inspector.

1.4.2.2 Inspection Documentation

The Contractor shall maintain on file all documentation relating to inspections and visits, to include the name(s), identification number(s), agency(s) of the inspector(s), reason for visit, and any remarks made during the visit. The Contractor shall include a copy of all reports received and, if samples are collected, similar samples. Samples shall be accompanied by a statement signed by the Contractor validating their authenticity.

1.4.3 SAFETY

All work shall be conducted in a safe manner and in compliance with OSHA, EPA, State, Installation safety office, and Corps of Engineer Safety Manual (EM 385-1-1) requirements. If the Contractor fails or refuses to promptly comply with safety requirements, the KO will issue an order stopping all or part of the work until satisfactory corrective action has been taken.

1.4.3.1 Safety Plan and Program

The Contractor shall develop a Safety Plan (CD 104R001) that delineates the processes and procedures the Contractor shall use to prevent accidents and preserve the life and health of Contractor and Government personnel and the public and protect work and property. The Contractor shall implement a Safety Program based on his Safety Plan on Contract start date. The Safety Program shall be implemented for the purpose of preventing accidents and preserving the life and health of Contractor personnel and Government personnel involved in performance of the Contract or receiving services provided under the Contract. The Contractor's safety program shall fully comply with the provisions of AR 385-10 and all other applicable Army, Installation, and Occupational Safety and Health (OSHA) regulations and directives. In cases where standards conflict, the stricter requirement shall apply. The Contractor shall designate qualified personnel to be responsible for administration of safety and OSHA programs. The Contractor shall submit the Plan to the KO no later than 30 calendar days prior to Contract start date. The Contractor shall update the plan as changes occur and shall submit a copy of the proposed plan to the KO for approval not later than 30 calendar days prior to the proposed effective date of the updated plan. The Contractor's Safety Plan shall include provisions for:

1.4.3.1.1 Safety Procedures. The Contractor shall develop procedures and practices that minimize accident risk. Any Contractor internal safety directives or standard operating procedures shall be submitted to the KO for review and approval.

1.4.3.1.2 Health Screenings. The Contractor shall provide necessary personnel health screening.

1.4.3.1.3 Safety Equipment Use. The Contractor shall furnish Contractor employees safety equipment, personal protective equipment, and safety devices. The Contractor shall ensure the use by employees of safety equipment and personal protective equipment and devices necessary to protect the individual. All safety equipment used shall be in accordance with OSHA standards, for example: safety boots, hard hats, safety glasses and goggles, ear plugs, rubber gloves for high voltage work, and insulated ladders.

1.4.3.1.4 Shop Safety. The Contractor shall have Contractor supervisors, particularly first line, ensure that workers observe Installation and shop safety precautions.

1.4.3.1.5 Flammables. The Contractor shall comply with AR 420-90 and with the National Fire Protection Association (NFPA) Codes and Standards Volume 3 for the storage and use of flammable mixtures which might constitute a fire hazard. The Contractor shall also address any relevant environmental concerns.

1.4.3.1.6 Smoking. The Contractor shall comply with AR 600-63 and any applicable Fort Benning policies for controlling smoking in Government buildings and facilities. Smoking is currently not permitted in any Government Building.

1.4.3.1.7 Incompatible Chemicals Storage. Incompatible chemicals shall be stored separately in accordance with OSHA and NFPA requirements as well as manufacturer recommendations. Incompatible chemicals shall be stored such that any vapors, liquids, or granular solids that are spilled or otherwise released from their containers do not come in contact with, nor react with, other incompatible chemicals. Information on chemical compatibility can be obtained from product material safety data sheets (MSDS), product labels, and other similar resources.

1.4.3.2 Inspections

1.4.3.2.1 Safety Inspections. The Contractor shall perform periodic inspections of safety equipment as required by Federal and State OSHA standards and other regulations. In addition, the Contractor shall inspect and survey Contractor work areas for potential safety hazards. The Contractor shall be subject to safety inspections by the Government.

1.4.3.2.2 Occupational Hygiene Inspections. The KO may conduct occupational and industrial hygienic surveys, evaluations, and inventories. The KO will notify the Contractor of any recommendations or evaluations which reveal actual or potential health hazards that require protective measures to be implemented. Contractor employees shall be instructed by the Contractor to notify the Contractor of any potential health hazards.

1.4.3.3 Unsafe Conditions

1.4.3.3.1 Violation Warnings. The Contractor shall post all notices of violations for hazards classified as Risk Assessment Codes 1 and 2 in the workplace or area where the hazards exist. The Contractor shall remove these notices when the hazardous condition no longer exists.

1.4.3.3.2 Hazardous Conditions

a. Previously Identified Hazardous Conditions. The Government shall provide a list of facilities that have been inspected for compliance with OSHA and for which any hazards have been identified. The Contractor shall correct these hazards at Government expense, or any hazards later discovered, in compliance with Government-recommended abatement measures, taking into account safety and health priorities. A higher priority for correction will not be assigned to the facilities provided under this Contract because of the award of this Contract. The identification of any hazardous condition(s) does not warrant or guarantee that no other possible hazards exist or that the procedures currently employed will be adequate to meet the responsibilities of the Contractor. Compliance with OSHA and other applicable laws and regulations for the protection of employees shall be the sole obligation of the Contractor, except as otherwise specifically stated in this Contract.

b. Contractor Identified Hazardous Conditions. If, at any time during the term of this Contract, hazardous conditions not previously identified are discovered by the Contractor or made known to it by any Federal or State agency responsible for enforcing OSHA, or if any workaround is determined to be inadequate by any Federal or State agency, the Contractor shall verbally report such conditions or determinations to the KO within 30 minutes after discovery and follow up with a written report within one working day. The KO may take any of the following actions, as he deems appropriate:

- recommend that the Contractor request variance from the pertinent standard if alternate safety measures will provide adequate protection;
- negotiate replacement of hazardous property with Contractor-owned property;
- replace hazardous property with suitable Government property; or
- authorize modification at Government expense.

1.4.3.4 Safety Training

The Contractor shall provide and document initial and annual instructions to enable employees to conduct their work in a safe manner and to recognize and report hazardous conditions. Initial indoctrination shall include instructions in safe practices; proper use, care, and maintenance of tools and equipment; accident reports and individual responsibility for accident prevention; and known hazards in work areas.
The Contractor shall complete initial indoctrination within 30 calendar days after Contract start date and shall maintain a file of Contractor employees, by name, who have attended the indoctrination. The Contractor shall conduct follow on training yearly, as a minimum, with notification to the KO within five working days after completion. The Contractor shall also develop procedures to ensure new hire employees are indoctrinated prior to work assignment. The Contractor shall conduct safety training for Contractor employees working in all applicable functional areas. The training shall include, but not be limited to, the following topics, as applicable:

- Petroleum, Oil, and Lubricants (POL) handling and storage
- Vehicle operations
- Packaging and shipping

- Hazardous materials (HAZMAT) storage, shipping, and handling
- Confined space entry
- Respiratory protection
- Hazard Communication (HAZCOM)
- Lock out/Tag out
- Personal Protective Equipment (PPE)
- Blood borne pathogens
- Radiation
- Trenching
- Lead
- Asbestos

1.4.3.5 Fire Safety

The Government will provide fire protection services on the Installation. However, the Contractor shall safeguard and maintain all Government property and provide for the safety of Government and Contractor personnel within the areas and facilities assigned to Contractor operations. The Contractor shall comply with NFPA Codes and Standards Volume 3. The Contractor may request advice in establishing a fire prevention program upon written request through the KO to the Installation Fire Prevention and Protection Division, Directorate of Public Safety. All Contractor personnel involved in the receipt, handling, storage, issue, and security of ammunition shall be thoroughly familiar with Contractor fire prevention and control procedures. The Contractor shall be subject to fire inspections by the Government.

1.4.3.5.1 Fire Prevention and Protection Plan. The Government will provide the Contractor with the Installation Fire Prevention and Protection Plan. In accordance with the Installation plan, the Contractor shall develop and implement a plan for each Government-furnished, Contractor-operated (GOCO) facility.

1.4.3.5.2 List of Responsible Personnel. The Contractor shall annotate on the Organizational Chart, personnel to be responsible for administering comprehensive Fire and Safety Plans in accordance with AR 385-10, The Army Safety Program, and AR 420-90, Fire Prevention and Protection. Notification of any changes in responsible personnel shall be submitted to the KO no later than five working days prior to the change.

1.4.3.6 Reporting Requirements

The Contractor shall prepare and submit or maintain the following reports and records.

1.4.3.6.1 Damage Reports. In all instances where Government property or equipment is damaged by the Contractor, the Contractor shall verbally notify the KO within two hours of occurrence. Upon request of the Government, the Contractor shall submit a full written report of the facts and extent of such damage within two working days of request (CD 104R002). The Contractor shall perform all work and operations to avoid damaging existing buildings, structures, pavements, equipment, and vegetation on Fort Benning. Repair or replacement of damaged property shall be performed as directed by the KO, and the corrective work shall be subject to inspection and acceptance by authorized Government personnel.

1.4.3.6.2 Emergency Notification Procedures. Contractor personnel witnessing a fire, an accident (to include train and aircraft incidents), a criminal act, a hazardous material release, a bomb threat, or other threatening act or condition, shall notify the proper authority on the Installation by dialing 911, or personally reporting the act or occurrence to the Installation Fire Department or Military Police.

1.4.3.6.3 Accident Reporting. The Contractor shall comply with AR 385-40, OSHA, and other regulatory agency requirements for record keeping and reporting of all accidents in the course of Contractor work which result in death, trauma, occupational disease, equipment damage, or environmental damage. The Contractor shall provide a verbal report to the KO within two hours of occurrence and a written follow-up report within two working days whenever an accident involving personal injury occurs (CD 104R003).

a. Serious Accidents. The Contractor shall call 911 immediately in cases of serious accidents or injury. The Contractor shall further report the accident, within 30 minutes, to the Installation Safety Office during regular duty hours or to the KO and Field Officer of the Day (FOD) after duty hours. Serious accidents and incidents include but are not limited to those defined in AR 385-40. Examples are as follows:

- One or more lives lost.
- One or more persons critically injured.
- Any persons hospitalized due to one incident.
- Fire causing major damage to structures, equipment or vehicles.
- HAZMAT and hazardous waste incidents.

1.4.3.6.4 Working Conditions Report. The Contractor shall make hazard reporting procedures available to Contractor employees in accordance with 29 CFR 1910 or 1960, as applicable.

1.4.3.6.5 Third Party Accident Claims. If any claims are made by a third party against the Contractor as a result of an accident which occurs in connection with the Contractor's performance, the Contractor shall submit, within 24 hours after the initiation of the claim, a full written report to the KO (CD 104R004).

1.4.3.6.6 APC Report. The Contractor shall submit a APC Report with every invoice submitted to the government. This report, (CD104R013), shall include all work
being performed by the contractor, Demand Maintenance Orders, PWOs, and OWO's. This report shall be submitted on TE 1-012 APC Report Form.

1.4.4 ENERGY AND UTILITIES

The Contractor shall conserve energy and utilities as stated herein in performing Contract operations. Upon request, the Contractor shall inspect facilities to determine compliance with applicable Federal, State, local, and Installation rules and regulations regarding energy conservation. Energy conservation data collection may include, but is not limited to, air flows, air temperature, chilled water temperature, high temperature water, and number of windows opened and closed.

1.4.4.1 Installation Energy Conservation Program

The Contractor shall coordinate energy requirements with the KO-designated Energy Manager. The Contractor shall comply with the Installation Energy Conservation Program, to include:

1.4.4.1.1 Energy Conservation Instruction. The Contractor shall instruct all Contractor personnel in energy conservation practices, and require them to operate utilities under conditions which preclude wasteful use of energy. Controls for heating, ventilation, and air conditioning (HVAC) systems shall not be adjusted by unauthorized workers. The Contractor will be notified, in writing, with regard to the start and stop dates of the summer cooling seasons and the winter heating season.

1.4.4.1.2 Temperature Control. During the summer cooling and winter heating seasons, the temperature control devices in all buildings shall be set to maintain a dry bulb temperature as designated by the KO.

1.4.4.1.3 Energy Level Adjustments. Upon KO request, the Contractor shall operate the water plant, sewer plants, lift stations, and boiler plants on emergency electrical generators when the Installation is reducing its commercial electrical consumption.
The Contractor shall be required to switch from electric to absorption chillers and adjust AC units in order to reduce the Installation's commercial electric consumption.

1.4.4.2 Contractor Energy and Utility Conservation Plan

The Contractor shall establish and implement an Energy and Utility Conservation Plan (CD 104R005) to support Government conservation programs. The Contractor's plan shall require the conservation of utility and energy usage as prescribed in AR 11- 27, 420-49, 420-54, TRADOC Supplements thereto, and TRADOC Regulation 420-11. At a minimum, the plan shall include the recovery and reuse of cleaning solvents, the closure of facilities or portions of facilities when not in use to conserve energy, the conservation of telephone usage, and the conservation of mobility fuels in accordance with AR 420-9, Energy Efficiency Program.

A comprehensive and detailed plan shall be submitted to the KO within 60 calendar days prior to Contract start date. The Contractor shall implement this plan in each Government-owned, Contractor-operated (GOCO) facility. The Contractor shall submit any revisions or changes to the KO for approval 30 calendar days prior to implementation. The Contractor shall instruct his employees in the principles and methods of conserving energy using the above listed publications as a guide.

1.4.4.2.1 Energy Officer. The Contractor shall annotate on the Organizational Chart, an Energy Officer. Notification of any changes in responsible personnel shall be submitted to the KO no later than five working days prior to the change.

1.4.5 ENVIRONMENTAL COMPLIANCE

The Contractor shall operate in full compliance with Federal, State, local, Army, and Installation environmental laws, regulations, and programs. A list of applicable state and federal environmental I permits, plans and policies can be found in the Technical Library. The Contractor shall be solely responsible for any penalties levied for noncompliance resulting from the action or inaction of the Contractor and his employees. Such fines and penalties shall not be cost reimbursable under this Contract in accordance with FAR 31.205-15. The Contractor and all Government facilities utilized by the Contractor shall be subject to environmental inspections by the KO, the Facilities engineering representatives, EPMB inspectors, State regulators, and other individuals approved by the KO. These inspections may be on an announced or unannounced basis. Applicable Records of Environmental Consideration (REC) and Environmental Baseline Studies (EBS) are available at EMD and Real Property. Technical Exhibit 1-010 contains a listing of record keeping required. Reporting requirements are set forth in the CD 104R006. Technical Exhibit 1-011 identifies those activities that have regulated activities under the Spill Prevention, Control and Countermeasure Plan (SPCC) and (Storm Water Pollution Prevention Plan) SWP3

These various procedures require reporting to various activities. A listing of these reports and activities are set forth in Technical Exhibits 1-010 and 1-011.

1.4.5.1 Interface with Environmental Regulatory Agencies

The EMD serves as the single point of contact (POC) with environmental regulatory agencies. All permit applications, mandatory notification requirements, mandatory reports, and proof of compliance actions required of the Installation will be submitted to regulatory agencies through EMD. Consequently, the Contractor shall submit copies of all information required of the Contractor by regulatory agencies to the EMD (CD 104R007).

1.4.5.2 Environmental Protection Plan

The Contractor shall establish and implement an Environmental Protection Program. Within 30 calendar days prior to Contract start date, the Contractor shall submit the KO a comprehensive Environmental Protection Plan that shall describe the policies and procedures of the Environmental Protection Program (CD 104R008). All environmental protection matters shall be coordinated with the KO and the Installation Environmental Coordinator.

1.4.5.2.1 Plan Content. The Contractor shall comply with AR 40-5, AR 200-1, AR 200- 2, AR 200-3, AR 200-5 and with all Federal, State, and local laws; wildlife, Jeopardy Biological Opinion (JBO), cultural, and historical regulations; and Installation plans regarding pollution control, clean air, clean water, natural resources, toxic substances control, resource conservation and recovery, other environmental matters, the Integrated Contingency Plan, Activity Specific Plans, the Installation Spill Contingency Plan(ISCP), Spill Prevention, Control and Countermeasure Plan (SPCC), Storm Water Pollution Prevention Plan (SWP3), Storage Tank Management, Risk Management Plan (RMP), , and the Installation Hazardous Waste Management Plan which can be found in the Technical Reference Library. The plan shall provide for minimization of the discharge of pollutants and all other influences on environmental quality, and address hazardous waste management and the handling, and disposition of industrial waste material. Note: Current Activity Specific Plan for DPWs facilities meets this requirement.

1.4.5.3 Citations

Citations and violation notices issued by regulatory authorities against GOCO facilities in noncompliance with environmental standards must be resolved between the KO and the EMD.

1.4.5.4 List of Environmental Training

Contractor employees shall be fully trained as required by local, State, and Federal regulations and permits in order to perform tasks as specified in this Contract. In addition, the Contractor shall have competent, trained, and experienced individuals to oversee the Contractor's environmental program. Individuals coordinating the environmental program shall have education, experience and training, at a minimum, equivalent to that required for an Environmental Specialist at the GS-11 Level as set forth in the Office Of Personnel Management Operating Manual of Qualification Standards for General Schedule Positions. Other typical environmental training, including initial and annual update, requirements for personnel include, but are not limited to:

Fort Benning Hazardous Waste Manager's Course for any individual handling hazardous waste

- Senior Environmental Compliance Officer (SECO),Environmental Compliance Officer (ECO) and/or Environmental Coordinator (EC) Training for Contractor's Environmental Compliance Officer
- Storm water Pollution Prevention Training Plan – Activity Coordinator
- Air Permit Training for operators of boilers
- Wastewater and Water Treatment Training for operators of these facilities
- Hazardous Communication and Handling of Hazardous Materials as required by OSHA, AR 200-1 and Ft. Benning Safety Office.
- Department of Transportation Hazardous Material Transportation Training for individuals transporting hazardous materials
- Spill Prevention, Control and Countermeasure Training – including training require for operators of SPCC facilities as per CFR 112.
- Emergency and First Responder Spill Response Training.

1.4.5.5 Scope of Environmental Compliance

The following information in not all inclusive and is included to give the Contractor a reasonable scope of services needed. The Contractor is expected to comply with the requirements of all of the following:

1.4.5.5.1 NEPA Fort Benning FB 144-R Process. The Contractor shall complete Records of Environmental Consideration (REC) for only their own activities (such as repair projects) as required by NEPA for all significant actions performed by the Contractor. Significant activities include, but are not limited to those actions involving construction and/or maintenance and/or repair and/or modification of buildings, roads and grounds; soil disturbance, modification to structures, disturbance of vegetation, tree management and landscaping; use of hazardous materials and hazardous waste, removal or remediation of toxic substances, and other physical modifications of structures and lands on Fort Benning. The government will review all REC's submitted by the contractor and issue concurrence/non-concurrence for all proposed actions. The Contractor shall comply with all comments provided by the environmental staff on reviewed RECs in completion of tasks required by this Contract. Generalized RECs
may be submitted for routine or ongoing maintenance tasks. Special areas of review in the REC process include:

a. Cultural Resources Coordination. Numerous archeological resources protected by Federal and State Laws are located on Fort Benning. The Contractor shall comply with the requirements of archeological laws, as detailed in reviewed RECs.

b. Historical Resources Coordination. Much of the Main Post of Fort Benning has been designated a Historic District by the Georgia State Historic Protection Office. As such, modifications and repairs to structures and components must adhere to SHPO requirements, as detailed in reviewed RECs Contractor & employees must have experience/training/skills to perform work on historic properties. The contractor shall be familiar with and comply with the Installation Cultural Resources Management Plan (ICRMP)

c. Threatened and Endangered Species Coordination. Fort Benning is home to numerous endangered and threatened species including the Red-Cockaded Woodpecker, the Gopher Tortoise, and the Relict Trillium. The Contractor detailed in reviewed RECs.

d. Other Environmental Programs. Comments on a REC may relate to other environmental areas including, but not limited to, solid waste, air, water, drinking water , hazardous materials, hazardous waste, toxic substances, lead-based paint, erosion control, storage tanks, Solid Waste Management Unit (SWMU),and storm water pollution prevention, etc.

1.4.5.5.2 Hazardous and Solid Waste, Used Oil/POL, and Recyclable Materials. The Contractor shall be responsible for characterizing, managing, transporting and disposal, reuse or recycling of all wastes and recyclable materials in accordance with State and Federal Regulations. General guidance for waste characterization, management and disposal is provided in the Fort Benning Hazardous Waste Management Plan. All hazardous waste as defined in 40 CFR 261 is subject to RCRA provisions. All such waste generated by the Contractor on the Installation from Government processes, equipment, or materials shall be stored in containers approved by Department of Transportation Reg. 49 CFR part 173, and AR 600-63 and disposed of in accordance with the Installation Hazardous Waste Management Plan available for review in the Technical Reference Library. All other wastes generated by the Contractor shall be disposed of by the Contractor in coordination with the Government. The Contractor shall implement a recycling program and coordinate with the Fort Benning Material Recovery Facility for recycling procedures.

a. The Contractor shall perform work in accordance with the following Salvage Rights to all materials belonging to Fort Benning:

   i. All materials not salvaged may not be sold. They shall be disposed in a landfill

   ii. Solid waste and demolition debris shall be properly disposed of in a permitted off-post landfill.

   iii. The Contractor shall provide all disposal records to the KO.

1.4.5.5.3 Hazardous Waste Disposal. The Contractor shall be responsible for ensuring all Hazardous wastes are managed and disposed of in accordance with the Fort Benning Part B Permit, State and Federal Regulations, and the Fort Benning Hazardous Waste Management Plan.

a. The Contractor shall be liable for all costs associated with Hazardous

Waste Management to include but not limited to spill response supplies, labels, DOT containers, waste sampling and analysis and disposal. Waste may be disposed of through the Installation's DRMO. Exceptions must be requested in writing and approved through the storage tanks, Solid Waste Management Unit (SWMU) Environmental Management Division (EMD).

b. DPW-EMD representative will sign disposal manifests as the co-generator for Fort Benning. Copies of all manifests will be forwarded to EMD for record keeping and tracking.

c. Hazardous Waste (HW) Accumulation Areas. Contractor activities generating hazardous wastes as defined in 40 CFR shall contact the EMD for assistance in setting up the accumulation area. The Contractor shall provide a point of contact (POC) to manage HW in accordance with Federal, State, Army, and Fort Benning Policies. The POC shall have training as set forth in the Fort Benning Hazardous Waste Management Plan. The POC shall provide DPW-EMD representative a listing of SAPs / CAPs inventory in December and June each year.

1.4.5.5.4 Clean Air Act.

a. CFC - training, certification, and reporting.

b. Title V - source notification, reporting requirements. All operations shall be performed in accordance with the permit, all records shall be maintained and reported to the EMD.

c. Section 112 (r) Risk Management Program – safe handling of all listed chemicals. The contractor shall implement and maintain a RMP approved by EPA and Ga. EPD.

1.4.5.5.5 Clean Water Act. The Contractor shall report and keep records for all petroleum spills over twenty gallons and any spills reaching public waterways to 911, immediately. In addition, the Contractor shall report all spills of hazardous materials/chemicals/substances/1 in accordance to the Ft. Benning ISCP and the Activity Specific Plan – Emergency Action Plan (ASP) 1 unless it meets the requirements of an incidental spill as defined by OSHA. All spill response planning shall be performed in accordance with the Fort Benning Integrated Contingency Plan and applicable Activity Specific Plans, to be provided to the contractor. All spill response training shall be performed in accordance with the Integrated Contingency Plan and cover the Installation Spill Contingency Plan (ISCP). The Contractor shall maintain a logbook of all spills reported. Spill record to be kept in Appendix A of the ASP. The logbook shall include, but not be limited to, the location of the spill, type of material spilled, and a description of the clean-up measures used. All reports and logs shall be submitted to the Environmental Management Division (EMD) on an annual basis

1.4.5.5.6 Spill Prevention, Control and Contingency (SPCC) Plan. The Contractor shall comply with the SPCC requirements as described on the Activity Specific plan for specific industrial areas and all areas where hazardous material are store in accordance with AR 200-1. SPCC Guidelines, inspections and updates shall be conducted and completed in accordance to the Integrated Contingency Plan and SPCC.

1.4.5.5.7 Storm Water Pollution Prevention. The Contractor shall comply with the Storm water Pollution Prevention Plan (SWPPP). The Contractor shall designate at a minimum, two storm water activity coordinators for each site included in the SWPPP. SWPPP Guidelines, inspections, updates and annual reports shall be conducted and completed in accordance to the National Pollutant Discharge Elimination System (NPDES), NPDES General Permit GAR000000 and the Ft. Benning Implementation SWP3 Guide. In addition, NPDES permit will be required under the Storm water General Permit for disturbance activities for large and small construction activities (Phase II).

1.4.5.5.8 Comply with the Georgia Waste Water Rules for Water Quality Control as applicable. Comply with the terms of the NPDES Permit for the 2 wastewater treatment plants at Fort Benning. Provide monthly wastewater discharge monitoring reports to the Georgia Environmental Protection Division with review, prior to submittal, by the Fort Benning DPW Environmental Programs Management Branch Clean Water Act project manager. Comply, as applicable, with requirements of Florida and Alabama for wastewater management and disposal. Comply with Code of Federal Regulations, Title 40, as applicable. Inspect wastewater treatment facilities and collection system components for proper operation. Inspect septic tank systems for proper operation and maintenance.

1.4.5.6 Safe Drinking Water Act

1.4.5.6.1 Comply with the Georgia Rules for Water Quality Control as pertains to drinking water and the Georgia Rules for Safe Drinking Water. Comply with the Code of Federal Regulations, Title 40 Part141. Comply with the terms of the Fort Benning Water Supply Permit, the Surface Water Withdrawal Permit, and the permits for the range water supply wells. Provide monthly drinking water reports (surface water and ground water) to the Georgia Environmental Protection Division with review, prior to submittal, by the Fort Benning DPW Environmental Programs Management Branch Safe Drinking water project manager. Inspect water supply activities for proper operation and maintenance.

1.4.5.8 Lead-Based Paint.

The Contractor shall comply with all applicable OSHA regulations.

1.4.5.8.1 Emergency Planning and Community Right-to-Know Act (EPCRA)/Superfund Amendment and Reauthorization Act (SARA) Title III. The Contractor shall maintain and provide the EMD and the Installation Safety Office the location of all hazardous material storage areas and materials stored. The Contractor shall inform the Central Hazardous Materials Control Center (CHMCC) of all hazardous materials purchased An annual report of hazardous materials/substances inventory, storage and use is required by the contractor at the beginning of each calendar year for the previous year. This data collection will be coordinate by the EMD Office; however, contractor is responsible to keep track of all records.

1.4.5.8.2 Erosion Control. The Contractor shall coordinate with the NRCS as required to minimize erosion of soils. The Contractor shall comply with the requirements of the Georgia Erosion Control and Sedimentation Act and Alabama Department of Environmental Management regulations. The Contractor shall obtain permits as required for disturbance of land larger than 1.1 acres. In addition, NPDES permit will be required under the Storm water Permit for disturbance activities for large and small construction activities (Phase II).

1.4.5.9 Hazardous or Toxic Materials

Hazardous materials are subject to OSHA Worker and Community Right- to-Know Laws. In addition, other hazardous materials may be encountered. Additional materials known to exist are listed in the Installation Integrated Contingency Plan and the applicable Activity Specific Plans available for review in the Technical Reference Library. The Contractor shall also comply with the Installation Hazard Communication Standard Program and Hazardous Communication Policy. These documents are available for review in the Installation Technical Reference Library.

1.4.5.9.1 Handling and Storage of Hazardous Material. The Contractor shall adhere to all applicable requirements of the Installation Hazardous Waste Management Plan, the Spill Prevention Control and Countermeasure Plan (SPCC), the Uniform Fire Safety Code, NFPA Life Safety Code 101, the Toxic Substances Control Act (TSCA), and of 29, 40, and 49 CFR and shall provide all materials, labor, training, transportation, and equipment for the handling and storage of hazardous materials. Employees handling hazardous or toxic materials shall be properly trained under 29 CFR 1910.1200 Hazard Communication (HAZCOM) and documentation of training shall be maintained on file. Personnel moving such materials in vehicles are required to have hazardous endorsements included in the CDL's.

1.4.5.9.2 Reporting. The Contractor shall provide an accurate consumption list and inventory of its hazardous materials to the EMD and the Installation Safety Office as required by the Community Right-to-Know Act.

1.4.5.9.3 Common Hazardous Material. The types of hazardous materials that may be found on the Installation and that the Contractor shall be responsible for removing include, but are not limited to:

a. Polychlorinated Biphenyls (PCBs). PCBs are acutely toxic and carcinogenic compounds. Even though the manufacture of additional PCB is prohibited by law, equipment with PCB insulating liquid is still in use and special handling procedures must be employed. Upon discovery of unknown PCBs and spills, the Contractor shall notify the KO.

b. Asbestos. Asbestos, used in the production of heat resistant materials, has been classified as a hazardous material. Insulation, wallboard, tile, vent pipe, mastic, doors, and other tangible items containing asbestos are still in use and may be encountered by the Contractor in the course of performing his duties. Upon discovery of asbestos, the Contractor shall contact the KO for further direction.

c. Lead. Lead-based paint has been classified as a hazardous material. Any work by the Contractor on surfaces where lead-based paint has been applied shall be performed in accordance with applicable OSHA regulations, EPA regulations, and DA policies.

d. Ozone Depleting Substances (ODSs). Venting of ODSs into the atmosphere is in violation of Public Law. ODSs shall be handled and stored in accordance with all applicable Federal, State, and local regulations.

e. Chlorine. Liquefied Chlorine gas is used for water treatment at the sewer plants, water plant, and swimming pools. The Contractor shall handle chlorine storage cylinders; operate and maintain chlorine dispensing equipment; provide handler and operator training; and shall adhere to all other Federal and State laws and regulations regarding this material.

1.4.5.9.4 Hazardous or Toxic Substance Exposure. As required by law, the Contractor shall notify the KO and 911 immediately upon discovery of air, land, or water being exposed to hazardous substances including, but not limited to: lead, asbestos, or other hazardous waste or substances. In the event that the exposure is the result of any actions by the Contractor, the Contractor shall alert the KO to this fact. The Contractor shall take all necessary steps to protect personnel and the environment immediately upon discovery of hazardous or toxic substances.

1.4.5.9.5 Hazardous and Toxic Substance Spills. The Contractor shall immediately report all spills to 911 and respond to all hazardous waste or toxic substance spills in accordance with the ISCP.. The Contractor shall provide equipment to any other unit or group involved in, or responding to, hazardous or toxic incident. The Contractor shall verbally report the spill to the KO within 30 minutes of the spill and provide a written report to the KO within five working days of the spill (CD 104R006). The Contractor shall take immediate action after a spill to protect the soils, surface, and groundwater at Fort Benning. The Contractor shall ensure proper handling of spilled materials, until disposition, in accordance with Federal and State laws.

1.4.5.9.6 Other Hazardous and Toxic Spill Requirements. The Contractor shall provide a Spill Standard Operating Procedure to the KO prior to the beginning of the Contract. Additionally, the Contractor shall take part in all hazardous spill drills conducted on the Installation.

1.4.5.9.7 Storage Tanks. The Contractor shall coordinate the establishment of all above and underground ground storage tank installations to the EMD and ensure that all tanks meet AR 200-1 requirements, including secondary containment. All record keeping shall be kept in accordance with the Clean Water Act, the Ft. Benning Storage Tank Guidance and reported to the EMD as required.

1.4.6 WARRANTY MAINTENANCE

1.4.6.1 Warranty Records

Prior to Contract start date, the Government will provide the Contractor with records of Government-owned property, vehicles, and equipment which are under warranty and used, managed, or supported under this Contract. Records will identify the item, the nature and expiration date of the warranty, and the name and location of the firm to contact about entitlement under the warranty. The Contract shall determine warranty requirements of subcontractors as well. The Contractor shall update and maintain all warranties on file and turn them over to successor contractors or the Government upon Contract termination.

1.4.6.2 Warranty Enforcement

The Contractor shall exercise existing manufacturers’ commercial warranties on all Government equipment, including warranties on existing equipment, equipment replacements, and new equipment installations under this Contract and by other contractors. The Contractor shall exercise the warranty to repair or replace any installed equipment and materials furnished by the Contractor that fail within the warranty period. The processing of all warranty claims shall be performed in accordance with DA Pam 750-8. The Contractor shall report to the KO difficulties encountered in the enforcement of warranties and instances in which the costs of enforcement would exceed the benefits derived. The Contractor shall monitor and track all Government-owned equipment falling under the maintenance responsibility of the Contractor.

1.4.6.2.1 Exceptions. The Contractor shall repair warranty items without recourse to the warranty if the failure or defect results from Contractor abuse, or from improper or inadequate Contractor maintenance, as determined by the Government, or if the Government directs such repair. Existence of a warranty does not alleviate the Contractor of his responsibility to perform work needed to prevent potential damage to personnel/property or to prevent unnecessary shutdown of facilities/functions.

1.4.6.2.2 Reporting. The Contractor shall inform the KO in writing of all warranty disputes (CD 104R009). The Contractor shall comply with standard Army execution and warranty claim action reporting programs as defined in AR 700-139 and DA PAM 750-8.

1.4.7 SECURITY

The Contractor shall comply with all Installation security requirements and the requirements specified herein. The Contractor shall establish and implement security plans to include processing security clearances, access to controlled areas, and control of classified material and equipment; security of vehicles, facilities, and equipment; and handling of ammunition, weapons, and sensitive and pilferable material. The following regulations should be used as a guide for developing these plans: AR 190-11, AR 190-13, AR 190-51, AR 380-5, AR 380-19, AR 380-40, AR 380-67, FM 19-30, DOD 5220-22M, and DOD 5220-22R. Security Plans shall include controls to ensure that only those contractor personnel with a valid security screening in accordance with AR 190-11, paragraph 2-11, subparagraph b, are given access to any controlled areas. The Contractor shall submit Security Plans within 30 calendar days of Contract start date for evaluation by the Government (CD 104R010). Contractor personnel performing work under this contract is not required to have a Security Clearance at time of the contract start date. The Contractor shall comply with Government personal identity verification procedures implemented in Homeland Security Presidential Directive (HSPD) – 12, Office of Management and Budget (OMB) Guidance M-05-24, and Federal Information Processing Standards Publication (FISP PUB) number 201. Contractor and all associated subcontractor employees shall comply with applicable installation, facility and area commander installation/facility access and local security policies and procedures provided by government representative. The contractor shall also provide all information required for background checks to meet installation access requirements to be accomplished by installation Provost Marshal Office, Director of Emergency Services or Security Office.
Contractor workforce shall comply with all personal identity verification requirements as directed by Department of Defense (DoD), Headquarters Department of the Army (HQDA) and/or local policy. In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes.

1.4.7.1 Facility Security Officer

The Contractor shall, in accordance with DOD 5220.22M, Industrial Security Manual for Safeguarding Classified Information, appoint a US citizen, who is required to be cleared as part of the Facility Security Clearance (FCL), to supervise and direct security measures for the application of Government-furnished guidance and specifications for classification, downgrading, upgrading, and safeguarding of classified information. The name of the person authorized to perform this task shall be submitted in writing to the KO as part of the Security Plans. Notification of any changes in responsible individuals shall be given to the KO within five working days of the change.

1.4.7.2 Personnel Security Clearances

Due to the nature and location of work to be performed under this Contract, and the requirement for access to restricted areas, classified equipment shipments, receipts, and certain classified data, specified Contractor personnel shall possess security clearances. Personnel security requirements are set forth in DD Form 254, Department of Defense Contract Security Classification Specification.

1.4.7.2.1 Positions Requiring Clearance. A personnel security clearance under the Defense Industrial Security Program (DISP) is neither a license for access to classified information nor a substitute for security measures designed to prevent unauthorized access. Therefore, security clearances under the DISP are only to be granted when there is a bona fide requirement for access to classified information in performance of duty assignment. The Contractor shall obtain security clearances for the following Contractor employees:

- Administrative personnel who process and control classified materials.
- Management personnel with a "need to know" in order to administer operations.
- Persons involved in the deployment or mobilization of forces.
- Persons involved in conducting evaluation of EDREs.
- Materiel Managers and other personnel requiring access to the Total Army Equipment Distribution Plan (TAEDP).
- Personnel having access to COMSEC, and other classified equipment to include personnel repairing equipment containing classified information.
- Personnel affecting control of automated system security.
- Personnel involved in providing transportation services for mobilization, deployment, or contingency movement requirements.
- C.E.P. Supervisor, Electronic Technicians (involved in EDREs, IS,SAV)

1.4.7.2.2 Employee In-Briefing. The Contractor shall perform an in-briefing of all new employees and annual refresher briefings in accordance with AR 380-5 which includes the completion of SF 312, Non-Disclosure Agreement Form, for those employees who will have access to classified material.

1.4.7.2.3 Employee Out-Briefing. The Contractor shall perform debriefing procedures for departing employees in accordance with DOD 5220.22M.

1.4.7.3 Physical Security Plan

The Contractor shall implement a Physical Security Plan in accordance with the requirements of DOD 5100, 76-M; ARs 190-11, 190-13, 190-40, 190-51, 380-20, 710-2; DA Pams 710-2-1 and 710-2-2; and the Installation Physical Security and Force Protection Plans and with TRADOC and FORSCOM Supplements thereto. The Contractor will be subject to unannounced physical security inspections by the Government.

1.4.7.3.1 Physical Security Officer. The Contractor shall annotate on the Organizational Chart a Physical Security Officer. Notification of any changes in responsible individuals shall be given to the Government no later than five working days prior to the change.

1.4.7.4 Information Security Plan

The Contractor shall develop an Information Security Plan in compliance with DOD 5220.22-M, Industrial Security Manual, incorporating DD Form 441-1 and AR 380-5. This plan shall provide for the control of classified information related to handling and accessing classified information and control of all computer security and communications security utilized within the scope of this Contract. The Contractor shall implement the plan in each GOCO facility. The Contractor shall comply with AR's 380-5, 380-19, 380-67, 380-40 and 530-1. The Contractor shall deliver an Information Security Plan to the KO for review 30 calendar days prior to Contract start date and implement the plan at Contract start date (CD 104R011) Security Officers. The Contractor shall provide an Information System Security Officer (ISSO) and an Assistant Information System Security Officer (AISSO) for each facility in which an automated information system (AIS) is located. A separate appointment letter shall be prepared for each AISSO and include the respective AIS serial number(s) and the facility number in which the AIS(s) are located. Notification of any changes in responsible personnel shall be submitted to the KO no later than five working days prior to the change. The content and structure of the appointment letter shall comply with AR 380-19.

1.4.7.4.1. Appointment. The Contractor shall appoint an ISSO for each standard and non-standard AIS and as many AISDMOs and alternates as required to provide adequate systems security to prevent unauthorized access, theft, and damage. The Contractor shall ensure that each ISSO, AISSO, and alternate comply with the responsibilities and duties identified in AR 380-19 and Installation SOPs.

1.4.7.5 Restricted Areas

The Contractor shall comply with restricted areas procedures and instructions. Contractor personnel working in restricted areas may be required to sign in and out and state the nature of their business at the entrance desk. A list of these facilities is available in the Directorate of Public Safety, Physical Security Office on an as required basis. See TE 1-009 for a general list of restricted areas. Work performed in restricted areas shall be coordinated with the respective restricted area Security Officer, and all work done on ranges will be coordinated with Range Control.

1.4.8 KEY CONTROL

The Contractor shall establish and maintain on file a written Key Control Plan to ensure that keys issued to the Contractor by the KO are not lost, misplaced, or used by unauthorized persons. Responsibilities and procedures shall be as specified in AR 190- 13, AR 190-51, and FM 19-30. The Contractor shall maintain records as required by AR 190-51 to ensure accountability of keys.

1.4.8.1 Master and File Keys

Master and file keys for buildings will only be issued to Contractor employees performing services under this Contract. Building numbers or room numbers shall not be placed on master and file keys. Keys shall be coded so as to identify buildings or room numbers. Keys to areas where fire alarms, panels, and extinguishing systems are located shall be provided to the Fire Department. Master and file keys shall not be checked out to anyone other than authorized personnel, i.e. to the Provost Marshal, the Facilities engineering Real Property Office, or the Fire Department, unless authorized by the KO.

1.4.8.2 Key Inventories

The Contractor shall provide two copies of the key inventories to the KO within five working days of the date of request.

1.4.8.3 Key Duplication

Government keys shall not be duplicated by the Contractor unless authorized by a valid work document.

1.4.8.4 Unauthorized Users

Use of Government keys issued to the Contractor by any person other than authorized Contractor employees is prohibited. The Contractor shall not permit entrance to locked areas to any person other than Contractor personnel engaged in performance of work in those areas without written authorization by the KO. The KO and his designated representative(s) will have access to any Government-owned property under the control of the Contractor.

1.4.8.5 Lost Keys

The Contractor shall verbally report any occurrence of lost master, grand master, and restricted keys to the KO within 30 minutes of discovery of the loss. The Contractor shall provide the KO a written report within one working day. The report shall contain the key number, location(s) accessed by the key, date the key was discovered missing, name of person signing for the key, and any other relevant details. If the Contractor is responsible for the loss of keys, the Contractor shall re-key affected buildings.

1.4.9 VEHICLES

Contractor personnel operating motor vehicles on Government installations shall have a valid State license to operate a motor vehicle and shall comply with AR's 190-5 and 385-55 regarding motor vehicle use and registration. All personal and Contractor-owned vehicles shall be properly registered, and insured in compliance with State and local laws.

1.4.9.1 Identification of Contractor-Owned Vehicles

Contractor-owned vehicles shall be identified with the firm's name, telephone number, and vehicle identification number on both sides of the vehicle with lettering or identifying signs as outlined in Chapters 8 and 9 of AR 58-1. Corporate markings are not permitted on Government-furnished equipment.

1.4.9.2 Vehicle Parking

Parking for Contractor personnel is available adjacent to most buildings on a first come, first served basis. Parking is subject to restrictions imposed by building occupants in buildings with joint Contractor and Government occupancy. The Government reserves the right to make changes in parking arrangements to accommodate new or changing requirements. Contractor employees shall park privately owned vehicles in designated parking areas and adhere to the Installation Vehicle Operating Codes.

1.4.9.2.1 POV Parking: Only privately owned vehicles with handicap stickers shall be allowed to park inside any fenced-in contractor shop area.

1.4.9.3 Vehicle Registration

The Contractor shall comply with all applicable requirements of Fort Benning USAIC Regulation No. 190-5, Motor Vehicle Traffic Regulation, in effect during performance of this Contract. All Contractor owned or operated vehicles as well as privately owned vehicles of Contractor personnel operated on the Installation shall be registered in accordance with USAIC Reg. 190-5. Vehicles must be registered within ten working days by making application to the Provost Marshal Vehicle Registration Section on DA Form 3626, Vehicle Registration/Driver Record. Evidence of vehicle ownership and liability insurance must be presented upon application.

1.4.9.4 Government Vehicles

Prior to operation of any Government-owned vehicle or heavy equipment (i.e., materials handling equipment) the operator shall have a current and valid State driver's license covering the type of vehicle or equipment being used, including CDL if required. The driver's license shall be obtained in accordance with applicable regulations and operating procedures as defined in Functional Area 5.12, Installation Transportation Services. Contractor shall comply with the requirements of FAR 51.201 and 51.202 for usage of Government-owned vehicles and heavy equipment.

1.4.9.5 Driver Safety Training

The Contractor shall attend classroom instruction in accident avoidance IAW AR 385- 55, Appendix B-4 to Contractor personnel who operate Government-furnished vehicles and equipment. The classroom instruction shall include refresher training on traffic rules, safe driving procedures, seasonal hazards, and related matters. The Contractor shall conduct the classroom instruction prior to operation of Government furnished vehicles and equipment by Contractor employees and shall attend refresher training every 4 years. The Contractor shall notify the KO, in writing, within five working days after completion of the training. All Contractor personnel who operate Government vehicles shall obtain a Defensive Driver Card (DDC).

1.4.10 WEAPON AND FIREARM CONTROL

In accordance with AR 190-11, Physical Security of Arms, Ammunition, and Explosives, Contractor personnel shall not transport, possess, or use privately-owned weapons on Fort Benning and its satellite locations.

1.4.11 SMOKING

Neither the Contractor nor Contractor personnel shall be allowed to smoke in government buildings or facilities IAW AR 1-8, Smoking in DA Occupied Buildings and Facilities. Smoking shall also be prohibited within a radius of 100 feet from any refueling/defueling operation.

1.4.12 ANTI-TERRORISM/OPERATIONS SECURITY REQUIREMENTS

1.4.12.1 AT Level I Training: All contractor employees, to include subcontractor employees, requiring access to Army installations, facilities and controlled access areas shall complete AT level I awareness training within 30 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. The contractor shall submit certificates of completion for each affected contractor employee and subcontractor employee, to the COR or the contracting officer, if a COR is not assigned within 30 calendar days after completion of training by all employees and subcontractor personnel. AT level I awareness training is available at the following website:

Https://atlevel1.dtic.mil/at. AT level I training is an annual requirement if option years are exercised.

1.4.12.2 Access and General Protection/Security Policy and Procedures: Contractor and all associated sub-contractors employees shall comply with applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by the government representative). The contractor shall also provide all information required for background checks to meet installation access requirements to be accomplished by Installation Provost Marshal Office, Director of Emergency Services or Security Office. Contractor workforce must comply with all personal identity verification requirements as directed by DOD, HQDA and/or local policy. In addition to the changes otherwise authorized by the changes clause of this contract, should the Force Protection Condition (FPCON) at any individual facility or installation change, the Government may require changes in contractor security matters or processes.

1.4.12.3 The contractor and all associated sub-contractors shall brief all employees on the local iWATCH program (training standards provided by the requiring activity ATO). This local developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training shall be completed within 30 calendar days of contract award and within 30 calendar days of new employees commencing performance with the results reported to the COR NLT 30 calendar days after contract.

1.4.12.4 Access to Government Information Systems: All contractor employees, and associated subcontractor employees, with access to a government information system must be registered in the ATCTS (Army Training Certification Tracking System) at commencement of services and must successfully complete the DoD Cyber Awareness Challenge training prior to access to the information system and then annually thereafter.

1.4.12.5 OPSEC Standing Operating Procedure/Plan: The contractor shall adhere to the Government's Operations Security (OPSEC) Standard Operating Procedure (SOP)/Plan during the entire period of performance.

1.4.12.6 Operation Security (OPSEC) training: In accordance with AR 530-1 all personnel, to include contractor employees and associated sub-contractor employees must complete Level I Operations Security (OPSEC) training which is composed of both initial and continual awareness training (annually). All personnel within the first 30 days of arrival in the organization must receive initial training to include a briefing on the organizations critical information (Essential Elements of Friendly Information) and read/sign the OPSEC Individual User Compliance Agreement form. The end state of initial and continual awareness training is that each individual should have the requisite knowledge to safeguard critical information. The following link is provided: https://www.iad.gov/ioss/index.cfm click on training, click OPSEC Fundamentals Course, then OPSE-1301 CBT.

1.4.12.7 Information assurance (IA)/information technology (IT) training: All contractor employees and associated sub-contractor employees must complete the DoD IA awareness training before issuance of network access and annually thereafter. All contractor employees working IA/IT functions must comply with DoD and Army training requirements in DoDD 8570.01, DoD 8570.01-M and AR 25-2 within six months of employment.

1.4.12.8 Information Assurance (IA)/Information Technology (IT) certification: Per DoD 8570.01-M, DFARS 252.239.7001, and AR 25-2, contractor employees and associated subcontractor employees supporting IA/IT functions shall be appropriately certified for each category and level upon contract award. The baseline and computing environment certifications are stipulated in DoD 8570.01-M must be completed upon contract award. Upon request by the Government, the Contractor shall provide documentation supporting the IA certification status of personnel performing IA functions. Contractor personnel who do not have proper and current certifications shall be denied access to DoD information systems for the purpose of performing IA functions.

1.4.12.9 Handling or Access to Classified Information: Contractor shall comply with FAR 52.204-2, Security Requirements. This clause involves access to information classified "Confidential," "Secret," or "Top Secret" and requires contractors to comply with— (1) The Security Agreement (DD Form 441), including the National Industrial Security Program Operating Manual (DoD 5220.22-M); any revisions to DOD 5220.22-M, notice of which has been furnished to the contractor.

Additionally, the contractor shall adhere and conform to all safety, security and attendance requirements that apply to the supported office. The contractor is required to attend and participate in all organizational meetings, training, and emergency preparedness events and is required to vacate the Government office work site when no sufficient Government oversight is available.

1.4.13 BACKGROUND CHECK REQUREMENTS

Contractor/Sub-Contractor employees providing services under contracts are subject to the following process:

1.4.13.1 Each employee (direct employee and sub-contractor employee) working under this contract who has contact with children under 18 years of age, shall be screened with the appropriate background checks as governed by the Army Directive 2014-23 "Conduct of Screening and Background Checks For Individuals Who Have Regular Contact With Children in Army Programs", dated 10 September 2014. A copy of the Army Directive can be found at http://www.apd.army.mil/pdffiles/ad2014_23.pdf.

The background check requirements for Contractors can be found in Enclosure 3 – Screening and background checks for appropriated and non-appropriated fund personnel, military personnel, foreign national employees overseas, and contractors.

1.4.13.2 Background checks will be initiated and paid for by the Government. The cost for the pre- employment screening is the responsibility of the Contractor. The Commander Designated Entity (CDE) at the Installation supports IMCOM by coordinating, submitting, tracking child related background checks, and reviewing and compiling the results to identify whether they contain derogatory information to be passed on for a suitability determination. The CDE will advise and assist CONTRACTOR and sub- CONTRACTOR employees completing all required documents required for the background check process. CDE contact information will be provided upon contract award.

1.4.13.3 The following is a list of the required screenings per the Army Directive 2014-23.

1.4.13.3.1 REGULAR/RECURRING contact with children

This is defined as contact with children more than an average of one (1) day a week for three (3) or more months OR daily for a period of one (1) month or longer. See required screening below:

i) Pre-employment screening: 1) Application, 2) Interview, 3) Reference check, 4) Statement of Previous Arrest or Charge, 5) Assessment of individual's eligibility, qualifications, temperament and suitability for work with children, 6) Tentative offer of employment (Proponent: Contractor is responsible organization);

ii) Preliminary investigation: 1) Installation Record Check (if applicable), 2) FBI Fingerprint check, and 3) Local Civilian Law Enforcement Check (Proponent: Government/Commander Designated Entity is responsible organization);

iii) CNACI – Child Care National Agency Check and Inquiries (Proponent: Government/Commander Designated Entity is responsible organization).

The CNACI will be initiated after successful completion of the preliminary investigation (1.3(a)(ii)). If deemed appropriate by the local Garrison Commander, a contractor can start working under Line Of Sight Supervision (LOSS) after successfully completing the preliminary investigation (1.3(a)(ii). If LOSS is not feasible, favorable CNACI (1.3(a)(iii) results are required BEFORE the contractor can start working under this contract.

1.4.13.3.2 SHORT TERM/OCCASIONAL contact with children

Contact with children that does not fall under the definition of Regular/Recurring contact. See required screening below:

i) Pre-employment screening: 1) Application, 2) Interview, 3) Reference check, 4) Statement of Previous Arrest or Charge, 5) Assessment of individual's eligibility, qualifications, temperament and suitability for work with children, 6) Tentative offer of employment (Proponent: Contractor is responsible organization);

ii) Preliminary investigation: 1) Installation Record Check (if applicable), 2) FBI Fingerprint check, and 3) Local Civilian Law Enforcement Check (Proponent: Government/Commander Designated Entity is responsible organization);

An employee must successfully complete the required screening (1.3(b)(i) and (ii) BEFORE working under this contract.

1.4.13.4 No person, regardless of circumstances, will be approved to provide child care services if the individual has been convicted of a sexual offense, a drug felony, a violent crime or a criminal offense involving a child. If the background check investigation reveals any other derogatory information a suitability determination has to be made by the Army. The adjudication of derogatory information is processed through an Army Program Review Board (PRB). Further details about the adjudication process can be found in Enclosure 6 of the AD 2014-23. At any time a Contractor can decide not to proceed with the PRB process.

1.4.13.5 Additional information on the background check process can be found at: http://www.opm.gov/investigations/background-investigations.

1.4.13.6 Re-verification (IRC and FBI fingerprint check) is required every 5 years.

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DEFINITIONS

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DEFINITIONS

2.1 GENERAL

As used throughout this Contract, the following terms shall have the meaning set forth below.

2.2 TERMINOLOGY

24-Hour Service: 24 hours is from time of receipt on one working day to the same time the following working day -- example: 0900, Friday to 0900, Monday.

72-Hour Service: 72 hours is from time of receipt on one working day to the same time three working days later -- example: 0900, Monday to 0900, Thursday.

Abandoned Property: Abandoned property is that lost, abandoned, or unclaimed personal property defined as any privately-owned personal property which has come into the custody or control of any Military Department and which is unclaimed by the owner (see DOD 4160.21-M).

Acceptable Quality Level (AQL): Maximum percent defective (or maximum number of defects per hundred units) that can be considered as a satisfactory performance average. Government will accept the great majority of lots provided percent defective (or defects per 100 units) in these lots be no greater than designated value of AQL. However, the Contractor shall not intentionally perform in a defective manner and shall re-perform any service found to be defective whenever possible. Decisions as to this possibility shall be made only by the KO.

Acceptance at Destination: The assumption of title to property by the Department of the Army (DA) at the specified delivery point. The term corresponds, generally, to the commercial term "f.o.b. destination".

Acceptance at Origin: The assumption of title to property by the DA at the point of shipment. The term corresponds, generally, to the commercial term "f.o.b. origin". Acceptance at origin does not imply that payment was made at the time title passed to the Army, nor does it necessarily mean that the Government, by assumption of title, forfeited the right to reject any article(s) not conforming with contract specifications.

Acceptance Sampling: A form of sampling used to determine a course of action. A procedure that gives a specified risk of accepting lots of given quality.

Account Processing Code (APC): A locally developed, four-position, alphanumeric code that abbreviates accounting classification and relates to Army Management Structure (AMS) and other codes used in computer processing.

Accountability: Accountability is the obligation to keep accurate and complete records of property, documents, or funds. Important data elements may include, but are not limited to, identification data, gains, losses, due-ins, due-outs, and balances on hand or in use.

Accountable Officer (AO): A person officially designated as such. An accountable officer is vested with accountability for property and maintains records in connection therewith, irrespective of whether the property is in his own possession for use or storage, or in the possession of others to whom it has been officially entrusted for temporary use or for care and safekeeping. Property accountability is not terminated until transfer to another accountable officer has been accomplished or until items have been dropped from accountability on valid credit vouchers in compliance with regulations. Specifically, accountability is not terminated by the disposition of property which merely places responsibility for its custody or safekeeping with another individual.

Activation: The designation of a new Army unit, organization, or activity and the provision of personnel, equipment, and supplies required to bring the unit to an authorized level of organization.

Activity Detail Cost Report: A cyclic Standard Financial System (STANFINS) listing displaying line item entries for obligations and expenses by APC and element.

Actual Cost of Damage (ACOD): The actual costs (parts and labor) incurred in the repair of an item.

Actual Deployment: The movement of troops off of a military Installation by surface or air.

Addition-Expansion-Extension: A change to a real property that adds to its overall external dimension.

Adequate Family Housing: Dwelling units which meet suitability standards set by Department of Defense (DOD) and for which military occupant forfeits his entitlement to basic allowance for housing (BAH) or are leased to civilians in accordance with AR 210-12. This includes Appropriated Fund, Capehart, and leased housing.

Advanced Wastewater Treatment: Those processes that achieve pollutant reductions beyond that achieved in conventional treatment (sedimentation, filtration, carbon absorption, and chemical precipitation).

Advisory Document: A directive which the contractor may use for information and guidance but is not binding for compliance.

Aerated Lagoon: A type of aerated pond, which is a natural or artificial wastewater treatment pond in which mechanical or diffused-air aeration is used to supplement the oxygen supply (see oxidation pond).

Aeration: The bringing about of intimate contact between air and water by one of several methods, such as spraying the liquid into the air or bubbling air through the liquid.

Agitator: Mechanical apparatus for mixing and aerating. A device for creating turbulence.

Air-Agitation: The process of agitating sand in a rapid sand filter during washing by injecting air under low pressure into bottom layers of the sand bed.

Air-Binding: The clogging of a filter, pipe, or pump due to the presence of entrapped air.

Air-Lift Pump: A pump, used largely for lifting water from wells, from which air under pressure is discharged into the water at the bottom of the well in fine bubbles. The bubbles mix with the water and reduce the apparent specific gravity of the air-water mixture, and the surrounding water causes the mixture to rise in the discharge pipe to the outlet.

Algaecide (Algicide): Any substance or mixture of substances intended for preventing, destroying, or mitigating any of numerous chlorophyll containing plants of the phylum Thallophyta.

Alien: Any person not a citizen or national of the United States of America.

Alkalinity: A term used to represent the content of carbonates, bicarbonates, hydroxides, and occasionally borate's, silicates, and phosphates in water. It is expressed in part per million (ppm) or calcium carbonate.

Alongside of aircraft: Delivery of fuel which requires the contractor to hand the delivery nozzle and bonding cable to the government personnel responsible for servicing the aircraft, thereby permitting the contractor to provide only the personnel required to operate the fuel delivery/defueling equipment.

Alteration: A change to interior or exterior facility arrangements to improve its current purpose. This includes installed equipment made a part of existing facility.

Anaerobic Lagoon: A natural or artificial wastewater treatment lagoon in which waste stabilization is brought about through the action of microorganisms in the absence of air or elemental oxygen.

Annual Recurring Requirements for Maintenance and Repair (ARR-MAR): Requirements for the maintenance and minor repair work required annually in order to maintain facilities at a level which minimizes the future need for major repairs and allows the facilities to be used for their original purpose.

Annual Work Plan (AWP): A prioritized work plan prepared prior to the start of each fiscal year, after receipt of the upcoming fiscal year's budget guidance, reflecting the use of resources to accomplish the mission.

Appointing Authority: An officer or civilian employee under the command of or on the staff of the approval authority. Individuals designated as appointing authority must be a lieutenant colonel or above, or a civilian employee in the grade of GS/GM-13 or above. (AR 735-5)

Appropriated Fund Property: Terms "Government Property," "Army Property," and "Property" includes all property under Federal Government control except property accounted for and owned by a non-appropriated fund activity.

Appropriated Funds: Moneys made available to Government agencies by an Act of Congress.

Approved Operating Budget (AOB): An approved financial plan for incurring expenses. It serves as a fund authorization document providing the authority to incur obligations and expenses for financed activities.

Approving Authority: An officer authorized to appoint a surveying officer and to approve reports of survey.

Appurtenance: Equipment attached to or connected to as part of a Real Property Facility (RPF) which allow a facility to function for its designated purpose. For purposes of defining appurtenances as it applies in defining Contractor work responsibilities under this specification, appurtenances shall be considered equipment of this type which is non-personal and is not readily movable. For example: an air compressor which is installed in a facility and is used to operate heating, ventilation, and air conditioning (HVAC) controls for facility would be considered an appurtenance to the RPF and would be Contractor responsibility. This is because the air compressor is necessary to properly operate facility HVAC system which is necessary for facility functional use. An office copier which is used in a facility by office personnel would not be considered an appurtenance to the RPF and would not be Contractor responsibility. This is because copier is not equipment which is necessary for facility functional use itself but rather it is movable equipment utilized by personnel working in facility.

Area Fire Marshal (AFM): Responsible person appointed in writing to manage Fire Prevention Program in accordance with AR 420-90, TRADOC Suppl 1, and any Fort Benning supplements thereto. Maintains a listing of all facilities under jurisdiction including Building Fire Wardens.

Area of Support: Units (TOE and TDA) assigned to Fort Benning and located within the confines of the reservation; Reserve and National Guard units at Fort Benning for training; and Reserve, National Guard, Reserve Officer Training Corps units as specified in AR 5-9, Intraservice Support Installation Area Coordination, and TRADOC and FORSCOM Supplement 1 to AR 5-9.

Army Audit Agency: An activity charged with auditing Army programs and activities.

Army Management Structure (AMS): A structure of accounts established by regulation to provide a single, uniform classification of the activities of the Army for use in programming, budgeting, accounting, and the reporting of cost, performance, and manpower data.

Army Oil Analysis Program (AOAP): A coordinated, Army-wide effort to detect impending equipment component failures through analytical evaluation of oil samples.

Army Regulations (AR): Publications issued by Department of the Army (DA) which are directive in nature and contain missions, responsibilities, policies, and administrative procedures necessary to insure uniform compliance with those policies.

Army Standard Information Management System (ASIMS): A computer network that accommodates standard management information systems that are processed Army-wide.

Army Stationing and Installation Plan (ASIP): A document projecting mission and troop strengths for definite periods on a long range basis.

As Directed, As Required, As Permitted, Approved, Acceptance: Where these words or words of similar import are used, it shall be understood that the direction, requirements, permission, approval, or acceptance of the Contracting Officer is intended unless stated otherwise.

As Shown, As Detailed: Where these words or words of similar import are used, it shall be understood that reference is made to the drawings, tables, or narrative comprising this specification, unless stated otherwise.

Attribute: The property a unit has of being either bad or good. That is, the quality characteristic of a unit is either within the specified requirement or it is not.

Audio Visual Equipment: Items used for optical and electronic information presentation; sound reinforcement; and visual and aural recording and reproduction such as cameras, video recorders and players, tape recorders, and public address sets.

Authorized Member: Personnel as defined in governing directives as being authorized the services set forth in that governing directive.

Authorized Stockage List: A listing of materiel (types and numbers of items) provided to support prescribed customers and their equipment.

Automated Information Systems: A method of accumulating, storing, and providing data through the application of computer hardware and software.

Automatic Voice Network: Worldwide network which has been established primarily for transmission of voice traffic.

Auxiliary ADP Equipment: Data processing equipment (except communications equipment) which directly supports or services a computer, including card punch machines.

Availability: A measure of the degree to which an item is in an operable and committable state at the start of the mission, when the mission is called for at an unknown (random) point in time.

Availability Objective: A factor used in stockage computations for operational readiness float. It is expressed as the percentage of time an operational readiness float serviceable asset is expected to be available for issue.

Available Man-hours (As Used In Labor Utilization Computations): Assigned direct labor man-hours intended to be expended on job orders in performance of maintenance tasks.

Available Time: The time that the equipment is on hand within an organization and is in full mission capable (FMC) condition.

Avionics Equipment: Electronic and electro-mechanical equipment designed primarily for installation in aircraft. The term applies essentially to communication and navigation equipment. It includes radio receivers and transmitters, radio compasses, marker beacon receivers, auto pilots, radar equipment, gyro-compasses, and intercoms, plus accessories, related wiring and antenna systems associated therewith.

Back-up Maintenance: That maintenance support required of the next higher level maintenance activity in support of a supported maintenance unit.

Backlog: Backlog is the overall measure of the direct labor resources required in terms of work to be done. Expressed in days, it is computed as follows:

A. Work to be done is the workload. Workload is the sum of the estimated remaining man-hours required to complete the open work orders. (Open work orders include those awaiting induction and those in progress; i.e., any status other than complete.)

B. For backlog computations, direct labor productive man-hours per day shall be computed as the direct labor assigned man-hours per day multiplied by 87.5% (seven direct labor productive man-hours for every eight assigned direct labor man-hours in the normal eight hour work day.)

C. Direct labor assigned man-hours are the assigned hands-on labor man-hours intended to be accounted for and reported on individual work orders (labor assigned to labor codes 01 and 06 in SAMS as of 4 May 90). Employees whose primary assigned duties are indirect labor (e.g., supply, drivers, clerks, supervisory) are to considered as assigned indirect labor and their man- hours will not be considered direct labor assigned man-hours.

D. The formula is: Workload (sum of the estimated remaining man-hours) / Direct labor productive man-hours per day = Days of Backlog (rounded to nearest tenth (.1)) This formula will be used to compute backlog by commodity, work center, or total backlog. Workload and direct labor productive man-hours will be grouped accordingly (commodity, work center, or total).

Backlog of Maintenance and Repair (BMAR): The end of Fiscal Year (FY) measurement of unfinanced maintenance and repair work, estimated at $50,000 or more, remaining on the AWP as a firm requirement of the FY, but due to lack of resources was not accomplished during the FY.

Backorder Reconciliation: The process of comparing open supply requests and requisitions between supporting and supported activities to identify and resolve differences in these records and to validate that a requirement continues to exist for the requested supplies.

Backwash: Process of reversing direction of liquid flow through a filter for purpose of cleaning filter media.

Bar Screen: A screen composed of parallel bars, either vertical or inclined, placed in a waterway to catch debris. Debris may be removed manually or mechanically from bar screen. Also called a rack or bar rack.

Basic Periodic Test: A test required to be performed on Field Artillery howitzers as specified in Technical Manuals.

Bench Stock: High usage repair parts, such as bolts, washers, and nuts, stored in the shop or work area for easy access.

Biennially: One time every two years.

Bill of Lading: Includes Government bills of lading issued by the Army (as defined in AR 55-355) and commercial bills of lading for transportation services administered by the Army (AR 735-5).

Bill of Materiel: A listing of supplies and materials required to complete a project.

Bi-monthly: One time every two months.

Biochemical Oxygen Demand (BOD): Quantity of oxygen required for biochemical oxidation in a given temperature usually for five days at 20 degrees Centigrade.

Bi-weekly: One time every two weeks.

Block Suspension: The suspension of one particular model of one kind of ammunition.

Blow Down: A procedure involving boilers where high concentrations of unwanted substances, such as solids, in boiler water are reduced by withdrawal of contaminated water and addition of fresh water.

Bona Fide Government Emergency: An emergency situation as expressed or declared by an authorized official of the U.S. Government.

Borescope Inspection: Examination of the internal surfaces of materiel, such as weapons and engines, using a periscope-like optical instrument.

Branch Circuit: Circuit conductors between final over-current device protecting circuit and outlet(s).

Breakdown: The stoppage or collapse of equipment, including installed equipment, or a component thereof, that requires immediate corrective action to restore it to an operating condition.

British Thermal Unit (BTU): A unit of energy measurement where one BTU is equivalent to the amount of heat required to raise the temperature of one pound of water one Fahrenheit degree.

Budget Execution Review (BER): An annual review of funds, at midyear, which enables the BASEOPS function to reflect changes that have taken place subsequent to the start of the fiscal year, to include updating unfinanced requirements and workload projections. It provides an assessment of the Engineer's resource posture for the remainder of the fiscal year.

Budget Manpower Guidance (BMG): A document which is distributed to the HQ TRADOC staff and all TRADOC Installations and activities. It includes specific narrative guidance outlining policies, priorities, and objectives developing in the Army and TRADOC which influence resources. Annexes cover workload, dollar, and manpower resource trails which represent the intended application of resources for accomplishment of the TRADOC mission/functions.

Building Fire Warden (BFW): A responsible person designated in writing by Area Fire Marshal to fulfill fire prevention duties for a specific facility in accordance with AR 420-90, TRADOC Suppl 1, and CASCOM&FL Reg. 420-3.

Buildings and Structures: Buildings, range firing sheds and positions, trailers, fencing, flagpoles, guard and water shacks and towers, grease racks, unattached loading ramps, training facilities other than buildings, monuments, grandstands and bleachers, elevated garbage racks, and other miscellaneous facilities and systems.

Calendar day: The time from midnight to midnight.

Calibration: The comparison of an instrument (measurement standard or item of test, measurement and diagnostic equipment) of unverified accuracy with an instrument of known and greater accuracy to detect and correct any discrepancy in the accuracy of the unverified instrument.

Calibration Recall System: A semi-mechanized system designed to indicate due date of next scheduled Test Measurement and Diagnostic Equipment (TMDE) calibration.

Call Back: A request for additional service following the initial service which has not provided the control required. Repeated call backs are possible.

Cancellation: A total or partial discontinuance of supply action requested of and confirmed by the supplier.

Cannibalization Point (CP): Supply source providing cannibalization support to authorized customers.

Cannibalize: To dismantle or remove serviceable parts from a machine, vehicle, or other equipment for use in the repair of another machine, vehicle, or equipment.

Cantonment Area: See map in Technical Reference Library.

Capital Decreases: Changes with a value of $1,000 or more ($100 or more in the case of Family Housing), which result in a decrease in value or deletion of real property.

Capital Equipment: Equipment with a value of $1,000 or more.

Capital Improvements: Changes costing $1,000 or more ($2.000 or more for Family Housing), regardless of source of funds, which provide additional items of real property and constitute an improvement which materially increases the value or substantially extends the useful life of the real property and increases the real property "units of measure".

Carrier: A railroad car, motor truck, ship, airplane, or other vehicle used for transporting supplies. Sometimes used to denote an entire rail, trucking, shipping, or air transport system.

Catalog: A uniform system of item identification and nomenclature to describe, classify, and number each item included in the system so that an item of supply is identified by a single stock number. Items cataloged are those items of personal property subject to stockage for supply support, repetitive procurement, distribution, and issue.

Causative Research: An investigation of discrepancies consisting of a complete review of transactions occurring since the last inventory. The purpose of this research is to assign a cause to the discrepancy so that corrective action may be taken.

Central Issue Facility (CIF): An Installation activity for the receipt, storage (to include maintenance), and issue of prescribed organizational clothing and individual equipment (OCIE) items.

Ceiling Fund: A fund limitation which cannot be exceeded.

Change Order: A written order signed by the Contracting Officer, directing the Contractor to make changes that are authorized by the Changes clause of the contract.

Check: To inspect, operate, and/or test for verification that the unit or item is in a fully operational condition or is performing its design function and to correct noted deficiencies in accordance with the requirements specified in this contract.

Class B Ammunition and Explosives: Class B ammunition and explosives are of a flammable hazard type. In general, they function by rapid combustion rather than detonation. Specific types and definitions thereof are further defined in "Hazardous Materials Regulation of the Department of Transportation", Tariff No 29, effective 14 January 1975.

Class C Ammunition and Explosives: Class C ammunition and explosives are of a minimum hazard type. In general, they contain Class A or Class B explosives or both, as components but in restricted quantities, and certain types of fireworks. Specific types and definitions thereof are further defined in "Hazardous Materials Regulation of the Department of Transportation", Tariff No 29, effective 14 January 1975.

Classes of Supply Legend:

Class I: Subsistence including gratuitous health and welfare items.

Class II: Clothing, individual equipment, tentage, tool sets and tool kits, hand tools, administrative, and housekeeping supplies and equipment.

Class III: POL: Petroleum fuels, lubricants, hydraulic and insulating oils, preservatives, liquid and compressed gases, bulk chemical products, coolants, deicing and antifreeze compounds, together with components and additives of such products, and coal.

Class IV: Construction: Construction materials to include installed equipment, and fortification or barrier materials.

Class V: Ammunition: Ammunition (including chemical, radiological & special weapons), bombs, explosives, mines, fuses, detonators, pyrotechnics, missiles, rockets, propellants, and other associated items.

Class VI: Personal Demand Items (Nonmilitary Sales Items).

Class VII: Major End Items: A final combination of end products which is ready for its intended use; e.g., launchers, tanks, mobile machine shops, vehicles.

Class VIII: Medical material including medical peculiar repair parts.

Class IX: Repair parts and components to include kits, assemblies and subassemblies, reparable and nonreparable, required for maintenance support of equipment.

Class X: Material to support nonmilitary programs; e.g., agricultural and economic development, not included in Classes I - IX.

Classification: The inspection of materiel to determine and record the existing state of serviceability or the correct identity of the items.

Classification (Freight): The process of changing an item noun into freight terminology to arrive at a shipment cost.

Classified Material: Documents, data, information, and items for which access is limited to those persons having a "need to know" and appropriate security clearance.

Clean: As used generally, means removal of dirt or impurities. As used for acceptance of work means gleaming, free from dirt, contamination, or impurities; unsoiled, unstained, recently laundered, fresh and unused, neat and tidy; having no flaws or roughness, clear, regular, or having few corrections.

Clearance: Authority permitting individuals cooperating in Department of the Army work, and having a legitimate interest therein, access to classified technical information, materiel, or equipment or admission to restricted areas or installations where such information or materiel is located.

Cold Storage: A facility where items may be stored, to include but not limited to perishable food, body remains, and batteries.

Collection Station: The designated points where solid wastes/refuse will be assembled in proper containers or bundles for collection by the Contractor. (May be referred to as collection points, pickup stations, or collection site.)

Commercial Equipment: Equipment which is also offered for sale to business and industry; may also be referred to as off the shelf equipment.

Commercial Publications: Non-Government published publications.

Commercial Traffic: Nonmilitary modes of transportation. Includes surface, air, and water transportation.

Comminutor: A grinder or shredder that converts bulky solid wastes into small particles.

Commitment of Funds: A firm administrative reservation of funds, based upon firm procurement directives, orders, requisitions, or requests that authorizes the creation of an obligation. Availability of funds is assured before a commitment is made.

Commodity Carrier: A commercial transportation medium for the shipment of specific commodities from one destination to another.

Commodity/Commodity Group: A grouping or range of end items with similar characteristics and applications, or end items that are susceptible to similar logistic management methods.

Common Access Card (CAC): A personalized smart card embedded with an integrated circuit chip for storing and processing data.

Common Service: Support performed by one activity for which payment is not required from the activity receiving the support.

Communal Other Real Property: Other real property used solely for support of family housing but not assigned to any one particular dwelling. This includes streets, exterior sidewalks, playgrounds, perimeter or security fencing, street lights, exterior utilities, and central heating plants.

Component Part: A component part as defined for the purposes of this contract is as follows: Facility: A component part is a part within a facility for which proper operation is necessary for facility to be utilized for its intended purpose. For example: A window in a facility would be considered a component part of that facility since proper operation of this component is necessary for facility to function for its intended purpose. System: A component part is a part within a system whose operation allows system to function in its intended manner. For example: A water control valve in a main water distribution line would be considered a component part of the water distribution system since operation of valve allows system to operate in its intended manner. Equipment: A component part is a part on an individual piece of equipment which is necessary for equipment to function in its intended manner. For example: A belt on an air handler would be considered a component part of the air handler itself since this part is necessary for unit to operate in its intended manner.

Component Replacement: The replacement of an entire component when it is more economical than to perform major repair. This includes installed equipment.

Components: Components of end items. Items identified in technical publications as part of an end item. Items troop installed or authorized (ITIA), and special tools, test and support equipment are not components. Components of assemblages. Items identified in a supply catalog component listing (SC/CL) as part of a set, kit, or outfit, or either assemblage.

Composite Labor Rate: The average combined-skill rate per hour to include labor, overhead, administration, handling, and profit.

COMSEC Custodian: A person designated as responsible for the security of COMSEC equipment and records.

Concealed Shortage or Damage: A shortage in, or damage to, the content of an original container or package detected after delivery, and contrasted with visible damage or shortages in the number of packages involved, not readily noticeable at the time of delivery.

Configuration: The functional or physical characteristics of hardware or software set forth in technical documentation and achieved in a product.

Conservation: The protection, improvement, and use of natural resources according to principles that will provide optimum public benefit and support of the military mission.

Construction: Erection, installation, or assembly of a new facility; addition, expansion, extension, alteration, conversion, or replacement of an existing facility; or relocation of a facility from one location to another. Includes equipment installed and made a part of such facilities, and related site preparation, excavation, filling, and landscaping, or other land improvements.

Container: A receptacle such as a bag, barrel, drum, box, crate, or package used in storage or shipment of a commodity to provide protection from physical damage or contamination of the commodity.

Container Marking: Numbers, nomenclature, or symbols stamped or painted on, or otherwise affixed to, items or containers for identification.

Contingency Plan: A document describing actions to be implemented or taken in the event of future occurrences.

Continuing Balance System - Expanded (CBS-X): An asset reporting system for items of equipment coded reportable item control code (RICC) 2 and 3.

Contract: All types of agreements and orders for the procurement of supplies or services.

Contract Administrator: An individual duly assigned by appropriate authority to administer a contract.

Contract Deliverable List (CD): Data required to be submitted by the Contractor to the Government. A proper and correct submission of a CD is evidenced by the following criteria: completeness, accuracy of data, preparation in accordance with applicable mandatory publication or other prescribing document, signature or initials by the certifying official/area supervisor, and correct and timely turn-in or distribution.

Contract Discrepancy: A failure of the contractor to perform in accordance with contract requirements and specifications. A contract discrepancy may result from a failure of the contractor to provide, or provide on time, the required contract products or services; or it may result because delivered products or services do not meet specific contract standards.

Contract Discrepancy Report (CDR): A formal, written documentation of Contractor nonconformance or lack of performance for contracted work. The CDR is initiated by the KO, or an authorized representative, whenever the performance as determined by the KO is unsatisfactory. The Contractor completes and returns the report to the KO.

Contract Maintenance: Any materiel maintenance operation performed under contract by commercial organization (including the original manufacturers of the materiel).

Contract Modification: Any written alteration in the specifications, delivery point, rate of delivery, contract period, price, quantity, or other contract provisions of an existing contract.

Contracting Office: The office which awards or executes a contract for supplies or services and performs post-award functions not assigned to a Contract Administration Office.

Contracting Officer (KO): An individual appointed in accordance with procedures prescribed by the Federal Acquisition Regulation (FAR) with the authority to enter into and administer contracts and make determinations and findings with respect thereto, or with any part of such authority.

Contracting Officer's Representative (COR): Any person who has been appointed in writing as the authorized representative of the KO acting within the limits of his authority.

Contractor: The term contractor as used herein refers to both the prime contractor and any subcontractors. Contractor's subcontractor shall comply with the provisions of the contract.

Contractor-Furnished Equipment (CFE): That equipment the Contractor is required to furnish in order to perform the requirements of the Contract.

Contractor-Furnished Property (CFP): That property the Contractor is required to furnish in order to perform the requirements of the contract. The Contractor retains title to all CFP. The term CFP includes Contractor-Furnished Equipment (CFE).

Contractor-Owned, Contractor-Operated (COCO) Equipment: That equipment which the Contractor has purchased or leased and which the Contractor uses and maintains to perform tasks under this Contract.

Controlled Exchange: The removal of serviceable parts, components, and assemblies from unserviceable economically repairable equipment and their immediate reuse in restoring a like item of equipment to a combat operable or serviceable condition.

Controller: A device or group of devices that serves to govern, in some predetermined manner, electric power delivered to apparatus to which it is connected.

Conventional Ammunition: This category of ammunition includes grenades, cartridges, projectiles, mines, pyrotechnics, bombs, warheads with fillers (e.g., high explosives, chemical, except nuclear), simulated nuclear weapons, bulk explosives, demolition materials, rockets without nuclear capability, propellant, and cartridge actuated component devices of aircrew escape systems such as ejection seats, extraction systems and capsules, missile systems, parachute airdrop and recover systems, and other special purpose munitions.

Conversion: A change to interior or exterior facility arrangements so that facility may be used for a new purpose. This includes installed equipment made a part of existing facility.

Corrective Action: Consists of those efforts required to correct reported deficiencies and determine that other products are not similarly defective.

Crew Served Weapons: Those weapons manned by a crew of two personnel or more.

Critical Care Areas: Areas of a hospital which include medical and surgical intensive care units.

Critical Equipment and Facilities: Items of equipment or facilities that must operate continuously or throughout the respective season in order to support critical missions. Failure of equipment or facilities in meeting design output requirements may affect the health and welfare of personnel or damage Government equipment or properties. Emergency or urgent service calls are often required to restore the critical equipment to optimum operating condition and provide the output required; examples are computer facilities, 24-hour operations (specify), fire prevention and protection facilities, medical facilities, electrical plants/systems, and water plants/systems.

Customer: Refers to units, individuals, or activities at and satellited around Fort Benning for support. Includes units or activities authorized support on an Interservice Support Agreement (ISA) or memorandum of understanding (MOU).

Deadline Rate: The percentage of items of equipment classified as "not ready for use" in a unit or activity.

Dedicated Support: Providing a driver with vehicle that will stay with the person or group being transported until the mission is complete or the driver is released by the senior officer being transported.

Defect: Any nonconformance of a unit of product with specified requirements or standards.

Defense Financial Accounting Service (DFAS): Agency responsible for payments to contractors.

Defense Joint Accounting System (DJAS): System used at the installation to generate various acquisition documents to include but not limited to purchase requests, travel, training, credit card purchases.

Defense Logistics Agency (DLA): Agency established in 1961 to provide centralized management in the procurement and distribution of supplies common to the various military departments.

Defense Switched Network (DSN): A Department of Defense telephone network.

Deferred Maintenance and Repair (DMAR): The end of FY measurement of unfinanced Family Housing Work estimated at $50,000 or more and validated by TRADOC, remaining on the AWP as a firm requirement of that FY, but due to lack of resources was not accomplished during the FY.

Deflagration: A rapid burning which generally does not produce a shock wave.

Delivery Service: Delivery of items to a customer's location.

Demand Maintenance Order (DMO) The order type created to execute the repairs to real property. Assets, and installed equipment that require demand maintenance, it consist of small minor maintenance and repair jobs for emergency work.

Demurrage: An assessment against the shipper or consignee as a penalty for the detention of a common carrier equipment beyond the period of free time allowed for loading or unloading.

Department of Defense (DOD): Comprises office of the Secretary of Defense and Military Departments.

Department of Defense Activity Address Code (DODAAC): A six-digit, alphanumeric code which identifies a customer, unit, or activity and the ship-to address of the storage facilities. The DODAAC is assigned by the Chief, Army Central Servicing Point (CSP) for each qualifying activity upon receipt of a request from the Major Command. DODAAC are identified in the DOD activity address directory (DOD 4000.25-6-M).

Director of Logistics (DOL): An Installation activity that provides supply, maintenance, and transportation services support for Installation customers.

Department of the Army Circulars (DA Cir): Publications issued by Department of the Army (DA) for guidance and which contain instructions relating to one-time actions, informational material of a temporary nature and procedures of limited duration. Each contains a statement indicating the date of expiration (normally one year or less).

Department of the Army Pamphlets (DA Pam): Publications issued by Department of the Army (DA) for guidance and which contain informational, instructional, or reference material of a continuing nature pertaining to administrative matters.

Depreciated Value: The dollar value of an item that is no longer new.

Desk (Preliminary) Estimate: An estimated cost of a project based on the estimator's personal experience and knowledge.

Detailed Estimate: An estimated cost of a project based on itemized equipment, material, and labor costs derived from established standards or historical data.

Detonation: A very rapid chemical reaction, characterized by production of a shock wave among other effects.

Device: A unit of an electrical system which is intended to carry but not utilize electric energy.

Direct Exchange (DX): A supply method of issuing serviceable material in exchange for unserviceable material on an immediate item for item basis.

Directorate of Public Works (DPW): An Installation activity that provides facility engineering support for Installation customers.

Direct Procurement Method: A method of shipping personal property by common (freight) carrier.

Direct Productive Labor (As used in Labor Utilization - Maintenance): Man-hours expended on job orders in performance of maintenance tasks (SAMS labor codes 01 and 06). The direct productive labor utilization rate will be computed by dividing direct productive man-hours by available man-hours rounded to the nearest tenth (.1).

Direct Support (DS) Maintenance: Performance of maintenance functions authorized to be accomplished at the lower level of maintenance and repair of equipment for return to user and repair of modules (components/assemblies) for the DX program. Repairs encompass the diagnosis and isolation of equipment/component malfunctions, adjustments, calibration, and alignment of modules which can be readily accomplished with easy-to-use tools and test-measuring and diagnostic equipment.

Discrepancies Incident to Shipment: When carrier fails to deliver to a consignee packages of property as listed on a bill of lading or a shipping document, or where the property is not delivered in as good a condition as when accepted for shipment by the carrier and the shortage or damage is discovered upon "checking-in operation" at the receiving point.

Discrepancy: A disagreement between the actual quantity or condition of property and that shown by the record of the property. A discrepancy usually is a disagreement between quantities or condition of property actually received in a shipment and that recorded on the shipping document. This type of discrepancy generally is referred to as a "discrepancy incident to shipment". Another form of discrepancy occurs when there is a disagreement between a stock record balance and the results of a physical count or inventory.

Discrepancy in Shipment Report (DISREP) (SF 361): A multiple use form, and the documentation in support thereof, for reporting transportation-type discrepancies as specified in Paragraph 1-5, AR 55-38. It is authorized for use as a Financial Liability of Property Loss, DD Form 200, with Checklist and Reporting Document for Financial Liability of Property Loss, DA Form 7531 to support claims against adjustment of inventory and financial accounting records.

Dispatch: The recording of data relating to vehicle utilization and the issuance of a written authorization for the vehicle to be used for a specific purpose.

Disposal: The processing of waste (at a facility approved for such processing by the appropriate State or Federal agency) in a manner that renders it no longer a hazardous waste as defined in 40 CFR. Some examples of these procedures are: chemical treatment, such as neutralization or detoxification; thermal treatment, such as incineration or pyrolysis; and recycling, reprocessing, or recovery.

Dress Code: Appearance appropriate

Dry Storage: A facility for storing nonperishable food.

Due-In: The quantities of material expected to be received under outstanding procuring and requisitioning instruments and quantities from other sources such as transfer, reclamation, and recovery.

Due-Out: That portion of stock requisitioned which is not immediately available for supply and which will not be referred to a secondary source of supply for supply action but will be recorded as commitment for future issue.

Dumpster: A container box, normally from eight to ten cubic yards in capacity, that empties into the hopper of a compaction-type refuse collection truck.

Durable: Items which are not consumed in use, retain their original identity and are not categorized as nonexpendable or expendable. Included are nonconsumable components of sets, kits, outfits, and assemblages, tools with a unit price in excess of $5.00 and any other nonconsumable item not otherwise coded nonexpendable.

Early Pick Up: A verbal or written request from a customer to deviate from a previously established schedule for transportation support.

Element of Expense: A fiscal code used to segregate cost data by type of cost; for example, personnel, supplies, and purchased services.

Elevated Storage: In any distribution system, storage of water in a tank supported on a tower.

Emergency: The reporting of sudden, usually unforeseen, occurrences where life or property are in immediate danger and require immediate action.

Emergency Deployment Planning: Planning associated with the immediate movement of a military unit in response to mission requirements, movement orders, major accident, or natural disaster.

Emergency Deployment Readiness Exercise (EDRE): A military exercise conducted to determine the combat preparedness of a deployable unit. Includes the provision of logistical support from the DOL and an evaluation of the deployable units' logistical training and capabilities. EDREs are further defined by category :

A. A Category I EDRE is the most realistic test of a deployment. Required actions will be fully performed with no simulations. Deploying unit will move to APOE for actual uploading of equipment. The Fort Benning Emergency Operations Center (EOC) will be operational 24 hours per day and supporting agencies operational or on call 24 hours per day.

B. A Category II EDRE is conducted with Installation support during duty hours only. Some requirements will be simulated, but FORSCOM Reg. 525-2 requirements will be met. A representative sample of unit personnel and equipment will be moved to a designated APOE or uploaded for shipment to SPOE.

C. A Category III EDRE is an internally controlled unit EDRE with limited Installation support. Requirements of FORSCOM Reg. 525-2 will be met. Units will not move to designated APOE but may utilize Fort Benning's aircraft simulator or rail load training facility as appropriate for loading a representative sample of unit equipment.

Emergency Dispatch: An unplanned, unprogrammed requirement for a vehicle to meet a situation which requires immediate response; e.g., deployments, EDREs, situations dictated by military training requirements, urgent civil requirements, and natural disasters.

Emergency Operations Center (EOC): A centralized facility where coordination and direction emanates from, in the provision of support to a bona-fide Government emergency, or a military exercise. Normally, an EOC is staffed with military or civilian expertise in every facet of support requirements.

Emergency Purchase: The purchase of an item or items when required delivery time cannot be met utilizing standard supply procedures.

Emergency Service Request: A request for service (with short response time) when health, safety, or military mission will be adversely affected if the situation is not abated as soon as possible.

Emergency Standby Stock: Emergency Standby Stock is non-demand supported items that normally have a long procurement lead time and are required in emergency situations.

Endangered Species: Those plants and animals which are in danger of extinction throughout a significant portion of their ranges as listed by United States Department of the Interior.

Engineered Performance Standards (EPS): The established number of work hours required for accomplishment of a certain unit of work of an acceptable quality. A set of standards established by the Army that establish the number of man-hours required for accomplishment of a certain unit of work of an acceptable quality. Specifically, EPS represent the time in man-hours it should take a trained worker, or group of trained workers, working at a normal pace, to produce a described unit of work, according to a specified method, under specific working conditions. They are derived from a complete objective analysis and measurement of each task. EPS are used for maintenance and repair work as well as new work and cover all of the various shops or elements workload.

Engineering and Housing: That activity which includes: All of general Real Property Maintenance Activity (RPMA) types of work and functions for functional categories of expenses described in Army Management Structure (AMS) (AR 37-100 Series) as H0000 Bachelor Housing (1910, 1920, 1930 accounts for Army Family Housing); .J0000 (Operation of Utilities); .K0000 (Maintenance and Repair of Real Property); .L0000 (Minor Construction); .M0000 (other Engineering Support to include custodial). Includes cost of family housing units and all services provided in operation of family housing facilities to include utilities, maintenance and repair, and Real Property incidental improvements, BP 1910/1920.

Environmental Pollution: Contamination of environment with man-made wastes.

Environmental Protection Agency (EPA): Federal agency that regulates through permits coordinated and effective Governmental action to assure protection of environment by abating and controlling pollution on a systematic basis. Basic organization consists of Headquarters at Washington, DC, and ten regional offices, all responsible to Administrator, Ref 40-Code of Federal Regulations (CFR) 1 and revisions thereof.

Equipment (As Used in Government-Furnished Property): An item of equipment owned by the Government, identified with a noun that is furnished to a Contractor for performance of contract requirements.

Equipment Improvement Report (EIR): A Government form used to document and recommend changes in the design or operation of a Government-owned item of equipment.

Equipment in Place (EIP): Nonexpendable property of a movable nature which has been fixed in place, or attached to real property but which may be removed without destroying the usefulness of the structure.

Equipment Logbook: A mandatory record of the events occurring during the life cycles of Government equipment made in accordance with TM 38-750, the Army Maintenance Management System (TAMMS). Contractor shall comply with TAMMS as applicable to each item of Government furnished equipment.

Equipment Publications: Those publications dealing with the installation, operation, maintenance, and repair parts support of Army materiel. Technical manuals, technical bulletins, lubrication orders, and modification work orders are the primary Department of the Army publications media used to provide these essential instructions for the major items of equipment.

Equipment Readiness Code (ERC) "A" Items: Primary weapons and equipment. Equipment essential to and employed directly in the accomplishment of assigned operational missions and tasks. ERC A items are designated on the unit MTOE.

Estimated Cost of Damage (ECOD): A written estimate of the dollar amount needed to repair an item, structure, or grounds to its configuration prior to being damaged. Expressed in terms of labor and materials.

Excess Stock: Portion of total quantity of an item on hand which exceeds authorized retention level for item.

Exercise: A military practice event.

Exodus: Mass movement of military troops during the Christmas to New Year Holiday period.

Expendable Operating Supplies: Items, other than repair parts, which are necessary for, and consumed in day-to-day functioning business.

Expendable: Items, regardless of type, classification, or unit price, which are consumed in use. This includes Class IX repair parts. Also included as expendable supplies are items not consumed in use that have a unit price of $100 or less and are not otherwise coded nonexpendable or durable.

Expenditure Limits: A percent of the acquisition cost of the end item or component for labor, assemblies, and repair parts that cannot be exceeded when effecting repair. The expenditure limits are reflected in the appendices of applicable TB-43 and TB-750 series.

Explosion: A chemical reaction or change of state effected in an exceedingly short space of time with generation of high temperature and pressure wave in the surrounding medium. The term includes both deflagration and detonation, which are differentiated by difference in rate of the reaction, with deflagration being the slower rate.

Explosives: Materials that either detonate or deflagrate. Any chemical compound or mechanical mixture which, when subjected to heat, impact, friction, shock, or other suitable initiation, undergoes a very rapid chemical change with the evolution of large volumes of highly heated gases which exert pressures in the surrounding medium.

Facilities: Buildings or structures, in whole or in part, furnished by the Government and assigned to the Contractor for contract performance. All items of Real Property other than land.

Facilities Engineering: That activity which includes all of the general types of work and functions for the functional categories of expenses described in the AR 37-100 Series as
.J1000 (Operation of Utilities); .K1000 (Maintenance and Repair of Real Property); .L1000 (Minor Construction); .M1000 (Other Engineering Support).

Facilities Engineering Management System (FEMS): The module designed to provide the tools that are necessary for the effective day-to-day management of FED operations at the Installation level, including cost accounting.

Facility Replacement: The replacement of an entire facility when replacement is more economical than major repair. This includes installed equipment.

Fair Market Value: The average selling price of an item or thing based upon condition and age.

Fair Wear and Tear (FWT): The average amount of deterioration of an item or thing based upon normal usage and age.

Family Housing (FH): Housing provided for eligible personnel with dependents and for chaplains without family members.

Family Housing Unit (FHU): Real property space used by one family. This includes the following: foundation, walls, roofs and other building components; interior utilities, fixtures, and equipment in place such as venetian blinds, cornices, furnaces and water heaters; utility connections at FHU building inside five foot line; carports, garages, storage and other facilities structurally connected to FHU; patios. The following are excluded: grounds; sidewalks and exterior utility lines beyond five foot line which are considered other real property; furnishings and authorized moveable equipment such as kitchen ranges and refrigerators; real and personal property acquired and installed with other than family housing funds.

Family Housing, Army Account: Includes cost of family housing units and all services provided in operation of family housing facilities to include utilities, repairs to areas beyond 5 foot line, and Real Property Facilities maintenance.

Feeder: All circuit conductors between service equipment or source of a separately derived system and final branch-circuit over current device.

Field Grade Family Housing: Housing provided for Field Grade Officers consisting of Majors (MAJ, 0-4), Lieutenant Colonels (LTC, 0-5) and Warrant Officers (CW4).

Files, Records, Documents: Books, papers, maps, photographs, notebooks, computer tapes, magnetic tapes, punched cards, floppy disks, **contents of hard drives, zip drives, and any other electronic mass media storage files,** holograms, bubble memories printouts, computer output, microfilm, microfiche, film slides, viewgraph transparencies, or other documentary materials, regardless of physical form or characteristics, created or received by the Contractor in pursuance of Federal law or in connection with the transaction of public business.

Finance Account Information: A combination of letters and numbers used in the governmental accounting system to describe types and purposes of expenditures.

Fire Control Items: Those items required and used to directly aim guns or controlled missiles at a particular target, to include instruments used in calculating and adjusting the proper elevation and deflection of missiles in flight or guns.

Fiscal Year (FY): A period of 12 months beginning 1 October and ending 30 September of the following year. Fiscal year is designated by calendar year in which it ends.

Flammable Gases: Liquefied petroleum and other compressed flammable gases. Example: Acetylene, butane, propane, hydrogen, and ethylene.

Floating Equipment: Those items listed in Section 1, Paragraph L "Floating Equipment," Appendix E, DA Pamphlet 738-750.

Flocculator: A mechanical device, usually in a tank, to enhance agglomeration of colloidal and finely divided suspended matter, after coagulation by gentle stirring of hydraulic means.

Follow-up: An inquiry of the status or action taken on a supply request previously submitted. The follow-up may take various forms including the use of single-line punch cards.

Fort Benning Emergency Communications System: Includes equipment located at the Fort Benning EOC and the Military Police Station.

Fringe Items: Items which do not meet the established criteria for stockage as prescribed by AR 710-2.

Fuel: For the purpose of this contract, "fuel" used herein applies to aircraft reciprocating engine fuel, aircraft jet engine fuel, diesel and automotive gasoline.

<u>Fuel Automated System: Automated system to track receipt of bulk fuel and issue to end-users.</u>

Fuel Allocation: The limitation placed on the amount of fuel which may be purchased. The allocation to Installations is made by U.S. Army Training and Doctrine Command, and may be sub allocated to using units or sections.

Fund Code: A two-digit code provided for the specific use of the requisitioner or Military Standard Requisitioning and Issue Procedures reimbursable requisitions to indicate to the distribution system that funds are available to pay the related charges and to identify the applicable funds.

Fund Limitation: A document specifying an amount of funds available for a specific period of time and restricting expenditures within the amount available and within the time frame specified.

Furnishings: Furniture, household equipment, and miscellaneous items procured under special authority, and furnishings authorized by CTA 50-909.

General Care Areas: All areas of a medical facility not classified as critical care areas.

General Ledger: A document used to record data and normally maintained in Army Installation Finance and Accounting Offices.

General Officer (GOQ) Quarters: Housing designated for or assigned to Brigadier Generals (BG, 0-7) and Major Generals (MG, 0-8)

General Services Administration (GSA): Agency established in 1949 with the Federal Supply Service as its major element for inventory management. Functions include supply management, procurement, quality control, cataloging, and supply distribution. Historically, primary concern has been with inventory management role in support of Federal civil agencies, with increasing support to DOD activities. (See DA Pam 700-1.)

General Site Plan: A plan showing what facilities exist, those proposed, and location of both.

General Supplies: SMA supplies in the Fort Benning SMA account, as identified on the Quarterly Stratification Reports, less those supplies identified as Medical (Material Category C2) and DPW (Material Category B9).

General Supply Account: Supply support that cannot be provided within the resources of units or activities; normally, a responsibility of the Installation logistics function.

General Support Maintenance (GS): Performance of maintenance above the direct support level which encompasses the diagnosis of failure and isolation of equipment faults as necessary for restoration of operational capability. GS maintenance also includes the repair of unserviceable modules in support of DX to lower category maintenance activities; the repair of unserviceable equipment/modules for return to Installation supply stocks or operation readiness float stocks; and repair and return to user.

Government: The term Government as used herein includes the Director of Logistics, the Directorate of Logistics Government-in-Nature (GIN) staff, the Contracting Officer, and the Contracting Officer's designated representatives.

Government Publications: Publications adopted or published by the agencies of the United States Government.

Government Reimbursable Work: Support furnished a tenant activity. All support furnished is paid for by the tenant activities command.

Government Representative: The Contracting Officer (KO), Contracting Officer's Representative (COR), Property Administrator (PA), and Quality Assurance Evaluators (QAE).

Government-Furnished Equipment: A term used in this contract to mean equipment in the possession of, or directly acquired by, the Government and subsequently made available for the sole use of the Contractor in the performance of this contract.

Government-Furnished Property: A term used in this contract to mean property in the possession of, or directly acquired by, the Government and subsequently made available for the sole use of the Contractor in the performance of this contract.

Government-Owned Property: A term used in this contract to mean property owned by or leased to, the Government or acquired under the terms of the contract and subsequently delivered to the Contractor for use by supported customers or on equipment of supported customers.

Government Travel Request (GTR): GTRs are accountable documents and the Contractor shall maintain storage, issue, and file in accordance with regulatory guidance.

Ground-level Storage: In any distribution system, storage of water in a shallow tank the bottom of which is below, or at the surface of, the ground. Booster pumps are ordinarily required to raise water from ground-level storage.

Ground-Fault-Circuit-Interrupter (GFCI): A device intended for protection of personnel that functions to de-energize a circuit or portion thereof within an established period of time when a current to ground exceeds some predetermined value that is less than that required to operate over current protective device of supply circuit.

Guard Services: The provision of people, animals, or items to provide security administration or law enforcement services.

Hand Carry: The delivery of items (documents and materiel) from one location to another via other than routine mail distribution (messenger) service, usually within a specified time frame and to meet an urgent requirement.

Hand Receipt: A signed document acknowledging acceptance of responsibility and liability for property.

Hazardous Materials (HAZMAT): Materials that are toxic, poisonous, corrosive, irritating, sensitizing, radioactive, biologically infectious, explosive, or flammable, and present a hazard to human health, safety, and environment. Special handling procedures and disposal facilities are required for their disposal in compliance with federal, state, and local regulations (reference C.6). Material that must be handled and disposed of in compliance with special provisions as outlined in safety and environmental publications.

Helbat: A special test of military equipment and systems.

Herbicides: Any substance or mixture of substances intended for use as a plant regulator, defoliant, or desiccant.

Herein: The word "herein" as used in this contract shall mean "in this contract".

High Priority Requisitions: Written requests for items or things that require delivery in a specified period of time. Requests are coded in compliance with AR 710-2.

High-Temperature Water (HTW): Water which has been heated above its boiling point but kept in a liquid state by applying pressure.

His: The word "his", as used in this Contract, is intended to mean his or her in a generic sense. It is not intended to denote gender.

Host: An Installation or activity that has management control over facilities and provides facilities, administrative, and logistical support (including base operation support) to another activity or unit that is dependent upon the providing Installation or activity for its administrative and logistical support requirements.

Hours of operation: That period of time from the hourly increment to the weekly increment whereas the contractor shall make the contractual services available to the government.

Household Equipment: Equipment which includes: cooking stove or kitchen range, refrigerator, clothes washer and dryer, freezer, portable dishwasher, portable fan, dehumidifiers, and fireplace accessories.

Housing Area: A group of family housing units which form an identifiable entity or community, defined by geographic features, or year of construction or grade of occupant, or other logical separation. Included are streets, drainage ways, open recreational areas and unused land. A housing area also includes common use areas serving units and other real property.

Housing Services Office (HSO): The office at the Installation level which operates the Housing Services Program.

Idle Days: Working days during which a vehicle is available for dispatch, but is not dispatched.

Igloo: An earth covered, arch-type magazine of concrete or steel designed for the storage of ammunition and explosives.

Improved Grounds: Grounds on which intensive development and maintenance measures are performed.

Improvement: Alteration, conversions, modernization's, renewals, additions, expansions, and extensions which are for the purpose of enhancing rather than repairing a facility or system associated with established housing facilities or areas.

Incidental Improvement: Minor improvements made within cost limitations of Army Family Housing (AFH) O&M Program. These are also referred to as alterations and additions by fiscal managers.

Includes: "Includes" to mean: includes, but is not limited to.

Inconsequential Discrepancies: When incident to shipment or stockage, damage not exceeding $25 in money value, or a damage which does not impair the usefulness of the article or render it unsuitable for use, may be determined to be "inconsequential" except when incorrect receiving practice or pilferage at the receiving station is indicated. The $25 valuation is applicable to discrepancies in a shipment covered by a single bill of lading. It may not be applied to loss or damage in individual transportation (different vouchers) shipped on the same bill of lading.

Inconvenience Claims: Claims submitted by individuals who feel that they incurred additional expenses through the fault of an individual or action of the Government.

Indirect Labor: That labor specifically required in performance of this contract and not identified as direct labor.

Indirect Productive Labor (As Used in Labor Utilization - Maintenance): Man-hours not identified as Direct Productive Labor; e.g., supervision, production control, supply (SAMS labor codes excluding 01, 06, 20 and higher). The indirect productive labor utilization rate will be computed by dividing indirect productive man-hours by available man-hours, rounded to the nearest tenth (.1).

Individual Job Order (PWO): DA Form 4283/4284; Directive for those maintenance, repair, and construction jobs for which detailed estimating of work hours and other resources is necessary. Maintenance, repair, construction, or alterations which exceed Demand Maintenance Order criteria. It may also be referred to as a Facilities Engineer work order or work request.

Industrial Plant Equipment (IPE): Plant Equipment with an acquisition cost of $15,000 or more; used for the purpose of cutting, abrading, grinding, shaping, forming, joining, testing, measuring, heating, treating, or otherwise altering the physical, electrical, or chemical properties of materials, components, or end items entailed in manufacturing, maintenance, supply, processing, assembly, or research and development operations. IPE is identified by federal supply class in Appendix 1A, AR 700-43. Also referred to as DIPEC equipment.

Input: Information transferred into the internal storage of data processing system, representing data to be processed for information to help control the process.

Inspect: Determination and identification of the condition, defects, or malfunctions of equipment, facilities, and systems with reference to established standards.

Install: To set in position and connect or adjust for total functional use equipment or materials.

Installation: The land and facilities occupied by TRADOC and its tenants.

Installation Fire Marshal: The Directorate of Public Safety is appointed the Installation Fire Marshal by AR 420-90.

Installation Planning Board: Consists of members (mostly comprised of major subordinate commanders and directors) concerned with the composition and development of the Installation.

Installation Property Record: DA Form 3329, database, or machine listing, for recording and accounting for all nonexpendable supplies, maintenance and services (M&S) equipment, shop machinery, radios, and items which are of a nature that special management and controls are required.

Installation Transportation Officer: An individual designated in writing to act for the Commander in the management and operation of Installation transportation activities.

Installed Equipment: Items of equipment or furnishings including materials for installing, which are required to make facility usable and are affixed as a permanent part of structure; these items include, but are not limited to, plumbing fixtures and equipment, fixed fire protection systems, elevators and escalators, overhead-crane runways, lavatory counters, cabinets, fans, air conditioners, furnaces, and similar fixed equipment. Machine tools, production and research equipment, and their foundations are excluded. (Appendix J, AR 420-17)

Integrated End Item: An item which is used with or installed in another end item (e.g., radio set or generator installed in a vehicle).

Integrated Facilities Data Entry Process (IFDEP): The minicomputer interactive preprocessor providing data entry for all IFS transactions and interactive status inquiries pertaining to work documents.

Integrated Facility System (IFS): A multi-command, automated information and evaluation system which encompasses life-cycle management of real property resources from conception through design, construction, operation, maintenance, and disposal.

Into aircraft: Where contractor personnel are required to perform all operations incidental to the refueling and defueling of aircraft, to include the actual operation of nozzles and the attachment of bonding cables.

Intra/Inter-Service Support Agreement (ISSA): A document wherein the participants, to preclude any misunderstanding, state clearly in writing, the agreement for the provision of support arrived at between the activities involved, especially the obligations assumed by each and the rights granted to each. An agreement, used for coordinating and providing support to component units, activities, and individuals located outside Installation real property boundaries. Intra/Interservice

In transit Inventories: Items or materiel in the process of movement as follows: From source of procurement to storage locations when the Government has taken title to the materiel at point of origin. Between storage locations. From customers to storage locations. Between storage locations and contractor plants.

In transit NonTactical Vehicles: Vehicles not assigned to any activity within the Fort Benning area of support but traveling through the Fort Benning area (only nontactical).

Invalid Cost: A cost that is determined by a Government official as not payable or authorized.

Inventory: A physical count of items located within an area.

Inventory Control: Phase of logistics which includes managing, cataloging, requirements determination, procurement, distribution, overhaul, and disposal of material.

Inventory Lot: A segment of the total items stored at an Installation which have been grouped for the purpose of inventory.

Investigative Personnel: Personnel designated to inquire into the circumstances surrounding an incident; e.g., surveying officers, AR 15-6 investigation officers, military police, and criminal investigators.

Issue Priority Designator (IPD): A two position numerical code which relates the Force Activity Designator (FAD) of the organization or activity requesting maintenance support with the Urgency of Need Designator (UND) for the support requested (AR 310-25).

Job Order, Work Order: Government forms used to justify and record data relative to requesting and performing work.

Joint Inventory: A physical count of items conducted by individuals representing separate interests for the purpose of establishing the quantities of property on hand.

Joint Service Interior Intrusion Detection System (JSIIDS): Intrusion detection system developed by DOD.

Julian Date: A four-digit numeric code as follows: The first digit will be the last digit of the current calendar year. The remaining three digits will be the consecutive day of the year. (052 represents 21 February; therefore, 0052 would represent 21 February 1990).

Key Account: A term used by the Army to segregate financial data in compliance with the Army management structure; normally the seventh digit in the accounting structure.

Labor Proficiency (Maintenance): Proficiency of maintenance labor. Labor Proficiency will be computed by dividing total estimated man-hours to perform job-ordered maintenance by total man-hours expended on job orders, rounded to the nearest tenth (.1).

Legal Public Holidays: Holidays in each calendar year identified as follows:

New Year's Day, January 1;

Martin Luther King, Jr.'s Birthday, the third Monday in January;

Washington's Birthday (President's Day), the third Monday in February;

Memorial Day, the last Monday in May;

Independence Day, July 4;

Labor Day, the first Monday in September;

Columbus Day, the second Monday in October;

Veteran's Day, November 11;

Thanksgiving Day, the fourth Thursday in November; and

Christmas Day, December 25.

Levels of Maintenance:

A. Direct Support: Maintenance normally authorized and performed by designated maintenance activities in direct support of using organizations. This category of maintenance is limited to the repair of end items or unserviceable assemblies in support of using organizations on a return to user basis. (This function was formerly known as 3rd echelon maintenance.)

B. General Support: Maintenance authorized and performed by designated TOE and TDA organizations in support of the Army supply system. Normally TOE and TDA general support maintenance organizations will repair or overhaul materiel to required maintenance standards in a ready-for-issue condition based upon applicable supported Army area supply requirements. (This function was formerly known as 4th echelon maintenance.)

C. Operator Maintenance: The authorized level and prescribed maintenance checks and services performed by an operator of Army owned or leased equipment. (Formerly known as 1st echelon.)

D. Organizational Maintenance: The authorized level and prescribed checks, services, and maintenance performed by organizational mechanics as prescribed in applicable Technical Manuals for each item of Army-owned or leased equipment. (Formerly known as 2nd echelon maintenance.)

Levy Group: Soldiers who have been alerted or placed on levy for an overseas tour.

Lifting Aids and Devices: Mechanical items that are designed to provide lift capabilities.

Line Item (Quantitative) Accountability: Term used to refer to the method of accounting for supplies and equipment whereby accountability is based on individual item and quantity thereof.

Load Test: A periodic test of lifting aids and devices to ensure their serviceability and safety.

Location:

A. Damp Location: Partially protected locations under canopies, marquees, roofed open porches, and like locations, and interior locations subject to moderate degrees of moisture, such as some basements and some cold storage locations.

B. Dry Location: A location not normally subject to dampness or wetness. A location classified as dry may be temporarily subject to dampness or wetness, as in the case of a building under construction.

C. Wet Location: Locations underground or in concrete slabs or masonry in direct contact with the earth, and locations subject to saturation with water or other liquids, such as vehicle washing areas, and locations exposed to weather.

Location Survey: A physical verification, other than actual count, between actual assets and recorded location data, to ensure all assets are properly recorded as to location, stock number, condition code, unit of issue, security/pilferage code, and shelf life code.

Location System: A record which shows the exact location of supplies stored within a storage activity.

Lock Out/Tag Out. Refers to the requirement to deactivate, disconnect, or otherwise render inoperative and tag to indicate this action has been accomplished any stored energy, such as electrical, hydraulic, or spring tension, while work is being performed on facilities or equipment affected by that stored energy.

Logistics. Areas related to transportation, materiel maintenance, base supply and associated actions.

Loss and Damage Claims: A written document submitted to the KO by persons or agencies in an attempt to recover dollars for losses or damages.

Lot: A quantity of material manufactured under identical conditions and assigned an identifying lot number, or a collection of service outputs from which a sample can be drawn and inspected to determine conformance with the standard.

Lot Size: The number of service outputs in a given lot.

Magazine: Any building or structure, except an operating building, used for the storage of ammunition, explosives, or loaded ammunition components; e.g., above ground magazine, igloo-type magazine, railroad car, and/or motor truck.

Main (as related to Water Treatment Plant Services): Water pipe from which domestic water supply is delivered to service pipe leading to specific premises.

Maintainability: A characteristic of design and installation which inherently provides for the item to be retained in or restored to a specified condition within a given period of time, when maintenance is performed in compliance with prescribed procedures and resources.

Maintenance: The recurring day-to-day, periodic, or scheduled work required to repair or maintain equipment and facilities in a specified condition, or to restore systems or components to initial or usable condition by overcoming the effects of breakdowns, wear and tear, damage, or deterioration. This includes work undertaken to prevent damage to a system or component which otherwise would be more costly to restore.

Maintenance and Service (M&S) Equipment: Items of mobile equipment or special purpose vehicles and/or equipment with an end item acquisition cost of $100,000 or more which are used to accomplish DPW functions.

Maintenance Capability: Availability of resources such as facilities; tools, test, measurement, and diagnostic equipment (TMDE); drawings; technical publications; trained maintenance personnel; engineering and management support; and repair parts required to perform maintenance operations.

Maintenance Capacity: A quantitative measure of maintenance capability usually expressed as the number of man-hours of direct labor that can be applied within a specific maintenance activity or shop, during a 40-hour week (one shift, five days).

Maintenance Categories: A designation within a system of maintenance of materiel which is based on the extent of capabilities, facilities, and skills required for the operation. Categories of maintenance are organizational maintenance, direct support maintenance, general support maintenance, and depot maintenance or variations of these in compliance with the maintenance concept for particular equipment systems and commodity groups.

Maintenance Contact Teams: One or more individuals possessing the technical skills, tools, and equipment to perform on-site repairs of equipment at a customer's location.

Maintenance Operations: The management and physical performance of those actions and tasks involved in servicing, repairing, testing, overhauling, modifying, calibrating, modernizing, and inspecting materiel in the operational inventory and the provision of technical assistance to equipment users' in support units of the Army Logistics System.

Maintenance Performance Data: Information relating to the use and results obtained from the application of maintenance resources (e.g., work force, equipment and funds) to perform maintenance operations on Army materiel.

Maintenance Records: Records used to control maintenance scheduling, inspections, and repair workloads which provide a standard method for recording repair action taken by responsible maintenance elements and are used in figuring the current status of equipment readiness, reliability of equipment utilization, and logistic requirements; and records designed to permit analysis of causes of equipment failure and mortality rates of components.

Major Component: An assemblage or combination of parts, subassemblies, and assemblies connected in such a manner as to be a self-contained unit which, although part of a larger item, is capable of operating independently of the larger item and is separately identified by type, model, and series. Examples are receivers or receiver-transmitters in radio sets and machine guns or other weapons in secondary armament subsystems of combat vehicles.

Major Inventory Discrepancy: A discrepancy which exceeds the dollar-value or percentage parameters as defined by the major command and entered in the code table file, symbolic name INVMAJOR.

Major Work Order: Work Orders which require a level of effort of more than $5,000 but less than $10,000 (total labor and materials costs).

Malfunction: The failure of an ammunition item to function as expected when fired or launched, or when explosive components function during a nonfunctional test. Malfunctions include hang fires, cook off, misfire, in addition to abnormal or premature functioning of explosive ammunition items, warheads, missiles, and rockets as a result of normal handling, maintenance, storage, transportation, and tactical deployment. They do not include accidents or incidents resulting from negligence, malpractice, or fires.

Management Control Number: A number, similar to a Federal Stock Number, assigned by the National Inventory Control Points under certain specific conditions for identification and accounting purposes. Consists of applicable four-digit class code number from the Federal Supply Classification, plus a letter to designate the assigning agency, followed by a six-digit number.

Manager Referral: SARSS output requiring manager review for information or action necessary to re-input for processing of supply documents.

Mandatory Document: A directive which the contractor is obliged to perform the effort strictly in accordance with the method specified in the directive, to meet the stated results of the directive.

Master Plan: An integrated series of documents which present in graphic, narrative, and tabular form the present composition of the Installation and the plan for its orderly and comprehensive development to perform its various missions in the most efficient and economical manner.

Material: Property which may be incorporated into or attached to an end item or which may be consumed or expended in performance of work. It includes, but is not limited to, clothing, raw and processed material parts, components, installed equipment, assemblies, small tools, and supplies which may be consumed in normal use in performance of work.

Material Safety Data Sheet (MSDS): A form which contains identification, handling, and hazard disclosure information over 75,000 chemical substances that require documentation by chemical manufacturers under the Hazard Communication and Label Standard of the Occupational Safety and Health Administration (OSHA). Forms contain chemical name; trade name; molecular formula; manufacturer's or importer's name, address, and telephone number; physical data (including description, boiling and melting points, specific gravity, evaporation rate, and solubility in water); fire and explosion data (including flashpoint, upper and lower ignition limits, and fire fighting techniques); toxicity and health effects (including first aid and antidotes); reactivity (including incompatibilities); decomposition; polymerization; handling, storage, and disposal conditions to avoid; and spill and leak procedures (including requirements for protective equipment).

Materiel: Consists of all tangible items (excluding real property, installed equipment, and utilities systems) necessary to equip, operate, maintain, and support military activities without distinction as to application for administrative or combat purposes.

Materiel Category: A division of supplies and equipment for processing through the Army supply system.

Materiel Fielding Plan (MFP): A document that describes new items of equipment or systems, dates to be available in units or activities and training requirements. MFPs will normally require a mission support plan to be developed at Installation level.

Materiel Maintenance: The function of sustaining materiel in an operational status, restoring it to a serviceable condition, or updating and upgrading its functional usefulness through modification or other alteration. It includes the sub function of maintenance engineering and maintenance operations.

Materiel Release Order Follow-up: An inquiry to an Installation storage follow-up relative to a previously transmitted Materiel Release Order.

May: Is permissive. However, the words "no person may..." mean that no person is required, authorized, or permitted to do the act prescribed.

Media and Status Code: A one-digit, alphanumeric code which denotes the activity designated to receive supply and shipment status for a given requisition. This code is a mandatory entry in card column 7 of DD Form 1348M by the document originator on MILSTRIP requisitions.

Mid-Year Review: A review of SMA budget program to analyze and determine problem areas, expenditures, balances, levels, and other items in order to prepare a report for higher headquarters, regarding the SMA for the remainder of the fiscal year.

Military Traffic Management Command Total Quality Assurance Program (MTMC TQAP): A Government method of evaluating shipments by commercial moving companies. TQAP evaluations of commercial moving companies are used as a basis of utilization and tonnage distribution to these companies by the Government.

Military-Unique (GFP Section): An item peculiar to military use, not normally obtainable through other than military procurement channels.

Minor Inventory Discrepancy: A discrepancy which does not exceed the dollar value or percentage-parameters as defined by the major command and entered in the code table file, symbolic name INVM.

Minor Maintenance: Minor maintenance of non-GSA leased vehicles includes tire repair, tune-up, lubrication, fluid change, repair part replacement not requiring diagnostic equipment and costing less than $100.00 per item, welding, body repair and painting costing less than $150.00, safety check, technical inspection, and as directed by the KO.

Minor Work Order: Work Orders which require a level of effort of more than $2,000 but less than $5,000 (total labor and material costs).

Mission Capable (MC): Equipment which is capable of performing its assigned mission.

Mission-Unique: Base operations support required by a tenant which is different from the kind of support services, supplies, and equipment normally provided or maintained by the host, e.g., specialized intelligence equipment or laboratory equipment for medical facilities.

Mobility Fuel: Fuels such as gasoline, kerosene, and diesel used in vehicles and aircraft.

Mobilization Exercise: Periodic tests or practices of a unit, organization, or activity's ability to respond to emergency or mobilization conditions.

Mobilization: A sudden buildup of military forces in the event of a national emergency.

Modernization: Improvement of a family housing unit or communal other real property by elimination of substandard conditions caused by obsolescence of otherwise serviceable constituent parts, materials, capacity and layout so that improved family housing unit or communal other real property facility will be at same standard as a new family housing unit or other real property facility currently being constructed by the Department of the Army.

Modification of Work Order: A DA publication providing authority and instructions for modification of Army materiel.

Mogas: Gasoline, regular unleaded for vehicles with gasoline engines.

Monetary Accounting: Term used to refer to the method of accounting for supplies whereby accountability is based on the monetary value of items of supply.

Motor Carrier Detention Charge: A fee charged to the consignee or consignor by commercial freight carriers when the time used for loading or unloading exceeds that allowed.

Munitions Area: Those designated areas where munitions and explosives are stored.

National Stock Number: The 13-digit stock number replacing the 11-digit Federal Stock Number. It consists of the four-digit Supply Classification code and the nine-digit National Item Identification Number. The National Item Identification Number consists of a two-digit National Codification Bureau number designating the central cataloging office of the NATO or other friendly country which assigned the number and a seven-digit (xxx-xxxx) nonsignificant number. The number shall be arranged as follows: 9999-00-999-9999.

Natural Resource Conservation Service: Agency under Department of Agriculture that has over sight of Government natural resource programs to include soil conservation.

New Construction: Erection, installation, or assembly of a new real property facility including equipment related site preparation, excavation, filling and landscaping, or other land improvements.

Nomenclature: Set or system of official names or titles given to items of material and equipment.

Nonappropriated Fund: A fund established by authority of the Secretary of the Army, for the purpose of administering moneys not appropriated by Congress for benefit of military personnel or civilian employees of the Department of the Army. Moneys for fund arise from profits derived from business enterprise activities operated by independent instrumentalities of the Department of the Army.

Nonappropriated Fund Property: Property accounted for and owned by a nonappropriated fund activity.

Noncritical Equipment or Facilities: Those categories of equipment or facilities that do not affect the health of personnel, cause damage to Government properties, or cause critical facilities such as ADP to shut down in case of equipment failure.

Nonexpendable: Items which are not consumed in use, retain their original identity during the period of use, and require that accountability be maintained throughout the life of the item. These include nonconsumable end items authorized by TOE, TDA, CTA, or other authorization documents.

Nonproductive Labor (As Used in Labor Utilization - Maintenance): Man-hours not identified as productive hours; e.g., vacation, TDY, Holidays, non-pay status (SAMS labor codes 20 and higher). The nonproductive labor utilization rate will be computed by dividing nonproductive man-hours by available man-hours, rounded to the nearest tenth (.1).

Nonstandard Item: Property which has civilian or commercial application but which has never been processed for standardization by an Army technical committee.

Nonstatic Dispatch: A dispatch made to a unit or section upon request, not on a recurring basis.

NonTactical: An item of equipment or event that does not require military unique standards.

NonTactical Vehicle (NTV): Nontactical motor vehicle; the term "nontactical" as distinguished from "tactical" denotes motor vehicles, of commercial design, which are used to provide transportation support to an Installation.

Nontemporary Storage: Storage authorized in lieu of shipment of personal property at Government expense. It is normally authorized for the period during which qualified service members are assigned overseas, but is sometimes used in connection with permanent change of station assignments when additional storage is authorized for periods of temporary duty enroute.

Normal Duty Hours: 7:30 a.m. to 4:30 p.m. Mondays through Fridays, excluding legal public holidays.

Notification Hour (N+Hour): The declaration start time for a series of required events occurring at pre-designated times in support of an exercise or deployment.

Not Mission Capable (NMC): Equipment which is not capable of performing its assigned mission.

Not Mission Capable Maintenance (NMCM): Equipment not capable of performing its assigned mission because of needed maintenance.

Not Mission Capable Supply (NMCS): Equipment not capable of performing its assigned mission because of a lack of parts needed to accomplish repair.

Nursery Stock: Plant materials such as trees, vines, shrubs, and hedges obtained from nursery suitable for transplanting.

Obligation: A legal reservation of funds recorded at time a legal binding agreement has been reached between an agent for the United States Government and a second party.

Obligation Funding Target: A goal established of the dollar amount specifically reserved against an appropriation or fund for expenditures in payment of an order placed, contract awarded or service received.

Off Post Dispatch: The recording of, and written authorization for, a driver of a Government-owned or leased vehicle to depart the Installation with the vehicle.

Off-Line Processing: A system and the peripheral equipment or devices in which transactions are processed separately, either manually or with peripheral equipment, without reference to the central files in the SAILS computer. For example, transactions with manager entry code (MEC) 8 or 9, which are processed manually, off-line, and are later entered into the SAILS computer as "post-post" transactions to establish a record in the Document History and Financial files.

Office Machines: Mechanical equipment used in offices such as typewriters, calculators, dictating machines, duplicating machines, office copiers, microfiche printers and readers, paper shredders, collators, sign printing machines, stencil cutting machines, form writers, decimal typewriters, facsimile (FAX) machines, and optical character recognition (OCR) typewriters.

Officers Quarters (Bachelor): Bachelor quarters for Lieutenants, Warrant Officers, and Chief Warrant Officers.

Oil Analysis: A test or series of tests which provide an indication of the condition of equipment components by applying a method of precision detection and quantitative measurement of wear metals in an oil sample. This includes physical property testing of the sample to determine contaminant content and condition of the used oil.

On-Call Dispatch Vehicles: On-call dispatch vehicles are U-drive vehicles requested on an unscheduled basis by separate requests. Some users will call a day in advance, others will request same-day assistance.

One Station Unit Training (OSUT): A training program integrating common skill (e.g., Basic Combat Training (BCT)) and selected MOS specific training (Infantry) into a single program, with one program of instruction, with the same cadre, in the same company sized unit, conducted at a single Installation.

On Post Fleet: Nontactical vehicles (NTV) that are dispatched on a recurring or nonrecurring basis to customers located on the Installation.

On-Site: Repairs or services performed at a customer's location.

Operation and Maintenance (O&M): The activities for the reliable operation of utilities, equipment, and systems; and the preventive measures required to assure continued trouble-free operation of utilities, equipment, and systems.

Operation and Maintenance, Army Reserve (OMAR): A subdivision of the operation and maintenance, Army Reserve appropriation, as reflected in the Army Maintenance Structure.

Operation Work Order (OWO) Type of order used when the scope of work is not within an SOO work Order.

Operational Performance Capability: The capability of a unit/formation, weapon system, or equipment to perform the missions of functions for which it is organized or designed.

Operational Readiness Float (ORF): End items of mission-essential maintenance-significant equipment authorized for stockage by maintenance support units or activities to replace unserviceable reparable equipment to meet operational commitments.

Organic Solvent: Organic fluids used for cleaning, thinning, calibration, or similar purpose. Solvents include hydrocarbons, halogenated hydrocarbons, oxygenated hydrocarbons, mixtures, and similar materials. Organic solvents have a wide variety of commercial and industrial uses which fall into three major areas: cleaners, diluents, and test fluids. Cleaning solvents include degreasing compounds for automotive and equipment parts, spot removers for fabrics, dry cleaning solvents, and corrosion removing compounds. Diluents are used in liquefiers or dissolvers and include the alcohol’s, ketone's, and esters, as well as thinners for paints and lacquers. Test fluids include heptane's and freon; an example of their use is for instrument calibration.

Organization Chart: Diagram showing the organization of units, offices, activities, or installations.

Outlet: A point on wiring system at which current is taken to supply utilization equipment.

Over Obligation: In excess of the dollar amount specifically reserved against an appropriation or fund for expenditures in payment of an order placed, contract awarded or service received.

Overhaul: The restoration of an item to a completely serviceable condition as prescribed by maintenance serviceability standards or to dismantle, examine, and restore equipment and equipment components to the original condition and manufacturer's specifications.

Overseas Pack: Package or method of packing designed to withstand rough handling of military transportation and distribution overseas.

Panel board: A single panel or group of panel units designed for assembly in the form of a single panel; including buses, automatic over current devices, and with or without switches for control of light, heat, or power circuits; designed to be placed in a cabinet or cutout box placed in or against a wall or partition and accessible only from the front.

Parent Account Customer: Units and activities authorized to draw supplies directly from the Installation Supply Account are designated Parent Account Customers.

Partial Mission Capable: Equipment which is capable of performing its assigned mission but not at the full level for a sustained period of time.

Performance Certificate: A written certification executed by a responsible official that the service called for in a contract, purchase order, or delivery order has been satisfactorily performed.

Performance Indicator: A characteristic of an output of a work process that can be measured.

Performance Requirements Summary (PRS): The PRS shows contract requirements, the component requirements related to each contract requirement, the price of each work requirement as a percentage of the associated contract requirement, the standard of performance, and the acceptable quality level (AQL) for each work requirement.

Performance Work Statement (PWS): The PWS consists of the definitive or descriptive words identifying the subject matter of the contract referred to as the specifications or work statement.

Personal Property (As Used in Transportation Services): Moveable and tangible items, such as automobiles, boats, household goods, and unaccompanied baggage, not to include real property.

Petroleum, Oils and Lubricants Facilities (POL): Grounds adjacent to structures and/or tanks that store or dispense petroleum, oils, and lubricants.

pH: A measure of hydrogen (H) ion concentration, indicating degree of acidity or alkalinity of a solution. Values below 7.0 indicate acidity and above 7.0, alkalinity.

Phase-in Period: The period of 150 calendar days beginning on Contract award date.

Physical Evaluation Test (Driver Testing): A brief test of vision and reaction time, administered to each driver tested by the facility.

Planned Sampling: Based on some subjective rationale and sample size arbitrarily determined.

Planograph: A scale drawing of a storage area showing the approved layout of the area, location of bulk, bin, rack and box pallet areas, aisles, assembly areas, walls, doorways, directions of storage, office space, wash rooms, and other support and operational areas.

Plant Equipment: Government-owned property of capital nature, consisting of equipment, furniture, vehicles, machine tools, and test and special test equipment, used or capable of use for any administrative or general plant purposes for Contractor to account for and maintain.

Policing Grounds: Pickup and disposal of paper, bottles, cans, cardboard, plastic, rags, and other refuse on grounds.

Policy: The general plan of operation.

Portable Structures: Small moveable structures without utility connections, which can be readily relocated intact such as bleachers, coal bins, dog kennels, and sentry boxes.

Post: A grouping of facilities, excluding family housing areas, located in the same vicinity, which support particular functions.

Post-Post: Transactions input into SAILS to update accounting records after an issue has been made to a customer.

Potable: Water which does not contain objectionable pollution, contamination, minerals, or infection, and is considered satisfactory for domestic consumption.

Premises Wiring (System): That interior and exterior wiring, including power, lighting, control, and signal circuit wiring together with all of its associated hardware, fittings, and wiring devices, both permanently and temporarily installed, which extends from load end of service drop, or load end of service lateral conductors, or source of a separately derived system to outlet(s). Such wiring does not include wiring internal to appliances, fixtures, motors, controllers, motor control centers, and similar equipment.

Prescribed Burning: Skillful application of fire to natural fuels under conditions of weather, fuel moisture, soil moisture, or other confining variables, that will allow confinement of fire to a predetermined area and at same time will produce intensity of heat and rate of spread required to accomplish certain planned benefits to one or more objectives of silviculture, wildlife management, grazing, hazard reduction, etc. Its objective is to employ fire scientifically to realize maximum net benefits at minimum damage and acceptable cost. Also known as controlled burning.

Preventive Maintenance (PM): Systematic and cyclic check, inspection, and repair of deficiencies, as well as reporting of deficiencies beyond scope of preventive maintenance. PM includes accomplishment of maintenance and repair.

Preventive Maintenance Order (PMO) The order type created to execute inspections, care, and servicing of equipment, utility plants, and systems, buildings, structures, and facilities for detecting and correcting developing failures and accomplishing minor maintenance.

Primary Equipment: Major equipment essential to, and employed directly in the accomplishment of assigned operational missions and tasks.

Prime Location: The main storage location from which an item is normally issued. In some cases this may be the only location for that item. Additional stocks may be stored in secondary locations.

Priority: Work which takes precedence over all other work and requires immediate attention. Such work is usually necessary for the immediate protection of health, life, safety, security, or property.

Privately Owned Vehicle (POV): A vehicle owned by a person or business and not the government.

Procedure: The step-by-step method or way that the policy or plan is to be carried out.

Procedure Plan: A comprehensive narrative description maintenance and repair methods prepared by the Contractor.

Process: A series of actions or operations that achieve an end or result.

Procurement Appropriation (PA) (Formerly Designated PEMA): A continuing (multi-year) appropriation providing funds for procurement, manufacture, and conversion of major items of combat and supporting equipment including ammunition, aircraft, missile systems, weapons, and combat vehicles. These funds are used almost entirely by the Army wholesale logistics system and the items procured are issued as unfunded items to the Installations and units.

Programmed Replacement: A systematic process of replacing Government Furnished Property (GFP) determined to be unsuitable for its intended use.

Project: A single undertaking or task involving maintenance, repair, construction, or equipment-in-place, in which a facility or group of similar facilities are treated as an entity with a finite scope.

Project File: A complete historical record of project from inception to completion. Correspondence and other documentation pertinent to project shall be included in project files at all appropriate levels. This shall include inspection reports and memorandums for record pertaining to decisions resulting from discussions, meetings and telephone conversations.

Property: Terms "Real Property", "Government Property", "Army Property", and "Property" include all property under control of the Department of the Army. Property includes but is not limited to land, facilities, equipment, supplies, parts, and accessories thereto, and alteration or installation of any of the foregoing. Not included is property accounted for and owned by a non-appropriated fund activity.

Property Administrator: An authorized representative of the Contracting Officer assigned to administer the contract requirements and obligations relating to Government property.

Property Book: Record establishing formal accountability for certain classes of non-expendable and expendable property and detailing all property transactions (authorizations, gains and losses) on each line item (AR 710-2).

Property Responsibility: Property responsibility arises from possession of property or from the obligation of command or supervision of others who are in possession of property.

Provide: As related to a specified Contractor responsibility, this word means that the Contractor shall furnish and install the item or furnish the service.

Purchase Request and Commitment (DA Form 3953): A document submitted to Directorate of Contracting for purpose of initiating local purchase action of supplies and equipment; also used on a monthly basis, citing a fund limitation in the situation where continuous purchase of commercial items are made from one vendor. In this instance, DA Forms 2765 (indicating noun and quantity of items purchased) are used as supporting documents.

Qualified Person: One having adequate knowledge, and thoroughly conversant in the installation, construction, or operation of apparatus or equipment and hazards involved. One who possesses knowledge, skill, and ability to competently, effectively, and safely accomplish task.

Quality: The composite of attributes or characteristics, including performance of an item or product.

Quality Assurance (QA): Actions taken by the Government to inspect or check goods and services to determine that they meet or do not meet requirements of the contract.

Quality Assurance Evaluator (QAE): That person responsible for surveilling Contractor performance.

Quality Assurance Program (QAP): A planned and systematic pattern of actions necessary to provide adequate confidence that the services conform to established contractual requirements.

Quality Assurance Surveillance Plan (QASP): An organized written document used by Government for quality assurance surveillance. Document contains sampling/evaluation guides, checklists, and the performance requirements summary (PRS).

Quality Characteristics: Properties of a unit of product that may be evaluated to the specific requirements of a technical manual, drawing, specification, model, or other standard.

Quality Control Program (QC): Contractor's system to control the equipment, systems, or services so that they meet the requirements of the contract.

Quality Deficiency Report (QDR): The authorized means for users of Army equipment to report, either by message or SF Form 368 equipment faults in design, operations, and manufacture.

Quality Management: A planned and systematic pattern of actions necessary to provide confidence that material, data, supplies, services, and products conform to established technical requirements and achieve satisfactory performance.

Quantity Shipped (Quantity Supplied): Quantity indicated as having been shipped by a source of supply. It may or may not agree with actual quantity received in the shipment.

Quarters: Government quarters of a fiscal year run as follows: 1 Oct -31 Dec, 1 Jan - 31 Mar, 1 Apr - 30 Jun, and 1 Jul - 30 Sep.

Raceway: An enclosed channel designed expressly for holding wires, cables, or bus bars, with additional functions as permitted in the National Electrical Code. Raceways may be of metal or insulating material, and the term includes rigid metal conduit, rigid nonmetallic conduit, intermediate metal conduit, liquid-tight flexible conduit, flexible metallic tubing, flexible metallic conduit, under floor raceways, cellular concrete floor raceways, cellular metal floor raceways, surface raceways, wire ways, and bus ways.

Railroad Trackage and Appurtenances: Term includes all Government-owned general purpose railroads used for transporting personnel and material, and standard and narrow gauge tracks used for the transport of targets on artillery, tanks, and other moving target ranges. Term includes roadbeds, tracks, bridges, trestles, culverts, other drainage structures, signs, signals, safety devices, and all other features and items necessary to meet operational and safety requirements, and maintain efficient use of trackage at prevailing weight, speed, and density of traffic.

Random Sample: A sampling method whereby each service output in a lot has an equal chance of being selected.

Real Property: Land, improvement to land, buildings, structures, and items permanently affixed to either land, building, or structure.

Real Property Facility (RPF): A RPF is a separate and individual building, structure, utility system, or other real property improvement identifiable in the three-digit Category Codes (Cat Code) listed in AR 415-28. Examples are as follows: Buildings. One enlisted personnel barrack (Cat Code 721) represents a single RPF. Utilities. A single (physically or geographically identifiable) water supply, treatment and storage facility (Cat Code 841) represents a single RPF. Items such as wells or water storage tanks may be considered as component parts of this RPF.

Real Property Inventory (RPI): (RPI) is defined as inventory of land, improvement to land, buildings, structures, and items permanently affixed to either land, building, or structure.

Rebuild: Restoration of an item to a standard as nearly as possible to its original condition in appearance, performance and life expectancy.

Receiving: The process of planning for the arrival and the handling of inbound supplies.

Receiving Report: A certified document listing the supplies and equipment and quantities received from a vendor in compliance with a particular contract, purchase order, or delivery order.

Recoverable Resources: Materials, to include metal scrap, scrap lumber, crating materials, empty barrels, boxes, textiles, bags, waste paper, cartons, kitchen waste and similar materials which retain useful, physical, chemical, or other property which retains reclaimable, recycling, salvage or salable value. These items are generally not subject to property accountability.

Recurring Dispatch: Assignment of a vehicle for the use of a specific unit or section on a daily basis.

Recurring Maintenance: The work classification of all maintenance work that is required to be performed on a cycle, such as daily, weekly, monthly, quarterly, semiannually, annually, or biennially.

Recyclable Material: Waste material which can be transformed into new products in such a manner that the original product may have lost its identity.

Recycling: The process by which recoverable materials are transformed into new or usable products.

Redesigned Army DUERS Data System: RADDS is a monthly report required by Army regulation 11-27, requiring all installations to report on energy consumption, fuel inventory, pricing and related information to have a current status per quarter on energy performance.

Refuse: All garbage, ashes, debris, rubbish, and other similar waste material.

Regional Storage Management Office Contract: A contract between the Government (Headquarters, Military Traffic Management Command) and storage warehouse companies for long-term (non-temporary) storage.

Regular Duty Hours: Those continuous hours designated by the Installation Commander or Contracting Officer for organizations and activities to perform work functions on a recurring day-to-day basis.

Reimbursable Work: Work performed for other agencies that provide funds to cover the costs upon completion.

Reject: A unit of product determined by quality control inspection to be unacceptable for its intended use.

Reliability Rate (RR): A measure of equipment performance. It is computed by dividing actual hours by required operating hours. Operating hours are considered acceptable only when the desired output is being maintained.

Remedial Maintenance: See Minor Maintenance.

Renovation: To restore to new or original designed condition.

Reorder Inventory: An inventory of subsistence supplies accomplished each accounting period (monthly) on warehouse stocks to determine reorder requirements.

Repair: Restoration of a RPF to such condition that it may effectively be used for its designated functional purpose. Repair may be overhaul, reprocessing, or replacement of deteriorated component parts, materials, or equipment. Repair includes correction of deficiencies in failed or failing components of existing facilities or systems to meet current Department of the Army standards and codes where such work, for reasons of economy, should be done concurrently with restoration of failed or failing components. Corrective work may involve incidental increases in quantities and/or capacities. Replacement parts or materials which are more durable and provide longer life may be substituted for original parts and materials. Intent is to provide the most durable, energy efficient, low maintenance, cost effective items. Repair may involve replacement of installed equipment should repair and reuse of existing equipment be determined by Government to be economically unfeasible or otherwise undesirable. Replacement of a complete RPF is construction, not repair.

Replacement Furnishings: Items procured to replace items in existing inventory which have become uneconomically repairable or unsuitable for their intended use.

Request for Issue or Turn-in: Forms authorized to be used by a unit, organization, or activity to request supplies and to turn in supplies to a supply officer, accountable officer, or property disposal officer.

Requisition: An authoritative demand or request especially for personnel, supplies, or services authorized but not made available without specific request.

Respond: Contractor workforce is at work site with adequate personnel, tools, parts, and equipment, prepared to begin work within established time frames.

Responsible Individual: A person entrusted with possession of, or supervision over, Government property (AR 735-5 and AR 735-11).

Restricted Area: Those areas designated by the Commander that require control of personnel for security reasons and/or equipment for protection of personnel and property.

Restricted Items: Items which require a higher degree of security because of their desirability or vulnerability to pilferage.

Rolling Stock: Items of railroad equipment that are designed to operate on railroads.

Romex: A brand name for non-metallic sheathed cable. The formal name is NM.

Salvageable: A piece of equipment or material (for example, rail road ties, guard rails, washers, etc.) that, despite a flaw in a portion of the equipment or material, could be used in construction or provide a mechanical service, such as a dishwasher, with a reasonable level of repair or fixing in terms of cost. A piece of equipment or material shall not be considered salvageable if it contains a flaw that compromises the safety of the item (i.e.; wiring which may create a fire hazard) or prevents the item from performing its intended use (i.e.; a guard rail which does not meet National Highway Standards and could not sustain impact).

Sample: A sample consists of one or more service outputs drawn from a lot, the outputs being chosen at random. The number of outputs in the sample is the sample size.

Sample Size: Number of units of product or of outputs in sample.

Sampling Guide: The part of the surveillance plan which contains the information needed to perform a random sample.

Sampling Plan: A plan which indicates AQL, number of units from each lot which are to be inspected (sample size), and criteria for determining acceptability of the lots (acceptance and rejection numbers). This plan is used to develop sampling guide.

Sanitary Sewer: A sewer that carries liquid and water-carried wastes from residences, commercial buildings, industrial plants, and institutions, together with minor quantities of ground, storm, and surface wastes that are unintentionally admitted.

Scope: The extent or limits of the policy.

Second Destination Shipments: The movement of Army supplies and equipment worldwide, after receipt from the depot or manufacturer.

Self-Help: Performance of minor maintenance tasks by facilities occupants to maintain or improve living conditions and appearance of facilities.

Self-Help Program: Self-Help Program involves military personnel and occupants of family/troop housing in accomplishing limited maintenance and repair work and minor improvements.

Semi-Improved Grounds: Grounds that require periodic maintenance, of a lesser degree than improved grounds.

Senior Enlisted Bachelor Quarters (SEBQ): Buildings or a portion of a barracks designated for occupancy by: Senior bachelor enlisted personnel (e.g., E-7, E-8 and E-9) and authorized comparable grade civilian employees not eligible for assignment to family housing. Visiting senior enlisted personnel and authorized comparable grade civilian employees on official temporary duty (TDY), regardless of marital status.

Senior Grade Officer Family Housing: Housing assigned to and occupied by Colonels (0-6).

Senior Officers Quarters (Bachelor): Bachelor quarters for Chief Warrant Officers (CW3), Captains (CPT), and above.

Sensitive Arms and Ammunition: Arms and ammunition requiring a high degree of protection and control because of their vulnerability to theft and their potential for use in civil disturbances. Sensitive arms and ammunition usually have an unpacked unit weight of 100 pounds or less.

Sensitive Items: Those items identified in the AMDF by a Controlled Inventory Item Code (CIIC) 1,2,3,4,5,6,7,8,9,N.P,Q,R,Y, and night vision devices.

Sensitive Material: That which required other than normal handling (e.g., hazardous, pilferable, refrigerated, controlled substance, and classified).

Service: Conductors and equipment for delivering energy from electricity supply system to wiring system of premises served.

Service Based Costing (SBC): A comprehensive identification of common installation services and resources consumed to produce the services, and output measures/workload associated with services.

Service Call: Unscheduled maintenance work that encompasses equipment installation, maintenance, and repair, and support requirements which require less than $2,000 of effort (total labor and material costs) and that do not require separate job planning or scheduling.

Service Conductors: Supply conductors that extend from street main or from transformers to service equipment to premises supplied.

Service Drop: Overhead service conductors from last pole or other aerial support to and including splices, if any, connecting to service-entrance conductors at building or other structure.

Service Equipment: Necessary equipment, usually consisting of a circuit breaker or switch and fuses, and their accessories, located near point of entrance of supply conductors to a building or other structure, or an otherwise defined area, and intended to constitute main control and means of cutoff of supply.

Service Lateral: Underground service conductors between street main, including any risers at a pole or other structure or from transformers, and first point of connection to service-entrance conductors in a terminal box or meter or other enclosure with adequate space, inside or outside building wall. Where there is no terminal box, meter, or other enclosure with adequate space, point of connection shall be considered to be point of entrance of service conductors into building.

Service Order (SO): DA Form 4287. Directive for those minor maintenance, repair, and construction jobs for which work hours and other resources are such that detailed estimating and scheduling are not economically justified.

Service-Entrance Conductors, Overhead System: Service conductors between terminals of service equipment and a point usually outside building, clear of building walls, where joined by tap or splice to service drop.

Service-Entrance Conductors, Underground System: Service conductors between terminals of service equipment and point of connection to service lateral. Where service equipment is located outside building walls, there may be no service-entrance conductors, or they may be entirely outside building.

Serviceable: Property which is in condition for use. (Group A -ready for use; Group B - ready for use with minor repair).

Shall: The word "Shall" is used in connection with the Contractor and specifies that the provisions are binding.

Scheduled Maintenance: The recurring day-to-day, periodic, or scheduled work (labor and material) required to preserve and maintain an RPF in such condition that it may be effectively used for its designated functional purpose. Maintenance includes work done to prevent damage which would be more costly to restore than to prevent. Maintenance may include repairs made to a facility, equipment, and/or system which will ensure dependable operation of facility, equipment, and/or system. These repairs may also include replacement of parts or components necessary keep equipment and/or systems operating effectively.

Sludge Pump: A pump capable of pumping wet sludge.

Sludge: Accumulated settled solids deposited from sewage or industrial wastes, raw or treated, in tanks or basins, and containing more or less water to form a semi-liquid mass.

SMA Obligation Authority (OA): The maximum amount of SMA dollars that can be obligated by material category in a specified period of time.

Solar Photovoltaic System: Total components and subsystems which in combination convert solar energy into electrical energy suitable for connection to a utilization load.

Solid Waste: Garbage, refuse, sludge, and other discarded solid materials resulting from industrial and commercial operations and from community activities.

Solvents: A broad category of organic chemicals (those made of carbon, hydrogen, and oxygen) capable of dissolving other substances and forming a uniformly dispersed solution. They are a concern because many are toxic and sometimes ignitable.

Spare Parts: Spare parts and PLL as used in this contract are synonymous. Spare parts and PLL are defined as supplies on hand or on requisition to repair equipment owned by the Government and accomplish required services.

Special Test: AMC directed testing of new weapons' systems or equipment.

Spill Kit: A collection of materials maintained to respond to hazardous material spills and contains absorbent materials, neutralizing chemicals, and protective gear.

Spotted, Spotting: The movement of railroad rolling stock from one location to another, or placement of railroad rolling stock at a specific place on the railroad.

Standard: An acknowledged measure of comparison.

Standard Army Maintenance System (SAMS): An automated system which maintains maintenance work orders, labor and parts costs transactions, and work order changes. The system provides reports which are used as tools for management.

Standard Level of Service (SLOS): To standardize services to soldiers throughout the installations; to provide tenants the expected SLOS; to develop a cost mechanism for services; to negotiate TRADOC/installation OMA contracts.

Standard Time: A unit time value for the accomplishment of a work task as determined by the proper application of appropriate work measurement techniques.

Standby Items: Items of supply, excluding repair parts, which are needed to safeguard health, to insure uninterrupted operation of Installation facilities, or to prevent destruction of property and for which requisitioning objectives cannot be established either because of stockage restrictions or insufficient demand.

Standing Operating Orders (OWO's): Standing Operating Orders shall be issued on a DA Form 4283/4284 for plant operation, operator maintenance, and recurring preventive maintenance services or other services where specific work and manpower requirements are relatively constant and predictable in advance. OWO's will be issued by Contracting Officer following review of Contractor work plans and schedules.

Standing Operating Procedures (SOPs): A set of instructions covering those features of operations that lend themselves to a definite or standardized procedure without loss of effectiveness. The procedure is applicable unless ordered otherwise.

Starships: Trainee barracks, category code 72181. See PWS TEs 1-003 and 1-003a.

Statement of Charges (DD FORM 362): Used in the processing of monetary charges against officer, enlisted, and civilian personnel (AR 735-5).

Subsistence Supply Activity (SSA): An activity organized and equipped to request, receive, store, issue, and account for subsistence used by military units.

Subsistence Supply Officer (SSO): The individual responsible for the administration of the SSA function.

Supply Management, Army (SMA): A budget program of schedules, statements, and exhibits based on experience, forecasts, and variables in determining Installation requirements for supplies and equipment for a specific time period.

Support System: Collectively, those tangible logistic support resources required to maintain a materiel system in an operationally ready condition. It is developed with the materiel system and merged with the ongoing logistic systems upon production and deployment. The following elements of integrated logistic support (ILS) constitute the support system: support and test equipment, supply support, transportation and handling, technical data, facilities, and trained personnel. The other elements of ILS are the means by which the support system is developed and implemented.

Supporting Documents: As used in Section G2 "Invoice Payment and Reimbursement Vouchers", supporting documents includes records, documents, and other evidence, regardless of form or type, sufficient to properly verify costs claimed to have been incurred in performing this contract. Sufficiency will be determined by the KO.

Surfaced Areas: Term surfaced areas covers all graded, paved, or stabilized (other than grass) areas used for vehicular, aircraft, track vehicle, or pedestrian traffic such as roads, streets, service drives, walks, parking areas, open storage areas, and airfield paved areas, including base and sub base courses.

Suspended (Ammunition): The withdrawal of munitions items from either issue, movement, test, or use, with or without qualifications, due to a suspected or confirmed unsafe or other defective condition. (See TB 9-1300-385)

Suspense Date: The date established by the originator of a document or requirement by which completed action or a reply is expected.

Switchboard: A large single panel, frame, or assembly of panels on which are mounted, on face or back or both, switches, over current, and other protective devices, buses, and instruments. Switchboards are generally accessible from rear as well as from front and are not intended to be installed in cabinets. (See "Panel board.")

System: A group of separately authorized items of equipment or components assembled to establish a single functional unit required for the successful performance of a task or mission. Reportable items which are part of a non reportable system are considered as separate items. Non reportable items which are part of a reportable system are considered as subsystems. A system, as used in the resultant contract, includes: all mechanical and electrical equipment; supporting structures; pneumatic, electrical and mechanical types of controls; and all auxiliary equipment required to provide a specific function and output requirements.

Tables of Distribution and Allowances (TDA): A requirements/authorization document which prescribes the organizational structure, personnel, and equipment authorizations and requirements.

Technical Bulletins: Publications containing information, procedures, and techniques of a technical or professional nature relating to equipment and general subjects.

Technical Cataloging: Editing of supply requests to insure sufficient information is available to complete the transaction and process it through the automated system.

Technical Inspection (TI): An inspection made to determine extent of maintenance required to restore the item, or to determine if the unserviceable condition is the result of other than normal wear, or to classify the item as economically or not economically reparable.

Technical Manual (TM): Publications containing technical instructions prepared on various subject areas, such as communications/electronics fundamentals, painting, and welding.

Telephone Service, Classes of:

A. Class A-1 service is provided to conduct official business of DA with direct access to the DSN and commercial circuits and access to long-distance service.

B. Class A-3 service is provided for the transaction of official business of DA, and is confined to the Installation and local commercial area, and is restricted from direct or operator assisted access to the DSN, toll, or long-distance circuits. It will not receive long distance calls.

C. Class C telephone service (official-restricted) is authorized for the transaction of official Government business as required throughout this Installation. This service is restricted from accessing local commercial or toll, long-distance, and DSN circuits. It also will not receive long distance calls.

D. Class C-2 telephone service is the same as Class C except C-2 numbers will receive long distance calls.

E. All Fort Benning numbers can receive incoming DSN and long-distance calls.

Tenant: Unit or activity of a different command satellited upon another Installation for support. All support furnished these tenants is government reimbursable type work. Examples are: U.S. Army Reserve Units, USACIDC, Commissary, MEDDAC.

Test: Procedure of obtaining, examining, analyzing, and evaluating data to determine conditions or verify performance capability.

Test, Measurement, and Diagnostic Equipment (TMDE): Any system or device capable of being used to evaluate the operational condition of a system or equipment to identify and isolate any actual or potential malfunction. TMDE includes diagnostic and prognostic equipment and calibration test and measurement equipment, whether identifiable as a separate end item or contained within an end item or system configuration.

Threatened Species: Those plants and animals which are likely to become endangered within the foreseeable future throughout a significant portion of their ranges as listed by the United States Department of the Interior.

Toxicity: The quality or state of being toxic or the degree to which a poison is toxic.

Trainfire Target Holding Mechanism: Electric motor driven mechanical linkage devices which are series connected to secure and activate targets in 20 to 50 target batteries (groups) used in quick-reaction firing of semiautomatic and automatic individual weapons.

Training and Doctrine Command (TRADOC): Located at Fort Monroe, Virginia, the next higher headquarters for Fort Benning.

Training Cycle: That period when individuals undergo Basic Combat Training (BCT) or advanced individual training (AIT).

Training, In-Service: Programs of instructions on specific subject matter which is offered to the general work force of the installation. The contractor may request space for contractor personnel.

Transition Period: Time preceding resultant contract start work date and is provided to Contractor for the purpose of observing existing Contractor operation. In order to have a smooth transition from Contractor operation to Contractor performance, no training or instruction will be provided by Government personnel to Contract employees. Prior Contractor will remain responsible for performance of all work. Transition period shall be that 60 day period prior to resultant contract start work date and shall end at 0000 hours on day preceding resultant contract start date.

Transportation Agent: An agent or agents of the Government appointed by, and authorized to act in the behalf of the Installation Transportation Officer.

Transportation Coordinator: An individual authorized in writing by a unit or activity to submit requests for transportation support and to which any questions relative to the requested support should be directed.

Troop Construction: Construction, maintenance, and repair projects accomplished by engineer troop units at Installation.

Troop Movement: The transportation of groups of military personnel by land, air, or sea carriers. May include personal baggage and equipment.

Trusted Agent: Initiate and approve applications for their assigned contractors for CAC issuance.

Turbidity: (1) A condition of a liquid due to fine visible material in suspension, which may not be of sufficient size to be seen as individual particles by the naked eye but which prevents the passage of light through the liquid. (2) A measure of fine suspended matter, usually colloidal, in liquids.

Turn-in: Refers to equipment that users return to the DPW or DOL due to the fact that this equipment is obsolete or that better equipment is available for the user. Turn-in equipment may be salvageable.

Turnaround Time: The overall measure of the duration of the maintenance cycle. It covers the period of time from acceptance of a work order to close out (final inspection). It does not include time awaiting customer pickup. Expressed in days, it is completed as follows: Total number of maintenance days as shown on DA Forms 2407 divided by the total number of items as shown on DA Forms 2407. DA Forms 2407 used solely for technical or classification inspections are not considered when computing turnaround time.

U-Drive-It-Vehicle: A vehicle belonging to the NTV motor pool but dispatched to and driven by a member of the unit or activity requesting the transportation support.

Unaccompanied Personnel Housing (UPH): Housing provided for those unmarried, divorced, or legally separated officers and senior enlisted personnel permanently assigned to Fort Benning and those military and civilian personnel who are temporarily assigned to Fort Benning without dependents.

Unconstrained Requirements Report (URR): An unconstrained budget report, prepared 18 months prior to the target fiscal year, designed to influence the DOD budgeting process. This report is developed without regard to either historical constraints such as prior funding and performance, or current data such as available manpower, manpower ceilings, unavailability of other resources, and similar constraints.

Uneconomically Reparable Items: Unserviceable equipment that cannot be economically repaired.

Unimproved Grounds: Grounds other than improved or semi-improved.

Unit Maintenance Officer: That person within a unit assigned the responsibility for determining maintenance and repair requirements for the operational readiness of vehicles and aircraft and for technically advising the commander on maintenance related matters.

Unit of Product: The item inspected in order to determine its classification as "defective" or "non defective" or to count the number of defects. The unit of product may be a single article, a pair, a dozen, a gross, or a set of stated quantity. It may be measured in terms of a length, an area, a volume, a weight, or any other suitable measurement. The unit of product may be a raw material, a material in process, a component of an end product, or the end product itself. It may be an operation within a specified area, an administrative procedure, a punched card, a document, or any record.

Unit Priority Designator: A numerical code used to establish orders of priority in Army supply and maintenance operations.

Uniform: Presenting an undiversified appearance of surface, pattern, or color.

Unscheduled Maintenance: As used in this contract, the term "Unscheduled Maintenance" shall be synonymous with nonrecurring maintenance, and unscheduled repair; and all these terms shall be understood as applicable to categories as defined herein.

Unserviceable: Items that cannot be used because they are obsolete, worn, damaged, or otherwise not suited for purpose intended.

Unserviceable Item: An item which, through normal fair wear and tear, misuse, mistreatment or neglect, can no longer be used for its intended purpose.

Utilities: Electricity, gas, water, sewage disposal, and steam are types of utilities used by this Installation. They can be provided by the Government or private companies under contract.

Utility Marking: The process of locating and marking underground utilities and the process of performing site investigation to determine if an underground utility exists.

Utilization Equipment: Equipment which utilizes electric energy for mechanical, chemical, heating, lighting, or similar purposes.

Visiting Officers Quarters (VOQ): Quarters maintained primarily to provide temporary housing for visiting Officers, Warrant Officers, and comparable grade civilian employees.

Wastewater Effluents: Liquid sludge withdrawn periodically from sedimentation tanks or clarifiers and backwash water from filters are principle sources of wastewater from a water treatment plant.

Weapons/Equipment System Designator Code (W/ESDC): The W/ESDC is a two-character code given to equipment of supply use. The code is recorded on DA Form 2407 and on supply requests.

Wildlife Management: Practical application of scientific and technical principles to wildlife populations and habitants so as to maintain such populations essentially for ecological, recreational, and/or scientific purposes.

Will: The word "Will" is used to express a declaration of purpose on the part of the Government.

Work Center: A grouping of personnel using similar machines, processes, methods, and operations and performing homogeneous type work usually located in a centralized area. The term is used to identify a relatively small activity within a broad functional segment. Personnel within a work center perform work that basically contributes to the same end product or results and their duties are similar or closely related.

Work Day: 7:30 a.m. to 4:30 p.m. Mondays through Fridays, excluding legal public holidays.

Work Instruction: Instruction for individual tasks within a procedure.

Work Order: Unscheduled maintenance work that encompasses equipment installation, maintenance, repair, and support requirements which require more than $2,000 level of effort (total labor and material costs).

Work Request: A work authorization document grouped into one to three categories: Individual Job Order, Standing Operations Order, or Service Order. Each work request is managed by techniques and procedures designed specifically for that category.

Work-Years: The total hours in a work-year, as commonly used in staffing guides and manpower surveys, is 1776.

Working Hours: The hours designated by the Installation Commander for an activity to provide a product or service.

2.3 ACRONYMS

For the purpose of resultant contract, the acronyms in AR 310-50 and those listed below shall apply:

A/C	Air Conditioner
AAA	Army Audit Agency
AAR	After Action Report
ACALA	Armament and Chemical Acquisition and Logistics Activity
ACIFS	Automated Central Issue Facility System
ACO	Administrative Contracting Officer
ACOD	Actual Cost of Damage
ADA	Americans with Disabilities Act
ADP	Automated Data Processing
ADPE	Automated Data Processing Equipment
AFARS	Army Federal Acquisition Regulation Supplement
AFH	Army Family Housing
AFM	Area Fire Marshal
AG	Adjutant General
AGE	Aviation Ground Equipment
AHA	Alert Holding Area
AHA	Ammunition Holding Area
AIS	Automated Information System
AISSO	Assistant Information System Security Officer

ALMS	Automated Load Manifest System
ALSE	Aviation Life Support Equipment
AMC	Air Mobility Command or Army Materiel Command
AMCOM	Aviation and Missile Command
AMEDDPAS	Army Medical Department Property Accounting System
AMMO	Ammunition
AMS	Army Management Structure
ANS	American National Standards Institute
AO	Accountable Officer
AO	Army Oil Analysis Program
AOB	Approved Operating Budget
AOR	Area of Responsibility
APC	Account Processing Code
APHA	American Public Health Association
APIC	Army Performance Improvement Criteria
APOD	Authorized Point of Debarkation
APOE	Authorized Point of Embarkation
AQL	Acceptable Quality Level
AR	Army Regulation
ARR-MAR	Annual Recurring Requirements for Maintenance and Repair
ASE	Aircraft Survivability Equipment
ASHRAE	American Society of Heating, Refrigeration, and Air Conditioning
ASIMS	Army Standard Information Management System
ASIP	Army Stationing and Installation Plan
ASL	Authorized Stockage List
ASMIS	Army Standard Information Management System
ASP	Ammunition Supply Point
ASP	Activity Specific Plan
AST	Aboveground Storage Tank
ASTM	American Society of Testing and Materials
ATC	See USAFATC
AUEL	Automated Unit Equipment List
AVIM	Aviation Intermediate Maintenance

AVUM	Aviation Unit Maintenance
AWOL	Absent Without Leave
AWP	Annual Work Plan
AWWA	American Water Works Association
BCE	Basic Commercial Equipment
BCT	Basic Combat Training
BDFA	Basic Daily Food Allowance
BDU	Battle Dress Uniform
BER	Budget Execution Review
BFW	Building Fire Warden
BII	Basic Issue Item
BMAR	Backlog of Maintenance and Repair
BMC	Barracks Maintenance Contract
BMG	Budget Manpower Guidance
BMP	Barracks Maintenance Program
BOA	Basic Ordering Agreement
BOD	Biochemical Oxygen Demand
BTU	British Thermal Unit
BUR	Built-Up-Roof
BWICS	Benning Weapons Inventory Control System
CAA	Clean Air Act
CAA	Clean Air Act
CAD	Chemical Agent Detector
CALMS	Computer-Aided Load Management System
CAM	Chemical Agent Monitor
CAP	Central Accumulation Point
CAPS-W	Computer Accounts Processing System-Windows
CARC	Chemical Agent Resistant Coating
CBL	Commercial Bill of Lading
CBS-X	Continuing Balance System-Expanded
CCA	Central Collection Authority
CCISO	Cryptographic Items Serialization Officer
CCISP	Control and Cryptographic Items Serialization Program

CCP	Central Collection Point
CCTV	Closed-Circuit Television
CDA	Catalog Data Agency
CDL	Commercial Driver's License
CDR	Contract Discrepancy Report
CD	Contract Deliverable
CE/COE	Corps of Engineers
CECOM	Communication Electronic Command
CF	Cubic Feet
CFA	Call Forward Area
CFC	Chlorofluorocarbon
CFE	Contractor Furnished Equipment
CFM	Cubic Feet per Minute
CFP	Contractor Furnished Property
CFR	Code of Federal Regulations
CHMSCC	Centralized Hazardous Material and Substance Control Center
CID	Criminal Investigation Division
CIF	Central Issue Facility
CIIC	Controlled Inventory Item Code
CIIP	Clothing Initial Issue Point
COCO	Contractor-Owned, Contractor-Operated
COE	Center of Excellence
COEI	Component of End Items
COMPASS	Computerized Movement Planning Status System
COMSEC	Communications Security (Classified) Equipment
CONUS	Continental United States
COR	Contracting Officer's Representative
COT	Commercial Off the Shelf (software)
CP	Cannibalization Point
CP	Cathodic Protection
CP	Chlorinated Polyethylene
CPVC	Copper Polyvinyl Chloride
CPW	Center for Public Works

CRC	CONUS Replacement Center
CSDP	Command Supply Discipline Program
CSPE	Chlorosulfonated Polyethylene
CT	Current Transformer
CTA	Common Table of Allowances
CTO	Commercial Travel Office
CTS	Copper Tube Size
CVE	Combat Vehicle Evaluation
CWA	Clean Water Act
CY	Calendar Year
DA	Department of the Army
DA CIR	Department of the Army Circular
DA PAM	Department of the Army Pamphlet
DAO	Defense Accounting Office
DB	Database
DDC	Defensive Driver Card
DDD	Desired Delivery Date
DFARS	DOD Federal Acquisition Regulation Supplement
DFAS	Defense Finance Accounting Service
DFSCH	Defense Fuel Supply Center Handbook
DIC	Document Identifier Code
DIN	Dining
DISP	Defense Industrial Security Program
DISREP	Discrepancy in Shipment Report
DITY	Do-It-Yourself
Div	Division
DJAS	Defense Joint Accounting System
DLA	Defense Logistics Agency
DMO	Demand Maintenance Order
DMAR	Deferred Maintenance and Repair
DMMC	Division Material Maintenance Center
DOC	Directorate/Director of Contracting
DOD	Department of Defense
DODAAC	Department of Defense Activity Address Code

DODIC	Department of Defense Identification Code
DODSASP	Department of Defense Small Arms Serialization Program
DOIM	Director/Directorate of Information Management
DOL	Director/Directorate of Logistics
DOS	Days of Supply
DOT	Directorate of Operations and Training
DOT	Department of Transportation
DPM	Direct Procurement Method
DPS	Defense Printing Service
DPW	Directorate of Public Works
DRM	Directorate/Director of Resources Management
DRMO	Defense Reutilization and Marketing Office
DS	Direct Support
DS/GS	Direct Support/General Support Maintenance
DSCP	Defense Supply Center Philadelphia
DSN	Direct Signal Network or Defense Switched Network
DTS	Defense Transportation System
DVD	Direct Vendor Delivery
DX	Direct Exchange
ECOD	Estimated Cost of Damage
ECOR	Estimated Cost of Repair
EDRE	Emergency Deployment Readiness Exercise
EDS	Electrical Distribution System
EIC	Equipment Identification Code
EIP	Equipment In Place
EIR	Equipment Improvement Recommendations
EMCS	Energy Management Control System
EMD	Environmental Management Division
EMD	Environmental Management Division
EMIS	Executive Management Information System
EMT	Electrical Metallic Tubing
Enl	Enlisted
ENT	Electrical Nonmetallic Tubing

EO	Executive Order
EOC	Emergency Operations Center
EOH	Equipment on Hand
EPA	Environmental Protection Agency
EPCRA	Emergency Planning and Community Right-to-Know Act
EPCRA	Emergency Planning Committee Right To Know Act
EPD	Environmental Protection Division
EPDM	Ethylene Propylene Diene Monomer
EPS	Engineered Performance Standards
ERC	Equipment Readiness Code
ERPSL	Essential Repair Part- Stockage List
ES	Equipment Serviceability
FA&O	Finance and Accounting Office(r)
FAA	Federal Aviation Administration
Fac.	Facility
FAM	Family
FAR	Federal Acquisition Regulations
FAS	Fuel Automated System
FCL	Facility Security Clearance
FEDLOG	Federal Logistics Data (on Compact Disc)
FEMS	Facilities Engineering Management System
FEWR	Facility Engineer Work Request
FH	Family Housing
FHA	Federal Housing Administration
FHU	Family Housing Unit
FINSFDLR	Finance Code Table File
FMC	Full Mission Capable
FOD	Foreign Object Damage
FOD	Field Officer of the Day
FORSCOM	U.S. Army Forces Command
FSC	Federal Supply Classification
FTE	Full-Time Equivalent
FTX	Field Training Exercise
FUP	Facility Utilization Plan
FWT	Fair Wear and Tear
FY	Fiscal Year

GAO	General Accounting Office
GATES	Global Air Transportation Execution Systems
GBL	Government Bill of Lading
GC	Garrison Commander
GCCC	Government Charter Coach Certificate
GCCS	Global Command and Control System
GFCI	Ground Fault Circuit Interrupt
GFE	Government-Furnished Equipment
GFF	Government-Furnished Facilities
GFI	Ground Fault Interrupt
GFOQ	General Flag Officer's Quarters
GFP	Government-Furnished Property
GIN	Government-in-Nature
GO	Government Owned
GOCO	Government-Owned, Contractor-Operated
GOGO	Government-Owned, Government-Operated
GOPAX	Group Operational Passenger System
GOQ	General Officer's Quarters
GPM	Gallons Per Minute
GS	General Support
GSA	General Services Administration
GTN	Global Transportation Network
GTR	Government Transportation Request
GTS	Government Travel Service
HAP	Hazardous Air Pollutants
HAZCOM	Hazard Communication
HAZMAT	Hazardous Material
HFC	Hydrofluorocarbons
HHG	Household Goods
HID	High Intensity Discharge
HM	Hazardous Material
HMIS	Hazardous Material Information System
HP	Horsepower
HQDA	Headquarters, Department of the Army
HSC	Health Services Command
HSG	Housing

HSMS	Hazardous Substance Management System
HSO	Housing Services Office
HTW	High Temperature Water
HVAC	Heating, Ventilation and Air Conditioning
HW	Hazardous Waste
HWMP	Hazardous Waste Management Plan
IAR	Inventory Adjustment Report
IAW	In Accordance With
IBSO	Infantry Branch Safety Office
ICAO	International Civil Aviation Organization
ICP	Integrated Contingency Plan
ID	Identification
IDS	Individual Deployment Site
IDS	Intrusion Detection Systems
IFA	Installation Food Advisor
IFDEP	Integrated Facilities Data Entry Processing
IFS-M	Integrated Facilities System-Mini/Micro
IG	Inspector General
PWO	Individual Job Order
ILAP	Integrated Logistics Analysis Program
IMDG	International Maritime Dangerous Goods
IMO	Installation Maintenance Officer
IMSA	Installation Medical Supply Activity
IPBO	Installation Property Book Office/Officer
IPD	Issue Priority Designator
IPE	Industrial Plant Equipment
IPR	In Process Review
IPS	Iron Pipe Size
IRAC	Internal Review and Audit Compliance
IRPRS	Integrated Requirements to Purchase Request System
ISA/ISSA	Intra/Inter-Service Support Agreement
ISB	Infantry Branch Safety Office
ISCP	Installation Spill Contingency Plan
ISCP	Installation Spill Contingency Plan
ISM	Integrated Sustainment Maintenance
ISM-X	Integrated Sustainment Maintenance-Expanded

ISO	International Standards Organization or Installation Supply Officer
ISSO	Information Systems Security Officer
ITB	Infantry Training Brigade
ITGBL	International Government Bill of Lading
ITO	Installation Transportation Officer
ITV	In-Transit Visibility
JCAHO	Joint Commission on Accreditation of Healthcare Organization
JCS	Joint Chiefs of Staff
JFTR	Joint Federal Travel Regulation
JMPTC	Joint Military Packaging Training Center
JOPES	Joint Operational Planning and Execution System
Jr.	Junior
JRT	Joint Readiness Training
JRTC	Joint Readiness Training Center
JSIIDS	Joint-Service Interior Intrusion Detection System
JULLS	Joint Universal Lessons Learned System
JUMPS	Joint Uniformed Military Pay System
KO	Contracting Officer or Duly Appointed Representative
L&E CARDS	Labor and Equipment Utilization Records
LAAF	Lawson Army Airfield
Lab & Vet	Laboratory and Veterinary
LAC	Logical Agency Check
LAN	Local Area Network
LIF	Logistics Intelligence File
LIN	Line Item Number
LO	Lubrication Order
LOI	Letter of Intent
LOW	Letter of Warning
LPR	Local Purchase Requests
LRDF	Locator Record Data File
LTC	Lieutenant Colonel
LUS	Local Unique Systems M&S
	Maintenance and Services
MAC	Maintenance Allocation Chart
MACH	Martin Army Community Hospital
MACOM	Major Command

MAJ	Major
MARKS	Modern Army Record Keeping System
MC	Mission Capable
MCN	Management Control Number
MEDDAC	Medical Department Activity
MEL	Maintenance Expenditure Limit
MFP	Material Fielding Plan
MG	Major General
MGD	Millions of Gallons per Day
MGT	Management
MHE	Material Handling Equipment
MIF	Menu Item File
MIG	Metal Inert Gas
MIL-HDBK	Military Handbook
MIL-SPEC	Military Specification
MIL-STAMP	Military Standard Transportation and Movement Procedure
MIL-STD	Military Standard
MIL-STRIP	Military Standard Requisitioning and Issue Procedure
MMO	Materiel Maintenance Officer
MOA	Memorandum of Agreement
MOC	Maintenance Operational Check
MOGAS	Motor gasoline, regular unleaded
MOI	Memorandum of Instruction
MOTO	Mobile Home One Time Only
MOU	Memorandum of Understanding
MP	Military Police
MPD	Maintenance Priority Designator
MPH	Miles Per Hour
MRE	Meals, Ready to Eat
MRO	Material Release Order
MROCS	Material Release Order Control System
MSDS	Material Safety Data Sheet
MST	Maintenance Support Teams
MTMC	Military Traffic Management Command
MWO	Modification Work Order
MWR	Morale, Welfare, and Recreation

N+Hour	Notification Hour
NBC	Nuclear-Biological Chemical
NCO	Noncommissioned Officer
NDE	Non-Destructive Evaluation
NDI	Nondestructive Inspection
NEC	National Electrical Code
NEOPRENE	Chloroprene Rubber
NEPA	National Environmental Policy Act
NEPA	National Environmental Policy Act
NET	New Equipment Training
NFPA	National Fire Protection Association
NFPA	National Fire Protection Association
NFS	National Forest Service
NICP	National Inventory Control Point
NIIN	National Item Identification Number
NIST	National Institute of Standards and Technology
NIULPE	National Institute for the Uniform Licensing of Power Engineers
NLT	Not Later Than
NMC	Not Mission Capable
NMCM	Not Mission Capable Maintenance
NMCS	Not Mission Capable Supply
NOK	Next of Kin
NPDES	National Pollutant Discharge Elimination System
NRCS	Natural Resources Conservation Service
NRMB	Natural Resources Management Branch
NSN	National Stock Number
NSS	Non-STAMIS System
NTS	Non-Temporary Storage
NTV	Nontactical Vehicle
O&M	Operation & Maintenance
OA	Obligation Authority
OCIE	Organizational Clothing and Individual Equipment
OCONUS	Outside Continental United States
ODS	Ozone Depleting Substance
OF	Official Form
OMAR	Operations and Maintenance, Army Reserve

OMC	Optical Memory Card
ORF	Operational Readiness Float
OSC	Objective Supply Capability
OSHA	Occupational Safety and Health Act
OSUT	One Station Unit Training
OWO	Operation Work Order
PA	Procurement Appropriation
PAM	Pamphlet
PBO	Property Book Officer
PCB	Polychlorinated Biphenyl
PCS	Permanent Change of Station
Pers	Personnel
PIB	Polyisobutylene
PL	Public Law
PLL	Prescribed Load List
PM	Preventive Maintenance
PM	Pest Management
PMCS	Preventive Maintenance Checks and Services
POC	Point of Contact
POD	Point of Debarkation
POE	Point of Embarkation
POI	Program of Instruction
POL	Petroleum, Oil and Lubricants
POM	Preparation of Overseas Movement
POP	Performance Orientated Packaging
POV	Privately Owned Vehicles
PPCIG	Personal Property Consignment Instruction Guide
PPE	Personal Protective Equipment
PPM	Parts per Million
PPPO	Personal Property Processing Office
PPSO	Personal Property Shipping Office
PPTMR	Personal Property Traffic Management Regulation
PRC	Passenger Reservation Center
PRS	Performance Requirements Summary
PSA	Port Support Activity PSI
	Practical Scaling Index

PSO	Physical Security Officer
PT	Potential Transformer
PT	Physical Training
PVC	Polyvinyl Chloride
PMO	Preventive Work Order
PWO	Project Work Order
PWS	Performance Work Statement
QA	Quality Assurance
QAE	Quality Assurance Evaluator
QAP	Quality Assurance Plan
QASP	Quality Assurance Surveillance Plan
QC/QCP	Quality Control Program
QDR	Quality Deficiency Report
QQR	Quarterly Quality Reviews
QTB	Quarterly Training Brief
QTRS	Quarters (housing)
QTY	Quantity
R & U	Repairs and Utilities
RAC	Risk Assessment Codes
RADDS	Redesigned Army DUERS Data System
RATTS	Radiation Testing and Tracking System
RCRA	Resource Conservation and Recovery Act
RCT	Repair Cycle Times
RDD	Required Delivery Date
REC	Records of Environmental Consideration
Recr	Recreation
REPSHIP	Report of Shipment
REQ	Requisition
REQVAL	Requirements Validation
RFI	Request for Information
RFMSS	Range Facility Management Support System
RIN	Routing Instructions
RMP	Risk Management Plan, under CAA
RO	Requisitioning Objectives
ROD	Report of Discrepancy
ROP	Reorder Points

RPF	Real Property Facilities
RPI	Real Property Inventory
RPO	Radiation Protection Officer
RR	Reliability Rate
RSMO	Regional Storage Management Office
RTCH	Rough Terrain container Handlers
RXA	Reparable Exchange Activity
SAACONS	Standard Army Automated Contracting System
SAMS-I/TDA	Standard Army Maintenance System - Installation/Table of Distribution
SAP	Satellite Accumulation Points
SAP	Satellite Accumulation Point
SARA	Superfund Amendment and Reauthorization Act
SARSS-O	Standard Army Retail Supply System Objective
SAS	Satellite Accumulation Sites
SASE	Self Addressed Stamped Envelope
SASSO	Small Arms Serialization Surety Officer
SATCOM	Satellite Communications
SAV	Staff Assistant Visits
SBC	Service Based Costing
SCR	Silicon Controlled Rectifier
SDWA	Safe Drinking Water Act
SEBQ	Senior Enlisted Bachelor Quarters
SEDRE	Sea Deployment Readiness Exercises
SF	Standard Form or Square Footage
SFDLR	Stock Funding of Depot Level Repairables
SHPO	State Historic Preservation Office
SIDPERS	Standard Installation/Division Personnel System
SIMS-X	Selected Item Management System-Expanded
SIT	Storage in Transit
SIV	Staff Inspection Visits
SLC	Stockage List Code
SLOS	Standard Level of Service
SM	Service Member
SMA	Supply Management, Army
SMACNA	Sheet Metal and Air Conditioning National Association

SO	Service Order
SOO	Standing Operating Order
SOP	Standing Operating Procedure
SPBS-R	Standard Property Book System Redesign
SPCC	Installation Spill Prevention Control and Countermeasure
SPCC	Spill, Prevention, Control and Countermeasure
SPLC	Standard Point Location Code
SPVI	Subsistence Prime Vendor Interpreter
Sr.	Senior
SRA	Special Repair Activity
SRA	Stock Record Account
SRP	Soldier Readiness Processing
SSA	Supply Support Activity or Subsistence Supply Activity
SSN	Social Security Number
SSO	Subsistence Supply Officer
SSSC	Self-Service Supply Center
STAMIS	Standard Army Management Information Systems
STANFINS	Standard Financial System
STARFIARS	Standard Army Financial Inventory, Accounting, and Report System
SWEAP	Severe Weather Emergency Action Plan
SWP3	Storm Water Pollution Prevention Plan
SWPPP	Storm water Pollution Prevention Plan
TACOM	Tank-Automotive and Armament Command
TAEDP	Total Army Equipment Distribution Plan
TAMMS	The Army Maintenance Management System
TAV	Total Asset Visibility
TB	Technical Bulletin
TCACCIS	Transportation Coordinators' Automated Command and Control
TCN	Transportation Control Number
TDA	Table of Distribution and Allowance
TDR	Traffic Distribution Record
TDR	Transportation Discrepancy Report
TDS	Total Dissolved Solids
TDY	Temporary Duty
TE	Technical Exhibit
TGBL	Through Government Bill of Lading

TI	Technical Inspection
TIG	Tungsten Inert Gas
TIM	Technical Information Memorandum
TM	Technical Manual
TMDE	Test Measurement and Diagnostic Equipment
TMP	Transportation Motor Pool
TOE	Table of Organization and Equipment
TOPS	Transportation Operations Personal Property Standard System
TQAP	Total Quality Assurance Program
TRADOC	Training and Doctrine Command
TRANSCOM	Transportation Command
TSCA	Toxic Substances Control Act
TV	Title V Under Clean Air Act (CAA)
UBL	Unit Basic Load
UDI	U-Drive-It
UIC	Unit Identification Code
UL	Underwriters Laboratories
ULLS	Unit Level Logistics System
ULV	Ultra-Low Volume
UMMIPS	Uniform Materiel Movement and Issue Priority System
UN	United Nations
UNACC	Unaccompanied military personnel
UPC	Uniform Plumbing Code
UPH	Unaccompanied Personnel Housing
URR	Unconstrained Requirements Report
USACHPPM	United States Army Center for Health Promotion and Preventative
USAIC&S	United States Army Infantry Center and School
USAMEDDAC	United States Army Medical and Dental Activity Command
USAR	United States Army Reserves
USC	United States Code
USR	Unit Status Report
UST	Underground Storage Tank
VA	Veterans Administration
VMR	Volume Movement Request
VOQ	Visiting Officers Quarters
VQM	Vacant Quarters Maintenance

VTC	Video Teleconference
WAR	Weekly Activity Report
WARCO	Warranty Control Officer/Office
WO	Work Order or Warrant Officer
WPCF	Water Pollution Control Federation
WPS	Worldwide Port System
WTP	Water Treatment Plant
WW	Waste Water
WWTP	Waste Water Treatment Plant
ZPP	Zinc Protoporphyrin

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GOVERNMENT-FURNISHED PROPERTY AND SERVICES

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List of Technical Exhibits

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GOVERNMENT-FURNISHED PROPERTY AND SERVICES

3.1 GENERAL

The Government will provide the property listed in the Technical Exhibits and described herein as Government-furnished property. Government-Furnished Property (GFP) consists of facilities, equipment, supplies and materiel. The Government will also provide certain services and required utilities to the Contractor. The Contractor shall not use GFP, utilities or services for any purpose other than execution of work under this Contract. The Contractor shall confine all operations, including the storage of materials, on Government premises to areas authorized or approved by the KO.

3.1.1 CONTRACTOR ACCOUNTABILITY

3.1.1.1 Transfer of Accountability

The Contractor shall become accountable for GFP when the KO transfers it from the Government accountable records to the Contractor via DA Form 3161 or DD Form 1149.

3.1.1.2 Property Management Plan

Within ten working days after Contract start date, the Contractor shall provide a written Property Management Plan to the Property Administrator (PA) for review and approval IAW FAR 52.245-1 (CD 301R001). The Contractor shall ensure that the PMP includes the requirements of this Contract and meets property requirements contained in Federal Acquisition Regulation (FAR) 52.245-1. The Contractor shall update the plan as required, based on changes to property regulations and requirements.

3.1.1.3 Property Administration

The Contractor shall perform property administration in accordance with FAR 52.245-1 as supplemented by Department of Defense (DFARS) and Department of the Army (AFARS), AR 71-32, AR 710-2, DA Pam 710-2-1 & 2, and AR 735-5.

3.1.1.4 Reports of Government Property

The Contractor shall prepare and submit to the KO and the PA a quarterly report, due in Jan, Apr, Jul, Oct, listing all Government Furnished Property accountable under this contract for the purpose of modifying and updating the Technical exhibits 3-001to this contract. The contactor shall prepare and submit to the PA (NLT 30 Sep) an annual Report of Government Property, DD Form 1662.

3.1.1.4.1 Contractor will develop a program for implementation of Item Unique Identification (IUID) Registry requirement for Government property in the contractor's possession. This system of identification and reporting is expected to replace the Report of Government Property, DD Form 1662.

3.1.1.4.2 DoD Unique Item Identification is required for:

(1) All delivered items for which the Government's unit acquisition cost is $5,000 or more;
(2) Items for which the Government's unit acquisition cost is less than $5,000, when identified by the requiring activity as serially managed, mission essential, or controlled inventory;
(3) Items for which the Government's unit acquisition cost is less than $5,000, when the requiring activity determines that permanent identification is required; and
(4) Regardless of value–
(5) Any DoD serially managed subassembly, component, or part embedded within a delivered item; and the patent unit contains the embedded sub subassembly, component or part.

3.1.1.5 Removal of Property

The Contractor shall not remove GFP from the Installation or other supported areas without written approval from the KO or the PA

3.1.2 INVENTORY MANAGEMENT

3.1.2.1 Initial Inventory Procedures

The Contractor shall attend a phase-in GFP transfer and inventory meeting with the KO and the PA. The meeting will be scheduled by the Government prior to Contract start date.

3.1.2.1.1 Not later than ten (10) working days after Contract award date, the Contractor shall begin a phase-in 100% joint inventory in accordance with AR 710- 2, DA Pam 710-2-1 and AR 735-5. This inventory shall include, but is not limited to, facilities, to include keys; equipment and materiel items of work in progress (e.g., equipment on active maintenance work orders). On-hand supplies shall be inventoried at a summary level. Activities to be inventoried will be designated by the KO. This provision does not preclude prior inspection of GFP by the Contractor. The operational or conditional status of all Government Furnished Property shall be determined. Any item found to be broken or not suitable for its intended purpose shall be recorded. The KO or the PA and the Contractor shall certify as accurate the joint inventory. The Contractor shall keep the inventory listing current for the duration of the contract.

3.1.2.1.2 All discrepancies shall be noted and may be corrected as approved by one or more of the following methods at the Government's option. A Government representative will determine validity.

a. The Government may correct noted discrepancies or

b. The Government may require the Contractor to repair discrepancies subject to reimbursement by the Government or

c. The Government may withdraw the item of property

3.1.2.1.3 The Contractor shall sign custody records. The Contractor shall prepare, certify, and submit to the KO a detailed final Government Property Inventory Report within 60 calendar days after Contract start date (CD 301R002). The Contractor shall ensure that the Government Property Inventory Report is jointly approved by the Government and the Contractor. The Contractor shall maintain property custody records in a current status.

3.1.2.2 Periodic and Special Inventories

3.1.2.2.1 The Contractor shall establish and maintain records of all Government Furnished Property. The records shall be maintained in accordance with FAR 52.245-1. The records system will be reviewed by the KO or the PA. The Contractor shall make all changes to the record system required by the KO or the PA.

3.1.2.2.2 Periodic and Special Inventories. The Contractor shall conduct periodic inventories in accordance with their property management plan and special inventories as requested by the KO or the PA.

3.1.2.3 Contract Expiration or Termination Inventory Procedures

The Contractor shall attend a phase-out GFP transfer and inventory meeting with the KO and the PA. The meeting will be scheduled by the Government prior to Contract completion or termination date.

3.1.2.3.1 The Contractor shall prepare, certify, and submit a detailed final inventory report (jointly approved by the Government and the Contractor) to the KO and the PA. The inventory shall include the same data as required for the initial inventory.

3.1.2.3.2 During the final inventory, all Government furnished GFP shall be jointly inspected by the Government and the contractor. All discrepancies shall be noted and may be corrected by one or more of the following methods at the Government's option. Validity will be determined by a Government representative.

a. The Contractor shall correct any discrepancies prior to Contract expiration unless waived by the KO or the PA, or

b. The cost of repair may be deducted from the final payment to the Contractor.

3.1.2.3.3 At the completion of the Contract, the Contractor shall return all Government furnished property or property equal in type, kind, quality, and quantity of items as originally furnished by the Government and accepted by the Contractor. Government furnished property shall be in the same or better condition as when originally furnished, less fair wear and tear.

3.1.3 PROPERTY SHORTAGES AND DAMAGES

3.1.3.1 Lost, Damaged, or Destroyed GFP

The Contractor shall report all loss (lost, theft, damage or destruction) of Government furnished property to the KO and the PA within two working days after the discovery. Loss (lost, theft, damage or destruction) of sensitive property will be reported to the KO and PA immediately upon discovery. Sensitive property is property for which the theft, loss, or misplacement could be potentially dangerous to the public safety or community security, and which must be subject to exceptional physical security, protection, control, and accountability.

3.1.3.2 Reporting Requirement

The Contractor shall investigate and submit a formal report of shortage, lost, theft, damage, destruction, or excessive consumption of GFP to the KO and the PA as soon as the facts become known or when requested by the Property Administrator. The Contractor shall report the specific property affected, including NSN, other identifying codes, property nomenclature; the circumstances surrounding the loss, theft, damage, destruction, or excessive consumption; the estimated cost of replacing the property, if required; and the expected impact on the Contractors ability to provide contracted services.

3.1.4 SECURITY

The Contractor shall ensure the physical security of GFP and installed equipment in accordance with AR 190-51, AR 190-11 and the requirements of this Contract. The Contractor shall secure all GFP when not being used or occupied by Contractor personnel. The Contractor shall maintain an active security checklist for each individual facility as part of the Contractor's Physical Security Program.

3.1.5 CHANGE OF STATUS FOR GOVERNMENT-FURNISHED PROPERTY

When GFP is no longer required or suitable for intended use, or has reached the end of its economic life, the Contractor shall request disposition instructions and approval from the PA or the KO. Upon approval from the PA or the KO for repair or disposal, the Contractor shall process the items as directed. All property furnished under and all scrap resulting from this Contract shall remain the property of the Government.

3.2 FACILITIES

3.2.1 GENERAL

The Government will furnish or make available to the Contractor facilities; areas within facilities; equipment, tools, furniture, materials, and supplies therein as specified in

Technical Exhibit3-001. The Contractor shall not relocate activities or operational units within assigned facilities unless approved by the KO. The Government will provide the Contractor with a marked set of keys to facilities being used by the Contractor in the performance of this Contract. The Contractor shall appoint a primary and alternate building custodian for each facility.

3.2.2 JOINT USE

The Contractor shall share some facilities (as indicated in Technical Exhibit 3-001) with the Government. In addition, other circumstances may arise which necessitate the sharing of additional facilities or equipment identified as GFP. The KO will coordinate necessary changes and facility access with the Contractor. The Contractor shall not mark or affix any decals, emblems, or signs portraying the Contractor's name or logo to Government Facilities or Real Property. Contractor may have use of facilities that are jointly used by other installation activities. In such cases, the contractor shall take care to secure property and equipment they are using in such a manner as to comply with contract requirements for Government furnished property and facilities.

3.2.3 MAINTENANCE AND REPAIR

3.2.3.1 The Contractor shall initiate and provide all maintenance and repair of Government facilities and installed equipment therein in accordance with the requirements of this Contract. The Contractor shall not make any alterations to facilities without the written permission of the KO. Any unauthorized alterations or improvements may be at the Contractor’s expense and become the property of the Government; however, the Government may require removal or dismantling of such improvements at any time.

3.2.3.2 The contractor shall perform facility grounds keeping for all Government Furnished Facilities identified in Technical Exhibit 3-001. This includes all areas within a compound enclosure, and the area from the compound enclosure or facility exterior to the nearest road. The contractor shall maintain grounds areas up to 200 feet from the facility or half the distance from the nearest building if it is within the 200 foot area. Where applicable, the area on all sides of the facility or compound should receive grounds keeping services. In addition, the contractor shall perform all grounds keeping functions for all jointly occupied facilities. General grounds keeping functions necessary to maintain a professional facility appearance include but are not limited to:

a. Removal of all debris/trash in or around the facility or compound.

b. Removal of all debris/trash within the facility parking areas.

c. Disposal of all debris/trash within the facility, near the facility, or located within the facility compound in accordance with installation refuse policies and the refuse contract.

d. Removal of snow and ice from walkways.

e. Routine maintenance of the landscaping or vegetation adjoining the facility, compound or within the 200 foot areas. Maintenance actions for landscaping include mowing or trimming all ground vegetation, trimming shrubbery, and collection and disposal of leaves and trimmings. This covers areas not included in the DPW grounds maintenance contract.

3.2.3.3 The contractor shall perform facility housekeeping for all Government Furnished Facilities. The contractor shall maintain all facilities in a professional, safe, organized, and clean condition.

3.2.4 FINAL CONDITION

The Government reserves the right to reallocate and relocate assigned facilities during the term of the Contract. Upon completion or termination of this Contract, or during such reallocation/relocations, facilities shall be returned to the Government in the same condition as they were provided, less fair wear and tear.

3.2.5 GOVERNMENT ACCESS

KO approved Government personnel shall have access to all Government facilities used by the Contractor. Government personnel may perform unscheduled visits during normal working hours. When required, Government personnel will check in and out with the Contractor, who may, at his discretion, escort them throughout the facility.

3.3 UTILITIES

The Government will furnish utilities as currently installed.. All facilities do not receive the same utility services. Types of utility services furnished include electricity, gas, water, sewage, steam, fuel oil, and LP gas. The Contractor shall not change or modify any utility system or component without prior approval from the KO. The Contractor shall not connect any Contractor-furnished equipment/system without prior approval from the KO.

3.4 EQUIPMENT

The Government will make equipment listed in the Technical Exhibit 3-002 available to the Contractor in “as is” condition at the Contract start date, ,The Technical Exhibit 3-003 also contains the listing of property for Camp Merrill which is located in Dahlonega, GA. and the Ranger Camp in Eglin AFB, Fl. The GFE listings shall not be construed as being sufficient to meet the requirements of this Contract.

3.4.1 ACCOUNTABILITY

3.4.1.1 Authorization

The Contractor shall prepare the forms required for justification, deletions, and changes to DA-controlled property items authorized on the Public Works Tables of Distribution and Allowance (TDA). The Contractor shall prepare DA Form 4610-R in accordance with the guidance provided in AR 71-32. AR 71-32, Chapter 8, Section IV, Appendix E, E-3 (MR Procedures) describes the type of items requiring submission of documentation for TDA changes.

3.4.2 USAGE

3.4.2.1 Government Furnished Vehicles and Mobil Equipment

The Government will provide equipment as listed in Technical Exhibits.3-002 The Government will furnish limited quantities of basic commercial equipment such as backhoes, loaders, and road sweepers for use on work specified under this Contract. Subject to availability and KO approval, the contractor may use on a one-time basis, vehicles from the Installation Transportation Motor Pool (TMP) for work specified under this Contract. Contractor's proposal shall assume all vehicles needed for this contract will be supplied by the government. However, for comparison with the cost of government furnished vehicles, the contractor shall provide as a separate bid item in the proposal for the cost of the contractor supplying all vehicles that are needed to execute this scope of work.

This cost will include all cars and pickups (including utility trucks) but does not include cost for heavy equipment such as backhoes, dump trucks, motor graders, etc. The cost should also include fuel, oil and maintenance for those vehicles. The cost estimate should also identify the paragraph of the contract for each vehicle that will be used.

3.4.2.1.1 Log Books. The Contractor shall maintain a log book for each vehicle and each item of material handling equipment. The log book shall include the manufacturers requirements for operator level maintenance, record of routine maintenance performed (e.g., daily operational checks, brake and clutch exercises, system checks and exercises, hydraulic exercises, activations, and other requirements) and any repairs performed. The Contractor shall perform maintenance in accordance with the manufactures instructions or the operator's manual as appropriate.

3.4.2.1.2 Safety Checks and Maintenance. The Contractor shall perform safety checks and first echelon maintenance at the start of each work day and shall record the result in the logbook.

3.4.2.2 Calibration Standards

The Government will provide Calibration Services for Government furnished property. The Contractor shall:

a. Provide a point of contact for each major functional area to coordinate calibration services.

b. Transport equipment to and from the Government's calibration site.

c. Ensure all equipment requiring test, measurement, and diagnostic equipment (TMDE) services in the Contractor's possession is labeled to indicate condition and latest calibration date.

3.4.3 REPLACEMENT OF GOVERNMENT-FURNISHED EQUIPMENT. Replacement of equipment which becomes unsuitable for its intended use shall be accomplished only as directed by the KO. The Contractor is not authorized to replace equipment without the prior approval of the KO. Replacement equipment shall in all instances be obtained by the method approved by the KO. The Contractor shall not be entitled to reimbursement for any cost associated with the procurement of replacement equipment, unless accomplished in accordance with the procedures set out in the following subparagraphs.

3.4.3.1 Reduction of Government Furnished Property

The Contractor is hereby given notice that it is the Government's intent to reduce, where economically feasible, the amount of Government-owned property currently in inventory. Therefore, when an item of non-expendable GFE becomes unsuitable for its intended use during the period of the Contract, the Contractor shall notify the KO. The written notice shall include a listing of all work requirements which cannot be performed until the failure (or failing) equipment is repaired or replaced. The written notice shall also include a list of alternate means of accomplishing the work and, if necessary, a method for obtaining suitable temporary replacement equipment. Each alternative shall include a cost benefit analysis and the contractor's recommendation.

3.4.3.2 Instructions on Replacement

Upon notification of failing non-expendable GFE by the Contractor, the KO will determine on a case-by-case basis, with input from the Contractor, whether the GFE shall be repaired or replaced and, if the item is to be replaced, the method which shall be used to obtain the replacement (e.g. lease or rental, Government purchase) Upon request of the KO, the Contractor shall also prepare a comprehensive proposal addressing in detail the method or methods being considered by the KO for the permanent replacement of equipment. In all cases, the final determination of the method of equipment replacement shall rest with the KO. .

3.4.4 TRANSFER

3.4.4.1 Withdrawal of Government Furnished Property

The Government retains the right to withdraw any Government furnished property at any time during the performance of the Contract. The Government will provide a written 30-day notice of the impending withdrawal of property provided for use on this Contract. If such property is still required to perform work required in this Contract, the Contractor, with KO approval, shall replace withdrawn property with Contractor-furnished property.

3.4.4.2 Turn-In of Government Furnished Property

The Contractor shall coordinate and process the turn-in of excess or unserviceable property as directed by the KO.

3.5 MATERIAL

3.5.1 GENERAL

Material includes, but is not limited to, supplies, parts, subassemblies, raw materials, and other components and end items utilized to accomplish work or services described in this Contract. This paragraph and associated subparagraphs, unless otherwise directed by the KO, It is the Government's intent to provide all materials and supplies on hand at the time of contract start date for use by the contractor.

3.5.2 GOVERNMENT SUPPLY SOURCES

The Contractor shall requisition through the wholesale logistics system those Government-furnished supplies, parts, and materials required for the performance of this Contract which cannot be purchased via the Contractor's own supply channels.

3.5.3 FUEL

The Government will furnish operating fuels for Government-Owned, Contractor-Operated (GOCO) vehicles TE-3-004 and mobile equipment used in performance of services specified in this Contract. The Government will also furnish operating fuels for Contractor-Owned, Contractor-Operated (COCO) vehicles and mobile equipment, but only if the vehicles and equipment are designated for use exclusively for performance of work under this Contract. The Government will not provide fuel used for transport of Contractor employees between their domiciles and their work. The Contractor shall pick up and dispense Government furnished fuels and supplies. Contractor must only use fueling points that are provided by the government. Once fueling point will be located near the contractor's shop on main post and another near the contractor's shop at Harmony Church.

3.5.4 EQUIPMENT MANUALS, SUPPLY CATALOGS, AND FORMS

3.5.4.1 Technical Reference Libraries

The Government will furnish Technical Reference Libraries. These Technical Reference Libraries will contain copies of all Government-unique regulations and publications cited in this PWS that are currently available at the Installation. Regulations and publications are identified in Section C-6 of this Contract. The Contractor shall maintain an account through the Publications Distribution Office and shall request supplements, updates, and other publications requirements in direct support of this Contract. Requests for official publications shall be prepared in accordance with AR 25-30, DA Pam 25-30, DA Pam 25-32, and DA Pam 310-10. When requesting publications requiring classified access, the Contractor shall submit the request to the KO. The Contractor shall submit all other requests as required in AR 25-30, DA Pam 25-30, DA Pam 25-32, and DA Pam 310-10.

3.5.4.2 Operating Manuals and Parts Catalogs

Equipment operating manuals and parts catalogs presently maintained by the Installation for work specified under this Contract will be turned over to the Contractor prior to start of work. Inventory of parts catalogs will not be taken since the catalogs are a disposable item and become obsolete within several years after issue.

3.5.4.3 Record Drawings

The Contractor is advised that existing record drawings of Installation facilities are not 100% accurate in showing current as built or actual status. Since some record drawings may not be completely accurate, the Contractor shall perform field verification of drawing accuracy prior to performance of work specified in this Contract. The KO will furnish, upon request, a copy of existing and available record drawings of Installation facilities that will be essential to perform the services specified in the

Contract. When discrepancies are found the Contractor shall update drawings and submit the updates to the KO.

3.5.4.4 Commercial Publications

Commercial publications required in the performance of this Contract shall be obtained by the Contractor. Subscriptions to newspapers, magazines, and other periodicals require approval of the KO.

3.5.4.5 Forms

3.5.4.5.1 Initial Supply. Initial supply of forms will be made available to the Contractor by the KO at Contract start date. Samples of standard Government forms required for the fulfillment of this Contract will be available for Contractor examination in the Technical Reference Library. These forms and logs are subject to change periodically.

3.5.4.5.2 Subsequent Supply. The Contractor shall establish an account for subsequent forms requirements, prepare requisitions for the new stock on DA Form 17, Requisition of Publications and Blank Forms, and submit the requisitions to the printing office. The Government will provide forms to meet the identified need throughout the Contract period. The Contractor shall use Government blank forms to the maximum extent possible to accomplish the requirements of this Contract.

3.6 SERVICES

3.6.1 EMERGENCY SERVICES

3.6.1.1 Emergency Vehicles and Personnel

Emergency vehicles and medical personnel will be provided in an emergency, on the job situation, when a Contractor employee suffers a serious or life-threatening injury. Government facilities and emergency treatment will be provided in these instances as the first point of medical care. Transfer to other than Government medical treatment facilities shall be effected as soon as possible and as determined by attending medical authorities. Based on Medical Center policies in effect, charges may be made to the employee.

3.6.1.2 Fire Inspections

The Government will perform fire inspections and fire suppression services.

3.6.1.3 Military Police

The Government will provide the services of the Installation Military Police for on-Installation emergencies.

3.6.2 COMMUNICATIONS

The Government (or through its designated separate contract) will install, maintain, repair, and remove, as necessary, all Government-furnished telephones, telephone instruments, and telephone distribution systems.

3.6.2.1 Equipment

Contractor personnel shall not relocate Government-furnished telephone communications equipment, nor tamper in any way with the telephone distribution system. Whenever changes to communication services are required, to include changing locations of extensions and adding and deleting phone lines, the Contractor shall prepare and submit DA Form 3938, Local Service Request, to the KO for approval or disapproval. The Contractor shall obtain prior Government review and written approval before connecting or disconnecting any Contractor-furnished equipment to Government-furnished communications systems, lines, or equipment.

3.6.2.2 Usage

The Contractor shall not utilize the Government-furnished communication equipment or services for any action not directly associated with the requirements of this Contract.

3.6.2.3 Services

The following official voice and message services will be provided to the Contractor in accordance with paragraph 1 9.2 of AR 105 23.

3.6.2.3.1 Voice. The Government will provide telephone service for official use only through the Government system for on-post, Defense Switched Network (DSN), and local and long distance commercial calls. Official telephone service, consisting of Class A (access to off-post locations), Class A-A (access to Direct Signal Network (DSN)), Wide Area Telephone Service (WATS), and Class C (restricted to on-Installation locations) will be provided by telephone instruments located in the Government-furnished facilities at Contract start date.

3.6.2.3.2 Message. The Government will provide Automatic Digital Network (AUTODIN) subscriber service.

3.6.2.4 Billing for Unofficial Calls

Bills for unofficial calls will be forwarded to the Contractor by the KO, for payment to the Defense Finance and Accounting Service (DFAS). Charges may include Government processing and handling fees.

3.6.2.5 Radio Equipment Frequency

Radio frequency assignments and authorizations will be controlled and furnished by the Government. Communication equipment operations shall be in accordance with AR 25-1, AR 105-23, AR 105-24, ACP 121 Communications Instructions-General, ACP

121 U.S. SUP-1 (CONFIDENTIAL) Communications Instructions-General (U), and ACP 131 Communications Instructions-Operating Signals.

3.6.2.6 Pagers and Radios

The Government will provide adios to the Contractor, as needed, to communicate with range control and emergency services.

3.6.3 PRINTING

3.6.3.1 Services

Printing support will be rendered to the Contractor by the Defense Printing Service (DPS), Detachment Branch Office pursuant to AR 25 30, Army Integrated Publications and Printing Program, with local supplements thereto.

3.6.3.2 Requisition

The Contractor shall requisition printing services on DD Form 282, DOD Printing Requisition/Order. The Contractor shall submit the completed request to the KO for approval in accordance with DPS time requirements.

3.6.3.3 Printing Control Assistant

The Contractor shall appoint one printing control assistant and no more than three alternates using DA Form 1687, Notice of Delegation of Authority Receipt for Supplies, indicating the individuals authorized to pick up printed material and submit DD Form 202 to the KO.

3.6.4 TRASH REMOVAL

The Government will furnish dumpsters at Government determined locations. The Contractor shall place trash, excluding recyclable material and hazardous waste, in the dumpsters. Dumpsters for construction materials will be specifically designated. The Government will provide trash pickup from dumpsters. The Government will provide recyclable material pickup or drop off points.

3.6.5 GENERAL SERVICES ADMINISTRATION (GSA) CREDIT CARDS.

The Government will provide tactical vehicle and General Services Administration (GSA) credit cards on hand at Contract start date for issue to authorized users of Government vehicles. The Contractor shall centralize and control GSA credit cards and credit cards for tactical vehicles. The Contractor shall replace GSA credit cards as set forth by GSA Regional Bulletin FPMR 7-G-171.

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CONTRACTOR-FURNISHED PROPERTY

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CONTRACTOR-FURNISHED PROPERTY

4.1 GENERAL

4.1.1 PROVISION, STORAGE, AND REMOVAL

The Contractor shall furnish all property not specifically identified as Government-furnished in Section C-3 of this Contract necessary to comply with the requirements of this Contract. Such property includes, but is not limited to, facilities, equipment, supplies, repair parts, vehicles, data processing equipment, safety clothing, identification system camera and badges, and timekeeping system and facilities.

4.1.2 SEGREGATION OR COMMINGLING OF PROPERTY

Government property shall be segregated and kept physically separate from Contractor-owned property. However, when advantageous to the Government and consistent with the Contractor's authority to use such property, and with the written approval of the KO, the property may be commingled. Property may be commingled at the following times:

a. When the Government property is special tooling, special test equipment, or plant equipment which is clearly identified and recorded as Government property;

b. When scrap of a uniform nature is produced from both Government-owned and Contractor-owned material and physical segregation is impracticable, or scrap produced from Government-owned material is insignificant in consideration of the cost of segregation and control; or

c. When otherwise approved by the KO.

4.1.2.1 Removal of Contractor Property

Within 30 calendar days after completion or termination of this Contract, the Contractor shall remove all Contractor-owned vehicles, equipment, tools, supplies, materials, and other items from the Installation. The Government shall not be responsible for any Contractor-owned property left after Contract completion or termination. If the Contractor does not remove said property from the Installation within the stated time, the Government may take possession of or dispose of the property.

4.2 FACILITIES AND UTILITIES

The Contractor may request additional Government-furnished facilities (GFF) upon determining a need for them. The Contractor shall include specifications and justification for the needed facilities in the written request to the KO. If the request is approved, the KO will determine the availability of suitable facilities. If no facilities are found, the Contractor may submit proposed locations.

4.3 EQUIPMENT

The Contractor shall furnish all equipment and materiel, including motor vehicles, Automated Data Processing Equipment (ADPE), and administrative equipment, not furnished by the Government but required for performance of work required under this Contract. Equipment condition shall not relieve the Contractor of any responsibility to provide services as required in this Contract. Tools and equipment acquired by the Contractor at Contractor cost to supplement those provided as Government-furnished shall remain the property of the Contractor upon termination or completion of the Contract with Government payment of usage fee based on equipment depreciation - with Contractor retaining title, except as otherwise specified herein.

4.3.1 MARKING

All Contractor-owned equipment (That equipment which the Contractor has purchased or leased and which the Contractor uses and maintains to perform tasks under this Contract.

and vehicles, including Basic Commercial Equipment (BCE) and Material Handling Equipment (MHE), shall be clearly and permanently marked with the Contractor's name, telephone number, and the vehicle identification number on driver and passenger sides.

4.3.2 UNSERVICEABLE CONTRACTOR EQUIPMENT

The Contractor shall repair and/or dispose of inoperable and unserviceable Contractor-furnished equipment or items IAW the approved Property Administration Plan, as set forth in CD 301R001, Property Control System Plan.

4.3.3 PERSONAL PROTECTIVE EQUIPMENT

The Contractor shall provide all protective clothing, protective equipment, and coveralls required by OSHA standards.

4.3.4 EQUIPMENT PLAN

Within 30 calendar days after Contract start date, the Contractor shall submit an Equipment Plan to the KO for modification and approval of capital equipment (CD 403R001). The Contractor shall update the equipment plan when capital equipment requirements or status of capital equipment changes, and shall submit proposals for all acquisition of new capital equipment.

4.3.4.1.1 The equipment plan shall include all equipment that the Contractor shall provide in order to perform work required under this Contract. The equipment plan shall be presented by major functional area.

4.3.4.1.2 For each piece of equipment to be acquired, the Contractor shall include the following information:

a. Equipment Type

b. Equipment Size or Capacity

c. Year of Manufacture

d. Manufacturer

e. Cost

f. Acquisition or Lease Timing

g. NSN/SN/Model Number

4.4 MATERIALS AND SUPPLIES

The Contractor shall furnish all supplies and materials necessary to meet the requirements of the Contract, except those specified as Government-furnished in Section C-3. Supplies and materiel provided by the Contractor shall be of equal or better quality than those being replaced by the maintenance, repair, or servicing action. All supplies and materiel shall meet applicable codes or manufacturer's specification.

4.4.1 STOCK LEVELS

4.4.1.1 Stock Levels

The Contractor shall maintain a sufficient quantity of on-hand materials and supplies to perform all work required under this Contract. Any failure on the part of the Contractor to provide sufficient quantities and quality of supplies and materials within the specifications of the Contract shall not be cause for reduction in any service or performance. The Contractor shall also maintain stocks to ensure continuous operation of mission critical systems. Mission critical systems are those that are directly related to health care, safety, and mission accomplishment. The Contractor shall provide a system for rapid procurement of items whose usage levels do not require maintenance of on-hand stocks. This system shall include a list of vendors for each such mission- critical repair item, information on availability, and expected delivery times.

4.4.1.2 Bench Stock Levels

The Contractor may keep bench stock, including bench stock stored on repair vehicles, at not more than a 15-day stockage level and shall prevent acquisition and retention of large quantities of inactive stock.

4.4.1.3 Small Tools and Equipment

IAW approved property administration plan, the Contractor shall furnish all hand tools, shop tools, special tools, and testing equipment, not furnished by the Government but required for performance of work required under this Contract, including replenishment of any initial stocks. Tools and equipment utilized by Contractor personnel shall be of type and grade required for the work to be completed.

4.4.2 QUALITY OF MATERIALS

Contractor-furnished, Contractor-replaced, or Contractor-acquired materials, supplies, and replacement parts shall conform to the following quality priorities.

a. The Contractor shall meet specifications or standards listed in this Contract.

b. The Contractor shall meet specifications and standards contained in applicable Military Specifications.

c. The Contractor shall ensure the same or better kind, grade, and quality as the item being replaced.

4.4.3 EMPLOYEE IDENTIFICATION

The Contractor shall provide an identification badge for each employee performing services under this Contract. These badges shall meet the stipulations as set forth in paragraph C.1.2.6.5 of this document.

4.4.4 FIRE EXTINGUISHERS

Each fuel servicing unit shall be required to have two (2) 20-pound capacity, compatible dry powder fire extinguishers mounted, one on each side of the vehicle so that they are readily accessible to personnel standing on the ground.

4.4.5 COMMUNICATION SERVICES

The Contractor shall obtain those communication services required to perform work specified in this Contract that are not Government-furnished. Contractor communication services will be subject to standard monitoring requirements of the Government telephone network.

4.4.6 HOUSEKEEPING

The Contractor shall perform housekeeping in Government-furnished, Contractor-occupied facilities and areas. Areas include, but are not limited to, interior facilities and ancillary work areas. The Contractor shall maintain GFF, supplies, and materiel in a safe, organized, and cleans condition.

4.4.7 GROUNDSKEEPING

The Contractor shall perform grounds keeping around Government-furnished, Contractor-occupied facilities. The Contractor shall maintain grounds areas up to 200 feet from the facility or half the distance from the nearest building if it is within the 200 foot area. In addition, the Contractor shall maintain all outdoor areas assigned to the Contractor for Contractor use free of trash, debris, vegetation, and leaves; to include removal of snow and

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TECHNICAL DESCRIPTION OF WORK

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Technical Description of Work

The work requirements described in this section are categorized into 13 different Functional Areas (Paragraphs 5.2 to 5.13). Paragraph 5.1 below presents workload projections, work management specifications, and other information which impacts all functional areas. Any additional tasks are reported in C-1 and it's supporting Technical Exhibits.

5.1 WORK CLASSIFICATION AND MANAGEMENT

The Contractor shall establish an organizational approach that ensures separate, distinct, and responsive lines of communication, control, and operations are maintained to react to the often-competing demands that exist between the Facilities engineering and its military operations support requirement.

The following paragraphs: (1) define terms used to classify work; (2) provide information on the level of effort that will be required at Fort Benning in terms of workload and trade distributions; and (3) describe the way in which work will be received, managed, and documented. Also included is work to be provided to installation customers that are part of Installation Support Agreements (ISA), Memorandum of Understanding (MOU), and Memorandum of Agreement (MOA). A listing of these documents in affect is set forth in Technical Exhibit 5.1-004.

5.1.1 WORKLOAD

Workload data is presented throughout this Contract to enable the Contractor to forecast the level of effort that may be anticipated during the performance of the required Facilities engineering functions. Workload data that is specific to a particular Functional Area is generally presented under that Functional Area. The following are some examples.

- Roads to be swept, areas to be mowed, and schedules of preventive maintenance requirements will all be found at the appropriate Functional Area level, such as Functional Area 5.7.

Maintenance and repair tasks will be documented through Operation Work Orders (OWO's), Demand Maintenance Orders (DMO's), Project Work Orders (PWO's) and Preventive Maintenance Orders (OWO's). Since these orders contain workload data that generally applies to more than one Functional Area, they are described below.

5.1.1.1 Work Coordination

Work Scheduling. The Contractor shall schedule all Government approved and ordered work requests assigned them to Contractor personnel, and perform them in accordance with the provisions defined in this Contract. The Contractor shall combine and group recurring work, scheduled work, and unscheduled work in a logical, cost effective manner, and shall formulate schedules of work that minimize wasted Contractor personnel time.

5.1.1.1.1 Customer Interface. As a part of the scheduling process, the Contractor shall coordinate all work with the facility where the work is to be done in a timely manner to preclude life-threatening situations, loss or damage of product or property, and operational problems at the facility. In instances where access to facilities could not be obtained by the Contractor because of a customer's failure to meet at the prearranged time, the Contractor shall leave, at the work site or facility, a notice and phone number and annotate the work document to acknowledge the attempt to perform work. The Contractor shall return to the customer's location after subsequent coordination and appointment by the occupant. If a second attempt fails due to customer unavailability, the contractor shall leave a notification POC for completion/rescheduling of work, and shall report the instances to the KO. When reporting such instances, the Contractor shall provide the KO with a copy of the work document and pertinent facts related to the incidents, such as the number of attempts made to obtain access. If work requires scheduled or unscheduled interruption, disconnect, or cut off of any utility to or within the facility, or that a facility be vacated, the Contractor shall (a) notify customers, facility users and the KO 15 calendar days in advance of a scheduled interruption and (b) minimize disruption of the activity's operation.

5.1.1.1.2 Work Status Inquiries. The Contractor shall respond to work status inquiries from the KO and all customers. The Contractor shall advise the customer of the status of the work within eight hours of receipt of request. In response to work status inquiries, the Contractor shall provide, at a minimum, the date work was started or is scheduled to begin, the last activity performed on the job, the projected completion date, and any problems that have arisen and are impacting the performance of the work.

5.1.1.2 Work Standards

The technical specifications below define the quality of maintenance and repair to be accomplished under this Contract. When a standard is not directed, the Contractor shall perform required work to meet trade customs and practices and manufacturer's recommendations. The Government will be the final authority to ensure these quality standards have been met by the Contractor under this Contract.

5.1.1.2.1 Job Preparation. The Contractor shall be responsible for planning equipment and material requirements; obtaining proper tools; laying out tools, material, and equipment; setting up to begin work; cleaning and storing tools and equipment; the cleanup of the job site; and considering all work and costs associated with receiving, planning, and performing a job assignment.

5.1.1.2.2 Occupational Health and Industrial Hygiene. The Contractor shall establish medical surveillance, industrial hygiene, and individual protective equipment programs sufficient to meet requirements delineated in Occupational Safety and Health Act (OSHA) standards

5.1.1.2.3 Repair Area. If floors, cabinets, appliances, foundations, and other items are removed in order to gain access to the system component to be repaired, the Contractor shall return any repaired area to a condition comparable to the original construction after work is completed. Any Government or occupant-owned property damaged by the Contractor shall be repaired or replaced immediately in accordance with the provisions of Section C-3.

5.1.1.2.4 Disposal. The Contractor shall dispose of materials, to include, but not limited to, doors, hardware, and other salvageable material such as compressors, condensing units, motors, garbage disposals, appliances, scraps, and other items at the Defense Reutilization and Marketing Office (DRMO). All refuse generated by this Contract shall be legally disposed of in accordance with all local, State, Federal, and Installation environmental codes for disposal (including disposal of hazardous materials and waste. Refuse shall be disposed of on a regular basis in order to ensure a neat and orderly appearance of the Contractor's area of responsibility. Roll-offs shall be used under a separate contract unless refuse is too large. The burning of debris, waste material, and other salvageable items is prohibited on the Installation.

5.1.1.2.4.1 Inert Refuse. The Contractor shall dispose of inert refuse material, to include, but not limited to, trees, stumps, uncontaminated soil, asphalt, and environmentally approved concrete debris at the inert landfill in accordance with Installation and State permits. Prior to disposal, the contractor must coordinate with the DPW EMD Solid Waste Program Manager.

5.1.1.2.4.2 Compost Refuse. The Contractor shall dispose of compost refuse material, such as leaves, at the compost site in accordance with Installation and State permits. The contractor must coordinate with DPW EMD prior to disposal.

5.1.1.2.5 Cannibalization. Removal of component parts from any Government-furnished equipment (GFE) resulting in cannibalizing shall not be allowed unless approved in writing by the KO.

5.1.1.2.6 Utility Clearances. The Contractor shall mark utilities as described in Functional Area 5.4, Utility Systems Operation and Maintenance.

5.1.1.2.7 As-Built Drawings. The Contractor shall provide updated as-built drawings to the KO for each instance that work performed impacts construction not later than five working days after final inspection of the work (CD 501R001). The KO will review the drawings for approval and direct the Contractor to make corrections as necessary. In addition, when discrepancies in drawings are found during work the Contractor shall update drawings and submit the updates to the KO. Drawings shall indicate all changes and modifications incorporated into the work to include, but not limited to, location and depth of subsurface lines, manufacturer and model number of all major items of equipment, and the location and type of all materials used hidden from view. In addition, the Contractor shall submit with the as-built drawings any shop drawings that constitute part of the design.

5.1.1.2.8 Policing of Work Site. The Contractor shall remove from the work area, at the end of each day and completion of the job, all waste, material, and by-products resulting from work performed. The Contractor shall return usable material to a designated storage area for reuse.

5.1.1.2.9 Installation Qualified Recycling Program. The Contractor shall participate in the Installation Recycling Program. The Installation Recycling Program is a joint resources conservation effort managed and operated by the Directorate of Public Works(DPW) in accordance with local regulations. The Program encompasses the collection of items and materials that are reusable in their present form, such as pallets, and items that are suitable for recycling, such as paper, glass, plastics, cardboard and metal cans. The contractor shall coordinate with the DPW EMD Solid Waste Manager to set up and maintain a recycling program.

5.1.1.2.10 Facilities, Systems, and Equipment Identified for Future Replacement. The Contractor shall maintain and repair equipment as outlined in this Contract, unless otherwise directed by the KO regardless of whether specific facilities, systems, or equipment have been identified for future replacement by the Government.

5.1.1.2.11 New Components. New components shall meet current industry standards. The Contractor shall consider aesthetics (i.e., color, texture, quality) in material selection. If existing aesthetics cannot be matched, the Contractor shall submit variations to the KO for approval prior to installation.

5.1.1.2.12 Proper Operation, Use, and Care of GFE and Contractor-furnished equipment (CFE). The Contractor shall:

a. Be responsible for the proper operation, use, maintenance, and repair of the GFE and CFE.

b. Not cannibalize or modify any GFE, including customers’ equipment being repaired, without prior written approval of the KO.

c. Load test GFE and CFE material handling equipment (MHE) as required by TB 43-0142.

d. Provide trained and licensed personnel to operate GFE and CFE as required by Federal, State and local laws and regulations.

e. Ensure that a "ROAD TEST" sign indicating a road test is authorized each time a vehicle or piece of equipment is road tested.

5.1.1.2.13 Warranties. The Contractor shall determine as part of each work request (SO, WO, etc.) whether the equipment or material requiring repair or replacement is currently under a warranty. The Contractor shall not perform any work that voids a warranty. Additionally, the Government may periodically provide the Contractor lists of new equipment or materials under warranty to further reduce the chance of voiding a warranty.

5.1.2 FACILITIES ENGINEERING

5.1.2.1 Operation Work Orders (OWO's)

OWO's represent work that is performed on a recurring basis, such as the daily operation of boilers; daily testing of water; monthly inspections of lift stations; quarterly preventive maintenance of heating, ventilation, and air conditioning (HVAC) units (e.g., filter replacements); and annual cleaning of cooling towers. SOO work is described under SCHEDULED TASKS in Functional Areas 5.2 through 5.13 .Scheduled work requires the pre-planning of resources resulting from a known requirement and encompasses the scheduling and execution of specific tasks. OWO's will be approved on DA Form 4283-1-E and issued on DA Form 4284-1-E. Contractor execution of OWO's may be pre-approved by the KO following the review of Contractor work plans and schedules. All OWOs will be loaded in to GFEBs for work Generation.

Paragraph 5.1.2.3 contains a discussion of the number of labor hours associated with SOO workload.

5.1.2.2 Demand Maintenance Orders (DMOs)

DMOs represent work that is generally corrective in nature (e.g., repairs and replacements) or those related services that are not generally considered to be maintenance activities. DMOs, therefore, are not known in advance. Examples include minor electrical, carpentry, and plumbing repairs; locating utility lines; and trimming trees following a heavy storm. SO work is described under UNSCHEDULED TASKS in Functional Areas 5.2 through 5.13. DMOs are issued on a computer-produced facsimile or printout.
An DMO may be initiated by the KO, by a customer via Demand Maintenance Order Reception (see Paragraph 5.1.2.5.2 for more information), by a customer through direct contact with Contractor personnel, or by the Contractor upon identification of a need for a corrective action.

Based on historical data, the number of DMOs projected per year is approximately 60,000. The distribution of DMOs by priority and the distribution of DMOs by craft.

To be classified as SO work, the task will not be expected to exceed $2,500 total cost. The total cost includes labor, materials, supplies, or other non-labor cost. If the total cost of a single SO is expected to exceed these thresholds, the task will be classified as a PWO. A SO can involve multiple crafts such as both electrical and carpentry work and can involve more than one related task at a common work site. The Contractor shall determine SO types and priorities according to the priority classifications in Technical Exhibit 5.1-002. The Government reserves the right to reclassify DMOs as it deems necessary. All DMOs will be loaded in to GFEBs for work generation.

5.1.2.2.1 Emergency (Priority 1) DMOs. DMOs are classified as Emergency (Priority 1) DMOs when immediate action is required to eliminate life threatening or serious injury hazards to personnel, prevent loss or damage to

Government property, ensure security of sensitive Government property, restore essential services, or respond to command priorities. The Contractor shall respond to Priority 1 DMOs within 1 hour after being notified during normal business hours (0700 to 1700 hours, Monday through Friday) and within two hours during non-business hours. The Contractor shall complete all work on Priority 1 DMOs within 24 hours of the request. However, once the emergency is arrested but cannot be completed, the contractor may reclassify the SO priority to a Priority 2 or 3, in which case the Contractor shall complete the work in accordance with Priority 2 or 3 requirements stated below.

5.1.2.2.2 Urgent (Priority 2) DMOs. DMOs are classified as Urgent (Priority 2) DMOs when the failure in service does not immediately endanger personnel or property, but would soon inconvenience or affect the security, health, or well-being of personnel. Priority 2 DMOs are also those that correct a condition which could become an emergency, respond to a command emphasis, or aid an activity in accomplishing its mission. The Contractor shall respond and complete Priority 2 DMOs within fifteen calendar days of being notified. If further labor and material are required to complete the SO, the Contractor, after approval by the KO, or the KO may reclassify the SO as a Priority 3 SO.

5.1.2.2.3 Routine (Priority 3) DMOs. DMOs are classified as Routine (Priority 3) DMOs when the work does not qualify as Emergency or Urgent. The Contractor shall complete Priority 3 DMOs within 30 calendar days of the request.

5.1.2.2.4 Completed DMOs/OWO's Documentation. Completed DMOs/OWO's shall be signed by the authorized requester on the SO/SOO, certifying that work was satisfactorily performed. If the authorized requester is not available, an authorized alternate's signature is acceptable, with a comment stating the reason that the authorized requester is not available.

5.1.2.3 Project Work Orders (PWOs)

PWOs represent project-oriented work such as repairs, modifications, replacements, or installations that exceed the thresholds defined above for a SO. PWOs can vary significantly in nature and scope and may involve multiple crafts and locations. PWO work is described under UNSCHEDULED TASKS in Functional Areas 5.2 through 5.13. An PWO may be initiated by the KO, by a customer, or by the Contractor upon identification of a need for a corrective action. An PWO is initiated by completing DA Form 4283 and submitting it to the Government designated project estimator for approval, assignment of priority, and scheduling. The maximum cost for an PWO shall not normally exceed $100,000 for this Contract. Any PWO which is initially estimated to exceed $50,000 shall be submitted to the KO for approval before execution of work. Any increase to scope of a previously issued PWO, regardless of original dollar value, shall be submitted to the KO for approval before execution of work. This authority is not delegatable. The functional manager will be responsible for any technical reviews.

5.1.2.3.1 The contractor shall provide a monthly report to the KO that provides the following information on all PWOs (CD 501R007). PWO number, work order number, description of work location, date issued, est. completion date, final completion date, status of work (in-progress, completed, cancelled, etc.), beginning estimated total dollars, material cost, ending actual total dollars. The KO will monitor the estimated quantities issued to ensure the estimated quantities for PWOs under the contract are not exceeded for the contract year. The KO will provide timely notice to the functional representative(s) and the contractor when the estimated quantities for PWOs have been met for the contract year. When the total estimated quantities

for PWOs have been reached for the contract year, no new PWOs shall be initiated without approval from the KO.

5.1.2.3.2 The contractor shall provide a monthly report to the KO on all PWOs that provides the following information (CD 501R008). Detailed description of the scope of work; estimated completion date or number of days to complete and estimated cost.

5.1.2.3.3 Based on historical data, the total number craft hours for OWO's, PWOs and other work projected per year is approximately 225,000 hours with total material costs of approximately $8,400,000 and services $4,800,000 . The distribution of PWOs by craft is presented in Technical Exhibit 5.1-003

5.1.2.4 Conversion of SO Work to PWO Work

The same day that the Contractor anticipates $2,500 cost of labor and materials threshold will be reached during corrective operations, the Contractor shall request KO approval for completion of the SO and conversion to an PWO. In the case of a Priority 1 SO, the Contractor shall continue work until any emergency condition has been arrested. The Contractor shall, at that time, request KO approval for completion of the SO and conversion to an PWO.

5.1.2.5 Work Reception

5.1.2.5.1 Sources of Work. Work can be identified by the KO, Units, other Government personnel, tenants or residents of the Installation, or the Contractor. Contractor-generated work is the result of requests made by operators and maintenance personnel, who identified deficiencies during inspections, preventive maintenance, or other work performance.

5.1.2.5.2 Demand Maintenance Order Reception. The Contractor shall receive DMOs for unscheduled maintenance and repair of buildings, structures, roads, grounds, utilities, HVAC equipment, and all other support services on a 24-hour, 7-day per week basis. This will include Demand Maintenance Orders for pest control not covered by this contract. All DMOs will be generated in GFEBs for work Generation. The Contractor should first attempt to provide technical assistance over the phone or refer the caller to the Self-Help Store for those services such as replacing a washer in a leaking faucet, replacing a fluorescent light bulb, and other basic household repair tasks that individuals may be able to perform on their own. If the caller cannot resolve the problem, the Contractor shall obtain the necessary information to initiate a SO, and assign a priority to the SO in accordance with the definitions in Technical Exhibit 5.1-002.

5.1.2.5.3 Unless specific clarification is required or additional approvals are necessary because of a change in scope, work items shall be performed by the Contractor without further input from the Government.

5.1.2.5.2.1 Number of Service Calls. Historically, approximately 210 service calls are received daily (Monday through Friday) between the hours of 0745 and 1615, 10 service calls between the hours of 1615 and 2400, 4 service calls between the hours of 0000 and 0745, and 30 service calls over an average weekend. Approximately,

90% result in the generation of DMOs. Historically, Demand Maintenance Orders for pest control averaged 500 per year.

5.1.2.5.2.2 Clarification of Work Requirements. If the work requirements are unclear, the Contractor shall contact the customer to clarify work request requirements.

5.1.2.5.2.3 Duplicate Work Requests. If a request is made by more than one customer representative for the same work at the same location or for repair of the same equipment, this work request shall be treated as a single work request and documented as such for reporting purposes.

5.1.2.5.2.4 Phone List. The Contractor shall provide an updated phone list as necessary to the KO of on-call and key personnel who can be notified for urgent matters.

5.1.2.5.3 PWO Reception. The Government will receive requests for work on DA Form 4283-1 that would exceed the threshold limits of a SO. The Government will assign priorities to these work requests and will determine which PWOs will be issued to the Contractor. 5.1.2.5.4 Work Approval

5.1.2.5.4.1 SOO Work Approval. The Contractor may receive blanket pre-authorization instructions from the KO to execute OWO's.

5.1.2.5.4.2 SO Work Approval. The Contractor may receive blanket pre-authorization instructions from the KO to execute DMOs.

5.1.2.5.4.3 PWO Work Approval. On receipt or initiation of a proposed PWO, the Contractor shall provide a brief definition of the work and a cost estimate in writing to the KO. The KO may approve the work at this point. Upon further request by the KO, the Contractor shall prepare a complete cost estimate using standard costing methods. This cost estimate shall be based on an actual site inspection and shall include: a completed detailed scope of work, a bill of materials using appropriate materials catalogues, with quantities, unit prices, and single-line drawings as necessary. If an estimate is not approved, the KO may request that changes be made to the scope of work, scheduling, or timing; a revised estimate be submitted; or that the project be canceled. If acceptable terms cannot be reached, the Government reserves the right to accomplish work by other means. After approval of an PWO by the KO, the work shall be scheduled, assigned to Contractor personnel, and performed in the order as received, in accordance with the work scheduling practices defined. The Contractor shall coordinate with the Installation Safety Office if the PWO poses a safety threat. PWOs shall not commence without the approval of the KO.

5.1.2.5.4.3.1 PWO Extension and Funding Increase. After initial approval of an PWO, all contractor request for changes of PWOs to include suspense date, not to exceed (NTE), or changes in scope of work, shall be submitted to the COR with written justification. Upon approval by the COR, the contractor shall submit the PWO to the functional manager for final approval.

5.1.2.6 Work Documentation

5.1.2.6.1 The Contractor shall purchase, maintain, record, and be able to correlate to GFEB, all Demand Maintenance Order and work order documents, plus receiving reports showing what, when, why and where, on each transaction.

5.1.2.6.2 Edit. The Contractor shall check each document for accuracy against the Daily Transaction Register. The Contractor shall correct any errors detected and reprocess them into GFEB. The Contractor shall stamp each closed document with the date it clears, the cost run, and the date it is closed. The Contractor shall hold delivery tickets and Number 7 copy of DD Form 1155 for purchases on BPA until they clear the Ledger History Report. The Contractor shall cross check issues on the transaction register when they clear the Ledger History.

5.1.2.6.3 Labor and Equipment (L&E) Forms. As a means to document all work performed, the Contractor shall ensure that each Contractor employee working on OWO's, DMOs, or PWOs, completes a L&E Form (DA Form 4288-1) at the conclusion of each working day. The Contractor shall be responsible for entering all data from these forms into the GFEBs. The Contractor shall maintain a file of these L&E Forms for Contract duration, and be able to provide these forms to the KO upon request. The Contractor shall also enter L&E forms for all Government-in-Nature (GIN) employees using the form.

5.1.2.6.3.1 Work Initiation for OWO's, DMOs, and PWOs. At work inception, the Contractor shall input the following information for OWO's, DMOs, and PWOs into the GFEBs systems:

a) Work classification

b) Priority

c) Work/problem nature

d) Work request number

e) Date and time requested

f) Location of work requirement (i.e., building/facility/activity code)

g) Name and telephone number of requester

5.1.2.6.3.2 On-Going Documentation for DMOs and PWOs. The Contractor shall input the following information for DMOs and PWOs directly into the management systems:

a) Job status (e.g., open, closed, on-hold, or canceled)

b) Date and time arrived at the job site

c) Material order date (other than shop stock), if applicable

d) Estimated material due date (other than shop stock), if applicable

e) Name of person who performed the work

f) Date and time completed

g) Description of actual work completed

h) Actual labor hours by craft

i) Material description and quantity

j) Remarks

5.1.2.6.3.3 Completed PWO Documentation. Completed PWOs shall be signed by the authorized requester on the PWO, certifying that work was satisfactorily performed. If the authorized requester is not available, an authorized alternate's signature is acceptable, with a comment stating the reason that the authorized requester is not available. After completion of each PWO, the Contractor shall input the following data into the automated maintenance management system:

a) Name(s) of the individuals who worked on the job

b) The trade of each individual

c) Total hours expended

d) Labor cost

e) Total material costs expended

f) Date and time completed

5.1.2.6.3.4 Final PWO Documentation. Contractor shall document the final PWO in accordance with TE 5.1-005 (PWO Completion Template) and will be filed with the completed PWO.

5.1.2.6.3.4 Variance Report. The Contractor shall provide the KO with a monthly Variance Report TE 5.1-006 summarizing the PWOs completed 60 days prior to report date on the 1st of the month. The cost estimates that were proposed, the actual costs, material cost, the percent variation for each job, and the total dollar and percent variance for the month (CD 501R002).Variance report will be due the 1st day of each month or on the first duty day of the month should the 1st fall on a weekend or federal holiday.

5.1.2.6.3.5 Reports. The Contractor shall provide adequate controls and reports to ensure effectiveness, efficiency, and prompt response to all workload requirements and provide controls to ensure that costs incurred are properly reported. On a weekly basis, the Contractor shall provide the KO with an updated Backlog Report by area of responsibility designating the backlog of DMOs and PWOs for that time period (CD 501R003).

5.1.3 WORK CONTROL

5.1.3.1 Contingency Service Plans

Not later than 30 calendar days prior to Contract start date, the Contractor shall submit Contingency Service Plans (CD 501R004) organized by Functional Area to the KO for approval. The Contingency Service Plans shall describe services to be performed and their frequencies as reflected by this Contract, standard operating procedures, and proposed staffing. The Contractor shall conform with established military priorities for supply as specified in AR 710-2 and AR 725-50 in the development of the Contingency Service Plans. These plans shall include specific maintenance tasks and frequencies, specific instructions for the accomplishment of tasks, required special tools and equipment, personnel requirements, safety requirements, emergency procedures, personnel training requirements, and documentation procedures. The Contractor shall revise and resubmit the plan annually, or portions of the plan as required. Revised or updated portions of the plans shall be approved by the KO prior to implementation. The Contractor shall review, modify as necessary, and resubmit any Government plans already in effect. In addition the contractor shall develop programs and procedures for accomplishing competing mission demands as set forth in CD 501R006.

5.1.3.1.1 Installation Spill Contingency Plan (ISCP). The Contractor shall develop and submit a Spill Contingency Plan, in accordance with AR 200-1, detailing response actions to remove or mitigate the effects of accidental spills or discharges of oil and hazardous materials and wastes stored at Fort Benning/Camp Merrill.

5.1.3.1.2 Fuel Contingency Plan. The Contractor shall develop and submit a Fuel Contingency Plan. The plan shall provide for the allocation of fuel oil by the Contractor should an interruption occur. The Contractor shall perform all applicable tasks listed in the plan in the event of an emergency.

5.1.3.1.3 Contingency Operating Plans. The Contractor shall submit a Contingency Operating Plans detailing how the Contractor shall use his personnel, materials, and equipment to respond to and accomplish emergency maintenance or repair work during all regular and non-regular working hours. The plans shall include contingencies for the following emergency services:

- Natural disasters
- Emergency work for the security or protection of Government property
- Emergency work due to loss of air conditioning or heating for the medical facilities, data processing equipment, and other temperature critical facilities
- Emergency work due to loss of operation of refrigeration and dining facility equipment
- Emergency work due to a malfunction in any Military Police 911 fire alarm system or fire sprinkler system
- Dam safety plan contingency
- Natural gas curtailment
- Chlorine leaks
- Leak at propane peak shaving plant

- Leakage or spill of hazardous material
- Hydrant Flushing Plan

5.1.3.2 Critical Systems

The Contractor shall be responsible for keeping the following critical systems operational 24 hours a day, every day. The Contractor shall obtain KO approval before implementing changes to existing back-up systems or procedures.

- Electrical Systems
- Air Conditioning Systems
- Refrigeration Units/Cold Storage Systems (As required)
- Security Devices
- Boiler Plants and Heating Systems
- Plumbing Systems
- Water Production and Distribution Systems
- Wash Racks
- Wastewater Treatment and Collection Systems
- Standby Generators
- Fire Detection and Suppression Systems
- Structural Integrity and Safety of Buildings
- Steam, High Temperature, and Chilled Water Distribution System
- Road Network and Bridges

5.1.3.3 Energy Management and Control Systems (EMCS)

The Contractor shall monitor the EMCS in Building 470, 24 hours per day, seven days per week and be responsible for remote monitoring of all utilities which are linked to the EMCS and are on the Operators Task Sheet. The Contractor shall report the status of the EMCS linked systems upon request. Using the EMCS, the Contractor shall make adjustments to the HVAC systems to regulate room or building temperatures and control designated lighting around the Installation. Conserving Government resources is a criteria in the evaluation of the Contractor's cost management actions. The Contractor shall also turn ON and OFF systems during maintenance actions as necessary. The Contractor shall also be responsible for the maintenance and repair of the EMCS with prior approval of all repairs or modifications requiring KO approval. Energy policy is the sole purview of the Government.

5.1.3.3.1 The Contractor shall provide all necessary qualified and trained personnel to utilize and monitor the EMCS system on a 24 hour per day, seven days per week basis. The EMCS operators shall maintain a twenty-four (24) hour daily operators log; the operators log shall contain the following items at a minimum: shift changes, system alarms, notifications, and Demand Maintenance Order numbers (note: all Demand Maintenance Orders generated shall be based upon the data received from the EMCS systems reports).

5.1.3.3.2 The Contractor shall utilize the EMCS to monitor and to obtain data with which it shall effectively and efficiently operate plants and buildings on the EMCS system. This shall include inputting, monitoring, retrieving, and interpreting operational data (e.g., equipment status, temperature, on/off historical data, duty cycling, alarm status, etc.).

5.1.3.3.3 The Contractor shall forward to the KO copies of reports and printouts to include Failure List Reports, System Program Reports, Programmed Start/Stop Operation Schedule Reports, and Elapsed Time Program Operation Reports as requested (note: some reports may not be available based on the building design/application).

5.2 BUILDINGS AND STRUCTURES MAINTENANCE

Specific tasks and other requirements for Buildings and Structures Maintenance can be found in Functional Area 5.2.

5.3 DINING FACILITY EQUIPMENT MAINTENANCE

Specific tasks and other requirements for Dining Facility Equipment Maintenance can be found in Functional Area 5.3.

5.4 UTILITY SYSTEMS OPERATION AND MAINTENANCE

Specific tasks and other requirements for Utility Systems Operation and Maintenance can be found in Functional Area 5.4.

5.5 HVAC SYSTEMS OPERATION AND MAINTENANCE

Specific tasks and other requirements for HVAC Systems Operation and Maintenance can be found in Functional Area 5.5.

5.6 GROUNDS MAINTENANCE

Specific tasks and other requirements for Grounds Maintenance can be found in Functional Area 5.6.

5.7 SURFACED AREA MAINTENANCE

Specific tasks and other requirements for Surfaced Area Maintenance can be found in Functional Area 5.7.

5.8 RESERVED

5.9 SELF-HELP SERVICES

Specific tasks and other requirements for Self-Help Services can be found in Functional Area 5.9.

5.10 REMOTE CAMP OPERATIONS, MAINTENANCE, AND SUPPORT

Specific tasks and other requirements for Camp Merrill Operations, Maintenance, and Support can be found in Functional Area 5.10.

5.11 RANGE MAINTENANCE

Specific tasks and other requirements for Range Maintenance can be found in Functional Area 5.11.

5.12 CEMETARY SERVICES

Specific tasks and other requirements for Cemetery Services at Ft. Benning can be found in Functional Area 5.12.

5.13 CHILLER MAINTENANCE AND REPAIR

Specific tasks and other requirements for Cemetery Services at Ft. Benning can be found in Functional Area 5.13.

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BUILDINGS AND STRUCTURES MAINTENANCE

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Functional Area 5.2

BUILDINGS AND STRUCTURES MAINTENANCE

5.2.1 INTRODUCTION

The Contractor shall maintain and repair buildings and structures on Fort Benning as specified in this Contract. Buildings and structures include, but are not limited to: offices, storage facilities, work areas, recreation facilities, training facilities, loading docks, medical facilities, bridge structures, and airfield facilities. The Contractor shall also perform alterations and minor construction as requested by the KO. The Contractor shall coordinate all work with the requester in order to minimize downtime or mission delays. All work shall be performed by qualified personnel as determined by the KO and Contractor management, and in accordance with applicable laws, regulations, and documents cited in Section C-6. Technical Exhibits provide expanded information for this Functional Area.

5.2.2 SCOPE OF SERVICES

5.2.2.1 Work Area and System Description

The Contractor shall perform inspections, preventive maintenance (PM), and repair of buildings, structures, and associated property and equipment encompassing a variety of trades including, but not limited to: carpentry, mechanical, floor covering, sheet rocking, plastering, plumbing, electrical, painting, metal working, roofing, structural, and masonry. Buildings and structures covered under this Functional Area include, but are not limited to, the following:

- Administrative buildings
- Training and range facilities
- Maintenance shops
- Warehouses
- Storage and equipment sheds
- Wholesale and retail outlets
- Transportation facilities
- Above ground tanks
- Petroleum, oil and lubricant (POL) points
- Medical facilities and clinics
- Airfield and heliport facilities
- Bridge structures
- Transient housing
- Recreational facilities

- Classroom facilities
- Barracks
- Billeting
- Chapels
- Flagpoles
- Obstacle Courses
- Traffic Lights

The Contractor shall maintain all buildings and structures including those constructed before and during the contracting period.

5.2.2.2 Work Management and Control

5.2.2.2.1 Reporting Requirements. The Contractor shall prepare, submit, and maintain all records and reports as specified herein and in accordance with the Contract Data Requirements Lists (CDs) in Technical Exhibit 1-004.

5.2.2.2.2 Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.2.2.2.3 Historical Structures. The Contractor shall be aware of and take adequate steps to maintain the historical structures and archeological sites located throughout the Installation. The Contractor shall abide by all applicable Federal and State historical structure and archeological laws. In the event damage or disturbance of a historical or archeological site occurs, the Contractor shall report such damage or disturbance to the KO immediately. Technical Exhibit 5.2-004 provides more detail on historic buildings and structures at Fort Benning, and location maps can be found at post publications. All maintenance and repairs to historical structures shall be in accordance with paragraph 1.4.5 of this Contract.

5.2.2.2.4 Services and Special Requirements for Army Medical Department Activities. The Contractor shall only provide those maintenance and repair services for the medical clinic, dental clinic, and other associated buildings that are beyond the capability of the Martin Army Community Hospital (MACH) maintenance shop, as directed by the KO or upon receipt of a valid work document.

5.2.2.2.5 Water and Gas Systems. The Contractor shall ensure that water, chilled water and hot water, natural gas, steam, and fire sprinkler systems are available at pressures and flow rates in accordance with system design specifications in all applicable buildings and structures. When repairs requiring a system outage are planned, the Contractor shall submit email notification to the Operations and Maintenance (O&M) Division Chief detailing the type of outage, scheduled dates, scheduled hours, areas, building number, a listing of all customers with their concurrence or non-concurrence to the scheduled outage, and any special provisions related to the outage. This letter shall be submitted no later than 15 calendar days prior to the scheduled outage date. The Contractor shall obtain concurrence from the customer as to the period when an outage would cause minimum impact on operations. Functional Areas 5.4 and 5.5 provide additional details in regard to these systems.

5.2.3 SCHEDULED TASKS

The Contractor shall perform the tasks described below on a recurring or scheduled basis and record the work under approved Operation Work Orders (OWO's). The Contractor shall maintain all facilities and buildings at Fort Benning during the Contract period in a safe and operational condition as specified herein or as directed by the KO. Routine and preventive maintenance shall be performed to a standard that is highly resistant to abuse. The Contractor shall be responsible for performing and scheduling all inspections and preventive maintenance actions in accordance with the paragraphs that follow and with current Federal, State, local, and Army regulations. The Contractor shall develop and implement a comprehensive Preventive Maintenance (PM) program for the requirements listed herein. A list of required scheduled tasks is provided in Technical Exhibit 5.2-001. The Contractor shall repair all defects found when performing preventive maintenance. If any defect is above the level of work expected to be performed as preventive maintenance, the Contractor shall notify the KO to obtain guidance. Contractor will use GFEBs to generate Preventive Maintenance Tasks. All PM's should be loaded in to GFEBs for Work Generation.

5.2.3.1 Electrical Preventive Maintenance

The Contractor shall perform scheduled inspection, testing, calibration, maintenance, and other services for electrical systems and components as specified below.

5.2.3.1.1 Emergency Lighting. Preventive maintenance of auxiliary lighting units and systems shall be performed by the Contractor annually for wet cell batteries and for maintenance free batteries. The emergency lighting system shall be maintained annually by the Contractor. This maintenance shall include: testing all systems and units for correct operation, cleaning battery compartments, and spot painting as necessary; maintaining normal battery electrolyte level; tightening loose connections; maintaining correct charging voltage; replacing defective indicator lamps; replacing defective illuminating lamps; and adjusting and focusing swivel head lamps. The Contractor shall annually check the emergency power pack in fluorescent lighting fixtures and replace if defective.

5.2.3.1.2 Additional Electrical Equipment. On a monthly basis, the Contractor shall perform inspections and preventive maintenance on traffic light controls, sirens, generator transfer switches, and electronic starters. Perform traffic light preventive maintenance semi-annually on cameras and light assemblies.

5.2.3.1.3 Smoke Detectors. The Contractor shall perform PM on smoke detectors on a semi-annual basis. The Contractor shall report missing or defective smoke detectors, building number, and point of contact to the Directorate of Public Safety Fire Department by the end of the first week of the following month, and replace or repair them as appropriate. There are smoke detectors in all sleeping areas, all kitchens, and some equipment rooms and administration spaces , and all dining facilities.

5.2.3.1.4 Warning Sirens. Directorate of Intelligence and Security will perform a test of the entire warning system and will advise the Contractor of any necessary repairs to the warning system identified during its test. The Contractor shall initiate required repairs via an approved work document.

5.2.3.1.5 Vent Hood System Cleaning. On a semiannual basis, the contractor shall clean vent hoods to prevent the accumulation of grease and dirt and thus fires in mess halls on post and Dahlonega Ranger Camp. Buildings, 40,200, 1680, 2745,2943,3009,3110,3200,3235,3310,3340,3400,3500,4320,4702,5021, 5140 & 9139. Cleaning of ventilation systems must be performed as required in Chapter 11 of NFPA 96.

5.2.3.1.5.1 Reporting Requirements. Semiannually the Contractor shall prepare, and maintain records of preventive maintenance, repairs and cleaning. Affix tag or decal with date and initials of technician, located in a conspicuous place on or adjacent to vent hood inside mess hall.

5.2.3.2 Fire Alarm System Checks

The Contractor shall notify the Directorate of Public Safety Fire Department, the Staff Duty Officer, and residential occupants of any building or facility in which any unfinished work cause the fire alarm system to be inoperable at the close of business on that day.

5.2.3.2.1 Daily Checks. On a daily basis, the Contractor shall check the Mastermind Accountability system (MAS) to ensure correct operation. As part of this check, the Contractor shall check all printouts and integrate the system. The Contractor shall make any necessary repairs based on the printouts. All fire alarm work shall be performed in accordance with the latest edition of the National Fire Protection Association (NFPA) National Fire Alarm Code.

5.2.3.2.2 Annual Checks. The Contractor shall perform a visual and operational check of all transmitters, control panels, heat detectors, pull stations, and audible and visual signals on an annual basis at Fort Benning and at Camp Merrill. The Contractor shall perform all maintenance in accordance with manufacturer's recommendations to ensure that each unit is operating correctly.

5.2.3.2.2.1 Alarms. The Contractor shall perform preventive maintenance on fire alarms. In addition, the Contractor shall respond to all "trouble" calls identified by the Fire Department.

5.2.3.2.2.2 Fire Sprinkler Systems. The Contractor shall charge fire sprinkler systems on an annual basis. The Contractor shall be responsible for the fire sprinkler systems found in the buildings identified in Technical Exhibit 5.2-023.

5.2.3.2.2.3 MAS Fire Alarm Schedule. TE 5.2-026 is for working document only, there will be changes made during this contract. TE 5.2-026 shall be submitted every six months unless there are no changes made to the document. If there is no changes made just state on the Document" no changes made".

5.2.3.3 Miscellaneous Building Maintenance

5.2.3.3.1 Annual Daycare Center Inspections. The Contractor shall provide annual inspections, floor-to-ceiling, for all daycare centers on the Installation in accordance with AR 608-10. There are a total of five daycare centers located at Fort Benning:

Main Post (Bldg 1366), Tot Town (Bldg 1051), Sante Fe (Bldg 9242), Middle School (Bldg 1056), and School Age Services (Bldg 2653). Technical Exhibit 5.2-007 shows the required inspection worksheet.

5.2.3.3.2 Roof Truss Inspections. The Contractor shall provide inspections to the buildings and facilities as specified in Technical Exhibit 5.2-005 and in accordance with AR 420-70. The Contractor shall document inspections on the form contained in Technical Exhibit 5.2-006.

5.2.3.3.2 Build 4 Systems. Contractor shall perform preventative maintenance on two fountains in front of building 4 and the rainwater collection system, and the rooftop solar panels in accordance with manufacturer’s technical manual. This requirement shall be done under a SOO.

5.2.3.3.3 Crane inspections: This requirement shall be done under a SOO. On annual basis, contractor shall obtain a certified Engineer of cranes. Any deficiencies will be noted and Demand Maintenance Orders and/or 4283’s will be generated as appropriate.

5.2.3.3.4 Geothermal systems. The contractor shall perform preventative maintenance checks for the building 17 geothermal system in accordance with the manufacturer technical manuals.

5.2.3.4 Winterization and Dewinterization of Buildings and Other Systems

The Contractor shall winterize and dewinterize non-heated facilities and other non-heated systems, to include, but not be limited to, various unheated water-supplied areas at Fort Benning. In addition, the Contractor shall be responsible for identifying and correcting any deficiencies noted during the performance of this work that present a potential for freeze damage or water leak damage. Other winterizations shall be performed as unscheduled tasks upon receipt of a valid work document. The Contractor shall complete the winterization or dewinterization of buildings and other systems within ten working days after receipt of a notice to proceed. These buildings are limited to vacant wooden buildings in Harmony Church, as well as bath houses and range buildings, and selected hose bibs.

5.2.3.4.1 Latrines and Bathhouses. The Contractor shall: turn off electrical power, fuel supplies, and water supplies to all appliances and fixtures in the non-heated facilities and underground systems; disassemble parts and equipment required to prevent freezing and property damage (items shall include, but not be limited to: shower valves, commodes and urinal flush meters, and p-traps); remove drain plugs; open hot and cold lavatory and shower faucets; drain water heaters; pump out commodes; and winterize drinking fountains.

5.2.3.4.2 Dewinterization. Dewinterization shall include, but not be limited to: reassembling removed parts or equipment of building plumbing, electrical, and other equipment; turning on utilities including, but not limited to, electricity and water; draining and flushing antifreeze from mechanical systems; checking mechanical systems including, but not limited to, the plumbing system, heating system, and building equipment for proper operation; repairing leaks and malfunctions due to freeze related damage; and turning water on at underground shutoff valves. The Contractor shall make all necessary repairs related to freeze damage to place these facilities and systems in proper operation within ten working days after the receiving notification to proceed from the KO.

5.2.3.5 Grounding Inspection and Testing

5.2.3.5.1 Ammunition Grounding Points. The Contractor shall check the grounding system once every five years at 2900 block (Sandy Patch), Parks Range, Dahlonega Bunkers (5 big and one small), Dahlonega Parking Pads, Wood Road (Used by 3rd Bge), Ochille Rail Head Ammo Holding Area.

5.2.3.5.1.1 Resistance levels. The contractor shall check resistance levels every five years to ensure that the maximum allowable resistance for each grounding resistance is no more than 10 ohms.

5.2.3.5.1.2 Reporting Requirement for Grounding Point Checks. The Contractor shall complete a Ground Safety Check Forms TE 5.2-013 thru TE 5.2-022, detailing the locations of ground points, the type of ground point, the reading taken, any discrepancies found, and the corrective action performed.

5.2.3.5.2 Ammunition Supply Point (ASP) Grounding Points. The Contractor shall check the grounding system at the ASP on an annual basis. The Contractor shall arrange with the supply functional areas for entry into the areas.

5.2.3.5.2.1 Resistance Levels. The Contractor shall check resistance levels on an annual basis to ensure that the maximum allowable resistance for each grounding resistance is no more than 10 ohms.

5.2.3.5.2.2 Reporting Requirement for Grounding Point Checks. The Contractor shall complete a Ground Safety Check Form TE 5.2-013 thru TE 5.2-022, (CD 502R003) detailing the locations of ground points, the type of ground point, the reading taken, any discrepancies found, and the corrective action performed. The Contractor shall submit the completed Ground Safety Check Form to the KO within five working days of the completion of the ground point checks.

5.2.3.5.3 Grounding Points at Lawson Army Airfield. (Beginning with 2004), grounding points at LAAF shall be tested; A Demand Maintenance Order will be generated to replace or repair any grounding points whose resistance is above exceeded requirements.

5.2.3.5.3.1 Resistance Levels in Hangars. The contractor shall check resistance levels annually to ensure that the maximum allowable resistance for each grounding resistance is no more than 25 ohms at each point.

5.2.3.5.3.2 Resistance levels on aprons. The contractor shall check resistance levels every five years to ensure that the maximum allowable resistance for each grounding resistance is no more than 10,000 ohms at each point on the apron.

5.2.3.5.3.3 Reporting Requirement for Grounding Point Checks. The Contractor shall complete a Ground Safety Check Form detailing the locations of ground points, the type of ground point, the reading taken, any discrepancies found, and the corrective action performed.

5.2.3.5.4 Inspecting Lightning Protection Devices on Ranges. On an annual basis, test and visually inspect air terminals, roof conductors, and overhead conductors, down conductors, grounding system, and other components. Identify such deficiencies as breaks in continuity, galvanic corrosion, damaged or missing components, and violations of NFPA 780. Verify resistance to ground not to exceed 1000 ohms; measure and record.

5.2.3.5.4.1 The contractor shall check resistance levels annually to ensure that the maximum allowable resistance for each grounding resistance is no more than 1000 ohms at each point. TE-5.2-025.

5.2.3.5.4.2 Reporting Requirement for Grounding Point Checks TE-5.2-018. The Contractor shall complete a Ground Safety Check Form detailing the locations of ground points, the type of ground point, the reading taken, any discrepancies found. Submit form measurements and deficiencies to the KO and the Range Control Maintenance Coordinator.

5.2.3.5.4.3 Identify on the form, all new or missed connection(s) and submit to the KO.

5.2.3.6 Traffic and Building Sign Inspection

5.2.3.6.1 Traffic and Building Sign Inspection. The Contractor shall inspect building and road signage at Fort Benning for damage and/or missing signs as part of this Operation Works Order.

5.2.3.6.2 The Contractor shall select one of the twelve areas listed on TE 5.2-009 to inspect each month, such that each of these twelve areas shall be selected once on an annual basis. As part of this inspection, the Contractor shall fill out and maintain a checklist showing the roads and buildings inspected and location of damaged and/or missing signs. The Contractor shall select one of the twelve areas listed on TE 5.2-009 to inspect each month, such that each of these twelve areas shall be selected once on an annual basis.

5.2.3.6.3 Traffic and Building Sign, Repair and Replacement.

The Contractor shall comply with Installation Design Guide, TM 5-807-10.

5.2.3.7 Welcome Ceremony Signs

5.2.3.7.1 The contractor shall install and remove “Welcome Home Ceremony” signs whenever notified by the COR, DPW, or O&M Division representatives. This requires placement and removal approximately 4 times per year. See TE 5.2-010 for locations, TE 5.2-011, and 5.2-012 for route maps.

5.2.4 UNSCHEDULED TASKS

The Contractor shall perform the tasks described below when initiated through either a SO or an approved PWO. All work shall be performed in accordance with the standards set forth in Section C-5, unless otherwise directed by the KO.

5.2.4.1 Carpentry

The Contractor shall perform carpentry services for construction, alterations, extensions, modifications, maintenance, and repairs on buildings and structures. The Contractor shall perform carpentry functions in all facilities to prevent or correct structural or non-structural damage. Maintenance, repair, and alteration of buildings and structures shall include, but not be limited to: all structural features such as foundations, walls, doors, door stops, door closures, windows, roof framing, roofing, floors and floor coverings, porches, gingerbread fencing and trim, stairs, fixtures, hardware, exterior and interior painting, and glazing. The Contractor shall maintain and repair other equipment including automatic entrance doors (including surface-mounted concealed, electronic eye, etc.), wood fencing, foot bridges, overhead doors, playground equipment, guard and watch towers, grease racks, unattached loading ramps, wash racks, training facilities, monuments, grandstands and bleachers, elevated garbage racks, sidewalks, platforms, rail ties, hand rails, interior and exterior woodwork trim, identification signs, bulletin boards, display cases, cabinets, partitions, ceilings, and waterproofing seals and flashing not included in other crafts and shop work as specified herein. Where structures of the type shown in this section are identified elsewhere in the specifications, this section shall be used as a supplement to the maintenance responsibilities shown elsewhere.

5.2.4.1.1 Quality Standards. Interior and exterior finishes, trim, and decor shall be maintained to match existing appearance and finished in a like new appearance in accordance with the requirements specified in paragraph 5.1.1.2. All carpentry work performed shall be consistent with the construction and appearance of the existing facility or structure, unless otherwise directed by the KO. The Contractor shall follow national building practices in all cases.

5.2.4.1.2 Treatment of Materials. Carpentry work shall be planned and accomplished to offer maximum resistance to fungus, mildew, termites, water absorption, and all other deleterious effects of the environment. The Contractor shall follow standard industry practices when choosing the proper wood treated materials. All woods in contact with concrete and masonry shall be preserved and treated by pressure methods and so marked in accordance with the American Wood Preservers Institute Standards. Wood treated with waterborne preservatives shall be air and kiln dried to the moisture content specified for lumber and marked with the word "dry." Treated wood shall be used in all exposed locations that lack protection from the weather. If a cut is made in treated wood, the cut shall be brush coated with a wood preservative.

5.2.4.1.3 Carpentry Functions. The Contractor shall perform multiple carpentry functions. This shall include, but not be limited to, repairing, replacing, installing, and removing items such as: framing; finish work; floor and wall coverings; ceilings; roofing; porches; cabinets; and hardware on buildings, structures, and facilities. The Contractor shall fabricate and repair shipping boxes, rooms of various sizes, and utility buildings.

5.2.4.1.3.1 Forms. The Contractor shall lay out, fabricate, install, and remove a variety of forms for concrete placement such as headwalls, pads, sidewalks, stoops, and curbs. The Contractor shall fabricate, install, and remove (strip) forms for footings, foundations, headwalls, sidewalks, manholes, concrete pads, freestanding steps, and all other tasks requiring form work. The Contractor shall install shoring in trenches, ditches, and over digs to support dirt banks.

5.2.4.1.3.2 Framing. The Contractor shall lay out, install, replace, repair, and remove interior and exterior walls, floors, roofs, platformed towers, steps and handrails, ceilings, and roofs. Components include: studs, ceiling and floor joists, diagonal bracing, cripples, trimmers, headers, fire blocks, sole plates, top plates, piers, girders, rafters, ridges, and bridging. The Contractor shall ensure that all structural members that contribute to structural integrity using wood or steel framing have climatic and environmental factors, as allowed by local fire and building codes. The Contractor shall layout, install, replace, and remove insulation in roofs, walls, floors, and ceilings.

5.2.4.1.3.3 Siding. The Contractor shall install and repair exterior siding and trim to include wood, metal, plastic, and compositions.

5.2.4.1.3.4 Windows. The Contractor shall install and repair windows, including storm windows and screens, sashes, lifts, locks and latches, balances, anchors, and trim. Windows may be fixed or movable, and may consist of glass or glazing. The Contractor shall also install and maintain weather-stripping, threshold windows (sliding, double-hung, jalousie, fixed and casement, window frames, window arm, and window crank operators), standard and non-standard windows, rails, frames and casings for windows, transoms, and ventilators.

5.2.4.1.3.5 Doors. The Contractor shall install and repair doors including, but not limited to: entrance, storm, interior, handicap accessible, overhead, sliding vehicle, roof hatches, gear driven, overhead chain, electronic eye, and other special access doors. Doors may be manual, power operated, or automatic, and may include hardware, glass, locks, latches, stop operators, automatic sensors, eyes, activators, kick and push plates, jamb, trim, anchors, butts, and hinges. The Contractor shall also install and maintain doors and door related equipment including, but not limited to, interior and exterior metal and wood doors (single, double, Dutch, storm, screen, pocket, cabinet, overhead, sliding, bi-folding, latrine partition, and accordion), door frames, and hardware (door closures, hinges, locks and knobs, barrel bolts, dead bolts, head and foot bolts, floor closers, overhead closers, pivot sets, thresholds, concealed door stop/holders, automatic/manual flush bolt, manual door stop and holder open, hasps, panic hardware, door bumpers, and door mounted transoms and ventilators). Fabrication of specific doors may be required.

5.2.4.1.3.6 Wall Coverings. The Contractor shall layout, install, repair, replace, and remove interior and exterior wall coverings such as wood paneling, wainscot, marmite, corium, Formica, drywall, plasterboard, stucco, plaster wire, plywood, fiberboard, shiplap, acoustical tile, and masonite. Maintenance and repair of ceilings and walls shall include patching or replacing broken or cracked areas as a result of wear and tear and touch-up painting of affected areas.

5.2.4.1.3.7 Walls. The Contractor shall install, repair, replace, and remove interior walls and partitions (to include insulation, wall covering, trim, and finish) consisting of drywall, plaster, wood, and paneling, trim (window, door, floor, and ceiling), and finishes (paint, vinyl, wallpaper, and pre-finished paneling).

5.2.4.1.3.8 Ceilings. The Contractor shall lay out, install, repair, replace, and remove ceilings such as sheetrock, acoustical tile, suspended ceiling, plywood, tin, and Celotex. The Contractor shall repair ceilings (to include insulation, furring, suspension systems, and anchors) consisting of plaster (slick or rough finish), drywall, wood, cellulose, concrete, and acoustical materials.

5.2.4.1.3.9 Floor Coverings. The Contractor shall lay out, install, repair, replace, and remove interior and exterior subflooring, underlayment, and floor coverings, including rubber, asphalt, or vinyl tile; tongue and groove; plywood; wood parquet; quarry tile; and mastipave. The Contractor shall install, repair, replace, and remove non-skid materials on areas such as stairs, steps, ramps, and docks. The Contractor shall install and repair floor covering and underlayment to include carpet, resilient tile and sheet goods, wood, quarry tile, terrazzo, quarter round, baseboard, rubber base, and monolithic placed composition.

5.2.4.1.3.10 Finish Carpentry. The Contractor shall fabricate, install, repair, replace, and remove finish carpentry work such as accessories (map boards, fire extinguisher brackets, towel bars, toilet paper holders, robe hooks, medicine cabinets, shades, traverse rods, mirrors, grab bars), gingerbread trim, molding, window screens, signs, and columns. Other items include pigeonholes, bulletin boards, storage bins, bookcases, various types of molding, pictures, crowns, shoes, coves, bat, and other endangered or protected species houses.

5.2.4.1.3.11 Cabinet Making. The Contractor shall layout, fabricate, install, repair, and replace hand made items to include, but not limited to: fine cabinetry, racks, shelves, showcases, storage units, file boxes, service counters, workstations, special shipping crates, kitchen and bathroom cabinets, Formica counter tops, molding and trim, signs, picture frames, plaques, specially constructed office furniture (which may have intricate, precise, and fancy features such as curved and contoured surfaces, scroll work inlays, complex joints), and laminated plastics. The Contractor shall fabricate cabinets by cutting, bonding, fitting, assembling, shaping, and contouring surfaces and shall make scrollwork using precise and intricate joining and decorating with power shapers, mortise, planers, jointers, routers, lathes, and various other woodworking tools, accessories, and machines. The Contractor shall devise fixtures, templates, and jigs to hold or guide items in woodworking machinery.

5.2.4.1.3.12 Attached Structures. The Contractor shall replace or repair porches, patios, decks, carports, covered walkways, and areas to include floors, decks, steps, risers, hand or other railings, columns and other support, roof structures, and roofing. Structures shall be maintained in a safe and serviceable condition. The Contractor shall repair or replace flooring, paint, and patch concrete; treat wooden or concrete patios with appropriate wood or concrete preservative to prevent rot and mildew; and use treated materials and primer to ensure maximum life.

5.2.4.1.3.13 Glass. The Contractor shall replace broken or cracked glass with new glass in buildings, structures, and facilities, to include cutting and installing glass for doors and windows. Types of glass shall include, but not be limited to: plate glass, Plexiglas, smoked glass, safety glass, block glass, and thermal pane, and shall match existing materials in use. The Contractor shall lay out, design, cut, drill, and grind glass as necessary. The Contractor shall cut, replace, and remove glass on different types of doors, windows, mirrors, display counters, bulletin boards, cabinets, and any other components requiring glazing. The Contractor shall secure windows and doors when glass required is not available. The Contractor shall ensure that all glass debris is picked up and disposed of properly.

5.2.4.1.3.14 Glazing. The Contractor shall glaze interior and exterior single paned doors and windows such as glazed aluminum sash, steel sash, channel aluminum, wood sash, and beaded aluminum. The Contractor shall use glazing compound, clips, angles, beads, splines, moldings, weather stripping, and spacer strips. The Contractor shall replace glass in doors and windows with glass of the same thickness.

5.2.4.1.3.15 Government-Owned Fencing. The Contractor shall perform the layout, fabrication, installation, repair, replacement, and removal of fencing.
Installations and repairs shall be accomplished to offer maximum resistance to fungus, decay, mildew, termites, water absorption, and other deleterious effects of the environment. Numerous fences that are on Government property or abut

Government property are privately owned and are not the responsibility of the Contractor.

5.2.4.1.3.16 Handrails, Steps, Ladders, and Machinery. The Contractor shall install, replace, repair, remove, and anchor handrails, steps, ladders, and machinery.

5.2.4.1.3.17 Weapon Racks and Security Safes. The Contractor shall install, replace, anchor, and remove weapon racks and security safes to wood, concrete, metal, block floors, and footers. The Contractor shall anchor security safes in accordance with the physical security guidelines.

5.2.4.1.3.18 Decorations. The Contractor shall lay out, fabricate, repair, replace, and remove various types of displays such as reindeer sleighs, wreaths, Christmas and Easter Nativity scenes, and other decorations for holidays, parade floats, and special event booths. A list of special events and operations for Fort Benning can be found in Technical Exhibit 5.2-003.

5.2.4.1.3.19 Beams and Columns. The Contractor shall repair, paint, or replace wood, concrete, and metal beams and columns as necessary to ensure structural soundness and mitigate rot, insect damage, and fire damage.

5.2.4.1.3.20 Steps. The Contractor shall maintain and repair wood, metal, and concrete steps, and all unattached steps often found on ranges as necessary to ensure their safety and structural integrity. Repair and maintenance shall include, but not be limited to: installing tread, applying wood preservative, repairing or replacing rotted wood, painting steps, replacing safety strip on tread, and tightening treads as necessary.

5.2.4.1.3.21 Ramps. The Contractor shall ensure that wood, metal, and concrete ramps are structurally sound and safe to use. The Contractor shall repair or replace wood; treat for rot; patch concrete; paint; install treads on ramps; and treat loading, unloading, mechanically operated, and access ramps with anti-slip abrasive coating. The Contractor shall perform all work on ramps in accordance with the Americans with Disabilities Act (ADA).

5.2.4.2 Roofing

The Contractor shall provide roofing services for all buildings and structures. These requirements will be charged to a OWO. These services include installing, repairing, replacing, and removing various types of roofing, gutters, and downspouts. On an annual basis as part of this PM, roofs will be inspected and all loose debris will be removed and gutters and downspouts will be cleared. This requirement shall be done on a SOO. The Contractor shall flash and re-tin or replace ventilators as necessary to repair damage and stop or prevent leaks. The Contractor shall cut roof penetrations, install roof jacks, and ensure that roof penetrations are sealed to prevent any kind of moisture leakage. In the course of roofing work, the Contractor shall also install and repair rafters, studs, sills, plates, joists, floors, subfloors, panels, siding, sheathing, building paper, insulating materials, dry and wood walls, acoustical tiles, step treads, roof drains, storm drain systems, and security attachments. Types of roofing systems (including flashings, penetrations, insulations, vapor barriers, and associated components) to be worked on shall include, but not be limited to: gravel asphalt, rolled, monoform, shingle, single ply membranes, tile, slate, clay tile, coal tar pitch, bituminous, raised rib metal, metal roofing, shingles, roof tiles, roll roofing, and foam.

5.2.4.2.1 Types of Flashing. The Contractor shall use various types of flashing, such as gravel stop, roof edging, capping, counter flashing, and gravel stop fascia with the appropriate roof applications in such a manner as to limit damage to Government property and minimize personal risk. All metal work shall be in accordance with Sheet Metal and Air Conditioning National Association (SMACNA) standards.

5.2.4.2.2 Temporary Repairs. The Contractor shall make temporary repairs to damaged roofing, as necessary, to keep facilities watertight until permanent repairs can be accomplished. The Contractor shall perform all temporary repairs in a safe manner so as to not place Government property or personnel at risk.

5.2.4.3 Locksmith Services

The Contractor shall install, maintain, adjust, and repair locks, safes, locking systems, latches, file cabinets, panic devices, electrical door locks, and strikes of different makes, sizes, and shapes. The Contractor shall set combination locks on vaults, open locks, and make new keys by code for buildings. The types of locks to be serviced include mortise, cypher, XO-7& 8 Mas Hamilton, arms room, and all historical locks up to current state-of-the-art locks such as electronic. The Contractor shall provide lock-out services for all Government vehicles, including key duplication and the re-keying of tool and storage boxes. The Contractor shall fabricate minor parts from raw stock and install, maintain, repair, clean, re-pack, and adjust all makes and types of door closures. The Contractor shall order and maintain an inventory of all materials necessary to complete the work order. Locksmith personnel must have security clearances equivalent to Government locksmiths to assure restricted access to areas containing secret and top secret materials as further specified in Section C-1. The Contractor must have a working knowledge of the National and State Life Safety Codes to assure physical security as well as handicapped accessibility.

5.2.4.3.1 Response Time. The Contractor shall provide 24 hour emergency services. The Contractor shall respond to all lockouts as a Priority 1 DMO. The Contractor shall also have the capacity to secure all soldier quarters, sheds, and other facilities in the event of deployment.

5.2.4.3.2 Key Control. The Contractor shall comply with Key Control practices in Section C-1 of this Contract. All requests for keys, re-keying, and change of locks shall be performed upon receipt of a valid work document. Upon completion, the Contractor shall document all work in the software system to provide key control tracking and historical records. The Contractor shall back up the system every three months and maintain a copy. All furnished locks shall fit into the existing master core system.

5.2.4.3.3 Keys. The Contractor shall make keys for all facilities and equipment on Fort Benning. All replaced locks shall have the security specifications equivalent to existing locks. New keys shall match existing keys to maintain a minimal number of keys to operate a housing unit or Government building as well as to enhance key control. Additionally, the Contractor shall re-key padlocks upon receipt of a valid work document. The Contractor shall provide technical and re-key services to the Military Police (MP) and Criminal Investigation Division (CID) in cases of theft and burglaries.

5.2.4.3.4 Safes and Containers. The Contractor shall install, repair, recombinate, and open all Government security containers and safes to include built-in vault doors, built-in safes, and free standing safes and containers. Drilled safes and containers shall be repaired to operable states following all repairs and shall maintain their original fire and burglary rating.

5.2.4.3.4.1 X-07,08, and 09 Mas-Hamilton Safes. The Contractor shall install X-07 locking systems and document the installation on an X-07,08, and 09 Lock Installation Form. All documentation shall be given to the TRADOC Security Officer within three working days of work completion (CD 502R004).

5.2.4.3.4.2 Modes of Safe Operation. The Contractor shall adhere to the mode of safe operation determined by the Physical Security Officer. The Contractor shall install and set the initial combination for the three modes of operation (three dial, two person integrity, and supervisor/subordinate).

5.2.4.3.5 Other Services. The Contractor shall install, repair, and maintain indoor viewers, door scopes, antique locks, and panic locks. The Contractor shall grind down useable keys, secure metal hinges on doors, plane doors with locks, stamp keys, remove logged in keys, and remove padlocks (both combination and keyed) from doors and other items. Additionally, the Contractor shall have the capacity to open, service, and repair all high security locking devices to include the Arms Room and other high security areas.

5.2.4.4 Masonry

The Contractor shall provide fabrication, minor construction, maintenance, and repair services for projects using brick, concrete, masonry block, cinder block, tile, piping, cement, mortar, marble, grout, and sealant traditionally associated with the masonry trades. Masonry work shall include, but is not limited to: the repair, replacement, and maintenance of foundations, fluid containment walls/barriers, retaining walls, walls, floor slabs, and chimneys (including deteriorated brick and tuck pointing); and the diversion of water from exterior walls, extending downspouts, and similar items. The completed work shall be consistent with the construction and appearance of existing facilities or structures.

5.2.4.4.1 Structural Components. The Contractor shall fabricate, maintain, and repair foundations, basement walls, and floor slabs to keep them sound and free of leaks. The Contractor shall correct conditions contributing to water in basement areas. The Contractor shall repair and replace underground foundations, seal utility entrances, and install sump basins. This work shall include diverting water from exterior walls at ground level, filling low spots, and extending downspouts as required.

5.2.4.4.2 Chimneys. The Contractor shall fabricate, maintain, and repair chimneys. Chimney work shall include replacing all deteriorated brick with like material, tuck pointing, and unplugging.

5.2.4.4.3 Floor Covering. The Contractor shall install and repair all types of ceramic and quarry tile, and install all types of shower stalls, walls, and floors. The Contractor shall lay out, install, repair, replace, and remove various types of ceramic tile (such as bull nose, sanitary base, sanitary quarry, base bull nose, and field); sanitary base corners on baseboards, floors, and walls; and other types of flooring (VCT, linoleum, wood, battleship, etc.). The Contractor shall repair and replace shower curtain rods, soap dishes, ceramic hand pulls, towel bars, and various ceramic fixtures.

5.2.4.4.4 Brick and Block Laying. The Contractor shall lay blocks and bricks to install all types of doors, fences, and windows in structure openings. The Contractor shall perform all types of masonry work as required on an emergency basis to maintain the security of the structure.

5.2.4.4.5 Concrete Work. The Contractor shall fabricate forms for steps, ramps, approaches, footers, storm drain manholes, concrete curbing footers, catch basins, sump basins, grease traps, concrete curbing, sanitary manholes, traffic islands, and retaining walls. The Contractor shall pour and finish concrete slabs, pavements, sidewalks, and stairs, to include placing reinforcements and other embedded items.

5.2.4.4.6 Walls and Ceilings. The Contractor shall fabricate, maintain, and repair all ceilings, walls, and plaster formations to match surrounding areas. This work shall include patching and replacing broken or cracked areas, and touch up painting affected areas.

5.2.4.4.7 Monuments and Memorials. The Contractor shall repair and maintain monuments, masonry decorations, or fixtures as required. This shall exclude pieces of equipment requiring allied trade work such as tanks and artillery pieces. Technical Exhibit 5.2-002 provides a list of monuments and memorials that the Contractor shall maintain on Fort Benning.

5.2.4.4.8 Swimming Pools. The Contractor shall perform masonry maintenance and repair services on Installation swimming pools to include, but not be limited to: re-plastering walls with correct replacement material, repairing concrete aprons and decks, replacing ceramic and quarry tile trim, and repairing vertical and horizontal expansion and control joints to prevent or control leakage of water.

5.2.4.5 Painting

The Contractor shall paint the interior and exterior of all buildings, structures, and facilities covered by this Contract. Painting shall be accomplished for appearance and for surface protection. The Contractor shall paint items such as walls, ceilings, railings, decks, floors, roofs, window frames, doors, doorframes, moldings, handrails, stairs, fire hydrants, gates, steps, stands, fences, carports, decking, equipment, ramps, towers, awnings, flagpoles, gutters, radiators, cabinets, bulletin boards, guy wires and guy wire guards, pools, outside equipment, reflectors, kitchen cabinets, conduit, and other constructed or manufactured items. Painting may be required prior to or after installation. Paints include, but are not limited to, all enamels, paints of all base types, varnishes, and stains. The Contractor shall paint different types of surfaces such as wood, drywall, plaster, metal, concrete, masonry, rocks, fiberglass, plastic, and glass. The Contractor shall maintain records of all painting performed.

5.2.4.5.1 Touch-up Painting. The Contractor shall perform touch-up painting on areas that have been damaged or soiled. Paint used for the touch-up of repaired surfaces shall blend with the color and texture of the surrounding areas. Surfaces include, but are not limited to, wood, metal, sheetrock, concrete, and plaster.

5.2.4.5.2 Surface Preparation. Prior to applying primers, paints, stains, shellacs, varnishes, or lacquers the Contractor shall prepare all surfaces, which include the removal of contaminants. The Contractor shall prepare all surfaces in a manner appropriate to the surface and the material to be applied to the surface. The Contractor shall use preparation methods such as sanding, sand blasting, pickling, stripping, and water blasting, and shall use heat guns and irons, solvents, acids, and paint removers to prepare surfaces. The Contractor shall repair and patch any imperfections on surfaces with wood putty, glazing compound, sealers, plastic wood fillers, spackling, joint compound, caulking, primers, and cement. Preparation includes, but is not limited to, taping, spackling, and repairing gypsum board surfaces.

5.2.4.5.2.1 Fixture Protection. Electrical outlets, covers, switch covers, windows, and fixtures shall be masked or removed before painting, and uncovered or reinstalled upon work completion.

5.2.4.5.2.2 Property Protection. The Contractor shall move, reset, and protect furniture and equipment from damage (e.g. paint dropping on Government and privately owned property during the performance of painting tasks).

5.2.4.5.2.3 Lead Abatement. The Contractor shall ensure the proper protection and removal of particles and chips from sites effected by lead paint removal in accordance with environmental regulations, TM 5-618, and the lead abatement standards specified in Section C-1 of this Contract.

5.2.4.5.3 Paint Application. The Contractor shall apply paint by brush, roller, or spray with prior approval by the KO, if spraying. Interior paint shall be applied in dust-free conditions in accordance with the manufacturer’s recommendations. All coats shall be prepared so that the quality and type of paint match and meet manufacturer’s recommendations.

5.2.4.5.3.1 Coat Application. The Contractor shall apply prime, intermediate, and finish coats as required. Each coat shall be sanded and dusted as prescribed by the paint manufacturer to produce a smooth finish that is free of runs, sags, or other surface preparation defects. The Contractor shall apply each coat of paint as a film of uniform thickness so that all surfaces including edges, corners, crevices, welds, and rivets receive a film thickness equivalent to that of adjacent painted surfaces. The Contractor shall allow sufficient time to elapse between successive coats to permit proper drying. This period shall be modified as necessary to suit adverse local weather conditions.

5.2.4.5.3.2 Finished Surface Quality. The Contractor shall apply paint to ensure that finished surfaces are free of runs, drops, ridges, waves, laps, brush marks, and any variations in color, texture, and finish. Paint surfaces adjoining other materials or colors shall be sharp and clean and shall not overlap

5.2.4.5.3.3 Paint Specifications

a. Paint Color. The color of paint for entire walls, rooms, and exteriors of buildings will be selected by the Government from samples furnished by the Contractor or based on the relevant Technical Manuals. The Contractor shall use tints to match existing colors and colors specified and approved on DA Form 4284.

b. Mixing and Thinning. Prior to application, the Contractor shall mix paints to ensure that they are completely blended and have no color variation, lumps, or foreign matter. The Contractor shall not mix paints from different manufacturers. The Contractor shall thin paint prior to application when necessary to suit conditions of surface, temperature, weather, or other application conditions. The Contractor shall not use more than one pint of suitable thinner per gallon, and the use of thinner shall not relieve the Contractor from the responsibility of completely hiding or covering surfaces.

5.2.4.5.4 Other Painting Tasks

5.2.4.5.4.1 Dry Wall. The Contractor shall tape, patch, and finish dry wall. The Contractor shall reset all nails, tape all joints, cracks, holes, and texture and paint dry wall. The Contractor shall use perforated tape and apply three coats of joint compound on all dry wall joints, cracks, and holes. The Contractor shall allow every coat of joint compound to dry completely before application of the next coat. The Contractor shall apply different types of wall textures such as smooth finish and roll on finish.

5.2.4.5.4.2 Finishing and Refinishing. The Contractor shall finish and refinish different types of surfaces and materials. The Contractor shall sand, stain, varnish, shellac, and lacquer picture frames, tables, book shelves, desks, paneling, wainscot, doors, benches, stairways, chairs, railing, wooden floors, and any other surfaces or materials.

5.2.4.5.4.3 Spray Painting. During all spray painting activities, the Contractor shall:

- Set gauges on spray guns
- Prime material to be sprayed
- Sand material
- Spray paint material to be painted
- Let dry and apply material's final coats
- Clean spray guns after each day's usage
- Clean and dispose of all empty paint cans
- Dispose of all aerosol cans properly
- Place all leftover paint in the appropriate barrels for turn in as Hazardous

Materials (HAZMAT) to the Substances Control Center

5.2.4.6 Sign Work

The Contractor shall repair, install, replace, remove, design, layout, paint, stencil, laminate, and mount permanent and temporary signage throughout the Installation, including directional, regulatory, interior, identification, informational, family housing residential name plates, and other types of signs. Signage work shall also include, but not be limited to, designing, fabricating, and painting posters and safety signs. The Contractor shall paint traffic signs in strict conformance with the standards set forth in the Manual of Uniform Traffic Control Devices for Streets and Highways and Fort Benning policies. The Contractor shall maintain signs on buildings, structures, appurtenances, and along streets so that they are properly positioned, clean, and legible at all times. Typical areas of installation include, but are not limited to: companies, battalions, brigades, chapels, training areas, gyms, and youth centers.

5.2.4.6.1 Traffic Signs. The Contractor shall repair, install, replace, remove, and inspect traffic regulatory, warning, guidance and yield, informational, reflective, reserved, and safety signs on paved or unpaved areas. The Contractor shall also replace, install, remove, and inspect delineators and bridge markers (e.g. height limit, weight limit, and stop on red).

5.2.4.6.1.1 Traffic Sign Material Specification. The Contractor shall use Telspar posts for high visibility traffic signs, due to suitability of their breakaway or yield design. The Contractor shall fabricate traffic signs in accordance with the Manual of Uniform Traffic Control Devices and the Installation Design Guide.

5.2.4.6.1.2 Repair of Traffic Signs. Traffic signs that are taken down for repair shall be replaced immediately so as to avert accidents. The Contractor shall repair or replace signs of a critical nature damaged by accidents or reported to be damaged beyond use or repair within one hour of discovery or notification of the matter. The Contractor shall notify the Military Police of all signs that are taken down which effect traffic and safety on the Installation. All repairs and installations shall be in accordance with the Manual of Uniform Traffic Control Devices and the Installation Design Guide.

5.2.4.6.2 Informational Signs. The Contractor shall fabricate special signs such as quarters numbers, street signs, directional signs, regulatory signs, individual group reserved parking signs, safety signs, boundary signs, fire prevention signs, physical and classified activity access signs, control signs, off limit signs, restricted area signs, medical activity signs, and special events signs.

5.2.4.6.3 Sign Design. The Contractor shall design, lay out, draw, letter, and paint murals and special artwork to maintain uniformity in accordance with the Fort Benning Design Guide, and Army Regulation 420-70, and the Installation Design Guide. The Contractor shall match existing style and lettering of current signage on the Installation. The Contractor shall lay out work in order to produce lettering and art features to scale, and shall perform free hand and gold leaf lettering. The Contractor shall fabricate silk screens, patterns, and templates. The Contractor shall design, fabricate, and paint or use a bonding machine for traffic signs, informational signs, directional signs, field training signs, and safety signs. The Contractor shall cut, punch, round corners, file, trim, assemble, install, and remove signs of different materials and gauges such as aluminum, galvanized sheet metal, plastic, wood, masonite, masonry, and black iron.

5.2.4.6.3.1 Type of Paint. The Contractor shall determine the type of paint best suited for the task involved and shall mix colors if necessary. Munitions and hazardous material storage signs shall be color coded as required.

5.2.4.6.3.2 Font. The Contractor shall use Series C (military block letter) on all Fort Benning Installation signs unless otherwise instructed by the KO.

5.2.4.7 Metal Work

The Contractor shall maintain and repair or replace the metal components of buildings and structures, installed building equipment, kitchen equipment, and sewer manholes and grates, and shall construct and install metal components in support of other maintenance activities. Metal working shall include heating and bending to form metal shapes, drilling, torch cutting, hammer forging, grinding, sawing, and fitting metal parts. The Contractor shall weld all types of metals. Processes include preheating, brazing, track welding, flame cutting, and heat-treating. Welding shall be performed on light, heavy gauge, and headlined metals using flat, vertical, horizontal, and overhead positions.

5.2.4.7.1 Material Specifications. The Contractor shall accomplish all work using materials that meet current standards. The Contractor shall match gauges and materials of existing components when components are being attached or replaced.

5.2.4.7.2 Examples of Work. The Contractor shall perform metal work and sheet metal work to buildings, structures, and facilities to include, but not be limited to, the following areas and tasks:

5.2.4.7.2.1 Metal Components. The Contractor shall be responsible for fabricating, installing, repairing, and replacing parts or complete assemblies as required. Types of repairs and fabrication shall include, but not be limited to: metal components of buildings, metal parts of installed building equipment, dining facility equipment, kitchen equipment, utility systems, roadway structures, drainage, and walks.

5.2.4.7.2.2 Security Equipment. The Contractor shall fabricate, install, and repair partitions and gates. The Contractor shall also fabricate, install, and repair security screens. For security screens, the Contractor shall imbed grid ends in structures or weld them to a frame on the inside.

5.2.4.7.2.3 Duct Systems. The Contractor shall design, lay out, fabricate, install, modify, repair, replace, and remove various types of duct systems such as heat, evaporative cooling, air conditioning, exhaust, collector, smoke stacks, and ventilating systems. The Contractor shall fabricate, install, modify, and remove accessories to duct systems such as mufflers, silencers, filter racks, flexible connections, insulation, control dampers, quadrants, fire dampers, turning vanes, extractors, straightening grips, mixing boxes, and gravity dampers.

5.2.4.6.2.4 Welding. The Contractor shall weld all types of metals to include, but not be limited to, copper, brass, aluminum, steel, galvanized steel, stainless steel, iron, cast iron, and cast steel. Processes to be used include arc, heli-arc, acetylene, inert gas welding (cutting oxy/acetylene brazing/grinding), and welding wire and Metal-Inert-Gas (MIG). Welding shall be performed on hardened metals of light and heavy gauges in all positions. Processes include preheating, brazing, bead welding, tack welding, spot welding, butt welding, frame cutting, pressure welding, and heat-treating. Portable welding shall be provided for repairs of metal doors, jambs, windows, handrails, and other components. Welding shall comply with Subpart Q of 29 CFR 1910, Occupational Safety and Health Administration (OSHA) General Industry Standards.

5.2.4.6.2.5 Fences and Cages. The Contractor shall repair all perimeter fences, transformer fences, gates, and other security cages to ensure that unauthorized entry to restricted areas does not occur. Repairs include, but are not limited to: repairing holes in chain link fencing and wire cages, stringing barbed wire, replacing or resetting fence support stanchions, replacing or repairing hinges and locking devices, removing rust and corrosion, and painting fences.

5.2.4.7.2.6 Gutter Systems. The Contractor shall design, fabricate, install, repair, replace, clean, and remove all types of gutters (e.g. build up gutters), downspouts, and other components of gutter systems. The Contractor shall match existing gutters and downspouts in design and gauge of materials such as ogee, K type, box, half round, and corrugated (round and rectangular) downspouts; scuppers; conductor heads; expansion joints; inside and outside miters; hangers, elbows, and strainers; spikes and ferrules; sumps; and splash pans. The Contractor shall use materials such as galvanized metal, aluminum, and copper.

5.2.4.7.2.7 Flashing. The Contractor shall design, fabricate, install, repair, replace, and remove all types of flashing, molding, and fascia including gravel stops, 90° standard drip edge, window and door lintels, coping, building set backs, sills, spandrels at grade and under partitions, roof valley, hip, ridge, and mansard roof flashing, counter flashing, parapet wall flashing, reglet, expansion joints, skylight flashing, drywall molding, tile molding, louvers, roof and wall penetrations, pitch pans, termite shields, mullion covers, and all special flashing, fascia, and moldings.

5.2.4.7.2.8 Doors and Frames. The Contractor shall install, repair, replace, and remove metal doors, metal kick plates to wooden doors, gussets, trim, tracks, skins, metal covering of doors for security, overhead doors (track and cylinder rolled), sliding doors (metal and wood), access doors, accordion doors, dutch doors, pocket doors, and frames. The Contractor shall install, repair, replace, and remove components such as cables, drums, springs, sprockets, rollers, track door panels, track hangers, gussets, handles, overhead door chains (pull chain and gear chains), guards, weather stripping, fastening devices, channels, hinges, eyebrows, personnel doors, and metal window frames.

5.2.4.7.2.9 Awnings. The Contractor shall design, install, repair, replace, and remove metal awnings on carports, porches, decking, walkway covers, doorway foyers, metal roofs, metal sheds, and other structures. The Contractor shall match gauges of metal and designs of existing components on the facility when replacing or installing new components.

5.2.4.7.2.10 Metal Containers. The Contractor shall design, fabricate, install, repair, replace, and remove containers such as storage bins, electrical wire troughs, cabinets, shelves, boxes, pans (splash and drip), tanks, pans with drains, troughs, scopes, and shields for equipment and machines.

5.2.4.7.2.11 Paneling. The Contractor shall design, fabricate, install, repair, replace, and remove metal paneling for different types of applications such as fire protection; to meet health, security, and safety requirements; and for general appearance improvements. The Contractor shall layout and fabricate panel designs such as quilting (square and diamond shape), and cross breaking, corrugated, ribbed, V beam, and other designs that match existing paneling. The Contractor shall use various gauges of stainless steel, copper, and galvanized materials to match materials on existing components to which the item is being connected. The Contractor shall also fabricate and install metal siding, fiberglass siding, soffits, and eave facings.

5.2.4.7.2.12 Lagging. The Contractor shall lag pipes, ducts, and equipment to preserve the insulation from weather deterioration. Lagging, used here, shall mean a covering, e.g., of metal, of polyurethane, placed over insulation.

5.2.4.7.2.13 Venting. The Contractor shall design, fabricate, install, repair, replace, and remove breechings, stacks, caps, spark arrestors, bird screens, exhaust, fresh air intakes, combustion air, relief, and all related fixtures on gas-fired equipment, fireplaces, or any type of system or equipment which requires venting.

5.2.4.7.2.14 Miscellaneous Metal Work. The Contractor shall design, fabricate, install, repair, replace, and remove a variety of miscellaneous metal work. This shall include, but not be limited to: hoods; kitchen equipment; walk-in cooler walls and panels; trailer skirting; curtain rods and extensions; TV brackets; alarm grids; prefab metal structures; road gates; sewer grates and manholes; playground equipment (swings, merry-go-rounds, slides, see-saws, and monkey bars); basketball and soccer goals; handicap ramps and other ramps (stairwell, dock and safety rails); motor covers; flag poles; handrails; exhaust fan guards; blackboards; volleyball posts in gyms; sign frames; swimming pools parts (ladders, grates, etc.); chimney caps and hoods; jail doors; and other metal work as required by valid work documents.

5.2.4.8 Electrical Systems

The Contractor shall maintain, repair, and install electrical system components including, but not limited to: wiring systems, conduit systems, cable systems, conductors, switches, receptacles, outlets, device plates, grounds, service equipment, motor control centers, control switchboards and consoles, lightning protection systems, fixtures, electrical kitchen equipment, kitchen and building ventilating fans, sump pumps, garbage disposal units, and electrical heaters in all applicable buildings, structures, and facilities. Additional electrical maintenance requirements for motors, motor controllers (e.g., starters, contactors, or variable speed controllers), equipment status monitoring systems, pump controls, and similar electrical apparatus that is appurtenant to plant and mechanical equipment are included in the applicable section for the maintenance of that equipment. The Contractor shall also perform other electrical work such as: providing temporary electrical service as required, providing emergency lighting systems, determining estimates, testing circuits and load, providing standby electricians for special events, and escorting uncleared personnel

5.2.4.8.1 Electrical Emergencies. The Contractor shall respond to all electrical emergencies as Priority 1 DMOs and restore power in the most expeditious and cost effective manner in compliance with applicable codes. This emergency work shall include: eliminating exposed energized wires, arcing in electrical circuits, tripped power circuit breakers, burning wire insulation, failure of fixtures and equipment, and dimming of security or building/facilities lighting. The Contractor shall have someone subject to notification on a 24 hours a day to turn off power in case of an emergency.

5.2.4.8.2 Work Coordination. The Contractor shall coordinate his interior electric work plan, as appropriate, with activities involved in the operation and maintenance of electrical facilities and distribution systems.

5.2.4.8.3 Temporary Repairs. The Contractor shall accomplish temporary repairs when it is essential to prevent damage to a facility or to minimize the impact to Government operations. In case of an emergency requiring temporary repairs outside normal scheduled working hours, the Contractor shall notify the KO within two hours after the beginning of the next scheduled work day and shall provide a description of the method of repair used, the reason for such repairs, and the proposed method and time to complete permanent repairs.

5.2.4.8.4 Materials Specifications. Electrical materials and equipment used by the Contractor shall comply with the latest edition of the codes of the NFPA and the National Institute of Standards and Technology (NIST). All work shall conform to the requirements of the National Electrical Code (NFPA No. 70 Most Recent Edition) where applicable.

5.2.4.8.5 Equipment. The Contractor shall use Test, Measurement, and Diagnostic Equipment (TMDE) expeditiously and accurately to complete testing and diagnostic procedures that are required to execute the work for which the Contractor is responsible. The Contractor shall maintain TMDE in accordance with industry standards.

5.2.4.8.6 Interior Electrical Systems (Including Electrical Equipment and Appliances). A building or facility receives electrical power at one or more points from which the power is delivered or made available to the various items of utilization by means of the building electrical distribution system. The term interior electrical system encompasses all components of the building distribution system including wiring, disconnects (switches and breakers), electrical panels, cabinets, load centers, transformers, capacitors, conduit, ducts, bus ways, meters and instrumentation; various types of controls and control systems; and lighting systems, as well as other systems and equipment and their related components, including fire alarm systems, auxiliary electrical generating systems, auxiliary lighting systems, and certain types of interior signaling systems. The Contractor shall repair, maintain, install, and test interior electrical systems and components. The Contractor shall perform electrical repairs, replacements, maintenance, and testing in a manner that ensures that all electrical systems are safe and reliable. All work shall conform to the requirements for the environment in which the equipment is located.

5.2.4.8.7 Exterior Electrical Systems. The Contractor shall be responsible for the exterior lights that are attached to buildings or towers, or that are installed at ground level and used to illuminate the facade of a building, as well as related system components and controls (e.g., photocells, time clocks, contactors, etc.). The Contractor shall respond to work directives for problems related to the operation of the system.

5.2.4.8.8 Wiring. The Contractor shall repair, maintain, install, and test interior electric wiring and wiring systems. Types of components and systems include: electrical conduit of all types including Electrical Metallic Tubing (EMT); rigid metal conduit; intermediate metal conduit; flexible metallic tubing; flexible metal conduit; liquid tight flexible conduit; and rigid nonmetallic conduit including all associated hardware and fittings, bending, cutting, and threading as applicable; wire, wiring, cabling, power cabling, open wiring, feeder circuits, branch circuits, fixture wiring, and flexible cabling and cords, and from single conductor to multi-conductor circuits; and all switches, outlets, junction boxes, terminal boxes, wire ways, under floor assemblies, bus ways, electrical floor assemblies, and multi outlet assemblies.

5.2.4.8.8.1 Replacement Wiring. The Contractor shall use wiring methods (e.g., concealed wiring, etc.) that are architecturally, electrically, and mechanically equivalent to the existing electrical systems design and installation. Replacement parts and components shall aesthetically match the existing installation and the decor of the facility. The Contractor shall obtain approval from the KO prior to using wiring methods, or parts and components that are not consistent with the existing installation.

5.2.4.8.8.2 Romex. The Contractor may use nonmetallic-sheathed cable (to include types NM, NMC, and UF), generically referred to as Romex, as permitted by the National Electrical Code (NEC), but only in temporary (World War II) facilities of wood frame construction and family housing units or other quarters that originally contained type NM cable. Romex shall not be installed, exposed, or used in other types of facilities without the prior approval of the KO.

5.2.4.8.8.3 Aluminum Conductors. The Contractor shall not install aluminum conductors as part of a building wiring system.

5.2.4.8.8.4 Electrical Nonmetallic Tubing. The Contractor may use electrical nonmetallic tubing (ENT) as permitted by the NEC

5.2.4.8.8.5 Rigid Nonmetallic Conduit. The Contractor shall not use rigid nonmetallic conduit within the interior wiring system of a structure, except for below grade or in slab construction.

5.2.4.8.8.6 Underground Utility Warning Tape. The Contractor shall provide an underground utility warning tape containing a locator detectable strip for all new or replacement underground electrical installations. The tape shall be installed a minimum of 12” above the raceway or cable, but in no case less than six inches below the surface.

5.2.4.8.9 Testing and Troubleshooting. The Contractor shall accomplish site investigation, testing, troubleshooting, and diagnostic procedures to properly diagnose the cause of a failure or malfunction of systems and equipment for which the Contractor is responsible. The Contractor shall use procedures that will minimize the time required to identify, isolate, and correct the failure or malfunction.

5.2.4.8.9.1 Electronic Equipment. The Contractor shall be responsible for all of the electronic equipment and components of the systems encompassed in the scope of this section. The Contractor shall repair or replace failed electronic components to include, but not limited to, capacitors, resistors, transistors, rectifiers, and silicon controlled rectifiers (SCRs). The Contractor shall repair printed circuit boards to include the replacement of failed discrete components (e.g., semiconductor devices or integrated circuits).

5.2.4.8.9.2 Monitoring Systems. The Contractor shall troubleshoot and repair electric or electronic monitoring systems (e.g., cold storage facility and equipment status monitoring and alarm systems, and leak detection systems) and fuel dispensing electronic monitoring and recording systems.

5.2.4.8.9.3 Other Equipment and Systems. The Contractor shall provide testing, measurement, and diagnostic reading services for electrical systems, services, feeders, circuits, and equipment for use in evaluations, design efforts, data for studies, load analysis, and similar requirements.

5.2.4.8.10 Lighting. The Contractor shall repair, maintain, install, and test interior electrical lighting and lighting systems. Types of components and systems include: switches and low voltage switching systems; dimmers and dimming systems; photo electric eyes and components; timers; magnetic contactors (lighting contactors); submersible and waterproof fixtures; explosion proof and vapor tight systems; temporary lighting and lighting systems, including Christmas lighting and special purpose lighting; and various types of lighting systems such as indoor lighting, outdoor lighting, security lighting, and emergency lighting.

5.2.4.8.10.1 Energy Conservation. The Contractor shall measure lighting intensity for adequacy of purpose and energy conservation purposes. The Contractor shall disconnect lighting systems or their components for energy conservation and reconnect lighting previously disconnected for energy or other purposes. All disconnections performed for the purpose of energy conservation will be approved prior to action by the Contractor.

5.2.4.8.10.2 Bulb Replacement. The Contractor shall replace light bulbs in areas other than those identified as a Self-Help requirement or in cantonment areas where the type of lamp is not available through the Self-Help program. The Contractor shall replace the lamps in fixtures requiring special tools or equipment for replacement in: all High Intensity Discharge (HID) type fixtures, fixtures above stairways or other places of difficult access, mechanical and equipment rooms, crawl spaces, and fixtures where lamp replacement cannot be accomplished safely from a step ladder six feet or less in height. The Contractor shall be responsible for marquee, sign, billboard, and similar lighting. The Contractor shall replace bulbs on the entrance signs on Benning Boulevard and the extension of I-185 to the Installation.

5.2.4.8.11 Directed Services. The Contractor shall perform directed work in conjunction with work being accomplished by other methods, and on facilities, systems, and equipment that have been installed, constructed, improved, or altered during the course of the Contract. The following are indicative, but not inclusive, of the types of services to be provided by the Contractor:

a. Provide temporary electrical service, circuits, and lighting for special events and other requirements.

b. Provide or remove raceways for Automated Data Processing (ADP) Systems, Intrusion Detection Systems, Signaling Systems, Communications Systems and similar installations. Raceways shall include "Tele-power poles". Raceways installed in support of other systems shall be provided with a poly pull line unless specified otherwise.

c. Install Government-furnished equipment (GFE) and system components. Remove equipment and system components that are to remain the property of the Government. The cost of GFE shall not contribute to the material cost in figuring the level of work.

d. Connect, disconnect, or provide circuits for kitchen equipment, equipment in place, and all other types of electrical utilization equipment.

e. Repair, maintain, and install electrical wiring in buildings, structures, and facilities to include, installing and repairing electrical wiring systems and associated switches, distribution panels, light sockets, and outlet boxes.

f. Measure, cut, thread, bend, assemble, and install conduit; and insert, splice, and connect wires to fixtures, outlets, switches, receptacles, and power sources.

g. Install wiring from blueprints, wiring diagrams, or sketches.

h. Locate, diagnose, and repair trouble occurring in power circuits, controls, switches, rheostats, thermostats, flow meters, stop controls, and motor control centers.

i. Test circuits and equipment.

j. Install, maintain, and repair lightning protection systems.

k. Install, maintain, and repair underground wiring to associated control panels and to trailer sites.

l. Inspect rides for special events.

5.2.4.8.12 Building Service Entrances. The Contractor shall repair, maintain, install, and test building service entrances. Types of components include: fused and non fused service disconnects; single phase to polyphase conduit, raceways, fittings, hardware, wiring, service masts and supports; weather heads; grounds; grounding systems and conductors; ground rods; fuses; circuit breakers; ground fault protection devices; surge arrestors; lighting protection devices; and both overhead and underground service entrance cables including all connections, drip loops, weatherproofing insulation, conduit, and fittings.

5.2.4.8.13 Load Centers. The Contractor shall repair, maintain, install, and test electrical distribution and load centers. Types of components include: main service power circuit breakers, including manual, automated, and ground fault interrupter (GFI) types; load distribution centers; both power and lighting; bus systems; raceways; distribution panels; branch circuits; and load circuit breakers varying in size, both single and polyphase.

5.2.4.8.14 Motors and Motor Control Centers. The Contractor shall repair, maintain, install, and test electrical motors and motor controls. Types of components include: single and polyphase motors, varying in operational voltages; continuous and intermittent duty motors; motor feeders; motor disconnects; motor circuit conductors; overloads; fuses; circuit breakers; motor controllers; contactors; coils; automatic door closures, and motors and controls; control circuits; motor control devices such as start/stop stations; pressure switches; gear driven limit switches; limit switches; mercoid switches; and control and power transformers. Motors and motor control centers are located indoors and outdoors. The Contractor shall repair, maintain, and install belts and shaft guards, motor pulleys, belts, couplers and couplings, and electric motor bearings. The Contractor shall also be responsible for the following electrical maintenance items:

- Installing, testing, maintaining, and repairing electric motors.
- Inspecting and replacing bearings.
- Cleaning, oiling, greasing, and painting motors.
- Rewinding and applying insulating paint to windings of electric motors (both single and three phase type) ranging from fractional to 800 horsepower (hp).
- Replacing defective capacitors, cutout switches, and brushes.
- Checking controls and components associated with motor controls and circuits.
- Troubleshooting electric motors on location as required.
- Fabricating sleeve bearings, washers, rings, bushings, and other parts related to electric motors; and cutting and under cutting motor generator commentators and metallizing of motor shafts.

5.2.4.8.15 Grounding Systems. The Contractor shall repair, maintain, install, and test interior electrical grounds and grounding systems. Types of components include: ground rods, grounding conductors, ground clamps, ground wells, ground lanes, helicopter tie downs, and fueling ground points at POLs and the helipad. The contractor shall check resistance levels to ensure that the maximum allowable resistance of each grounding electrode is no more than 23 Ohms to earth for all types of grounding check points TE-5.2-018.

5.2.4.8.16 Furniture Wiring Repairs. The Contractor shall be responsible for wiring to the outlet for modular furniture, work centers, and moveable partitions that are electrified. When directed, the Contractor shall install or remove the wiring from the outlet to the point of the electrical service connection to modular furniture, and shall accomplish electrical repairs to the modular furniture's electrical wiring, devices, and lighting fixtures.

5.2.4.8.17 Raceways. The Contractor shall be responsible for ADP Systems, Intrusion Detection Systems, Signaling Systems, Communications Systems, and similar Installation raceways, including "Tele-power poles". The Contractor shall locate and mark the location of raceways and wiring system components, trace circuits, and identify feeder and branch circuit protective devices.

5.2.4.8.18 Clock and Bell Systems. The Contractor shall be responsible for the maintenance and repair of integrated clock systems and bells.

5.2.4.8.19 Traffic Devices. The Contractor shall be responsible for the operation, maintenance, and adjustments to traffic signal controllers (computerized), to include traffic flow sensors, timing controllers, signaling devices, and cable. Any changes to the timing sequence of the traffic devices will be approved by the Traffic Manager prior to change.

5.2.4.8.20 Electrical Requirements at Fort Benning Helipads, Airfields, and Related Maintenance Areas. The Contractor shall be responsible for Fort Benning Helipads, Airfields, and related maintenance areas. The Contractor shall maintain the following navigational equipment:

a. Navigational lighting to include appurtenant wiring, controls, and equipment. The operational voltage is 120 - 240 single phase at 60Hz.

b. Auxiliary lighting (e.g., wind cone and wind tee) and pole mounted lights.

c. Aviation obstruction lighting, except for those installed on steel structure towers that are dedicated to communication equipment or those that are installed on utility pole structures.

5.2.4.8.21 Warning Sirens for Post Alert System. The Contractor shall ensure that the Post Alert System and its associated transmitter are operational at all times.

5.2.4.8.22 Lawson Army Airfield (LAAF) Lighting and Electrical Systems. All work performed at the airfield shall be coordinated with the airfield operations staff.

5.2.4.8.22.1 The Contractor shall inspect, maintain, and repair the underground cables at LAAF to ensure they are operational.

5.2.4.8.22.2 The Contractor shall maintain and repair the airfield lighting at LAAF upon the receipt of a valid work document. The Contractor shall replace all defective lighting, and perform all other necessary corrections identified in the inspection.

5.2.4.8.22.3 The Contractor shall inspect, maintain, and repair the rotating beacons to ensure correct operation upon the receipt of a valid work document.

5.2.4.8.22.4 Maintenance of power conditioners for Lawson Army Air Field (LAAF) Lighting: Contractor shall perform preventative maintenance of Liebert power conditioners associated with the LAAF lighting vault in accordance with manufacturer's recommendations. This requirement shall be charged to a OWO.

5.2.4.8.23 Industrial Air Compressors. The Contractor shall inspect, maintain, and install air compressors including storage tanks, piping, controls, and accessories. The Contractor shall repair, clean, make adjustments to, and replace components of compressors and mechanical systems on an as required basis to ensure proper operation. Work shall include, but is not limited to, troubleshooting and testing compressors and associated devices such as air dryers, purifier piping, distribution piping, valves, and auxiliary equipment.

5.2.4.8.24 Appliance Standards. The Contractor shall repair and replace Government-furnished major appliances including range hoods and ducts as directed by the KO in facilities other than family housing units with new government owned, Contractor-furnished appliances that are similar in function and quality.

5.2.4.8.25 Miscellaneous Responsibilities. The Contractor shall operate, maintain, repair, install, and alter all interior electrical systems, alarm systems, auxiliary power and lighting systems, electrical systems and controls used to operate targets on training ranges, and a variety of other electrical fixtures, tools, and appliances. The Contractor shall ensure that electrical systems, system components, appurtenant parts, and equipment are reliable, efficient, functional, safe, and continually available for their intended purpose as specified herein. The Contractor shall operate, maintain, repair, install, and alter all interior electrical systems and related systems to include, but not limited to:

a. The installation, maintenance, and repair of electrical wiring including the tightening or reinstallation of loose wires and connections, replacement of bare conductors, convenience outlets, switches, and dimmers.

b. The operation, maintenance, repair, and installation of electrical disconnects such as switches and circuit breakers; including connections; replacing tripping elements; adjusting and cleaning contacts and operating mechanisms; and testing and adjusting relays.

c. The installation, maintenance, cleaning, and repair of electrical cabinets, panels, and load centers; balancing electrical loads; labeling to identify circuits and feeders; maintaining and repairing disconnects contained therein; and the replacement of fuses.

d. The installation, maintenance, and repair of transformers, capacitors, and power filters.

e. The installation, maintenance, and repair of conduits, ducts, and bus ways, to include checking to ensure that entrances and fittings are properly installed to prevent a means of access for moisture, dirt, oil, and other foreign material. The installation, maintenance, and repair of equipment, such as, power loading ramps, electrical hoists, conveyers, battery chargers, door operators, motors, motor controls, well pumps and controls, cardboard/paper compactors, form presses, steam irons, score boards, and car wash equipment.

f. The installation, maintenance, repair, and calibration of installed meters, instruments, and recorders such as ammeters, voltmeters, kilowatt meters, flow meters, temperature and pressure meters, recorders, frequency meters, and hour meters.

g. The installation, maintenance, repair, and calibration of various electrical and electronic control systems and their related components, including electronic control systems for clocks, combustion controls on boilers, and control systems (to include computerized systems) on heating and cooling systems.

h. The installation, maintenance, and repair of lighting systems, to include replacing various types of lamps, ballasts on fluorescent lights, and complete lighting fixtures. The fabrication of extension cords, power cords, and lighting sets.

i. The installation, maintenance, and repair of interior signaling systems such as lamp and buzzer type call systems; bell, buzzer, carillon, and chime systems; magnetic signaling and lamp-type paging systems; and individual smoke alarms.

j. The installation, maintenance, and repair of auxiliary and emergency lighting systems used for interior illumination and exit lighting. These systems are primarily battery powered and are automatically operated upon loss of electric power to the primary lighting system.

5.2.4.8.25.2 The Contractor shall install, maintain, and repair electrical circuits and conduits supplying power to J SIIDS Control Units, monitors, and transmitters, and their associated telephone terminal boxes, and between the monitor consoles and their associated telephone terminal boxes. The Contractor shall also install conduit connecting the system components. The Contractor shall install commercial intrusion detection systems.

5.2.4.8.25.3 The Contractor shall maintain all ceiling fans on Fort Benning. This shall include, but not be limited to, wiring, switches, and associated lights.

5.2.4.8.25.4 The Contractor shall repair and maintain Installation ice machines. This shall include, but not be limited to, ensuring correct wiring, and eliminating excessive vibrations, and other electrical problems.

5.2.4.9 Plumbing and Steamfitting

The Contractor shall be responsible for all plumbing systems and their associated equipment and components, including systems on the various ranges. The Contractor shall provide for the proper installation, maintenance, repair, and replacement of all plumbing, potable and non potable water, heating, gas, steam, hot water, high temperature high pressure hot water and chilled water systems and their associated equipment and components, fittings, and piping as required. The Contractor shall maintain, repair, install, inspect, remove, replace, and modify all plumbing equipment and components for water, sewer, swimming pools (indoor and outdoor), inert gas, natural gas, air vacuum, compressed air systems, steam, and POL equipment. The Contractor shall install, maintain, and repair all types of special equipment or fixtures and replace old equipment, components, and fixtures requiring plumbing connections. All material, equipment, components, fixtures, and workmanship associated with plumbing systems shall be in accordance with applicable sections of the Uniform Plumbing Code (UPC), Army Regulations, and Technical Manuals.

5.2.4.9.1 Plumbing and Steamfitting Services. The Contractor shall install, maintain, repair, replace, modify, relocate, remove, and inspect plumbing and steam fitting equipment, components, systems, fixtures, instruments, valves, and other associated equipment. These services include, but are not limited to, the following types of work:

a. Removing blockages from drain lines, interior and exterior sewers, floor drains, fan coil units, air handling units, stacks, vents, grit chambers, and storm drainage systems.

b. Adjusting, repairing, or replacing leaky joints, faucets, and other outlets as necessary to maintain proper and efficient operation of all related plumbing and steam fitting systems including, but not limited to: hot water systems, chilled water systems, potable water systems, sprinkler systems, compressed and vacuum air systems, valves, inert and natural gas systems, and other associated systems, equipment, and components.

c. Installing and repairing domestic, industrial, and commercial water heaters, hot water storage tanks, and plumbing fixtures, equipment, and components, including commodes, urinals, kitchen and utility sinks, water fountains, frost-proof hydrants, showers, bathtubs, lavatories, shower pans, garbage disposals, dishwashers, fan and air handler coils, valves, pumps, and associated equipment and components.

d. Installing, maintaining, repairing, and replacing pipe insulation including, but not limited to, fiberglass and armorflex insulating materials. The Contractor shall install protective metal covering to the insulation if directed to do so by the KO.

e. Placing and connecting water, air, natural gas, sewage and water/storm fixtures to sewage lines, water closets, lavatories, showers, sinks, gas heaters, stoves, air compressor equipment, shop hydraulic vehicle lifts, cooling towers, grease traps, garbage disposals, dishwashers, floor drains, condensate and drip pans, circulation systems for heating and cooling systems, and associated equipment and components.

f. Cutting or drilling holes and openings in walls, ceilings, floors, chases, or slots, and setting sleeves, thimbles, or inserts to provide passage and supports for pipe, fittings, equipment, and components. The Contractor shall install access panels in cases where openings are cut to work on pipes to facilitate future repairs.

g. Measuring, cutting, and threading pipe, assembling pipe sections, pipe welding and brazing, and hanging or laying assemblies in position.

h. Laying and joining concrete, clay, polyvinyl chloride (PVC), copper, chlorinated polyvinyl chloride (CPVC), Quest, ward flex, brass, steel alloy, and piping in accordance with proper trade methods to ensure joints are properly aligned, graded, and sloped to provide necessary and proper drainage and services.

I. Maintaining and repairing interior and exterior plumbing waste lines, stacks, vents, mixing, ball check, backflow, shower, waste, overflow valves, water coolers, commodes, sinks, lavatories, and fire department connections, sprinkler systems (14 current systems), fan coil units, air handler coils, steam, hot water, chilled water, traps, valves, drainage systems, commercial, industrial, domestic hot water heaters, and storage tanks.

j. Replacing bolts, pipe hangers, strainers, drain covers, insulation, traps, valves, fittings, equipment, and components that are damaged or missing to maintain continuous customer services.

k. Checking and inspecting equipment for proper operation, maintenance, repair, and replacement at required intervals in accordance with applicable technical manual.

l. Performing repair and overhaul work on pumps (e.g., disassembling pumps, replacing worn impellers, shafts, bearings, gland seals, ream bearings, and reassembling). Repairing or replacing sump pumps; connecting all interconnecting piping from pumping systems to service lines; replacing and servicing check valves, gate valves, globe valves, flow control valves, circuit setting valves, motorized and pneumatic valves, and pressure gauges. Removing, repairing, and installing entire pumping systems and making modifications to piping systems.

m. Performing repair and overhauling work on equipment and components found in pumps, (e.g., boiler feed water systems, condensate return pump systems, hot water circulating systems, chilled water systems, potable water systems, swimming pool circulation (on an as needed basis) and chemical feed systems, and compressed air systems).

n. Installing, repairing, modifying, and adjusting all types of gas and oil fired appliances and equipment such as hot water heaters, furnaces, stoves, and burners.

o. Connecting steam, gas, water, and sewer drainage lines to kitchen, dining facilities, domestic, commercial, and industrial facilities located on the Installation.

p. Maintaining, installing, and repairing mixing/temperature control valves for x-ray equipment and film processing equipment and components.

q. Inspecting all buildings with delineation of remedial measures.

r. Eliminating existing and potential cross connections.

s. Relocating propane tanks.

t. Assisting units in keeping oil and antifreeze drains open to prevent spillage of hazardous material. Troubleshooting, repairing, replacing, inspecting, and servicing oil skimmers in motor pool areas as needed.

5.2.4.9.2 Piping Systems. The Contractor shall be responsible for all domestic water, wastewater, natural gas, and vent piping to include, but not limited be to, piping located inside buildings; underground water, gas, storm drainage, and sewer piping; and POL lines. The Contractor shall install, maintain, replace, relocate, modify, and repair all piping distribution systems above and below ground, in tunnels, or elsewhere. The Contractor shall provide potable water lines as necessary with approved backflow prevention devices.

5.2.4.9.2.1 Types of Piping. Types of piping the Contractor is responsible for shall include, but is not limited to: concrete, galvanized, copper, cast iron, clay, PVC, iron piping, CPVC, iron pipe size (IPS) and copper tube size (CTS) gas pipe, asbestos concrete, ductile iron, black iron, coated and wrapped steel, cast iron, and brass.

5.2.4.9.2.2 Waste Lines. The Contractor shall install, maintain, replace, relocate, modify, and repair all air, gas, potable and non-potable water, wastewater drain lines, vent lines stacks, commercial, domestic, and industrial water lines.

5.2.4.9.2.3 Piping Obstructions. The Contractor shall locate and remove obstructions using trade practices, methods, and equipment from water lines, waste lines, storm sewers, and gas lines. The Contractor shall eliminate sluggish drainage in sinks, wash basins, tubs and showers, floor drains, urinals, commodes, garbage disposals, dishwashers, water coolers, and other drains.

5.2.4.9.3 Plumbing Fixtures. The Contractor shall be responsible for all types of domestic, commercial, and industrial water heaters, hot water storage tanks, and plumbing fixtures, to include but not be limited to: commodes, urinals, kitchen and utility sinks, garbage disposals, dishwashers, drinking fountains, showers, bathtubs, grinders, spouts, traps, faucets, sprayers, water closets, flushometers, automatic flushometers, lavatories, hot water temperature control devices, and associated equipment and components.

5.2.4.9.3.1 Water Heaters. The Contractor shall be responsible for gas fired, steam generated, and electric water heaters. The Contractor shall install, maintain, and repair all steam and high-temperature high-pressure hot water generators, gas fired water heaters, electric water heaters, electric steam generators, booster heater, and hot water storage tank facilities.

5.2.4.9.3.2 Fixtures. The Contractor shall install, maintain, and repair all plumbing fixtures, equipment, and components, including showers, bathtubs, lavatories, water closets, water coolers, urinals, kitchen sinks, laundry sinks, utility sinks, disposal units, grinders, and all associated plumbing fixtures.

5.2.4.9.4 Sump and Circulating Pumps. The Contractor shall be responsible for all sump pumps, hot water circulating pumps, chilled water pumps, dual temperature pumps, and potable water pumps. The Contractor shall clean sand, dirt, and other debris from pits and sump pump pits. The Contractor shall inspect and perform repair and overhaul work on pumps (e.g. disassemble pumps; replace worn impellers, shafts, and bearings for perfect fit; and reassemble pump and proper align pump couplings for proper operation). The Contractor shall inspect, repair, or replace sump pumps and medical vacuum centrifugal pumps; connect interconnecting piping from pumping systems to service lines; and replace check valves, gate valves, globe valves, ball valves, flow control valves, circuit setting valves, motorized and pneumatic valves, pressure gauges, and temperature control devices. The Contractor shall inspect, repair, install, and replace pumping systems when required and make alterations to piping systems. The Contractor shall remove and properly dispose of debris collected in sump systems and pits. The Contractor will ensure that all flood alarm systems, located in the basement facilities, are in proper operating order where applicable. The Contractor shall clean strainers on circulating pumps on an as needed basis. The Contractor shall operate pumps to ensure the accessibility of areas for maintenance and repairs.

5.2.4.10 Swimming Pool Maintenance

The Contractor shall maintain the swimming pools, as requested, in accordance with Army Technical Manual 5-662 and accepted industry standards. All required cleaning and maintenance of pools that necessitates a pool shutdown shall be approved by the KO. Fort Benning currently has one indoor swimming pool, eight adult outdoor pools, and two baby outdoor pools. The outdoor pools are maintained on a seasonal basis opening from Memorial Day until Labor Day or as directed by the KO. Tasks associated with the maintenance of the pool include, but are not limited to, the following:

5.2.4.10.1 Circulation Pumps, Boilers, and Motors. The Contractor shall maintain, repair, or replace circulation pumps, valves and other plumbing systems, boilers, and motors as necessary to ensure proper operation.

5.2.4.10.2 Chlorine Cylinders and Equipment. The Contractor shall install chlorine cylinders and calibrate chlorinator equipment as necessary to ensure its safe operation.

5.2.4.10.3 Metal Components and Unit Exteriors. The Contractor shall remove dirt and grime from bare metal components and unit exteriors to prevent rusting.

5.2.4.10.4 Bath and Pump Houses. The Contractor shall paint structures and piping, maintain concrete, and repair and replace plumbing. The Contractor shall maintain bathrooms within bath houses.

5.2.4.10.5 Life Guard Stands and Diving Boards. The Contractor shall repair, replace, and relocate all lifeguard stands and diving boards as necessary.

5.2.4.11 Wash Rack Facility Maintenance and Repair

The Contractor shall maintain and repair the wash rack facilities and their associated equipment specified in Technical Exhibit 1-003. The Contractor shall dispose of all contaminated sand from the grit chambers at these facilities as directed and in accordance with all Federal, State, and local regulations.

5.2.4.12 Fire Alarm and Suppression Systems

The Contractor shall perform maintenance and repair of DFAC Vent Hoods, and suppression systems, and all associated components. All work shall be in accordance with the NFPA National Fire Alarm Code, current edition. In addition, the Contractor shall inform the Fire Department of any non-operational systems prior to undertaking repairs. TE 5.2-024

5.2.4.12.1 Government Access. The Contractor shall note that the Government has access to suppression systems in case of emergencies or for other Fort Benning Fire Department purposes. The Contractor shall contact the Fort Benning Fire Department point-of-contact, to be determined upon Contract award, prior to and upon completion of all work. The point-of-contact shall also be used by the Contractor to report any special problems or circumstances that require Fire Department inspections. The Fire Department will reset alarm systems to operational status due to water surges, accidental activation, or fires.

5.2.4.13 PWO Project Accomplishment

The Contractor shall accomplish maintenance, repair, and minor construction projects with an estimated cost of $100,000 or less. Projects will be ordered by the Government through the issue of a valid work document for maintenance, construction, alteration, or repair (including dredging, excavating, and painting) of buildings, structures, or other real property. The terms "buildings, structures, or other real property" include, but are not limited to, improvements of all types, such as dams, plants, streets, sewers, mains, power lines, cemeteries, pumping stations, and railways. All construction projects with an estimated cost in excess of $100,000, regardless of characterization or funding source, will be the responsibility of the Government and will be accomplished through other means available to the Government. The Government retains the right to accomplish by separate contract any project that is determined by the KO to be beyond the Contractor's resources or otherwise not appropriate for Contractor performance.

5.2.4.13.1 Project Assistance. The Government will select and approve projects or portions of projects to be accomplished under the Self-Help program or troop construction program. Projects selected will be desk estimated for funded and unfunded costs by the Government prior to approval of the DA Form 4283. The Contractor shall prepare drawings, detailed estimates, and bill of materials for these projects when directed by the Government. The Contractor shall, upon official tasking, requisition, receive, and issue materials necessary for the completion of the project.

5.2.4.13.2 Subcontracting. For projects tasked for execution by subcontract, the Contractor shall forward subcontract submittals to the KO for review and approval prior to project start date.

5.2.4.14 Demolition

The Government will perform all State Historic Preservation Office (SHPO) coordination required for the demolition of buildings. The Contractor shall demolish facilities and portions of facilities identified by the KO according to the standards specified herein, OSHA Volume III, Construction Standards and Interpretations, and US Army Corps of Engineers EM-385-1-1 Safety and Health Requirements, current edition. Asbestos removal shall be the responsibility of the Contractor.5.2.4.14.1 Protection and Safety.

5.2.4.14.1.1 Protection of Existing Work. The Contractor shall protect existing work that is to remain in place or be reused by using temporary covers, shoring, bracing, and supports. The Contractor shall repair or replace all such items damaged during demolition. The Contractor shall not overload structural elements and shall provide new supports or reinforcement for existing construction weakened by demolition or removal work.

5.2.4.14.1.2 Protection from Weather. The Contractor shall protect building interiors and all materials and equipment from the weather at all times. Where removal of existing roofing is necessary to accomplish work, the Contractor shall provide materials and workers immediately to provide adequate and approved temporary covering of exposed areas. The Contractor shall provide monitoring and maintenance to temporary coverings as required to ensure effectiveness and to prevent displacement.

5.2.4.14.1.3 Traffic Restriction. The Contractor shall use traffic barricades with flashing lights and other similar warning and area restriction devices to prevent traffic through areas where demolition is occurring. The Contractor shall notify the KO prior to work initiation when traffic will be impeded by work.

5.2.4.14.1.4 Use of Explosives. The Contractor shall not use explosives in the course of any work specified herein.

5.2.4.14.2 Structures, Walls, and Partitions. The Contractor shall remove structures, walls, partitions, and slabs as required to properly accomplish maintenance, repair, alteration, or removal of facilities directed by the KO.

5.2.4.14.3 Paving and Slabs. The Contractor shall remove concrete and asphaltic concrete paving and slabs including aggregate base to the depth necessary to perform required work, including patching, repairs, and demolition.

5.2.4.14.4 Roofing. The Contractor shall remove built-up roofing to effect connections with new flashing or roofing or as required in demolition. The Contractor shall cut existing felts and insulation along straight lines and remove gravel surfacing from existing roofing felts for a distance of not less than 18 inches back from the cut. The Contractor shall remove gravel without damaging felts down to the top ply of felt. Unless otherwise directed by the KO, the Contractor shall remove minor roofing and insulation without damaging the roof deck.

5.2.4.14.5 Masonry. The Contractor shall remove masonry carefully so as to prevent damage to surfaces that will remain and to facilitate the installation of minor masonry new work.

5.2.4.14.6 Concrete. The Contractor shall remove minor concrete structures by sawing along straight lines to a depth of not less than 2 inches. The Contractor shall make all cuts in walls perpendicular to the wall face and in alignment with cuts in the opposite face. The remainder of the concrete shall be broken out, provided that the broken area is concealed in the finished work, and that the remaining concrete is sound. At locations where the broken face cannot be concealed, it shall be ground smooth or the saw cut shall be made through the entire slab.

5.2.4.14.7 Filling. The Contractor shall fill holes and other hazardous openings

.5.2.4.14.8 Disposition of Material. Title to all materials and equipment to be removed, except as otherwise directed, will remain with the Government and shall be handled in accordance with the directions found in PWOs or DMOs. The Contractor shall remove and store carefully all materials and equipment to be reused or relocated, and reinstall equipment as required.

5.2.4.15 Miscellaneous Labor and Support

5.2.4.15.1 Ceremony and Special Event Support. The Contractor shall provide all services required for ceremonies and special events. A list of special events and operations at Fort Benning can be found in Technical Exhibit 5.2-003. These ceremonies and special events necessitate work including, but not limited to: lighting for trees, stages, area lighting, and spotlighting; cleanup; decorating areas; setting up and removing chairs or bleachers; electrical support for the duration of the event; power requirements for bands, vendors, and sound equipment; fabrication, delivery, painting, and assembly of stages, platforms, fencing, and other wood work; hanging flags, banners, pennants, and bunting; and setting up and removing signs.

5.2.4.15.2 Interstation and Intrastation Moves. The Contractor shall provide services for the movement of furniture and technical and non-technical equipment resulting from the relocation of Government operations. The Contractor shall perform work that includes, but is not limited to, moving furniture or bulky items and providing labor support to Government activities.

5.2.4.15.3 Staging and Scaffolding. The Contractor shall perform staging and scaffolding services to support work as required herein and as requested by the KO. The Contractor shall install and remove prefabricated metal scaffolding. All wooden staging shall be fabricated, installed, and removed.

5.2.4.15.4 Hurricane and Storm Support. The Contractor shall perform inspections and work during Hurricane weather and other storms as directed by the KO. The Contractor shall follow the Directorate of Public Works (DPW) Standing Operating Procedure (SOP) on Weather Requirements. Inspections and work includes, but is not limited to: the operation of sump pumps; electrical distribution lines; and critical generators; inspection of loose gutters, siding, and possible flying objects; boarding up or taping large glass enclosures; and sand bagging.

Functional Area 5.3

DINING FACILITY EQUIPMENT MAINTENANCE

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Functional Area 5.3

Dining Facility Equipment Maintenance

5.3.1 INTRODUCTION

The Contractor shall operate and manage the kitchen equipment related to the dining facilities, at Fort Benning and Dahlonega Georgia. The work shall only be accomplished by a valid work order. Qualified personnel in accordance with applicable laws, regulations, documents, and Government-developed annual and long-range plans shall perform all work.

5.3.2 SCOPE OF SERVICES

5.3.3 Dining Facility Equipment Maintenance

The Contractor shall perform maintenance and repair of Government-owned dining facility equipment at Fort Benning and Dahlonega Georgia. The Contractor shall perform services as required herein to provide complete maintenance of dining facility equipment. Fort Benning's current inventory of dining equipment totals 2,345 pieces located in various buildings on the Installation and Camp Merrill. The Contractor shall replace parts and assemblies as necessary to ensure that all equipment is safe and operable. Demand Maintenance Orders on Refrigeration equipment, dishwashers and garbage disposals in dining f a c i l i t i e s will be Priority 1. Food Prep, Equipment will be a Priority 2, with a 5 day completion time. Other Dinning Facility equipment will be Priority 3.Contractor will record cost of repair parts on the service order

5.3.4 Work Area and System Description. Dining facility equipment covered under this Contract includes, but is not limited to, the following:

- Coffee makers, urns, beverage dispensers, and ice machines
- Braising pans and steam kettles
- Bread slicers, food saws/slicers, and electric mixers
- Toasters
- Broilers, cookers, grills, steamers, ovens, ranges, fryers, and microwave ovens
- Dishwashers and racks

- Garbage disposals
- Refrigerators and freezers

An inventory of all dining facility equipment covered under this Contract is provided in Technical Exhibit 5.3-001.

5.3.5 Preventive Maintenance of Dining Facility Equipment. The Contractor shall perform preventive maintenance on dining facility equipment on a semi-annual basis unless otherwise specified. All PM shall include an operational check and a check for rusted components that could cause leakage or contaminate food. Preventive maintenance shall be scheduled so as to minimize interference with dining facility activities. The Contractor shall prepare and submit a DA Form 4283 within 24 hours after discovering any work that exceeds the scope of a service order or the PM requirements listed below, and the repairs shall be made via Project Work Order.

5.3.5.1 Preventive Maintenance Records. The Contractor shall maintain a current record of all dining equipment receiving PM. The Contractor shall update the record on the same working day that the PM is performed and include the following information:

- Dining facility name/number
- Date PM performed
- Equipment name/number receiving PM
- Hours expended on PM for that equipment
- Description of work performed
- Parts Cost on Service Order

5.3.5.2 Ice Machine Preventive Maintenance. The Contractor shall perform the following PM tasks on a semi-annual basis for ice machines:

- Check oil condenser fan motor
- Clean condenser (air and water cooled)
- Clean water circuit with ice machine cleaner
- Check capacity output
- Check controls (bin thermostat and ice size thermostat)
- Check refrigerant pressure
- Check ice bin for cleanliness
- Observe operation if each unit through one complete cycle of operation

5.3.5.3 Reach-in Refrigerator and Freezer Preventive Maintenance. The Contractor shall perform the following PM tasks on a quarterly basis for reach-in refrigerator and freezers:

- Clean condenser (air and water cooled)
- Oil condenser fan motor
- Oil evaporator fan motor
- Clean evaporator coil
- Change temperature charts
- Check doors for proper alignment
- Check door gasket condition
- Check thermostat for proper setting
- Check pressure controls for proper setting
- Check fan cycling control
- Check defrost heaters (freezers only)
- Check defrost time clock
- Check thermometer for actual temperature reading
- Check condition of cabinet
- Check condensate drain lines
- Check drain line and pan heaters
- Check electrical wiring
- Check inside light fixture
- Observe operation if each unit through one complete cycle of operation

5.3.6 Repair of Dining Facility Equipment. The Contractor shall adjust, maintain, and repair dining facility equipment. Work shall be done in accordance with the manufacturer's instructions and applicable regulations.

5.3.7 Types of Equipment Repair. The following paragraphs contain types of equipment and potential repairs that may be required as part of this Contract.

a. Refrigeration Equipment Repair. The Contractor shall repair domestic and display case refrigerators (including sandwich units); walk-in refrigerators and refrigerator/freezers with external condensers; ice cream cabinets, freezers, and machines; cold food counters; ice making machines; fluid coolers; and all other refrigerated units in covered facilities. Repairs shall include, but not be limited to, the following:

- Repair pumps and spray assemblies.
- Repair floats and incorrect water levels.
- Repair or replace fan blades, fan belts, vibration pads, refrigerant leaks, compressors, crankcase heaters, metering devices, cabinet parts, valves, defective piping, pulleys, and insulation.
- Repair or replace improperly operating starter panels and controls, burned or loose contacts, and loose connections.
- Repair, adjust, or replace damaged door gaskets.

b. Ice Machine Repair. The Contractor shall repair ice machine equipment to include, but not be limited to, the following:

- Repair/replace condensers (air and water cooled)
- Repair/replace thermostats and controls
- Repair/replace all hoses and valves
- Repair/replace electrical wiring
- Repair/replace pressure gauges.

c. Beverage Dispenser Equipment Repair. The Contractor shall repair beverage dispenser equipment to include, but not be limited to, the following:

- Repair hoses and valves to prevent leakage.
- Repair electrical wiring for both safety and operational reasons.
- Repair/replace pressure gauges.
- Repair/replace broken door latches.
- Repair/replace fans and belts.

d. Baking Equipment Repair. The Contractor shall repair baking equipment, including electric floor and counter mixers, electric cookie makers, dough dividers, dough rollers, automatic proofers, and electric pie makers. Repairs shall include, but not be limited to, the following:

- Repair belts, gears, motors, and motors bearings.

- Repair switches, controls, and conveyors.
- Repair electrical wires.
- Repair cutters and chain drives.
- Repair motors, thermostats and heating elements.

e. Deep Fat Fryer. The Contractor shall repair deep fat fryers (conventional, pressurized, and automatic conveyor belt) and ensure repairs conform to NFPA Standard 96. The Contractor shall check and set high limits semi-annually. Repairs shall include, but not be limited to, the following:

- Repair compartments, valves, and piping for leaks and tighten as required
- Repair/replace pilot, igniter, and jets on gas operated units
- Repair thermostat and calibration
- Repair elements, switches, controls, contacts, and wiring
- Replace broken knobs

f. Broiler (Conveyor-type gas broiler and electric). The Contractor shall repair broiler equipment. Repairs shall include, but not be limited to, the following tasks:

- Pilots, igniters, and jets
- Replace air filters, chains and counterweights
- Repair and replace piping and valves
- Repair wiring defects
- Repair/replace thermostats
- Replace/repair doors for proper functioning

g. Ovens (Gas, Electric, Convection, Microwave, Rotary, etc.) The Contractor shall repair ovens. Repairs shall include, but not be limited to, the following tasks:

- Repair doors, seals, hinges, broken knobs
- Repair piping, valves, fan blades and the fan motor (where appropriate)
- Repair pilot, igniter, and jets
- Repair/replace elements, switches, controls, and wiring
- Repair doors, door gaskets hinges and latches

- Repair/replace motors, chain drive mechanisms, and bearings
- Repair/replace thermostats

h. Gas and Steam Cookers. For gas and steam cookers, repairs shall include, but not be limited to, the following tasks:

- Repair doors, seals, hinges, broken knobs
- Repair piping, valves, fan blades and the fan motor (where appropriate)
- Repair pilot, igniter, and jets
- Repair/replace elements, switches, controls, and wiring
- Repair doors, door gaskets hinges and latches
- Repair/replace motors, chain drive mechanisms, and bearings
- Repair/replace thermostats
- Replace/repair steam traps, strainers and pressure gauges

I. Grills, Griddles, and Ranges. For gas and electric grills, griddles, and ranges, the Contractor shall repair the following, but not be limited to:

- Repair piping and valves
- Repair pilots, igniter, and jets
- Repair/replace elements, switches, controls, and wiring
- Repair doors, door gaskets hinges and latches

k. Steam Kettle. Contractor repairs shall include, but not be limited to, the following tasks:

- Repair piping and fittings for leaks
- Repair the electric water valve
- Repair gear mechanisms

l. Steam Table. Contractor repairs shall include, but not be limited to, the following tasks:

- Repair water compartments, steam coil, valves, and piping for leaks
- Repair pressure regulating valves and gauges
- Repair/replace pilots, igniter, and jets
- Repair insulators, connections, and wiring
- Repair thermostats and temperature gauges

m. Electric Floor and Counter Mixers. The Contractor shall perform all repairs that are required to mixers. Repairs shall include, but not be limited to, the following:

- Repair, align or adjust belts and gears
- Repair motors, anchor bolts, and lift mechanisms

n. Dishwashers. The Contractor shall perform all repairs that are required to dishwashers. Repairs shall include, but not be limited to, the following:

- Repair electric insulators, connections, and wiring
- Repair motor and bearings
- Repair dish conveyor and chain
- Repair spray coverage, drainage and feeder lines
- Repair doors, wash, rinse, and drain valves, and pumps
- Repair thermostatic probe, gauges and thermometers

o. Garbage Disposal. The Contractor shall perform all repairs that are required to garbage disposals. Repairs shall include, but not be limited to, the following:

- Repair motors and drive shafts, electrical wiring, and cutters
- Repair supply and drain connections, grinders, drives, and motors

p. Trash Compactor. The Contractor shall perform all repairs that are required to trash compactors. Repairs shall include, but not be limited to, the following:

- Repair rams, guide block retainers, and ram walls
- Repair safety switches, contacts, and disconnects
- Repair motor exteriors, lines, and mechanical areas

q. Miscellaneous Equipment Repair. The Contractor shall perform all repairs that are required on miscellaneous pieces of equipment, including electric choppers, electric food slicers and saws, electric bread slicers, electric vegetable peelers, dough dividers, dinnerware dispensing equipment, etc. Repairs shall include, but not be limited to, the following:

- Repair pumps, spray assemblies, floats, and water levels and devices
- Repair blades, levers, handles, and safety devices
- Repair or replace fan blades, fan belts, vibration pads, refrigerant leaks, compressors, crankcase heaters, metering devices, cabinet parts, valves, defective piping, pulleys, and insulation

Repair or replace improperly operating starter panels and controls, burned or loose contacts, wi

5.3.8. Equipment Removal and Installation. Upon receipt of a valid Service Order, the Contractor shall remove and install dining facility equipment.

5.3.8.1. Replacement. Upon determining that a piece of equipment requires replacement, the Contractor shall submit a written recommendation to the KO Substantiating data to support the recommended replacement shall be included. Substantiating data includes, but is not limited to, written results of the technical inspection and documentation of replacement cost versus repair cost. The Contractor shall include documentation that utility connections are available, that the recommended equipment will fit in the available space, and that the equipment will fill the required function.

5.3.8.2. Assembly Instructions. New dining facility equipment shall be assembled according to the manufacturer's instructions and applicable regulations. Cooking equipment shall be installed in accordance with NFPA Standard 96.

5.3.8.3. Connection. The Contractor shall connect new dining facility equipment to available utilities connections. Where connections require modification or installation, the Contractor shall arrange for the work to be performed in accordance with Functional Area 5.2, Buildings and Structures Maintenance, and Functional Area 5.4, Utility Systems Operation and Maintenance.

5.3.8.4. Start-up Inspection. The Contractor shall perform start up inspection and services before new, reconditioned, or inactive dining facility equipment is placed in service. Start-up services include, but are not limited to, inspection of all parts, accessories, and connections for correct adjustment before the equipment is started, and testing and repair, if necessary, to ensure proper operation. The Contractor shall leave dining equipment in a configuration that allows it to be used immediately for food preparation.

5.3.8.5. Shutdown Services. The Contractor shall perform shutdown services as necessary to protect equipment that is to be inactivated or placed on standby use and to conserve materials. Such services include inspecting and servicing equipment to ensure that it is properly secure, protected against deterioration, and are in stand-by condition. Stand-by services shall include, but not be limited to, the following:

- Shut off water, gas valves, and electrical current at wall or unit control box outlet
- Drain all water and cooking compartments
- Drain and clean peel traps for vegetable peelers
- Drain circulating pumps for dishwashing machines and coffee urns

- Drain all water, steam, and drain piping that forms a part of the equipment
- Relieve belt tension on all belt-driven equipment

5.3.9. Removal Services. The Contractor shall remove and dispose of dining facility equipment as directed by the KO. The Contractor shall close off utility connections and properly shut down, remove, and transport equipment to designated disposal sites. Gas shall be secured to the facility, and lines shall be capped or disconnected. The Contractor shall drain refrigerant and other chemicals and dispose of them in accordance with Installation procedures

Functional Area 5.4

UTILITY SYSTEMS OPERATION AND MAINTENANCE

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LIST OF TECHNICAL EXHIBITS

Functional Area 5.4

UTILITY SYSTEMS OPERATION AND MAINTENANCE

5.4.1 INTRODUCTION

The Contractor shall perform maintenance and repair of the water distribution systems, storm water collection systems, fuel distribution systems, dual temperature pipes, steam pipes, and associated structures and appurtenances at Fort Benning. The Contractor shall perform services as required herein to provide complete maintenance of all utility systems. Qualified personnel in accordance with applicable laws, regulations in Section C-6 of this Contract, shall perform all work and Government-developed annual and long range plans. Technical Exhibits provide expanded information for this Functional Area.

5.4.2 SCOPE OF SERVICES

5.4.2.1Work Area and System Description

The Contractor shall be responsible for connecting and disconnecting all Government-provided utilities. Utility systems to be maintained under this Contract are partially identified in the maps provided in the Technical Library. All distribution lines, water mains, and associated facilities shall be maintained as specified herein. The Contractor shall all Government-owned utilities located on properties owned by Fort Benning. This includes all training areas and ranges.

5.4.2.1.1 Potable Water Storage and Distribution Systems. The potable water distribution system includes, but is not limited to, water mains, water storage tanks, pump stations, booster pump stations, fire hydrants, shut-off valves, pressure reducing valves, backflow assemblies, meters, and water reservoirs. . The Government will submit approved change requests to the State of Georgia.

5.4.2.1.2 Storm water Collection System. The Contractor shall maintain and repair the storm water collection system that includes, but is not limited to, culverts, grates, catch basin inlets, manholes, outlets, swales, gutters, channels, terraces, underground pipe and conduits, inlets, junction structures, drop structures, chutes, energy dissipaters, and storm sewers. On annual basis starting in November, and ending on 31 January of each year, the contractor shall inspect storm drains and gutters along all roads in the Main Post Historic district twice per month and unclog as necessary to prevent flooding on those roads. Contractor is to pump trash and sediment as needed from the storm drain systems on Main Post, Kelley Hill, Sand Hill, Harmony Church, and family housing. Most of the vortex chambers installed are in the new armor school area of Harmony Church and Sand Hill. There is one in Kelley Hill and one at Building 4.This requirement shall be coved under OWO.

5.4.2.1.3 Electrical Systems. The Contractor shall maintain and repair electrical systems which include, but are not limited to, traffic lighting systems, area lighting (e.g. PT fields and ball fields), school crossing lights, navigation and airfield lighting, security lights on exteriors of buildings, and roof mounted hanger lights.

5.4.2.1.4 Fuel Storage and Distribution Systems. The Contractor shall maintain and repair the fuel distribution system that includes, but is not limited to, fuel distribution lines and underground and aboveground fuel storage tanks.

5.4.2.2 Work Management and Control

5.4.2.2.1 Reporting Requirements. The Contractor shall prepare, submit, and maintain all records, reports, and submittals as specified herein and in accordance with the Contract Data Requirements Lists (CDs) in Technical Exhibit 1 -004.

5.4.2.2.2 Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6 of this Contract.

5.4.2.2.3 Compliance. While performing any work on utility systems, the Contractor shall comply with all applicable Federal, State, and local laws, Army regulations, standard industry codes, and manufacturer's recommendations.

5.4.2.2.4 Cross-Connection and Backflow Prevention Program. The Contractor shall develop and implement a comprehensive cross-connection control and backflow prevention program in accordance with American Water Works Association (AWWA), Federal, State, local, and Code Compliance regulations. The program shall, at a minimum, prevent the connection of potable and non-potable water systems and ensure that approved protective devices are operational to prevent backflow between systems. The Program shall be implemented as part of the Contingency Service Plans within 60 calendar days of Contract start date and shall provide a systematic approach to accomplishing plumbing work in the first year of the Contract (CD 504R001).

5.4.2.2.4.1 Initial Surveys. The Contractor shall perform initial surveys of new facilities to be added to the building inventory at Fort Benning to ensure that the proper and approved type of backflow prevention devices were installed and meet AWWA, Environmental Protection Agency (EPA), State, and local compliance codes.

5.4.2.2.4.2 Records. The Contractor shall maintain documentation of all records and make them available for Government inspection within one hour of being notified. The Contractor shall maintain accurate as-built drawings of new and modified backflow prevention devices. The Contractor shall maintain records of the required quarterly meetings and ensure these records are available for inspection within one hour of being notified by the Georgia Department of Health, the EPA, or local authorities.

5.4.2.2.5 Notice of Water Service Disruption. The Contractor shall plan and coordinate all service outages with the KO for all Installation buildings. The Contractor shall notify the affected parties at least 15 calendar days prior to the scheduled outage. The Contractor shall notify the KO and all activities involved within one hour after an emergency outage requires discontinuation of potable water service. The

Fire Department shall be notified immediately when any section of the potable water distribution system is out of service and when the system has been returned to normal service. The Contractor shall make every attempt to utilize existing loop systems and reinforced connections to minimize the impact on normal service operations. The contractor shall provide temporary water service, where applicable, while the system is down.

5.4.2.2.6 Notice of Steam Service Disruption. The Contractor shall plan and coordinate all service outages with the KO for all Installation buildings. The Contractor shall notify the affected parties at least 15 calendar days prior to the scheduled outage. The Contractor shall notify the KO and all activities involved within one hour after an emergency outage requires discontinuation of steam service.

5.4.2.2.7 Notification of Other Utility and Telecommunications Providers. The Contractor shall notify utility providers regarding any work that could impact the utility lines owned by these providers. Examples of utility providers include, but are not limited to, Southern Bell, AT&T, and Atmos Energy. The KO will notify the Contractor of any changes or additional utility providers.

The Contractor shall also notify each potential utility that may require interaction with the Contractor and provide a point of contact.

5.4.2.2.8 Lock Out/Tag Out. The Contractor shall furnish and use Lock Out/Tag Out devices when performing maintenance of electrical systems. Devices shall indicate the employee name, shop, telephone number, and date of maintenance

5.4.3 SCHEDULED TASKS

The Contractor shall perform the tasks described below on a recurring or scheduled basis and record the work under approved Operation Work Orders (OWO's). Preventive maintenance and other scheduled task frequencies that apply to this Functional Area are presented in Technical Exhibit 5.4-001. If a requirement for further repair work on any equipment is identified during the performance of scheduled tasks, the Contractor may be required to accomplish such work on an approved Demand Maintenance Order (DMO) or Project Work Order (PWO).

Contractor shall provide temporary water service, where applicable, while the system is

5.4.3.1 Water Distribution Systems

The Contractor shall read water meters on a monthly basis and report the readings to the KO. The Contractor shall replace, repair, and maintain water meters as required.

5.4.3.2 Electrical Systems

The Contractor shall maintain Electrical Systems to provide efficient and appropriate service to the Installation. The Contractor shall handle all non-critical requirements in accordance with emergency, urgent, or routine work, as applicable to the occasion.

5.4.3.2.1 Engine Generators. The Contractor shall perform a monthly inspection of each auxiliary engine generator to test for start-up and verify full load, voltage, and current. The Contractor shall note any pertinent information on the inspection performed in a log book, including the date, generator tested, time and duration of the test, load carried, condition of equipment, name of operator, and any problems encountered. A list of engine generators to be load tested is presented in Technical Exhibit 5.4-002.

5.4.3.2.2 Logging hours for Emergency Generators. The contractor shall log the hours for each generator and the reason for operation, i.e., Emergency lighting operation, maintenance, monthly inspection, power outage etc. Contractor shall record the size of the associated fuel tank and whether or not it is a UST or AST. This should be a one-time data call, as the information should not change unless the generator changes or a new tank is installed. Contractor shall ensure that periodic oil and filter changes, air filter, hose, and belt inspections are performed and the maintenance completed is recorded on TE 5.4-002a.

5.4.3.2.2 Electric Meters. The Contractor shall replace and install instrument transformers, current transformers (CTs), and potential transformers (PTs). Both primary and secondary (less than 750 volts) meters are included in this requirement.

5.4.3.2.3 Weather Sirens. The Contractor shall be advise of any repairs that are needed during normal checks of operation of the Installation weather sirens.

5.4.3.3 Fuel Distribution Systems

The Contractor shall maintain facility fueling systems to include, but not limited to: propane tanks, underground and above ground heating fuel tanks, underground and above ground propane and diesel piping, and regulators. Technical Exhibits 5.4-003 and 5.4-004 list the locations and capacities of various fuel and propane tanks on Fort Benning. Contractor shall record all facility fuel deliveries and usage on a monthly basis. Technical Exhibit 5.4-004a contains the same data for Camp Merrill. In addition, the Contractor shall maintain vehicle-fueling systems including, but not limited to, dispensers, meters, piping, controls, tanks, and calibrating dispensers. The Contractor shall monitor fuel consumption, specifically fuel oil, JP8, diesel. The contractor shall track fuel usage per vehicle/equipment using TE 5.4-006. Bulk vehicle storage and transfer tanks shall contain identification data permanently attached and fuel log for each tank. Signage indicating emergency contact information (same as LP tanks). Maintain record of fuel dispensed and delivered to each individual tank. Fill openings and dispensers shall be secured with locking devices and maintained key control. Equipment transfer tanks shall meter dispensed in DPW equipment. DPW fuel tanks shall not dispense fuel to GSA, leased or other non-authorized vehicles. Technical Exhibit 5.4-003 provides a list of the fuel tank locations and tank capacity, while Technical Exhibit 5.4-004 provides a list of the fuel and propane tank locations and capacities.

5.4.3.3.1 RADD Reports. The Contractor shall prepare a monthly facility fuel report. This report shall contain a record of all facility fuel (fuel oil, gasoline or diesel) delivered for the month (CD 504R002).

5.4.3.3.2 Propane Fuel Distribution Systems. Fort Benning uses propane as its primary fuel with a total storage capacity of 172990 gallons. On an annual basis, the Contractor shall inspect all propane distribution lines for possible leaks or damage. The Contractor shall perform maintenance or repairs to lines upon discovery of any damage or leaks. In addition, a minimum of every three years, the Contractor shall paint approximately 199 propane tanks and perform all required maintenance to ensure the proper operation of the tanks. TE5.4-004 provides a list of the propane tank locations.

5.4.3.3.3 Signage and Marking of Propane Tanks. Signage and markings of propane tanks, equipment, instruments and appurtenances are required by NFPA regulations. The Contractor shall insure that all current and future NFPA regulations are met.

5.4.3.5 Peak Shaving Propane Plants

Fort Benning has two propane plants used for peak shaving natural gas to maintain gas costs at the lowest possible level. The plants have a total of 25 30,000-gallon tanks and two buildings containing mechanical equipment to include, but not limited to, unit heaters, and air compressors. The Contractor shall be responsible for operating the plants whenever called upon to do so by the gas company (currently Atmos Energy), which has historically varied from one to fifteen occurrences per year. The Contractor shall commence operation within 30 minutes of receiving notification from the gas company. Last sentence deleted.

5.4.3.5.1 Start-up. The Contractor shall begin operation of the Peak Shaving Plants within 30 minutes of notification from the KO. Start-up shall include creating the proper Wobbie index mix and the suggested operation of the plants in accordance with specifications in Technical Exhibit 5.4-005 and in further detail in the Technical Library.

5.4.3.5.2 Gas Boiler. On an semi-annual basis, the Contractor shall: check, grease, and operate gas valves; clean gas strainer; check diaphragm in regulators and adjust regulator; check all gauges; check gas lines for leaks; verify operation of gas pressure safeties; clean and service the burner and burner assembly; check operation of flame rod and check milivolt output and adjust as necessary; drain boiler; inspect and clean refractory and tubes; check pilot assembly and gas delivery medium; check, clean, oil, and grease all controls, operators, and linkages; calibrate air to fuel for all combustion ranges; check all safeties, high limits, low limits, and pressure reliefs; drain and remove water feeder; inspect float assembly; check operation of internal switches; check blow- down on water feeder; remove all dielectric unions and internally clean; check all electrical connections; check all contacts and replace as necessary; calibrate barometric draft damper; open and close all valves, repack valves, and grease as necessary; drain and clean expansion or surge tank; clean mechanical room; observe operation of each unit through one complete cycle of operation; and repair all deficiencies.

5.4.3.5.4 Air Compressors and Receiver. On a semi-annual basis, the Contractor shall: drain water from receiver; check for air leaks on receiver and lines; operate all valves serving receiver; check all gauges and relief valve on receiver; clean or replace intake air supply filter; clean or replace discharge air filters; check all protective controls including high discharge air temperature, high bearing temperature, main drive motor, starter overload, lubricant filter indicator, and inlet air filter indicator; check alignment of mounting bolts and tighten as necessary; check for excessive vibration; lubricate motor and bearings; check belts for wear and replace as necessary; check air pressure control, regulators, and pressure reliefs for proper setting; check compressor oil and filter/separator; adjust oil pressure to the proper level; check air-cooled after cooler and oil cooler coolant for proper level; check lines for leaks; check moisture separator; check all components of control panel; check all electrical connections contacts, relays, starters, and heaters; check current draw on motors; check pump up times;

observe operation of each unit through one complete cycle of operation; and repair all deficiencies.

5.4.3.5.5 Air Dryer. On a semi-annual basis, the Contractor shall: check air dryer cabinet for damage; check oil filter; check condensation drain and oil indicators for proper operation before and after filter change; check refrigerant pressure; observe operation of each unit through one complete cycle of operation; and repair all deficiencies.

5.4.3.5.7 Propane Pumps. On a semi-annual basis, the Contractor shall: check frame, support and pad for excessive vibration; check pipe and supports to ensure no undue stresses; check flanges, bolts, gaskets, gauges, unions, and valves for leakage; check pump seals and bearings and tighten, repack, and lubricate as needed; check alignment shafts and couplings for slippage; check condition of pump housing; check condition of pump housing and impeller; check impeller for proper rotation; check electrical connections, relays, starters, and heaters; check control system and system safeties; observe operation through a complete cycle; and repair all deficiencies.

5.4.3.5.8 Propane Tanks. On a semi-annual basis, the Contractor shall: check temperature, pressure, and capacity gauges; check for rust and paint as necessary; check foundation and supports to include piping supports; check grounding of each tank; check for leaks in piping and tanks; check all valves for proper operation; check safety reliefs; and repair all deficiencies.

5.4.3.5.9 Specific Gravity Instrument. On a semi-annual basis, the Contractor shall: check for proper calibration; check chart pen and drive mechanism; lubricate motor and bearings; check belt and motor pulley; observe operation through a complete cycle; and repair and correct all deficiencies.

5.4.3.5.10 General Operation and Maintenance. The Contractor shall test run the Propane plant semi-annually during the preventive maintenance periods. Additional maintenance items include: check all valves, including those used in main gas yard; check piping, igniter, and controls at burn-off station observe operation through a complete cycle; and repair and correct all deficiencies.

5.4.3.6 Cross Connection and Backflow Prevention

5.4.3.6.1 Perform Testing: Contractor shall perform testing of all backflow and cross connection retentions devices with the plan referenced in paragraph 5.4.2.2.4

5.4.4 UNSCHEDULED TASKS

The Contractor shall perform the tasks described below when initiated through either a DMO or an approved PWO.

5.4.4.1 Water Testing and Treatment of Swimming Pools

The Contractor shall take samples from the one indoor pool located at Fort Benning upon the receipt of a valid work document or as directed by the KO to test for pH, calcium, and chlorine levels. The Contractor shall replenish the pool with the appropriate approved chemicals (calcium, chlorine, or sodium bicarbonate) as necessary to maintain proper water quality. The Contractor shall notify the Government when chemicals for the indoor pool need to be replenished.

5.4.4.2 Utility Marking

The Contractor shall locate and mark all underground utilities maintained under this Contract, perform site investigations to determine if an underground utility line exists, and coordinate all utility marking requirements with the Utilities Protection Center, Inc. All excavation work performed under this Contract shall have a utility spotting accomplished prior the start of excavation.

5.4.4.2.1 Utility Marking Requests. The Contractor shall maintain an office that shall be staffed during normal duty hours at which utility marking requests can be made either by phone or in person. Routine utility markings shall be completed within three working days of request and emergency markings shall be accomplished within two hours of request, 24 hours per day. The Contractor shall maintain a log of all utility marking requests.

5.4.4.2.2 Markings for Multiple Customers. Utility markings shall be performed not only for the work under this Contract, but also for other work performed on the Installation to include, but not limited to, phone line installation and repair, cable TV installation and repair, and support for other contractors.

5.4.4.2.3 Marking Locations of Utilities. The Contractor shall locate and mark underground utilities with spray paint or other suitable markers that are approved by the National Utilities Code of Color for individual underground cables.

5.4.4.2.4 Coordination. The Contractor shall coordinate all utility marking requests directly with the Directorate of Information Management (DOIM), the cable TV provider, and the Utilities Protection Center (800-282-7411).

5.4.4.3 Storm Drainage Systems

The Contractor shall maintain, install, modify, and repair the storm drainage system and associated equipment. Any entry into a storm water collection system requires a Confined Space Entry Permit, in accordance with the Confined Space Entry Regulations.

5.4.4.3.1 Storm water Collection Main Repair. The Contractor shall repair, install, modify, or replace all storm water collection mains to preserve the integrity of the storm water collection system. The Contractor shall be responsible for locating storm water collection main leaks, repairing the pipe, and returning storm water lines to serviceable condition. The Contractor shall check the storm water collection system during excessive rainfall to determine the problem areas.

5.4.4.3.1.1 Storm Sewers. The Contractor shall remove, install, and replace storm sewers, grates, drop inlets, and covers as necessary.

5.4.4.3.1.2 Replacing Pipe. The Contractor shall repair or replace damaged, broken, collapsed, or clogged drainage pipes by reopening pipe trench and repairing or replacing damaged or worn pipes. The Contractor shall excavate pavement to gain access to storm sewers or drainage facilities where required. The backfill to the excavation shall be compacted and surfaced area returned to its original condition.

5.4.4.3.1.3 Manholes. The Contractor shall maintain and repair sewer manholes including repairing or replacing caps, ladders, rings, covers, and interior linings; uncovering those covered by dirt and debris and raising top to ground level; and removal of grit, sand, and debris from manholes.

The Contractor shall, after rodding or cleaning sewer lines, remove all waste and debris that surrounds the manhole clean out area.

5.4.4.3.2 Excavation and Backfilling. If excavation and backfilling is required, the Contractor shall erect barricades prior to commencement of work and remove them after the work is complete. Backfilling shall be done with dry sand or compactable dirt and placing components on sand or dirt compacted every 2’ at 80% compaction. No gravel or rock larger than 1/2” diameter shall be allowed to contact utilities.

The Contractor shall dispose of waste from excavation at the sanitary landfill or as directed by the KO. The Contractor shall restore the area to its original condition.

5.4.4.3.2.1 Catch Basins. The Contractor shall clean catch basins, drop inlets, open flumes, manholes, and similar structures based on rate of silting or clogging with debris in order to ensure proper runoff. The Contractor shall maintain headwalls to prevent erosion or scour of the embankment adjacent to culvert inlets or outlets. No catch basin shall remain out of service longer than seven calendar days.

5.4.4.5 Steam Distribution Systems

The Contractor shall inspect, maintain, and repair all steam distribution lines to ensure no visible sign of steam leaking; check to ensure that steam air vents are not blowing steam; check insulation for corrosion, rusting, or pitting that all insulation is intact; check condition of ground to ensure there are no signs of underground leaks; and observe operation of each unit through one complete cycle of operation.

5.4.5.6 Electrical Systems

Exterior Lighting. The Contractor shall repair the exterior lighting system as required. The exterior lighting system includes, but is not limited to, walkway area lighting; perimeter and security lighting; windsocks and lights around windsocks; PT and recreational fields; exercise tracks; fire watch and communications towers; water tanks; 250 ft. training jump towers; obstruction markings; helipad and airfield lighting; and encompasses bases, poles, fixtures, and controls. Cleaning of light fixtures shall coincide with relamping.

5.4.5.6.1. Traffic and Street Lighting. The Contractor shall repair, replace, install, maintain, and test all traffic and street lighting systems in accordance with the National Electrical Code (NEC) Manual and operation manuals.

5.4.5.6.2 Airfield Lighting. The Contractor shall repair, replace, install, and maintain runway approach and obstruction lights.

5.4.5.6.3 Emergency Engine Generators. The Contractor shall maintain and repair emergency engine generators. AR 420-43 requires that the Contractor shall comply with maintenance schedules published in TM 5-811-3 and TM 5-684. The Contractor shall comply with all such mandatory maintenance schedules. In addition, AR 420-43 specifies special schedule requirements for auxiliary equipment used in certain applications.

5.4.5.7 Fuel Distribution Systems

The Contractor shall repair, locate, modify, or replace all lines associated with the fuel distribution system to include, but not limited to, the bulk petroleum, oil and lubricants (POL) facility, motor pool fuel tanks, and dispensers in accordance with 49 CFR Part 191-192.

5.4.5.7.1 Fuel Distribution. The Contractor shall respond to all fuel leaks in accordance with Priority I work. Fuel leaks are defined as those leaks causing evacuation of facilities and the response of the Fire Department and Environmental Department to all complaint calls. The Contractor shall notify the Fire Department by telephone and radio, within five minutes after the discovery or notification that there is a fuel leak for any of the underground or above ground storage tanks.

5.4.5.7.2 Refilling Fuel Tanks. The Contractor shall coordinate fuel deliveries with the service contractor. The Contractor shall complete DA Form 2765-1 and send copies of the form to the Fort Benning Budgeting Office and DOL Base Supply. The Contractor shall escort the service contractor during fuel deliveries. The Contractor shall submit a completed DD Form 250, within two working days after a fuel delivery, to DOL (three copies), to the Budgeting Office (one copy) and to the Fort Benning Contracting Office (one copy). The Contractor shall retain a copy of the DD Form 250 for one year.

5.4.5.8 Natural Gas Distribution Systems

Response to Natural Gas Service Calls. The Contractor shall respond to all gas leaks, odor complaints, or as directed by the KO in accordance with Priority I work. All gas leaks are defined as those leaks causing customer complaint of an odor; leaks caused by broken or ruptured piping, valves, or gaskets that produce a odor of gas; and any other gas leaks determined to be Priority I work by the Installation Fire Marshal, the KO, the customer, or any person that suspects or smells what is believed to be a gas leak, inside and outside of structures, along roads and trails where gas lines are located throughout the Installation properties. The Contractor shall notify the Fire Department, within five minutes after the discovery or notification that there is a gas leak or a suspected gas leak. The Contractor shall use approved gas leak detection equipment and shall maintain this equipment in peak operating condition at all times for the detection and location of gas leaks. The contractor shall notify Atmos Energy if the repairs fall within their contract. Atmos owns the natural gas distribution system. The line of demarcation of ownership is the outlet side of the pressure-reducing regulator for the facility.

5.4.5.9 Peak Shaving Plant Equipment.

5.4.5.9.1 Repair and Maintenance. The Contractor shall maintain and repair all components of the Peak Shaving Plant to include storage tanks, piping, valves, fittings, safety valves, mixing valves, air compressors, heaters, boilers, burners, pumps, all electrical controls and relays, meters, electric motors, wiring starters, electrical safety devices, and interlocks in accordance with TM 5-551K and manufacturer maintenance manuals. The Contractor shall adjust and calibrate controls, mixers, specific gravity controls, meters, electrical and electronic controls, and devices as needed.

5.4.5.9.2 Operation and Calibration. The Contractor shall calibrate equipment, adjust all controls in accordance with operating standards, and operate the peak shaving plant during its periods of operation.

5.4.5.9.3 Treatment of hot water boilers. Contractor shall add corrosion protection to all hot water boilers as shown on Technical Exhibit 5.4.xxx. Contractor shall conduct quarterly analysis of the boilers and add chemicals as needed to maintain proper concentration of the corrosion inhibitors.

5.4.5.9.3 Signage and markings on propane plant, equipment and instruments are required by NFPA. The contractor shall insure that all current and future NFPA requirements are met.

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Functional Area 5.5

HVAC SYSTEMS OPERATION AND MAINTENANCE

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Functional Area 5.5

HVAC SYSTEMS OPERATION AND MAINTENANCE

5.5.1 INTRODUCTION

The Contractor shall operate, maintain, repair, and replace heating, ventilating, and air conditioning (HVAC) equipment and systems. This shall include, but not be limited to, the handling of all fuels, lubricating oils, chemicals, auxiliary machinery, and boiler water treatment; attending all boiler fires to maintain steam pressure as required; and controlling the operation of chillers and cooling towers, as required by the KO. The Contractor shall be responsible for operating, maintaining, repairing, and replacing all air distribution systems (e.g. air handling units, exhaust fans, and fan coil units), evaporative cooling equipment, vacuum systems, chilled water pumps, and hot water pumps. The Contractor shall also be responsible for installing and inspecting new equipment and systems in accordance with manufacturer recommendations and all other applicable codes and regulations. Qualified personnel in accordance with Fort Benning's Title V Air Operating Permit, and all other applicable laws, regulations shall perform all work, and Government-developed annual and long range plans. Technical Exhibits provide expanded information for this Functional Area.

5.5.2 SCOPE OF SERVICES

5.5.2.1 Work Area and System Description

Systems and equipment covered under this Functional Area include, but are not limited to, the following:

- Air conditioners
- Air curtains
- Air dryers
- Air handling units
- Blowers
- Chilled water coils
- Chillers and associated components
- Cold storage facilities (walk-in and reach-in)
- Combination units (cooling/gas fired heating)
- Compressed air and compressed gas equipment
- Compressors (reciprocating or hermetic)

- Condensate pumps and tanks
- Condensers
- Building Controls (electrical, mechanical, and pneumatic)
- Coolers
- Cooling towers and associated components
- Evaporative condensers (open or closed-tube)
- Evaporators
- Exhaust fans
- Expansion tanks
- Fan coil units
- Filters (air and coolant)
- Guards, casings, hangers, supports, platforms, and mounting belts
- Heat recovery equipment
- Heaters
- Heating coils
- Hot water boilers and associated components
- Humidifiers
- Induction units
- Instruments
- Mechanical ventilation and exhaust equipment
- Mixing boxes
- Motors and drive assemblies
- Oil tanks
- Paint spray exhaust equipment
- Pumps
- Sand/grit blast and dust collection equipment
- Steam boilers and associated components
- Thermostats
- Vacuum equipment
- Wiring for HVAC systems

5.5.2.2 Work Management and Control

5.5.2.2.1 Compliance. While performing any work on HVAC systems, the Contractor shall comply with all applicable manufacturer recommendations and American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE); Occupational Safety and Health Administration (OSHA); Environmental Protection Agency (EPA); National Fire Protection Association (NFPA); 29 CFR; 49 CFR; American National Standards Institute (ANSI); Uniform Plumbing; and National Electrical codes, regulations, and standards.

5.5.2.2.2 Reporting Requirements. The Contractor shall prepare, submit, and maintain all records and reports as specified herein and in accordance with Contract Deliverable Lists (CDs) in Technical Exhibit 1-004.

5.5.2.2.3 Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.5.2.2.4 Conservation of Energy. The Contractor shall observe and test the efficiency of the operating equipment for the conservation of energy. The Contractor shall maintain HVAC system outputs as required by ASHRAE for each type of equipment. The Contractor shall ensure that minimum air changes are effected in accordance with ASHRAE requirements and based on comparisons of the temperatures of the outside air, the return air, and the mixed air.

5.5.2.2.4.1 The Contractor shall be responsible for programming Energy Management and Control System (EMCS) computers (including the FM Load Management system), damper controls, pressure, and air quality in such a manner as to ensure energy efficiency, the health of building occupants, and in accordance with Fort Benning energy policies.

5.5.2.2.5 Equipment Not in Use. The Contractor shall maintain all boilers and chillers that are not currently in use in a ready condition. If any boiler or chiller cannot be placed on line within eight hours, notification, including the probable cause, shall be given to the KO.

5.5.2.2.6 Emergency Notification. Operational emergencies, such as ruptured mains, loss of boilers, and loss of power, which reduce the steam rate by 15% or boiler pressure by 20% or more for a period extending beyond 30 minutes, shall be reported within 30 minutes of occurrence to the KO, together with identification of the probable cause and the estimated time required before full pressure and steam rate can be restored.

5.5.2.2.7 Noted Defects. The Contractor shall correct defects noted while maintenance is performed, but shall ensure minimum downtime when more complex work is required.

5.5.2.2.8 Use of Lubricants. The Contractor shall use types, grades, or weights of lubricants that are generally acceptable within the trade to perform all lubrication of equipment unless special types, grades, or weights are specified.

5.5.2.2.9 Chlorofluorocarbons (CFCs). The Contractor shall control all CFCs and prevent the unintentional release of CFCs to the atmosphere. Personnel with the appropriate certification to handle CFC materials shall only accomplish CFC exchange. CFC control shall be in accordance with EPA regulations. Additional instructions are shown in Technical Exhibit 5.5-002.

5.5.2.2.10 Marking. The Contractor shall mark and color code all new or unmarked exposed piping or conduit for chilled water, heating water, domestic water, oil, electric, natural gas, refrigerant, and all related HVAC systems. Any new marking or renewal of obscure or faded markings shall conform to Installation policies.

5.5.2.2.11 Thermostat Setting. The Contractor shall set the thermostats in non-exempt buildings at no lower than 78°F for the cooling season, and no higher than 68°F for the heating season, unless otherwise directed by the KO.

5.5.3 SCHEDULED TASKS

The Contractor shall perform the tasks described below on a recurring or scheduled basis and record the work under approved Standing Operating Orders (OWO′s). Preventive maintenance and other scheduled task frequencies that apply to this Functional Area are presented in Technical Exhibit 5.5-001. A partial list of installed serviceable equipment located at Fort Benning and its properties is provided in Technical Exhibit 5.1-001. If a requirement for further repair work on any equipment is identified during the performance of scheduled tasks, the Contractor may be required to accomplish such work on an approved Demand Maintenance Order (DMO) or Project Work Order (PWO).

5.5.3.1 Maintenance and Operation of the Energy Management and Control System (EMCS) and the FM Load Management System. Contractor shall provide comprehensive maintenance for the UMCS and all of its associated monitor and control points, up to but not including final controlled elements. Final elements such as fans, fan coil units, control valves, heat exchangers, pumps, variable speed drives etc. will be the responsibility of others.

5.5.3.1.1 On a semiannual basis, the Contractor shall inspect the head-end of the EMCS (DDC) Controller at each building and/or facility as follows: verify operation of diagnostic LED's for equipment status, visually inspect all electrical connections, test for correct voltage levels and continuity (as applicable); inspect for corrosion and correct voltage in the battery back-up system; inspect equipment for physical damage (repair/replace as applicable); verify IP addresses to include verifying proper network communication (if applicable) and test for proper operation between controller and the EMCS control center;

and check communication cables on all controllers and monitored equipment for damage. Note that some maintenance performance items may not apply for some controllers based upon the controller's design/purpose.

5.5.3.1.2 The contractor shall operate and monitor the Frequency Modulation (FM) Load Management System that is a computer based system located in Bldg 470. This system includes a computer/operating system, interface processor, transmitter site and FM receivers integrated into controls systems throughout the installation. This system by operator interface has the ability to run time of schedules and is the installation's primary controller during electrical load shedding periods. Method of controlling is by transmitting a unique code to a receiver (s) that can decode this signal to activate or de-activate the device (s) being controlled. The FM Load Management System currently controls air conditioning systems, heating systems, school crossing lights and field lighting and can be interfaced with all types of control systems where a "off-on" (binary) controller is required. Any problems experienced with the operation of the system are to be brought to the attention of the KO immediately.

5.5.3.1.3 Contractor shall utilize EMCS equipment resources and provide information as required to assist other service technicians in troubleshooting malfunctions and failures in equipment controlled or monitored by the EMCS. This shall include field visits, configuring the EMCS software, and producing logs, trends, and graphs. Maintenance and repair of the UMCS shall include, but not be limited to, host computer(s), printers, communications equipment, communications cable, control panels, sensors, transmitters, signal wiring, relays, and actuators used to enable the EMCS to effectively control or monitor HVAC equipment. After completing repairs that affects the integrity of the UMCS system, the contractor shall test the equipment and/or system. The Contractor shall correct deficiencies that are found before bringing the equipment or system on line. Repair and testing shall be accomplished during normal duty hours unless directed otherwise by the Contracting Officer. Where performance of the equipment is critical, the system shall be tested and proofed at the time of the repair.

5.5.3.1.4 The contractor shall develop a list of maintenance duties/tasks. It shall be incorporated in a Standard Operating Procedure (SOP) and be submitted and approved by the Contracting Officer. The list and SOP shall be updated periodically as needed and resubmitted for approval. Contractor is expected to perform all action necessary to maintain the EMCS.

5.5.3.1.5 Contractor shall perform maintenance and repair consisting of minor and major, routine, urgent, and emergency repairs to the UMCS as necessary for optimal efficiency and performance. Contractor shall provide, install, and configure updates and additions to software/firmware in EMCS equipment as required.

5.5.3.1.6 Contractor shall repair the EMCS devices upon notice of failure by operators or others, and when identified during scheduled maintenance. This includes, but is not limited to, servers, computers and peripherals, software maintenance, transmission equipment, control panels and all wiring, sensors and control devices.

5.5.3.1.7 Contractor shall verify communications from the local control devices to the control room, calibration, and proper operation of end devices in buildings. This includes buildings incorporating interfaces. Calibration shall be accomplished by comparing the actual value measured in the field with that displayed on an terminal. Proper operation verification shall be accomplished by observing that the final element or controlled equipment state is consistent with commands issued from the control room. A written report of discrepancies shall be furnished the Contracting Officer.

5.5.3.2 Boiler Plant Operations

The Contractor shall maintain and operate three steam boilers located at Building 9202 24 hours a day, 7 days a week in accordance with AR 420-49 and the Clean Air Act. On an hourly basis, the Contractor shall perform a visual inspection of all equipment to ensure proper operation, appropriate pressures, and the absence of apparent defects. In addition, on an hourly basis, the Contractor shall read the gauges and charts on all equipment and record the results in the approved log book to include, but not be limited to, gas and water consumption for each 24 hour period. Every eight hours, the Contractor shall blow down the sight glass, water columns, and low water cut out to ensure that low water safeties are working; check the oil and grease in all operating equipment; and check the boiler flame safety control by shutting the fire off. Class A and C inspections are required for high-pressure boilers greater than 15 psi. Buildings 2491, 9202, 3025, 3105, 3210, 3240, 3245, 3305, 3335, 3405, and 3425 have a total of 14 boilers greater than 15 psi.

5.5.3.2.1 Weekly Inspections. On a weekly basis, the Contractor shall inspect the air compressors and perform an inventory of boiler chemicals; grease the valves and operating equipment; complete combustion efficiency tests; and check the levels in the fuel oil tanks. The Contractor shall record results in the logbook.

5.5.3.2.2 Monthly Inspections. The Contractor shall inspect and maintain the following units located in the boiler plant on a monthly basis: surge tank; central plant valve; air compressors; large dual fuel boilers; chemical pumps and tanks; de-aerator tank; electric feed water, electric fuel oil service, and pumps; forced draft fan; and fuel oil strainers. In addition, on a monthly basis, the Contractor shall prepare the boiler water samples to be sent for testing to an independent testing laboratory.

5.5.3.2.3 Water Testing of Boilers. Every eight-hour period, the Contractor shall sample and test the boiler water and condensates. These test results shall be documented in the appropriate operator log. Based upon these tests, the Contractor shall adjust the chemical treatment of each boiler.

5.5.3.2.3.1 Annual Overhaul and Class "A" Boiler Inspection. The Government will designate a third party to conduct the Class "A" Inspection, and the third party will issue its report directly to the KO. The third party and the contractor shall coordinate the inspection schedules. The Contractor shall perform the following work on all boilers to prepare for the Class "A" Inspection.

Remove boiler from service.

Cool down boiler.

Open fire side.

Clean tubes, fire box, and tube sheets.

Remove manhole covers.

Clean waterside.

Dismantle and clean low water cutoffs.

Remove piping to operating and safety controls.

Remove plugs from cross connections.

Place boiler back in operation.

Check boiler operation.

Check boiler safety and operating controls.

Perform a hydrostatic test.

The Contractor shall then overhaul boiler components as necessary so that the boiler meets start-up specifications. Overhaul shall be performed via SO or PWO, upon receipt of a valid work document.

5.5.3.2.3.2 Annual Class "C" Boiler Inspection. The Government will designate a third party to conduct the Class "C" Inspection, and the third party will issue its report directly to the KO. The Contractor shall support the third party as necessary to assist in the inspection. The Contractor shall repair or replace worn or otherwise defective components upon receipt of an approved SO or PWO.

5.5.3.2.4 Feed water Pump Preventative Maintenance. The Contractor shall perform the following tasks on electric feed water pump systems at the Boiler Plant at the frequencies specified:

Visual inspection on each shift.

Grease pumps each week.

Perform amp test each month.

5.5.3.2.5 Burning of Oil. The Contractor shall burn oil in the boilers in accordance with State permits.

5.5.3.2.6 Notification of Gas Curtailment. Fort Benning is subject to natural gas curtailments imposed by the Installation's natural gas provider. Upon notification of gas curtailment, the Contractor shall switch to No. 2 Fuel Oil by the specified time. The Contractor shall maintain all equipment in a ready state to ensure an efficient transition. The Contractor shall continue to burn fuel oil until directed by the KO.

5.5.3.2.7 Record keeping. The Contractor shall maintain daily records of fuel used in each boiler and the propane peak shaving plant. From these daily records, the Contractor shall prepare the Fuel Consumption Report to meet state permit requirements (CD 505R001). This report shall be submitted to the KO by 10 Jan (for records from 1 Jul – 31 Dec) and 10 Jul (for records from 1 Jan – 30 Jun).

5.5.3.2.7 Treatment of Hot Water Boilers. Contractor shall add corrosion protection to all hot water boilers. Contractor shall conduct quarterly analysis of the boilers and add chemicals as needed to maintain proper concentration of the corrosion inhibitors. This requirement shall be completed on a SOO.

5.5.3.3 Water Testing and Treatment

The Contractor shall conduct water tests of HVAC equipment as designated in the following paragraphs or as required by the KO. If tests show abnormalities, the Contractor shall take corrective action. For all systems, the chemical feed rate and bleed-off shall be adjusted as necessary, a record of chemicals used, test results, and an analysis of findings shall be kept up to date, and chemical feed equipment shall be maintained. The Contractor shall conduct sample analysis on-site.

5.5.3.3.1 Condensate Returns. Condensate samples shall be taken daily for boilers in operation. The Contractor shall test pH and conductivity. The Contractor shall maintain condensate pH levels between 7.5 and 12.5. If conductivity is greater than 20 to 25 micro ohms, the sample shall be tested for calcium hardness. Iron shall be tested at condensate returns annually. Only a generic neutralizing amine shall be used.

5.5.3.3.2 Steam Boilers. The following parameters shall be tested and analyzed on steam boilers: hydrate alkalinity, phosphate, specific conductance (conductivity), total dissolved solids (TDS), silica, make-up and feed water hardness, causticity (23 to 200 ppm), phosphate (30 to 60 ppm), sulfite (20 to 40 ppm) on boilers with de-aerators, and polymers (2 to 5 ppm). Hydrate alkalinity shall be tested by barium chloride precipitation or determined by the (2p)-M calculation form conventional alkalinity. The Contractor shall also test de-aerators to ensure that oxygen is less than seven parts per billion. Unless otherwise stated, all steam boilers in operation shall be tested and treated weekly. All chemicals used in boilers shall be generic.

The Contractor shall record the results and take appropriate actions if deficiencies are discovered.

5.5.3.3.3 Reserved.

5.5.3.4 Air Conditioning to Heat and Heat to Air Conditioning Changeover

The Contractor shall maintain a report of all work performed for changeover services, including date, hours expended, equipment name and number, and facility name and number, and shall indicate any needed repairs submitted via SO. Technical Exhibit 5.5-003 provides a partial list of the buildings requiring changeover at Fort Benning. See para. 5.2.2.2.4

5.5.3.4.1 Heat to Air Conditioning. The Contractor shall begin start -up of all applicable cooling systems and shutdown of all applicable heating systems, historically on approximately 15 April to 15 May, and complete the start-up within 15 calendar days. Climatic variations may require the KO to direct changeover of seasonal equipment at dates earlier or later than those specified. During changeover, the Contractor shall take corrective action via SO or PWO if any operational deficiencies are identified.

5.5.3.4.1.1 Shutdown of Hot Water Boilers. At the end of the operating season, the Contractor shall add dispersant several days prior to shutdown, drain the system, flush the system, and lay the system up with treated water.

5.5.3.4.2 Air Conditioning to Heat. The Contractor shall begin shutdown of all applicable cooling systems and start-up of all applicable heating systems, historically on approximately 15 October to 15 November, and complete the changeover within 15 calendar days. Climatic variations may require the KO to direct changeover of seasonal equipment at dates earlier or later than those specified. During changeover, the Contractor shall take corrective action via SO or PWO if any operational deficiencies are identified.

5.5.3.4.3 Additional Tasks. The Contractor shall: switch over valves from air conditioning to heating and vice versa as specified; check EMCS controls; turn on computer controls; run systems through a complete cycle; activate thermostats; verify operation of all safety and operating controls; check systems for water and refrigerant leaks; check pressure gauges, dampers, and linkages; and check cooling/heating output and overall system operation.

5.5.3.5 Cooling Plants and Systems

The Contractor shall operate and maintain cooling systems and related auxiliary equipment including, but not limited to: chillers, cooling towers, air conditioning units, refrigeration systems, pumps, compressors, and motors. All equipment shall be maintained to meet operational requirements except when off-line for maintenance or secured for energy conservation. All scheduled maintenance shall include a visual inspection of all components to ensure proper operation and to check for leaks or other deficiencies. The Contractor shall take corrective action as necessary. The Contractor shall also ensure control systems operate and function in such a manner so as to maintain the specified output of the mechanical system it controls. The Contractor shall maintain logbooks and a building checklist following completion of all maintenance activities. The Contractor shall comply with Title VI(Clean Air Act)Stratospheric Ozone Protection Requirements on a monthly basis, (CD 505R-002). See para. 5.2.2.2.4

5.5.3.5.1 Semi-Annual Maintenance. The Contractor shall perform the following maintenance on a semi-annual basis on all cooling systems. Those tasks performed during the cooling season while the systems are in operation shall be external in nature. Those tasks performed during the heating season while the systems are shutdown shall be internal in nature.

5.5.3.5.1.1 Direct Expansion Cooling/Dehumidifiers. Maintenance tasks shall include, but are not limited to:

Cleaning coils.

Changing filters.

Checking for refrigerant leaks.

Checking expansion valves.

Cleaning drain and condensate pans.

Calibrating the system.

5.5.3.5.1.2 Air Compressors. Maintenance tasks shall include, but are not limited to:

Replacing air intake filters.

Checking the belt tension and wear.

Checking the air pressure control and pressure relief for proper setting.

Checking compressor oil.

Cleaning air-cooled condenser on air dryer.

Checking the air dryer cabinet and the oil filter with the automatic drain. Checking the automatic condensate drain.

Repairing and replacing pneumatic air lines.

Change oil on condenser pump electric motors.

5.5.3.5.1.3 Refrigeration Equipment. Maintenance tasks shall include, but are not limited to: Checking vessel connections, relief valves, coils, and lines for leaks.

Cleaning coils, drain pans, drain lines, fan blades, and cabinets.

Checking refrigerant oil, dryers, sight glasses, and charge; and calibrating the system.

5.5.3.5.2 Air Filters. Selected air filters shall be changed monthly. In conjunction with this task, the Contractor shall clean the equipment, identify other needed repairs, properly dispose of used filters, and maintain a record of where and when filters were changed. All new filters will be annotated with the technicians initials and date of filters changed. (All initials and dates will be legible). The buildings for which this requirement applies are listed within Technical Exhibit 5.5-006. The contractor will be responsible for keeping an updated building list monthly.

5.5.3.6 Heating Plants and Systems

The Contractor shall operate and maintain heating systems and related auxiliary equipment, including electric boilers, oil boilers, gas boilers, steam boilers, hot water boilers, boiler feed pumps, condensate return pumps, air compressors, and related equipment. All equipment shall be maintained to meet operational requirements except when off-line for maintenance or secured for energy conservation. All scheduled maintenance shall include a visual inspection of all components to ensure proper operation and to check for leaks or other deficiencies. The Contractor shall take corrective action as necessary. The Contractor shall also ensure control systems operate and function in such a manner so as to maintain the specified output of the mechanical system it controls. The Contractor shall maintain logbooks and a building checklist following completion of all maintenance activities. Technical Exhibit 5.5-005 contains a building list for heating plants and systems. The contractor will be responsible for keeping an updated building list monthly. The Contractor shall perform the following tasks on steam boiler systems and hot water boiler systems annually: See para. 5.2.2.2.4.

Pull, clean, and service burner and burner assembly.

Clean and inspect boiler water level control assembly (including fleat).

Check proper operation of flame management system

Verify operation of all safeties, high limits, low limits, and pressure relief.

Service and calibrate boiler controls for proper operation.

5.5.4 UNSCHEDULED TASKS

The Contractor shall perform the tasks described below when initiated through either a SO or an approved PWO.

5.5.4.1 Unscheduled Repair

The Contractor shall perform unscheduled repair on systems including, but not limited to: cooling systems, heating systems, air conditioning units, air compressors, vacuum systems, mechanical ventilation equipment, air distribution systems, EMCS and automatic temperature control systems. The Contractor shall inspect, test, clean, adjust, calibrate, and repair or replace all parts, or components, necessary to restore the equipment, or system to a condition to perform the function for which it was designed.

5.5.4.1.1 Cooling Equipment. The Contractor shall adjust, modify, replace, and repair all cooling systems and cooling towers, to include, but not limited to: motors, fans, drains, valves, and control devices. The Contractor shall locate and remove piping obstructions from fan coil units, air handler units with coils, and other cooling equipment. Trade practices, methods, and equipment shall be used.

5.5.4.1.2 Heating Equipment. The Contractor shall maintain and repair all steam fitting, low and medium pressure steam, and high temperature, high pressure hot water heating lines for heating and processing boilers, converters, pumps and associated controls, piping, and equipment. The Contractor shall maintain and repair all steam, hot water, gas fired, oil fired, electric generated heating equipment, and components.

5.5.4.1.3 HVAC Testing and Balance. The Contractor shall occasionally test and balance HVAC systems. The Contractor shall make all necessary adjustments to HVAC equipment based on the results of the tests. Test and balance would be performed to ensure proper airflow, water flow, and static pressure drop.

5.5.4.1.4 Water Softeners. The Contractor shall periodically check all boilers that have a zeolit water softener for hardness using the Total Hardness Test. When excess hardness is present in the softened water, the Contractor shall regenerate the softener. Hardness shall not exceed 2 PPM at any time. If necessary, the Contractor shall add sodium chloride to the brine tank as needed.

5.5.4.1.5 Air Conditioning Units and Air Cooled Condensers. Repair, replacement, and maintenance tasks shall include, but are not limited to: repairing air cooled condensers; coils; fan blades; fan bearings; fan belts; compressor oil; pressure controls; electrical connections; relays; starters; and heaters.

5.5.4.1.6 Air Compressors. Repair, replacement, and maintenance tasks shall include, but are not limited to: changing the compressor oil; motors; and all electrical contacts.

5.5.4.1.7 Cold Water Dispensers. Repair, replacement, and maintenance tasks shall include, but are not limited to: checking water lines for leaks; checking fan motor oil; and performing fan motor repairs.

5.5.4.1.8 Steam Boiler and Hot Water Boiler Systems. The Contractor shall perform repair, replacement, and maintenance tasks on steam boiler systems and hot water boiler systems. These shall include, but are not limited to: repair electrical equipment in accordance with TM 5-650; maintain soot blowers for proper operation; adjust pressure, temperature, and volume of boiler; repair or replace condensate pumps; repair or replace tanks; repair or replace float switches, couplings, valves, gauges, drains, seals, fittings, and strainers; and repair or replace electrical connections, starters, and contacts.

5.5.4.1.9 Gas Boiler. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing gas valves; gas strainers; regulators; gauges; gas lines; burner and burner assembly; refractory and tubes; pilot assemblies; controls; linkages; water feeders; float assemblies; barometric draft dampers; valves; surge tank; and insulation.

5.5.4.1.10 Oil Fired Hot Water Boiler. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing fuel filters and strainers; gauges and sight glasses; flame rod; milivolt output; controls and linkages; barometric back draft damper and air to fuel ratios for all combustion ranges; high and low limits; pressure reliefs; internal switches; float assembly; dielectric unions; and electrical connections and contacts.

5.5.4.1.11 Water Heater. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replace burner assemblies; thermocouples; gas lines; air and gas pressure; flue baffles; temperature and pressure relief valves; automatic flue dampers; and elements.

5.5.4.1.12 Steam and Condensate Trap. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing steam and condensate traps for normal operation; temperature sticks; internal components; bellows; seats; floats; plugs; and other mechanisms.

5.5.4.1.13 Oil Fired Furnace. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing fuel tanks; fuel filters; fuel strainers; fuel pumps; gauges and sight glasses; fuel lines for leaks; burner fans; burner assemblies and nozzles; milivolt output; ignition transformers; fireboxes and exchangers; safeties and electrical connections; barometric back draft dampers; filters; and ductwork.

5.5.4.1.14 Gas Fired Furnace. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing gas valves; gas strainers; diaphragms in regulators; gauges; gas lines; gas pressure safeties; burner assemblies; ignition transformers; igniters; electrical connections; and filters.

5.5.4.1.15 Dehumidifiers and Humidifiers. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing fan motors; evaporators; condensers; humidifier strainers; drains and drain pans; float assemblies; spray nozzles; and electrical connections, relays, and contacts.

5.5.4.1.16 Air Handling Units, Fan Coil Units, Unit Heaters, and Unit Ventilators. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing coils; electrical connections; motor mounts and supports; and check belts.

5.5.4.1.17 Exhaust Fans and Ventilators. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repair, clean, adjust or replace motors; electrical connections; motor mounts and supports; belts; fan blades; bird/insect screens and covers; frames; leaks; starters; contacts; louvers; and grills.

5.5.4.1.18 Heat Exchangers. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing leaks; strainers; safety devices; temperature controls; steam traps; and hand valves.

5.5.4.1.19 Circulating Water Pumps. The Contractor shall perform the following tasks on an as needed basis. These tasks include, but are not limited to: repairing, cleaning, adjusting, or replacing frames; supports; pads; flanges; bolts; gaskets; gauges; unions; valves; and air vents.

5.5.4.1.20 Chilled Water Distribution System. The Contractor shall maintain and repair approximately 96,120 linear feet of chilled water distribution lines. Maintenance tasks shall include, but not be limited to: lubricating valves; checking mechanical couplings; checking the operation of sump pumps, piping, and electric power to sump pumps; and cleaning pits of debris. In addition, on a monthly basis, the Contractor shall check the condition of the ground for signs of leaks and the condition of the insulation. Plant piping and equipment shall be painted as needed.

5.5.4.1.21 Closed-loop A/C Systems. As required, the Contractor shall add anti-freeze to replenish safe levels in the systems, prevent freeze-up, and bring the freezing point to -12°F. There are approximately 65 closed-loop water systems on Fort Benning.

5.5.4.1.22 Paint Booth. The Contractor shall repair and maintain all safety controls, operating controls, and heat and ventilation equipment on all paint booths.

5.5.4.2 New Equipment Installation and Inspection

The Contractor shall install new HVAC equipment and systems (to include EMCS/FM Load Management Systems) upon receipt of a valid work document. The Contractor shall inspect and evaluate new equipment for compliance with the manufacturer's specifications and shall notify the KO of all deficiencies noted during this inspection.

5.5.4.3 Equipment Modifications

The Contractor shall modify HVAC equipment and systems upon receipt of a valid work document and shall evaluate the modification of the equipment for compliance with the manufacturer's specifications. As detailed in Section C-1, the Contractor shall not perform any modification that voids an existing warranty.

5.5.4.4 Equipment and System Removal

The Contractor shall remove HVAC equipment and systems upon receipt of a valid work document.

5.5.4.5 Reduced Power Response

When required for reduced Electrical Distribution System (EDS) loading, the Contractor shall shut down non-critical air conditioning systems during reduced EDS loading within one hour after notification from the KO.

5.5.4.6 Unscheduled Air Conditioning to Heat Changeover

The Contractor shall change HVAC systems from air conditioning to heat or heat to air conditioning due to fluctuations in temperature or as directed by the KO.

5.5.4.7 Freezing Weather Maintenance

The Contractor shall perform maintenance, repairs, and checks during freezing weather conditions to prevent damage to HVAC equipment. All work shall be done in accordance with the freeze-up plan submitted as part of the Contingency Service Plans.

5.5.4.7.1 Boiler and Piping. The Contractor shall drain and blow out all idle equipment; check service water lines for possible freezing; check that insulation is not damaged; check tracing around control lines and transmitter boxes; and check steam traps for proper operation.

5.5.4.7.2 Compressor Air Systems. The Contractor shall drain air tanks; inspect air dryers, after coolers, and intercoolers; and drain, blow out, and flush with glycol all idle compressor jackets.

5.5.4.7.3 Air Conditioning and Refrigeration Systems. The Contractor shall drain, blow out, and flush all seasonal equipment, condenser lines, tubing, and piping. The Contractor shall inspect outside dampers and cut back cooling tower fan speeds.

5.5.4.7.4 Pressure Vessel Vents. The Contractor shall drain loops and bends and correct low spots in long runs. The Contractor shall protect vessel vents from ice or snow accumulation.

5.5.4.7.5 Relief Valves. The Contractor shall protect relief valves from snow or ice accumulation, clear relief valve body drains, protect valve discs from freezing shut, and relocate relief valves to heated areas wherever possible.

5.5.4.7.6 Mechanical Equipment. The Contractor shall drain all idle pumps and compressors. The Contractor shall ensure that jackets are vented, provide proper lubrication for cold weather protection, provide heated enclosures around operating equipment wherever possible, and install no-flow switches and alarms in cooling water lines.

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Functional Area 5.6

Grounds Maintenance

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List of Technical Exhibits

Exhibit Number	Title
5.6-001	Required Scheduled Tasks
5.6-002	List of Memorials
5.6-003	Leaf Removal Areas

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Functional Area 5.6

GROUNDS MAINTENANCE

5.6.1 INTRODUCTION

The Contractor shall perform maintenance and repair of grounds, landscaping, unimproved areas, associated structures and appurtenances at Fort Benning and supported satellite locations. The Contractor shall perform mowing, trimming, edging, aeration, and fertilization; weed and brush control; flower bed services; tree and shrub pruning; erosion control; debris, drain, and ditch cleanup; inspecting and correcting damage caused by vehicles, and storms; landscaping operations; and other services as required herein to provide complete grounds maintenance. All work shall be performed by qualified personnel in accordance with applicable laws, regulations, and documents in Section C-6 that include, but are not limited to, AR 200-1, AR 200-2, AR 200-3, TM 5-630, and Government developed annual and long range plans. Level and frequency of grounds maintenance will be in accordance with IMCOM Enterprise Municipal Grounds Maintenance Performance Standards. Technical Exhibits provide expanded information for this Functional Area

5.6.2 SCOPE OF SERVICES

5.6.2.1 Work Area and System Description

Grounds areas are segregated into the following categories: improved grounds and unimproved grounds. Improved grounds are those areas that encompass buildings and facilities and include lawn areas, ornamental bushes, and trees. Unimproved grounds are those areas that are located about the perimeter of the Installation and include natural grass, brush, and trees. There are approximately 173,000 acres of unimproved grounds on Fort Benning. Unimproved areas shall be maintained only to the level necessary to prevent fire hazards or obstructions that would cause traffic hazards, and to prevent problems arising from excessive dust, and erosion. Grounds areas covered under this Contract include, but are not limited to, the following:

- Post Cemetery inside Brick wall around Cemetery including Columbarium
- Sidewalks, driveways, curbs, shrubs, and hedges
- Common high profile grounds
- Maintenance areas
- Contractor’s shop/work areas
- Soldiers Park

5.6.2.2 Work Management and Control

Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.6.3 SCHEDULED TASKS

The Contractor shall perform the tasks below on a recurring or scheduled basis and record the work under approved Operation Work Orders (OWO's). Preventive maintenance and other scheduled tasks that apply to this Functional Area are presented in Technical Exhibit 5.6-001. If a requirement for further work on any area beyond the scheduled requirements below is identified, the Contractor may be required to accomplish such work on an approved Demand Maintenance Order (DMO) or Project Work Order (PWO).Level and frequency of grounds maintenance will be in accordance with IMCOM Enterprise Municipal Grounds Maintenance Performance Standards

5.6.3.1 Operation of Inert Landfill

The contractor shall have personnel available to operate the inert landfill from 0745 through 1615 Monday through Friday in accordance with Installation and State permits for allowable Dumping items when requested by an authorized user of the inert landfill. This shall include, but not be limited to: pushing inert debris such as tree, stump, and concrete debris into designated areas with a bulldozer; covering the debris with soil and ensuring proper drainage; inspecting trucks entering to ensure only inert material is dumped; opening and closing the main gate to prevent inappropriate dumping. Lumber shall be disposed of at the Installation Recycling Point and not at the landfill. The State permits for the operation of the inert landfill are contained in the Technical Library.

5.6.3.1.2 Operation of the Compost Site: The Contractor shall operate the compost site Monday thru Friday, January thru December. Operations shall include, but not be limited to: turning and spreading compost material with appropriate equipment to release gases and expedite decomposition; and visually noting that only compost material is being dumped in the areas.

5.6.3.2 Mowing

5.6.3.2.1 Mowing Standards. Grass shall be mowed when grass height reach 4" back to 2"in accordance with the IMCOM Enterprise Grounds Maintenance Performance Standards. An acceptably cut grassed area shall be cut to a uniform height. Grass cutting is to be accomplished in a manner such that it is not uneven or rough and is free of scalping, rutting, and bruising. The Contractor shall ensure that mower blades are sharpened to prevent shredding of grass ends during mowing operations. In lieu of collecting and disposing of grass clippings, the Contractor may use mulch-type mowers when grass height does not exceed 3 1/2 inches in unimproved areas only. The Contractor shall ensure that all ruts are repaired within two working days of lawn mowing completion. Grass in ditches shall be maintained according to the maintenance level specified for the surrounding area and cut such that the flow of water will not be impeded during storms and heavy rainfall. Normally, the Fort Benning grass growing season extends approximately eight months from early March through late November.

5.6.3.2.2 Other Responsibilities. Prior to mowing, the Contractor shall pick up and remove all rubbish, debris, and trash (which includes, but is not limited to, leaves, rocks, paper, pine cones, limbs, and other portable objects) within the maintenance area and all trash, papers, leaves, and limbs lodged in shrubs, hedges, fences, and along foundation walls. All clipping rubbish, debris, and trash removed shall be properly disposed of as directed by the KO. In conjunction with mowing services, the Contractor shall inspect grounds for trash and debris buildup, and possible fire hazards. Corrective action, if required, shall be initiated in accordance with the DMO or PWO provisions of Paragraph C-5.1.

5.6.3.3 Memorial Maintenance

The Contractor shall maintain official Government-maintained memorials year-round. The Contractor shall inspect the plants and trees semi-annually, or more frequently if necessary, and perform needed maintenance of plants and trees to keep them in a healthy, growing condition. For those memorials maintained by private associations or active military organizations, the Contractor shall be responsible for maintaining the grounds A list of those memorials to be maintained by the Contractor can be found in Technical Exhibit 5.6-004.

5.6.3.4 Specific Requirements for Buildings 4, 6 and 70

5.6.3.4.1 Buildings 4 and 6. The Contractor shall maintain shrubbery (approximately1800 bushes) and the parking lot that is adjacent to the buildings. The curbs and sidewalks must be trimmed and free of all debris, grass clippings, and weeds. Weeding and trimming shall be performed monthly from March through October. Shrubbery shall be maintained in such a fashion that pine straw is removed along with any excess and unintended vegetation. Shrubbery shall be trimmed monthly from March through October. During the winter months, leaves shall be cleared from the building and its surrounding areas, dead shrubbery removed, trees pruned, grounds fertilized, and grass thatched.

5.6.3.5 Leaf Removal

5.6.3.5.1 Mechanical Leaf Raking. The Contractor shall perform mechanical leaf raking and removal on a weekly basis from high visibility areas on the Main Post for ten weeks beginning approximately at the end of November. Technical Exhibit 5.6-005 contains the leaf removal schedule for these designated areas.

5.6.3.6 Plowing Jump Towers and Fryar Field

On a quarterly basis, the Contractor shall plow the two active jump towers at Fort Benning and Fryar Field. This consists of leveling the dirt in the areas mechanically or by manual means if required. The Jump Towers are approximately 5. 4 acres each and the area to be plowed at Fryar field is approximately 130 acres

5.6.3.7 Dredging of Ditches

On an annual basis, the Contractor shall dredge ditches from Bradshaw Street to LoBauch Street on 82nd Airborne Division Road.

5.6.3.8 Fencing Inspection

A minimum of twice each year, the Contractor shall inspect the fencing along the Ammunition Supply Point (ASP) and at Lawson Army Airfield (LAAF) to identify any holes or gaps between the fencing and the ground. Upon discovery of any holes or gaps, the Contractor shall fill the holes to close off the open area.

5.6.3.9 Culverts and Ditches

On an annual basis at approximately 12-month intervals, the Contractor shall clean sand and debris from Laundry Creek going through LAAF. The Contractor shall remove growth along the ditches of Laundry Creek and the ditch immediately south of the abandoned runway near Gate 20.

5.6.3.11 Debris and Limb Pickup.

Weekly, the contractor will inspect all major thorough fares (including the R i v e r walk) and remove deadfall and debris. In addition, the contractor shall remove leaves from the Riverwalk from Marne Road to Baltzell Street once per year during the month of January." This requirement shall be done on a OWO.

5.6.3.12 Maintenance of Edwards Street

On a semi-annual basis, the contractor shall trim trees and bushes, remove weeds, repair fence and/or fence fabric, and install new pine straw along Edwards Street between Wold Ave. and building 228

5.6.4 UNSCHEDULED TASKS

The Contractor shall perform the tasks described below when initiated through either a DMO or an approved PWO. The Contractor shall provide miscellaneous labor support to various agencies, such as installing lines, windsocks, handrails, machinery, and recreational equipment. The Contractor shall perform unscheduled grounds maintenance, to include mowing, vegetation removal, earth fill for erosion control, and other services as requested by the KO.

5.6.4.1 Trimming

The Contractor shall perform trimming on an as needed basis and as directed by the KO. The grass shall be trimmed from around trees, shrubs, cultivated areas, fences, poles, guard posts, fire hydrants, buildings, structures, parking lot bumper blocks, walls, sprinkler heads, valves, and other similar objects to match the height and appearance of surrounding vegetation without causing damage to desirable vegetation. String trimmers shall not contact the bark of the tree, shrubs, or building skirting. After trimming, all cuttings and other debris shall be removed from the sidewalks and paved areas and disposed of properly. Additionally, the Contractor shall trim vegetation from ditches and dredge ditches on an as needed basis.

5.6.4.2 Mulching

The Contractor shall mulch throughout all plant and shrub beds and around designated ornamental and shade trees for weed and grass control, soil conservation, and to minimize moisture evaporation. The mulch cover shall consist of bark mulch or other materials approved by the KO and shall be free of grass and weeds. Mulch will be added to a minimum depth of three inches and a maximum depth of four inches and shall extend beyond the perimeter of beds to established borders such as sidewalks, pavements, curbs, or walls. In the absence of established borders, mulch shall extend to the drip line of perimeter plants and shrubs. Mulch around designated ornamental and shade trees that are not growing in plant or shrub beds shall extend out from the base of each tree for a distance of 18 inches.

5.6.4.3 Trees and Shrubs

5.6.4.3.1 Tree Removal. The Contractor shall remove dead trees and trees which are a hazard or nuisance because of location (e.g., near electric lines or roads). The work shall include complete removal of trees, stumps, and debris; filling of hole with topsoil; and the seeding and mulching of the bare ground area. Logs, debris, and wood chips shall be deposited in the inert landfill. Bare ground areas shall be seeded as directed by the KO, as appropriate. Tree disposal shall be carried out at the inert landfill.

5.6.4.3.2 Stump Removal. The Contractor shall remove existing stumps to (24" depending on stump removed and all top roots) below ground level, fill the hole with topsoil, and seed and mulch bare ground as directed by the KO. All wood chips and debris shall be utilized on the Installation or deposited at the inert landfill. The Contractor shall chip removed stumps.

5.6.4.3.3 Pruning and Trimming. The Contractor shall trim and prune shrubs, ground covers, native plants, and trees planted in improved grounds. The Contractor shall trim hedges, bushes, and shrubs. All wood chips and debris shall be utilized on post or deposited at the inert landfill.

5.6.4.3.3.1 Shrub Pruning. All shrubs, bushes, hedges, and other cultivated plants are to be pruned according to their natural growth habit for proper health, attractive appearance in the appropriate season, and to prevent interference with pedestrian and vehicular traffic. Shrub pruning shall be accomplished throughout the year. Pruning is to be done in a manner so as to:

a. Prevent growth in front of windows, over entranceways or walks, and in all areas in which growth will obstruct vision at street intersections. Remove and dispose of dead, damaged, or diseased wood. If a shrub is removed, the Contractor shall fill in the hole created with topsoil and level that particular area to match the surrounding areas.

b. Evenly form and balance the shrub, bush, or plant.

c. Maintain the established hedge shape and appearance.

5.6.4.3.3.2 Tree Pruning. Pruning shall evenly form and balance the tree, promote proper health and growth, and prevent interference with pedestrian and vehicular traffic. Trees are to be pruned to a uniform height; tree growth around the base of the tree shall be removed. Pruning is to be done in a manner so as to:

a. Remove dead, damaged, or diseased wood, parasitic vegetation, and structurally weak limbs that may cause a safety hazard.

b. Remove branches to provide clearance over sidewalks, parking lots, driveways, buildings, and roofs; eaves; and windows that are in danger of damage.

c. Remove branches to provide clearance for buses, moving vans, and similar vehicles along streets and road systems.

d. Cut back branches that overhang or grow into power lines.

e. Shape the entire tree rather than notch the top.

f. Prevent growth of small trees in front of windows, over entranceways or walks, and in areas in which growth will obstruct vision at street intersections and within a six-foot radius of facilities.

g. Promote proper health, growth, and appearance.

5.6.4.4 Fertilizer and Lime Application

The Contractor shall also fertilize plants, grasses, and ground covers at the time of planting and transplanting on an as needed basis. The Contractor shall apply a variety of chemical and organic fertilizers in combination or in single application to correct soil and plant deficiencies, but not burn root system. When directed by the KO, the Contractor shall apply weed free organic manure.

5.6.4.5 Grass Planting and Sodding

5.6.4.5.1 Seeding. Upon the receipt of a valid DMO or PWO, the Contractor shall prepare a proper seed bed. The Contractor shall seed designated areas to ensure a full even lawn growth throughout seeded areas with no bare or uneven spots after six weeks of growth from the time of seeding. Bare spots exceeding 1/2 square foot will be re-sodded with the prevalent sod. Grounds areas where seed will not germinate shall be sodded to establish turf. The Contractor shall apply topsoil to new lawn areas at a depth of 4 inches. The Contractor shall spread topsoil to an even and level appearance in conjunction with contours specified for the area and shall fill holes and ruts in the area prior to being dressed with topsoil.

5.6.4.5.1.1 Seed Areas. The Contractor shall grade, remove debris, cultivate, seed, and mulch disturbed areas with straw. Grass shall be applied at the rate of .75 lb. hulled seed per 1,000 sq. ft.. The type of grass seeds used shall depend on the area of Fort Benning.

5.6.4.5.1.2 Fall Seeding's. Fall seeding's shall take place between 1 September and 31 March. The type of grass seeds used shall depend on the area of Fort Benning.

5.6.4.5.1.3 Maintenance of Cool Season Grass Cover. Upon receipt of a valid work document, the Contractor shall establish and maintain an annual cool season grass cover for improved areas during the fall and winter growing season and replace these areas with perennial warm season grass as scheduled. Grassed areas damaged by vehicular traffic shall be graded, seeded, or sodded and irrigated as specified in TM 5 630, until the desired grass is established to conform to adjacent turf areas. On high pedestrian traffic areas and road intersection shoulders, roping off is required until the grass is established. The Contractor shall accomplish grass planting and sodding year round.

5.6.4.6 Debris Removal

The Contractor shall clean, remove, and dispose of debris from various areas such as remodeling jobs, new construction sites, sewage ponds, shop hoppers, improved grounds, and unimproved grounds. The Contractor shall dispose of debris at the inert landfill or as directed by the KO.

5.6.4.7 Storm Damage

The KO will notify the Contractor when the Fort Benning Severe Weather Emergency Action Plan is in effect after which time the Contractor shall clean up wind and storm damage. The Contractor shall remove as priority work all fallen trees, limbs, and debris deposited by water runoff on improved grounds after periods of heavy wind or rainfall. The Contractor shall dispose of storm debris at the inert landfill or as directed by the KO. The Contractor shall dispose of shingles and other building debris in locations designated by the KO. Routine maintenance work shall be suspended during periods deemed necessary by the KO for storm damage cleanup.

5.6.4.8 Establishing New Landscape Areas

The Contractor shall establish new landscape areas as specified on a valid work document. The Contractor shall layout and prepare new landscape areas in accordance with the above paragraphs on soil preparation, seeding, re-vegetating, irrigating, and fertilizing.

5.6.4.8.1 Drainage. The Contractor shall layout, excavate, and install various types of drainage systems such as intercepting, french, blind, perforated pipe, and diversion drainage to facilitate proper water drainage as specified on a valid work document. Work shall be accomplished in accordance with CEGS 02710, Sub-drainage System, CEGS 02711 and CEGS 02720, Storm Drainage System.

5.6.4.9 Soil Conservation

The Contractor shall root plow and seed various areas such as ranges, firebreaks, forest areas, and drainage ways for soil conservation. The Contractor shall seed various types of grass to the local area as directed by the KO. The Contractor shall consult the EMD and the Soil Conservation Service when conducting erosion control and soil conservation tasks or when performing tasks that result in the disturbance of soils in the area.

5.6.4.9.1 Soil Erosion. The Contractor shall implement the soil erosion program as prescribed by the EMD at Fort Benning. The Contractor shall take an active role in identifying any developing erosion areas not already identified in the plan. This shall include, but not limited to, erosion on road leading to and from ranges and training facilities, erosion on main and back thoroughfares, and erosion in residential areas.

5.6.4.9.1.1 All activities to occur in an area greater than 1.1 acres require NRCS Service permits which must be obtained through the EMD.

5.6.4.10 Excavation Support

The Contractor shall support other public works functions, such as the pipefitting and electrical trades, by providing equipment and manpower for excavation support. The Contractor shall perform various tasks such as removal of debris, providing backfill materials, backfilling excavated areas, and dressing areas to match existing landscape (e.g., adding topsoil, reseeding, planting, and adding gravel). The Contractor shall dispose of debris at the inert landfill in accordance with all Federal, State, and local laws.

5.6.4.10.1 Excavation. The Contractor shall mark the intended dig with white paint and contact the designated point-of-contact in charge of utility locating before starting any excavation of 6” or greater below the surface on any area. All utility lines are to be marked within 72 hours after the call. The Contractor shall notify the KO within one hour when utility lines or components have been damaged or unearthed during excavation. The Contractor shall remove any shrubs or trees necessary to excavate, unearth, and remove any debris to reach the work area. Upon completion of repairs or installation, the Contractor shall fill the hole, grade, replace landscaping, and seed/straw the excavation area, as required, to return it to its original condition. The excavated area shall be secured with barricades when unattended.

5.6.4.11 Stockpile Area

The Contractor may use the fenced area behind Building 330 to stockpile needed materials, including topsoil, brick, sand, and crush and run.

5.6.4.12 Hauling Materials

The Contractor shall deliver various types of materials, to include, but not be limited to, topsoil, crushed stone, brick, sand, river rock, bark mulch, logs, tree debris, asphalt (hot and cold), fill dirt, and trash.

5.6.4.12.1 Construction Materials. The Contractor may use the unfenced area behind Building 330 to temporarily store construction materials and any other materials approved by the KO.

5.6.4.12.2 Miscellaneous Debris. The Contractor shall dispose of debris at the appropriate disposal area.

5.6.4.12.3 Waste and Spoil Materials. The Contractor shall respond with an Emergency Response Team to remove all hazardous waste upon call 7 days a week and 24 hours a day if needed. The Contractor shall haul the waste or spoil materials, usable materials, equipment, machinery, supplies, and water to and from such areas as storage areas, job sites, ranges, and approved off-Installation dumpsites. The Contractor shall also haul waste materials such as asphalt, concrete, lumber, tree and shrub trimmings, sludge, dirt, rocks, trash, mud, snow, and building materials. All debris shall be disposed of in accordance with all Federal, State, and local laws. Specific details of waste and spoil materials handling and disposal is addressed in Section C-1 of this document.

5.6.4.13 General Ground Structure Repair

The Contractor shall perform unscheduled repair services for ground structures as directed by the KO. Ground structures include, but are not limited to flag poles, benches, Government-owned fencing (including gates), goal posts, planters, drainage culverts and systems, and athletic tracks and field equipment.

5.6.4.13.1 Outdoor Recreation Areas

5.6.4.13.1.1 Athletic and Recreation Fields. The Contractor shall haul in fill dirt and tractor grade as necessary to maintain athletic and recreation areas for their designated use. The Contractor shall grade, level, sand, and build pitcher's mounds, and repair backstops to ready athletic fields for use. Fences surrounding these areas shall be repaired with like materials by the Contractor as needed. The Contractor shall mow grass during the growing season at the request of the KO. The user shall accomplish requested trimming.

5.6.4.13.1.3 Flagpoles. The Contractor shall be responsible for repairing, maintaining, replacing, installing, and removing flagpoles, including ropes, snaps, and pulleys. The majority of these repairs and replacements occur in troop and range areas.

5.6.4.13.2 Government-Owned Fencing and Gates. The Contractor shall perform repair as required on all types of wire fencing and gates, including security chain link, non-security chain link fences, barbed wire posts and gates, razor ribbon, concertina wire, and various steel gates. Chains and protective wire barriers around improvements or hazardous areas shall be maintained. The Contractor as required shall accomplish installation of new fence. Fencing of secure areas shall be maintained to the requirements of AR 190-13 and AR 190-51.

5.6.4.13.2.1 The Contractor shall repair fences to prevent rust or corrosion, keep fencing taut, keep ties and braces in place, and keep fence posts straight and firmly anchored in the ground; replace and tighten wire or mesh fencing; adjust and properly rehang gates so they operate without sagging; keep holes under fencing filled and keep the fence free of both vegetation and trash.

5.6.4.13.2.2 The Contractor shall be aware that numerous boundary fences are not the property of the Government and should not be repaired. Numerous schools, and the golf course fences are not Government property. In order to avoid repairing non-Governmental fences, the Contractor shall clear repairs with the KO.

5.6.4.13.3 Security Chain Link Fencing. The Contractor shall repair, replace, install, and remove security chain link fencing and gates. The Contractor shall repair all security fencing in accordance with Priority 1 work. The Contractor shall enclose restricted areas, interior security buildings and enclosures, motor pools, Game Management fisheries, water catchments, structures, ammunition dumps, and other limited and excluded access areas with security chain link fence. The Contractor shall maintain drainage along fences.

5.6.4.13.4 Non-Security Chain Link Fence. The Contractor shall repair, replace, install, and remove fences at tennis courts, backstops, dugouts, dog kennels, boundary fences, and cemeteries.

5.6.4.13.5 Barbed, Razor, and Concertina Wire Fences. The Contractor shall repair, replace, install, and remove barbed wire fences, gates, fence line/end posts, wire stays, and all fencing hardware for perimeter boundaries, interior enclosures, and other facilities. The Contractor shall repair, replace, install, remove, and inspect razor ribbon, concertina wire, and hardware along the roof edges of buildings and towers that form the perimeter of secured areas in conjunction with fencing and gates.

5.6.4.13.6 Cable Wire. The Contractor shall install, replace, and remove cable wire used for gates and as added security to perimeter fences.

5.6.4.13.7 Gates. The Contractor shall repair, replace, install, and remove swing or roller type metal gates, aluminum gates, scissors gates, gate hinges, gateposts, and gate stops of various sizes and gauge 1 to 6 in round stock pipe. Gates range from 2 to 8 feet high and 2 to 50 feet wide.

5.6.4.14 Repair of Culverts

The Contractor rebuilds and repairs culverts and all storm drains as needed to maintain the free flow of storm water.

5.6.4.15 Special Events

The Contractor shall provide support, as required, for special activities or events on the Installation, such as holidays, Armed Forces Day, and other events deemed necessary by the Government.

5.6.4.16 Decorative Materials

The Contractor shall install, repair, replace, and remove a variety of decorative materials upon receipt of a valid work document.

Functional Area 5.7

Surfaced Area Maintenance

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Functional Area 5.7

SURFACED AREA MAINTENANCE

5.7.1 INTRODUCTION

The Contractor shall perform maintenance and repair of surfaced areas, including roadways, parking lots, trails, widened shoulders, gravel and earth surfaced areas, bridges, and associated structures and appurtenances at Fort Benning. The Contractor shall perform maintenance and repair of approximately 23 million square yards of surfaced areas, including approximately 4.7 million square yards of paved surfaces. Qualified personnel in accordance with applicable laws, regulations, and documents in Section C-6 shall perform a l l work, and any Government developed plans. Technical Exhibits provide expanded information for this Functional Area. Data and maps related to the paved surfaces are available in the Technical Library.

5.7.2 SCOPE OF SERVICES

5.7.2.1 Work Area and System Description

Surfaced areas covered under this Contract include, but are not limited to, the following:

- Paved and unpaved roadways
- Paved and unpaved parking areas
- Concrete tank trails
- Sidewalks and Curbs
- Vehicle storage and staging yards
- Maintenance yards
- Supply yards
- Bridges
- Airfield runways and taxiways (including McKenna and Decker Air Strips)

5.7.2.2 Work Management and Control

5.7.2.2.1 Reporting Requirements. The Contractor shall prepare, submit, and maintain all records and reports as specified herein and in accordance with the Contract Data Requirements Lists (CDs) in Technical Exhibit 1-004.

5.7.2.2.2 Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.7.2.2.3 Safety. Slow moving equipment used on roads that may create an unsafe condition shall be equipped with rotating or flashing amber lights. The Contractor personnel performing services on or adjacent to roadways shall be equipped with reflectorized vests. The Contractor personnel acting as road guides shall carry an 18-

inch square flag of reflectorized material, yellow-orange in color. During hours of darkness, road guides shall also be equipped with a red light capable of being seen by approaching motorists from a distance of at least 100 yards. Guides shall be instructed in the correct method of rendering hand signals.

5.7.2.2.4 Traffic Flow. The Contractor shall maintain the free flow of traffic during paving repairs in accordance with Part VI of the Manual on Uniform Traffic Control Devices and through the use of traffic control methods or devices, including flagmen, traffic cones, detour signs, signal devices, and flashers. The Contractor shall notify the KO in writing a minimum of 14 working days in advance of any scheduled or potential disruption of traffic flow, including closure of streets (CD 507R001). This would include notices as set forth in Section C-1.

5.7.2.2.5 Equipment Operators. All Contractor personnel operating mobile equipment and vehicles shall possess a valid state operator's license, to include Commercial Driver’s Licenses (CDLs) with special permits to include air brakes and hazardous endorsements, if required for the equipment and vehicles operated. Contractor motor vehicle drivers and equipment operators shall be appropriately qualified, certified (HAZMAT), and licensed. Operators will perform operator maintenance prior to the use of equipment and report deficiencies on a daily DA 2404 form.

5.7.3 SCHEDULED TASKS

The Contractor shall perform the tasks below on a recurring or scheduled basis and record them under approved Standard Operating Orders (OWO′s). Preventive maintenance and other scheduled task frequencies that apply to this Functional Area are presented in Technical Exhibits 5.7-001. If a requirement for further work on any area beyond the scheduled requirements below is identified, the Contractor may be required to accomplish such work on an approved Demand Maintenance Order (DMO) or Project Work Order (PWO).

5.7.3.1 Paved Surface Sweeping and Cleaning

The Contractor shall provide sweeping services of leaves, dirt and debris from paved vehicular surfaces, parking areas, and pedestrian walks throughout the Installation. The Contractor shall sweep all specified paved surfaces (e.g., roads, road shoulders, parking lots, and pedestrian walkways) with a broom-type sweeper. A properly swept surface is that resulting from a single-pass of a properly operating mechanical broom-type sweeper over an area receiving normal sweeping maintenance and not covered with spills or debris. Technical Exhibit 5.7-002 lists the areas to be swept with their frequencies, and additional sweeping as required by the KO

5.7.3.2 Road Inspections:

On a monthly basis, the contractor shall inspect all surfaced roads at Ft. Benning for potholes and damaged guard rails. As potholes or damaged guard rails are identified, they will be repaired on Demand Maintenance Orders. A DA4283 will be initiated for any repairs that are beyond the scope of a Demand Maintenance Order. Any other abnormal conditions may be noted during this inspection (e.g. sinkholes, excessive growth along the right-of-way etc.). As part of this inspection, the contractor will schedule road inspection areas monthly using the locations described in TE 5.7-003. The contractor shall fill out a checklist showing the roads inspected, location of potholes or damaged guard rails (and any other problems noted), to include corresponding Demand Maintenance Order numbers or DA4283 numbers for all needed repairs.

5.7.3.3 Pop-up Barriers and Bollards: Contractor shall perform preventative maintenance on all pop-up barrier devices at Harmony Church and Custer Road checkpoints in accordance with manufacturer's recommendations, this requirement shall be charged to a SOO.

5.7.4 UNSCHEDULED TASKS

The Contractor shall perform corrective actions as needed to surfaced areas, including, but not limited to: adding dirt and gravel to roadways and parking areas, repaving shoulders, regrading, repainting, repairing potholes, clearing drainage areas, and repairing guardrails. The Contractor shall maintain and repair all paved surfaces, including flexible and rigid pavements, paved roads, paved parking areas, sidewalks, recreational areas, road shoulders, drainage systems, ramps, bridge surfaces, curbs, gutters, manhole surfaces, erosion control areas, and appurtenances. The Contractor shall perform the tasks described below when initiated through either a SO or an approved PWO.

5.7.4.1 Paved Surface Sweeping and Cleaning

The Contractor shall provide sweeping services of leaves, dirt and debris from paved vehicular surfaces, parking areas, and pedestrian walks throughout the Installation. The Contractor shall sweep all specified paved surfaces (e.g., roads, road shoulders, parking lots, and pedestrian walkways) with a broom-type sweeper. The Contractor shall also manually clean sidewalks, gutters, storm drains, and other areas inaccessible to mechanical sweepers associated with the above listed paved surfaces to remove rock, sand, dirt, and debris. All debris and litter collected by the street sweeper shall be disposed of by the end of the same working day at the compost site. The Contractor shall use high-pressure water wash- down methods when necessary to meet appearance requirements.

5.7.4.2 Painting of Paved Surfaces

The Contractor shall perform street and road painting and striping upon the receipt of a valid work document or as designated by the KO.

5.7.4.3 Maintenance of Gravel and Earth Surfaced Areas

The Contractor shall perform maintenance, grading, and other services for gravel and earth surfaced areas. The Contractor shall perform maintenance and repair as required on approximately 550 miles of paved roads and 378.7 miles of tank trails at Fort Benning, and approximately 3 square miles of parking lots and roads.

5.7.4.3.1 Roads and Lots. The Contractor shall maintain and repair gravel and other earth surfaced roads, access roads, driveways, and parking lots by blading or dragging the surface until all ruts, holes, and washboards are filled. The surface shall be shaped to maintain a smooth riding surface and a uniform crown that permits proper drainage, without ponding of water. The Contractor shall maintain earth
shoulders and cut and fill down slopes to allow surface drainage and to protect road edge and control erosion. Soil stabilization and vegetation practices shall be performed as required to prevent erosion. Undesirable vegetation shall be controlled within 15 feet of the pavement edge.

5.7.4.3.2 Tank Trails. Tank trails consist of gravel and or dirt trails running to, from, and through various wheeled and tracked training areas as detailed in Functional Area 5.11, Range Maintenance. The Contractor shall maintain and repair gravel and dirt tank trails through grading, filling with gravel or dirt, cutting new trails, repairing culverts, and other steps required to ensure the tank-training mission is not impeded. During heavy usage periods or periods with heavy rain, daily repair may become necessary.

5.7.4.4 Storm Drain Systems

The Contractor shall repair storm grates, manholes, concrete storm sewers, asphalt drainage ditches, brick drainage areas, and all associated paved and unpaved grounds associated with storm systems. This shall include the fabrication of manhole covers, plates, grates, and all type of drain covers.

5.7.4.5 Maintenance and Repair of Bituminous and Cement Concrete Pavements

All repair work shall be performed in accordance with established approved plans, AR 420-72, TM 624, and TM 5-822-7. Items to be maintained include, but are not limited to, plant mix cold laid, plant mix hot laid, sand asphalt, sand tar, rock asphalt, and bituminous penetration macadam. The Contractor shall maintain paved surfaces of all types in a good state of repair with uniformity in appearance, at the original alignment and elevation, and free of damage, vegetation, potholes, scaling, spalls, surface breaks, and cracks. In order to provide smooth and safe surfaces, the Contractor shall maintain surfaces to be free of potholes and depressions deeper than 3/4” that pond water.

5.7.4.5.1 Spot Repair. The Contractor shall repair potholes, upheavals, and eruptions of asphalt around fireplugs, in asphalt drainage ditches, and on all bituminous surfaced areas. The Contractor shall repair, replace, and clean up cracks, grade depressions, upheavals, utility cut depressions, potholes of various sizes, raveling, bleeding, channel ruts, corrugation and shoving, fuel spillage, poor drainage, weathering, construction deficiencies, poor bituminous materials, aggregate deficiencies, overheating, improper mixing of material, poor proportioning, placement errors, plant mixed materials, and aggregate surfaces, subgrades, sub bases, and base.

5.7.4.5.1.1 Shoulders. The Contractor shall maintain and repair road shoulders to protect the basic pavement structure, eliminate traffic hazards, and ensure proper drainage. The Contractor shall accomplish this requirement by leveling ruts and washes, filling in low areas, and cutting down high areas to achieve proper grade and slope. The Contractor shall revegetate the disturbance in accordance with the Fort Benning Integrated Natural Resources Management Plan. The Contractor shall maintain all roadsides and intersections to ensure unobstructed visibility of all signs, markers, guardrails, and fire hydrants.

5.7.4.6 Utility Cuts

The Contractor shall perform utility cuts in pavements as required to repair existing utilities or to install new service. Appropriate lighted barricades and construction fencing shall be used on open cuts. The Contractor shall repair utility cuts using materials with the same physical qualities as adjacent undisturbed areas in accordance with TM 5-818-4.

5.7.4.7 Repair and Replacement of Barriers

5.7.4.7.1 Guardrails. The Contractor shall install, fabricate, repair, and replace damaged guardrails and guardrail posts with new materials or with salvaged, undamaged guardrails that meet National Highway Standards. The Contractor shall install guardrails along roads where hazardous conditions exist.

5.7.4.7.2 Barricades and Lighting Devices. The Contractor shall install, replace, remove, and inspect permanent and mobile barricades (types I, II, III) cones, drums, vertical panels, and lighting devices which warn drivers of hazards created by construction, maintenance, and storm damage in or near traveled areas and which guide and direct drivers safely past the hazards.

5.7.4.7.3 Bumper Blocks. The Contractor shall install, replace, remove, and inspect concrete, plastic, wood, and metal bumper blocks used in parking areas, roadways, hazardous areas adjacent to buildings, signs, and fences to keep vehicles from protruding past the designated parking boundaries.

5.7.4.8 Construction of New Surfaces

The Contractor shall construct new surfaced or graveled areas to KO-directed specifications. Work shall include, but not be limited to, the following:

- Widening or lengthening surfaced areas
- Installing additional appurtenances and associated structures (bridges, drainage structures, subgrade, sub base, base, and compaction)
- Adding curbs
- Adding sidewalks
- Drainage structures

- Ford construction (low water crossings) in approximately 7 locations

5.7.4.9 Special Maintenance and Repair Requirements at Infantry Schools

The Contractor shall assist the infantry schools with heavy equipment on an as needed basis. This shall include loading and unloading equipment with cranes, and other tasks.

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Functional Area 5.8

Materials Recycling Facility Operations and Maintenance

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Functional Area 5.8

Materials Recycling Facility Operations and Maintenance

5.8.1 INTRODUCTION

The Contractor at the Materials Recycling Facility (MRF) Building, shall provide laborers, tools, vehicles, and equipment, necessary to collect, process, and prepare for sale recyclable materials. In addition, the contractor is to assist the MRF personnel operate, maintain, and replace equipment listed in TE 5.8-003 that is essential to the operation of the MRF and other related activities as outlined in this document. The contractor is also responsible for the access road into the MRF area. Other activities include chipping scrap wood, grinding waste asphalt and concrete, and managing the composting area. All work will be performed by qualified personnel in accordance with applicable laws, regulations, and documents in Section C6.

5.8.1.1 Overview:

5.8.1.1.2 A Qualified Recycling Program (QRP) was established at Ft Benning in the mid-1990's as a response to the Resource Conservation and Recovery Act of 1976. A DOD Memorandum issued on 28 Jan 83 establishes policy for the sale of recyclable materials at Ft Benning. The memorandum stated that all installation and satellite activities will actively participate in the installation's QRP in order to reduce waste streams, prevent pollution, and conserve natural resources. The Materials Recycling Facility (MRF), Building 4000, in Harmony Church was built to accommodate the administrative offices, equipment, and storage needs to operate a recycling program.

5.8.1.1.3 The value of recycling was further emphasized when Ft Benning closed its' municipal solid waste landfill in the late 1990's and the cost of collecting and transporting Ft Benning's solid waste to an off-post landfill substantially increased. Domestically produced recyclable materials, primarily consisting of packaging materials, potentially represent a considerable source of income to the installation. This significant portion of the recycling program will be captured through voluntary participation using satellite collection points to be judiciously located throughout the installation.

5.8.1.1.4 Currently 51 industrial and administrative facilities participate in recycling at Ft Benning, as well as hundreds of troop housing and training units. This is a very small percentage of the Ft Benning community. There is potential for additional pickup points to be added based upon user requirements and generation. In addition, units are encouraged to bring recyclable materials to the material recovery facility. Sorting and disposal of non-marketable materials into roll-off containers located adjacent to BLDG is the responsibility of the units.

5.8.2 SCOPE OF SERVICES

5.8.2.1 Work Area and System Description

5.8.2.1.1 The recycling facility operation and related activities include the following work categories: collecting, sorting, processing, baling, and temporary storing all recyclable materials intended for sale; as well as chipping, grinding, and composting of yard materials intended for Installation use.

5.8.2.1.2 The collection-work category includes picking-up recyclables at buildings throughout the installation that are on scheduled pick-up. All but two facilities are on a weekly schedule. Only bldg 4 and 6 are on a daily pick-up schedule. See Technical Exhibits 5.8-001 for list of See Technical Exhibits 5.8-001 for a list of buildings served and Technical Exhibits 5.8-004 for a location map of trailer sites and served buildings.

5.8.2.1.3 The processing-work category will be the Recycling Center (Building) in Harmony Church. This facility includes a 12,000 sq ft steel building, covered loading and storage dock. Administrative duties, and the sorting, processing and storage of recyclables take place in this area. The roll-offs are also located in this area.

5.8.2.1.4 The outside-yard work category consists of operating the large outdoor facility located east of the building. Wood chipping and concrete/asphalt grinding as well as storing and loading of processed materials take place in this 3-to-4 acre area. The bioremediation facility is located in the southeast corner of the yard. This facility will be used to remediate soils contaminated with petroleum products.

5.8.2.2 Work Management and Control

5.8.2.2.1 Publications and Forms.
Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.8.2.2.2 Compliance

5.8.2.2.2.1 While performing any maintenance work or performing routine work with the above listed equipment, the Contractor shall comply with all applicable manufacturer recommendations and Ft Benning Environmental Plans, Occupational Safety and Health Administration OSHA codes, EPA, and GA EPD regulations, and standards.

5.8.2.2.2.2 The contractor shall be knowledgeable of and comply with all applicable Federal, State, and Local Laws, regulations, and requirements regarding environmental protection. In the event environmental laws and regulations change during the term of this contract, the Contractor shall be required to comply with such laws when they come into effect.

5.8.2.2.2.3 The MRF personnel shall implement the Activities Specific Plan for all Environmental Requirements at this facility.

5.8.2.2.2.4 It is the Contractor's responsibility to obtain and adhere to all industry standards. The Contractor shall always refer to the most current version of all referenced publications listed in Section C-6. The Contractor shall comply with all federal, state and local codes, regulations and statutes related to this specific task order which are in effect at the time work is performed.

5.8.2.2.2.5 The Contractor is responsible for complying with Ft Benning and OSHA Safety Regulations and hold regular safety meetings to ensure that the equipment is operated by qualified personnel in a safe manner.

5.8.2.2.3 Notification of Environmental Spills

5.8.2.2.3.1 In the event of an environmental spill or release of substance, the MRF personnel shall immediately notify Ft Benning's Spill Response Team by calling 9-1-1. Immediate containment of the release or spill using approved spill control measures must be carried out to reduce environmental impact.

5.8.2.2.3.2The recycling facility (MRF and associated activities) is considered an industrial facility under the GAR 000000 NPDES Permit. An Activity Coordinator shall be appointed on order. Duties of the Activity Coordinator will include implementing the Storm Water Pollution Prevention Plan 3 (SWP3).

5.8.3 SCHEDULED TASKS

The contractor shall provide laborer's to perform the tasks below on a recurring or scheduled basis and record the work under approved Operation Work Orders (OWO's). If a requirement for further work on any area beyond the scheduled requirements below is identified, the Contractor may be required to accomplish such work on an approved Project Work Order (PWO).

5.8.3.1 Operation of the Materials Recycling Facility (MRF)

5.8.3.1.1 The Contractor shall have laborers available to assist in the operation of the MRF from 0745 to 1615 Monday through Friday and record the work under approved Operation Work Orders (OWO's). Duties at the MRF include but are not limited to performing all assigned recycling operations; to include loading and unloading recycling materials, baling cardboard, sorting and baling paper, de-bind books and manuals for sorting and baling, and sort and bale plastic. Metal cans are to be sorted as follows; Aluminum cans are to be baled and steel/tin cans will be placed in the scrap metal roll-off. The glass will be sorted by color and stored outdoors until sold. All outdoor open storage bins for glass and plastic will be covered with tarps to protect the materials from the weather. The volume of all stockpiled material for recycling must be reduced by 60% every 90 days in accordance with the State of Georgia Solid Waste Rule.

5.8.3.1.2 As sufficient material is collected it will be processed (sorted and baled) to ready it for sale. Technical Exhibit 5.8-002 shows the minimum quantity of items required to be on hand before an item can be submitted to Directorate of Moral, Welfare and Recreation, (DMWR) for bid. Baled and un-baled material will be placed on the loading dock area to protect it from bad weather.

5.8.3.1.3 Customer-units operated vehicles carrying mixed materials shall be met and inspected prior to unloading into the roll-offs. Soldiers and other customers shall be instructed to sort their waste to capture as much recyclables as possible and reduce the amount of waste filling the roll-offs. Proper paperwork from DRMO showing that the items were rejected for resale must be shown prior to disposal.

5.8.3.1.4 Pallets brought to the MRF are to be sorted by condition for resale or chipping. No pallets or other scrap wood shall be disposed of in the roll-offs.

5.8.3.1.5 Preventive maintenance on equipment is to be done according to manufacturer's instruction or as recommended in operating manuals. Other scheduled task frequencies that apply to this Functional Area are presented in Technical Exhibit 5.8-001. A list of installed and other serviceable equipment located at Materials Recycling Facility and its properties is provided in Technical Exhibit 5.8-003.

5.8.3.2 Building Collections

5.8.3.2.1 Route/Schedules Submittals. The MRF contractor personnel shall establish collection schedules for the individual buildings shown in Technical Exhibit 5.8-001 that are on a regular pick-up schedule. This list may be modified as buildings are added or subtracted based on need and personnel. All changes to the schedule need to be made at least one month in advance to permit time to inform building occupants. Every effort should be made to service high density areas (i.e. Buildings 4 and 6) as a priority.

5.8.3.2.2 Satellite-Sites Collections: Recycling trailers will be placed in eleven designated areas listed in TE 5.8-004. The contractor shall bring the recycling trailers to the MRF for emptying and returned to their location. Trash under or around the recycling trailer shall be the contractor's responsibility to pick up when the trailers are picked up for emptying. The contractor shall keep the recycling trailers clean. The outside of the trailer and removable bins are to be rinsed off (or washed) as needed to maintain their new like appearance and for health purposes. This need not be on a set schedule but determined by the MRF contractor's supervisor. The goal being that these trailers will be sitting in high profile areas and residents will complain if the trailers become unsightly. Dumpsters will be placed nearby and are part of the Solid Waste Contract. The Solid Waste contractor is solely responsible for the cleanliness of the area around the dumpsters

5.8.3.4 Operation of the wood chippers and concrete/asphalt crusher

The contractor shall provide sufficient specialized labor to operate the wood chipping and concrete/asphalt crushing equipment. The contractor shall coordinate these activities in conjunction with other recycling activities. The quantity of materials sold and/or utilized internally on Installation projects and wood and concrete chipped or ground shall be included in the monthly activities report submitted on CD 508R001. The volume of all stockpiled material for recycling must be reduced by 60% every 90 days in accordance with the State of Georgia Solid Waste Rule.

5.8.3.4.1 Pallets: The contractor shall sort pallets brought to the recycling facility according to their condition. Pallets in good condition that can be sold or reused will be stacked neatly in a designated area to minimize damage. Broken pallets will be ground in the wood chipper for use in erosion control or placed in the inert landfill. Careful handling of the pallets should be encouraged to keep as many as possible for resale. Pallets will not be disposed of as waste in the roll-offs. To prevent the pallets from becoming a waste management problem pallet grinding should be scheduled on a bi-weekly basis as a minimum.

5.8.3.4.2 Concrete/Asphalt Crushing: The concrete grinder is available for the purpose of recycling concrete and asphalt for recycling as road-base material in accordance with equipment manufacturer's specifications. Training is required for personnel responsible for the operation and maintenance of the grinder to ensure safe and proper use. The contractor must provide this training to the employees. Waste asphalt and concrete must be in small enough pieces (operator's manual) to allow proper feeding by the auger into the hammers. To prevent the concrete/asphalt from becoming a waste management problem concrete crushing should be scheduled on a monthly basis as a minimum. The volume of all stockpiled material for recycling must be reduced by 60% every 90 days in accordance with the State of Georgia Solid Waste Rule**.**

5.8.3.5 Oil Igloo

The MRF contractor personnel shall also maintain the used oil igloo, ensuring that the area is kept clean and free of improper containers. Oil spill equipment will be kept near the igloo for quick response to spills. The Contractor will comply with installation spill contingency plans for HW storage.

5.8.3.6 Oil Storage ASTs

5.8.3.6.1 The MRF personnel shall be responsible to monitor rain water on containment slab at the Oil Storage ASTs during rain events and open drain valve when necessary to drain water off the slab. The contractor shall also report any spills to the Environmental Management Branch (EMB), and not to open the valve and drain off water when the presence of oil is evident. The MRF is considered an Industrial NPDES facility and is permitted under a GA EPD general storm water permit # GAR 050000.

5.8.3.6.2 The MRF personnel shall monitor the loading and unloading of oil in the ASTs and report any spills to the spill response coordinator in the Environmental Branch. The MRF personnel shall inspect the secondary containment area after a rain event for sheen on the standing water. If there is no sheen the water shall be released and the valve shall be re-closed. If a sheen is observed the oil hauling contractor shall be called to remove the sheen before releasing the water.

5.8.3.7 Management of Yard Waste

5.8.3.7.1 All yard waste will be managed at the Waste Water Treatment Plant (WWTP) site on Marne Road. Customers and the solid waste contractor will be directed to take yard waste to that location.

5.8.3.7.2 Operation of the Compost Site: The Contractor shall operate the compost site Monday thru Friday, January thru December. Operations shall include, but not be limited to: turning and spreading compost material with appropriate equipment to release gases and expedite decomposition; and visually noting that only compost material is being dumped in the areas.

5.8.3.8 Roll-off Waste Disposal Areas (General Waste and Scrap Metal)

5.8.3.8.1 The Contractor shall keep areas leading to the roll-off staging area free of debris and swept for nails and other sharp objects. The area where the roll-offs are located that are not a part of this contract, are as follows: 67ft (from edge of concrete near scale to the fence) x 390ft (from glass storage bin to fence). Management of the Roll-off area is the responsibility of a separate Solid Waste contractor. Overflowing roll-offs must be reported immediately to the COR. Grass will be kept mowed during the growing season and trim vegetation around the buildings, parking areas, and other areas as necessary.

5.8.3.8.2 Unmarketable Materials: Disposal of non-marketable materials into the available roll-offs at the MRF is acceptable; however, the contractor is not responsible for the management of the roll-off containers.

5.8.3.9 Equipment Maintenance

5.8.3.9.1 The contractor is responsible for maintenance on all equipment used to perform the duties outlined in this work statement. Equipment servicing outside of routine maintenance shall be performed by personnel trained on such equipment.

5.8.3.9.2 The Contractor shall use types, grades, or weights of lubricants and the proper fuels that are shown in owner's manual and service manuals/bulletins when performing maintenance and refueling equipment. All work is to be performed following OHSA Regulations applicable to this subject area. This shall include, but not be limited to, the handling of all fuels, lubricating oils, chemicals, the sorting and bailing machinery, wood grinders, forklifts, and operating the forklifts and tractors.

5.8.4 UNSCHEDULED TASKS

The Contractor shall perform the tasks described below when initiated through an approved PWO.

5.8.4.1 Special Projects and Events: The contractor shall provide manpower and equipment for "special" collections the earliest possible date. Special collections may be scheduled in support of PCS moves, Mob/Demob Activities, Help the Hooch, Spring Clean-up, 4th of July, Concerts, Festivals, Soldiers Day, etc. Reschedule collection as required.

5.8.5 REPORTING AND RECORD KEEPING

5.8.5.1 The MRF personnel shall provide a monthly activities report showing the gross weight of each recyclable material processed for sale or utilized on the installation during the previous month, (CD 508R001). Bill of Ladings shall be submitted on CD 508R002 as processed materials are picked up by buyers.

Plastic	Newspaper
White & Mixed Office Paper	Scrap Metal
Pallets	Glass
Cardboard	Aluminum Cans

5.8.5.2 The MRF personnel shall provide a quarterly efficiency report on the MRF operation no later than the10th day of the month following the month covered in the report., (CD 508R003).

5.8.5.3 If the recyclables collected by the solid waste contractor are contaminated with other refuse to a degree that affects the marketability of the recyclables, the contractor shall notify the COR and Environmental Management Division (EMD), Bldg 6, Room 310. The location of the recycling dumpster or recycling container should be noted so a letter of non-compliance or follow-up visit to the location can be scheduled.

5.8.5.4 All CDs shall be submitted no later than the 10th day of each month following the month covered in the report.

5.8.6 Vehicle Equipment Operation

This task requires Contractor personnel to operate Government vehicles, therefore a valid (state issued) driver's license and DDC card is required by the contract individuals.

Functional Area 5.9

SELF-HELP SERVICES

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SELF-HELP SERVICES

5.9.1 INTRODUCTION

The Contractor shall operate a Self-Help program under the guidance of the Government. The Self-Help program provides Installation military and civilian units with the training, tools, and supplies needed to maintain, repair, and perform minor construction on buildings. The intent of the program is to involve occupants in the preservation and upgrade of their premises. Qualified personnel in accordance with applicable laws, regulations, and documents in Section C-6, including TRADOC Regulation 420-5, shall perform operation of the Self-Help program. Technical Exhibits provide expanded information for this Functional Area.

5.9.2 SCOPE OF SERVICES

5.9.2.1 Work Area and System Description

Self-Help services shall include the training of potential users of the Self-Help program in basic skills required to utilize the materials and tools available through the program. At any given time, the Self-Help Center maintains approximately $85,000 of expendable stock items.

5.9.2.2 Work Management and Control

5.9.2.2.1 Reporting Requirements. The Contractor shall prepare, submit, and maintain all records and reports as specified herein and in accordance with the Contract Data Requirements List (CDs) in Technical Exhibit 1-004.

5.9.2.2.2 Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.9.2.2.3 Self-Help Guides. Guides for Self-Help repair and maintenance tasks are shown in TM 5-610, Preventive Maintenance for Facilities, Engineering, and Buildings and Structures. The term "customers" throughout this Functional Area shall mean Installation Self-Help program customers. The Contractor shall release only those items authorized for Self-Help issue to qualified customers who are in the Self-Help Center database.

5.9.2.2.3.1 Building Coordinators. Unit Commanders will designate individuals on DA Form 1687, Signature Card, individuals who are authorized by the Government to request and accept Self-Help materials for buildings and units. The Government will provide the names and signatures of these individuals to the Contractor. The Contractor shall ensure that the using unit's building coordinator signs all non-housing requests for Self-Help items.

5.9.2.2.4 Self-Help Center Warehousing. The Contractor shall maintain warehouse space for Self-Help supplies and materials in accordance with TM 743-200-1 and TM 743-200-2. The Contractor shall keep an updated central locator file of all supplies and materials. The central locator file shall include the following information for each item in stock: nomenclature, current location(s), unit cost, and quantity on hand.

5.9.2.2.4.1 Labeling of Supplies. The Contractor shall prepare, apply, and maintain item storage identification labels, tags, or placards to provide item identification of supplies. The Contractor shall use an approved locator and bin Card, for all small item locations of supplies and materials.

5.9.2.2.4.2 Center and Warehouse Environment. The Contractor shall maintain the Self-Help Center and warehouse in a clean, secure, hazard-free storage environment to preclude pest infestation, water damage, fire hazard, and wind or storm damage to supplies and materials.

5.9.2.2.5 Pest Control Report for Units. The Contractor shall prepare and submit the monthly Pest Control Report for Units to the KO (CD 509R001). The report shall detail the military units that have been issued insect and rodent control items, the quantity of items, description of item type, and the unit building number.

5.9.3 SCHEDULED TASKS

5.9.3.1 Operation of the Self-Help Center

The contractor shall provide high quality customer service at all times. The Contractor shall stock and operate the Self-Help Center 52 weeks a year, Monday through Friday, except Federal holidays, from 0800 to 1600 or as directed by the KO. Operating hours shall be displayed prominently both outside and inside the facility. Any changes to the hours of operation shall be approved by the KO and published in the Fort Benning Bayonet no later than one week prior to implementation.

5.9.3.1.1 Store Stock. The Center shall supply parts and materials for users of the Self-Help program. A recommended stockage list of Self-Help Center items is provided in Technical Exhibit 5.9-001.

5.9.3.1.1.1 Stockage List. The Contractor shall develop a list of Self-Help Center items, to include stockage levels, and shall submit the list to the KO for approval prior to 30 calendar days before Contract start date (CD 509R002). Unless otherwise approved by the KO, stockage levels shall not exceed ten working days and shall be based on historical customer demand. In addition, at any given time, the Contractor shall be able to provide 90% of the requested items at the time of the request.

5.9.2.1.1.2 Stockage Levels. The Contractor shall maintain the stockage level for required items in the Self-Help Center at the level approved in the stockage list, unless otherwise approved or directed by the KO.

5.9.3.1.2 Database Software. The Contractor shall use contractor provided software to maintain a database of customer files, account for all issues and returns, monitor equipment repair, monitor customer usage, control Self-Help Center inventory, and generate reports as required.

5.9.3.1.2.1 Customer Files. The Contractor shall maintain current files of all authorized Self-Help program users. A database of Self-Help customer files will be turned over to the Contractor at the start of the Contract. The Contractor shall use this database as the basis for initial Self-Help Store customer files and shall update the database with all new Self-Help program users upon completion of Self-Help program training. On a weekly basis, the Contractor shall monitor the individual Self-Help limits of each customer. The Contractor shall return the files (including new Self-Help customers) to the Government at Contract completion or termination.

5.9.3.1.3 Issue of Items from Self-Help Center. The Contractor shall issue Self-Help supplies and materials to authorized customers upon request. If the requested item is not available for issue, the Contractor shall inform the customer of the approximate date the item will be available, and notify the customer when the item is available for issue. Type of paints and pest control items must be coordinated with the KO as to what type of item should be stocked.

5.9.3.1.3.1 Required Identification. Only authorized users with current Self-Help cards may sign for issues of materials. Self-Help facility personnel are not permitted to sign for persons that do not possess a Self-Help card.

5.9.3.1.3.2 Issue of Material. The Contractor shall only issue materials on the approved Self-Help Center item list.

a. Items not Approved. The Contractor shall not issue items that are not approved for across the counter issue until the customer presents a FB(DPW) Form 17-2 which has been submitted to and approved by the KO.

b. Construction Material and Material Exceeding $100. Material for construction projects and any project that exceeds $100 requires a Government-approved DA Form 4283.

5.9.3.1.3.3 Transaction Records. The Contractor shall maintain records by facility number of all items issued. Records shall reflect facility number, date of issue, individual item name, individual item number, number of items issued, quantity of issue, initials of authorized user receiving the materials, and user's self-help card number, TE 5.9-002.

5.9.3.1.3.4 Self-Help Supplies and Materials Inventory. At the end of each fiscal year, the Contractor shall perform an inventory of Self-Help supplies and materials, to include reconciliation, as required by AR 420-18 and AR 735-5. The Contractor shall submit a copy of the inventory to the KO within 15 calendar days of the end of the fiscal year (CD 509R003). The Contractor shall prepare adjustment documents to inventories as required by AR 735-5 and as directed by the KO.

5.9.4 UNSCHEDULED TASKS

5.9.4.1 Troop Self-Help Training

The Contractor shall provide Troop Self-Help training to authorized units in accordance with United States Army Infantry Center (USAIC) Regulation No. 11-27, Energy Conservation Program, TRADOC Reg. 420-5, and all other applicable regulations and standards. Additionally, Troop Repairs and Upgrades (R&U) training will consist of one or two classes a month, for troop self-help classes. These sessions would be arranged at the unit level. The classes shall consist of instructions on painting preparatory work and type of paint and primer to be used in different cases interior, as well as exterior surfaces, how to replace filters and what types, e.g. pleated or medal. Also, subjects covered would be how to replace floor tile, cover base and how to replace ceiling tile and ceiling grids. If needed, how to hang dry wall, measure and cut dry wall accurately for switches, receptacles, and other obstacles. How to finish seams from taping through final sanding for a professional looking results, as well as repair holes in existing dry walls using various methods. Such things as replacing a faucets and toilets, unclogging drains, replacing light switches, caulking tubs, repairing shower heads and leaky faucets will also be taught. How to lie out and build wall will also be covered. These would be supplemented on an as needed basis when requested by the unit.

5.9.4.1.1 Course Content. Participating units and troops shall be trained in areas to include, but not limited to, the following:

5.9.4.1.2 Troop Self-Help Training. The Contractor shall provide all troops with an introduction to the Troop Self-Help program at Fort Benning. As part of the introduction the Contractor shall instruct troops on subjects to include, but not limited to:

- interior and exterior painting
- Repairs and Upgrades (R&U)
- minor carpentry
- minor construction
- minor electrical and plumbing repairs
- minor phases of insect and rodent control.

5.9.4.2.1.2 Kitchen Equipment Training. The Contractor shall provide troops with detailed training in the operation and function of all kitchen equipment located in mess halls at Fort Benning when requested by the KO.

5.9.4.2.1.3 Energy Conservation Training. In accordance with USAIC Reg. 11-27, the Contractor shall provide troops and units with training in energy conservation to reduce energy consumption at Fort Benning and at supported facilities to ensure the availability and supply of energy consistent with established missions, priorities, and combat readiness goals when requested by the KO.

5.9.4.2.1.4 Certification of Training. The Contractor shall prepare and present cards to those personnel who attended and successfully completed training to certify that they are qualified to perform Self-Help tasks in these areas.

Functional Area 5.10

REMOTE CAMP OPERATION, MAINTENANCE, AND SUPPORT

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Functional Area 5.10

REMOTE CAMP OPERATION, MAINTENANCE, AND SUPPORT

5.10.1 INTRODUCTION

The Contractor shall maintain and repair buildings and structures; operate and maintain the water treatment and wastewater facilities; maintain and repair the heating, ventilation, and air conditioning (HVAC) systems; maintain and repair utility systems; perform grounds and surfaced area maintenance and repair unless otherwise specified in this Contract. Various locations will require different work to be performed as described below. All work shall be performed by qualified personnel as determined by the KO and Contractor management, and in accordance with applicable laws, regulations, the requirements of Section C-1, and documents in Section C-6. Technical Exhibits provide expanded information for this Functional Area. The Contractor shall provide Facilities Engineering support at Camp Merrill.

5.10.2 FACILITIES ENGINEERING SCOPE OF SERVICES

5.10.2.1 Work Area and System Description

Operation and maintenance areas covered under this Contract include, but are not limited to, the following:

- Buildings and structures
- Utility systems
- HVAC systems
- Grounds maintenance (e.g. mowing, edging, and trimming)
- Surfaced area maintenance
- Water and wastewater treatment

5.10.2.2 Work Management and Control. All other facilities related work orders and Demand Maintenance Order reception at Camp Merrill will be handled IAW the provisions set forth in paragraph 5.1.2.

5.10.2.2.1 Reporting Requirements. The Contractor shall prepare, submit, and maintain all records and reports as specified herein and in accordance with the Contract Data Requirements Lists (CDs) in Technical Exhibit 1-004.

5.10.2.2.1.1 The Contractor shall maintain on site reports of all work performed at Camp Merrill.

5.10.2.2.2 Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.10.2.2.3 Buildings and Structures. The Contractor shall maintain buildings and structures located at Camp Merrill in accordance with all Federal regulations and professional standards to include, but not limited to, the National Electric Code (NEC), Uniform Plumbing Code (UPC), American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE), and the American National Standards Institute (ANSI).

5.10.2.2.4 Utility Systems. The Contractor shall maintain, operate, and repair utility systems to include, but not limited to, water distribution systems, plumbing systems, and fuel distribution systems. All work on utility systems shall conform to Environmental Protection Agency (EPA) regulations and State of Georgia permits and mandates.

5.10.2.2.5 HVAC Systems. The Contractor shall maintain and repair HVAC equipment and systems to include, but not limited to, boilers, refrigeration, chillers, and cooling towers. All work shall be performed in accordance with all applicable manufacturer recommendations and ASHRAE, Occupational Safety and Health Administration (OSHA), EPA, National Fire Protection Association (NFPA), 92 CFR, 49 CFR, ANSI, UPC, and National Electric codes, regulations, and standards.

5.10.2.2.6 Grounds Maintenance. The Contractor shall perform mowing, trimming, edging, aeration, and fertilization; weed and brush control; flower bed services; tree and shrub pruning; erosion control; debris, drain, and ditch cleanup; inspecting and correcting damage caused by pest infestation, vehicles, and storms; landscaping operations; and other services as required herein to provide complete grounds maintenance.

5.10.2.2.7 Surfaced Area Maintenance. The Contractor shall perform maintenance and repair of surfaced areas, including roadways, parking lots, trails, widened shoulders, gravel and earth surfaced areas, bridges, and associated structures and appurtenances.

5.10.2.2.8 Water Plant and Sewage Processing

5.10.2.2.8.1 Camp Merrill Water Treatment Plant. The Contractor shall be responsible for the operation of the Micro Floc Water Treatment Plant located on Camp Merrill. A schematic of the water treatment plant at Camp Merrill can be found within the Technical Library.

5.10.2.2.8.2 Sewage Processing at Camp Merrill. The Contractor shall process sewage and maintain the sewage lagoon, pump house, and associated equipment at Camp Merrill.

5.10.2.2.8.3 Operation. The Contractor shall operate the water treatment plant located at Camp Merrill. The Contractor shall perform sewage processing at Camp Merrill. In addition, the Contractor shall respond to emergencies 24 hours per day, seven days per week, including Federal holidays.

5.10.2.2.9 Equipment Operators. All Contractor personnel operating mobile equipment and vehicles shall possess a valid state operator's license, to include Commercial Driver's Licenses (CDLs) with special permits to include air brakes and hazardous endorsements, if required for the equipment and vehicles operated. Contractor motor vehicle drivers and equipment operators shall be appropriately qualified, certified, and licensed.

5.10.3 FACILITIES ENGINEERING SCHEDULED TASKS AT CAMP MERRILL

The Contractor shall perform the tasks described below on a recurring or scheduled basis and record the work under approved Operation Work Orders (OWO's). The Contractor shall be responsible for performing and scheduling all inspections and preventive maintenance actions in accordance with the paragraphs that follow and with current Federal, State, local, and Army regulations. A list of required scheduled tasks is provided in Technical Exhibit 5.10-001. The Contractor shall repair all defects found when performing preventive maintenance. If a requirement for further repair work on any equipment is identified during the performance of scheduled tasks, the Contractor may be required to accomplish such work on an approved Demand Maintenance Order (DMO) or Project Work Order (PWO).

5.10.3.1 Buildings and Structures Maintenance

5.10.3.1.1 Electrical Preventive Maintenance

5.10.3.1.1.2 Smoke Detectors. The Contractor shall perform preventive maintenance on smoke detectors on a quarterly basis. The Contractor shall report missing or defective smoke detectors, building number, and point of contact to the KO/QAE in writing. Monthly by the end of the first week of the following month, and replace or repair them as appropriate. There are one each in all sleeping areas, kitchens, and equipment rooms in all other buildings such as barracks, administration buildings, and dining facilities.

5.10.3.1 .1 .3 Grounding Point Checks. The Contractor shall perform grounding point checks on the ammunition supply area, fences, and the airfield on an annual basis. The Contractor shall submit a PWO (4283) to correct all discrepancies found within three working days. The Contractor shall complete a Ground Safety Check Form (CD 51 0R002) detailing the locations of ground points, the type of ground point, the reading taken, any discrepancies found, and the corrective action performed. The Contractor shall submit the completed Ground Safety Check Form to the KO within five working days of the completion of the ground point checks.

5.10.3.1.2 Fire Alarm System Checks

5.10.3.1.2.1 Alarms. The Contractor shall perform preventive maintenance on fire alarms located at the Troop Barracks and all other buildings identified in Technical Exhibit 5.10-001 on a quarterly basis. In addition, the Contractor shall respond to all "trouble" calls identified by the Fire Department.

5.10.3.1 .2.2 Fire Sprinkler Systems. The Contractor shall charge fire sprinkler systems on an annual basis. The Contractor shall be responsible for the fire sprinkler systems found in the buildings identified in Technical Exhibit 5.10-002.

5.10.3.1 .3 Winterization and Dewinterization of Buildings and Other Systems.

5.10.3.1.3.1 The Contractor shall winterize and dewinterize an average of 4 non-heated facilities and other non-heated systems per year, to include, but not limited to, various unheated water-supplied areas at Camp Merrill. In addition, the Contractor shall identify and correct deficiencies that present a potential for freeze damage or water leak damage. Other winterizations and dewinterizations shall be performed as unscheduled tasks upon receipt of a valid work document. The Contractor shall complete the winterization or dewinterization of buildings and other systems within 14 calendar days after receipt of a notice to proceed from the KO.

5.10.3.1 .3.2 Heaters. Prior to the start of the heating season, the Contractor shall perform preventive maintenance on heaters and heating systems to ensure that units are operating and burning properly. The Contractor shall specifically inspect the heaters located in the buildings identified in Technical Exhibit 5.10-003. In addition, on a quarterly basis, the Contractor shall remove and replace filters in the buildings contained in Technical Exhibit 5.10-003.

5.10.3.1.3.3 Gas-Fire Water Heaters. Prior to the start of the heating season, the Contractor shall perform preventive maintenance on gas-fired water heaters to include the circulating pump, if applicable, to ensure that units are operating and burning properly. The Contractor shall inspect the gas-fired water heaters in the buildings identified in Technical Exhibit 5.10-004.

5.10.3.1 .3.4 Ventilation Systems. Prior to the start of the heating season, the Contractor shall perform preventive maintenance on ventilation systems. The Contractor shall maintain and repair the ventilation systems identified in Technical Exhibit 5.10-005.

5.10.3.1 .3.5 Chilled Water Systems. During the cooling season, the Contractor shall check and perform preventive maintenance on chilled water systems at Camp Merrill to ensure proper operation.

5.10.3.1 .3.6 Winterization. Winterization shall include, but is not limited to: disassembling parts or equipment of building plumbing, electrical or other equipment; turning off utilities including, but not limited to, electricity and water; adding antifreeze to mechanical systems; checking mechanical systems including, but not limited to, the plumbing system, heating system, and building equipment for proper operation; repairing leaks and malfunctions; and turning water off at underground shutoff valves. The Contractor shall make all necessary repairs to place these facilities and systems in proper operation within 14 calendar days after the receipt of notification to proceed from the KO.

5.10.3.1 .3.7 Dewinterization. Dewinterization shall include, but is not limited to: reassembling removed parts or equipment of building plumbing, electrical or other equipment; turning on utilities including, but not limited to, electricity and water; draining and flushing antifreeze from mechanical systems; checking mechanical systems including, but not limited to, the plumbing system, heating system, and building equipment for proper operation; repairing leaks and malfunctions due to freeze related damage; and turning water on at underground shutoff valves. The Contractor shall make all necessary repairs related to freeze damage to place these facilities and systems in proper operation within 14 calendar days after the receipt of notification to proceed from the KO.

5.10.3.1.4 Safety

5.10.3.1 .4.1 Test: Conduct semi-annual stress and drop tests at Camp Merrill Rappelling Tower/Lower Mountain Rappelling Wall, and IAW FM 21-20, Technical Exhibit 5.11-004. The drop test and static test shall be conducted on all fall/climbing nets. A static load test will be conducted on all other anchor points at these courses. Examples of anchor points include eyebolts used for rappelling anchor points, eyebolts used for vertical climbing ropes, and anchor points on other climbing structures. The drop and static tests shall be conducted with a sand bag weighing at least 400-pounds.

5.10.3.1 .5 Dining Facility Equipment Maintenance

5.10.3.1.5.1 The Contractor shall perform maintenance and repair of Government-owned dining facility equipment at Camp Frank D. Merrill. The Contractor shall perform services as required herein to provide complete maintenance of dining facility equipment. The Contractor shall replace parts and assemblies as necessary to ensure that all equipment is fully mission capable (FMC). See Functional Area 5.3 for equipment maintenance associated with the dining facility at Camp Merrill, Ga.

5.10.3.1 .5.2 Work Area and System Description. Typical dining facility equipment covered under this Contract is shown in Technical Exhibit 5.3-002

5.10.3.1 .5.3 Preventive Maintenance of Dining Facility Equipment. The Contractor shall perform preventive maintenance on dining facility equipment in accordance with manufacturers' operator and maintenance manuals on a semi-annual basis unless otherwise specified. Technical Exhibit 5.2-010 lists the PM tasks associated with the various types of equipment. The Contractor will maintain preventive maintenance records in accordance with AR 750-1 and DA Pam 750-8. The Contractor shall prepare and submit a Demand Maintenance Order within one working day after discovering any work that exceeds the scope of the PM requirements. Those repairs shall be made via that Demand Maintenance Order vice the PM work order.

5.10.3.2 Utility Systems Operation and Maintenance

5.10.3.2.1 Water Tower Lighting. On a monthly basis, the Contractor shall perform preventive maintenance on one water tower and tank lighting. This maintenance shall include, but is not limited to, testing all systems and units for correct operation; replacing defective bulbs, fixtures, or components; and testing the cathodic protection systems. See TE 5.2-001

5.10.3.2.2 Deleted

5.10.3.2.3 Propane Fuel Distribution Systems. Camp Merrill uses propane as its primary fuel with a total storage capacity of 38,000 gallons. On an annual basis, the Contractor shall inspect all propane distribution lines for possible leaks or damage. The Contractor shall perform maintenance or repairs to lines upon the discovery of any damage or leaks. In addition, a minimum of every three years, the Contractor shall paint approximately 34 propane tanks and perform all required maintenance to ensure the proper operation of the tanks. TE 5.4-004a

5.10.3.3 HVAC Systems Operation and Maintenance

5.10.3.3.1 Air Conditioning to Heat and Heat to Air Conditioning Changeover. The Contractor shall perform changeover of the HVAC systems in the Troop Barracks located in Building 61. The Contractor shall maintain a report of all work performed for changeover services, including date, hours expended, equipment name/number, and facility name/number.

5.10.3.3.2 Heat to Air Conditioning. The Contractor shall begin start-up of all applicable cooling systems and shutdown of all applicable heating systems, historically on approximately 15 April to 15 May, and complete the start-up within 15 calendar days. Climatic variations may require the KO to direct changeover of seasonal equipment at dates earlier or later than those specified. During changeover, the Contractor shall take corrective action via a SO or PWO if any operational deficiencies are identified.

5.10.3.3.3 Shut-down of Steam Boilers. At the end of the operating season, the Contractor shall add dispersant several days prior to shutdown, drain the system, flush the system, and lay the system up with treated water. TE 5.10-006

5.10.3.3.4 Air Conditioning to Heat. The Contractor shall begin shutdown of all applicable cooling systems and start-up of all applicable heating systems, historically on approximately 15 October to 15 November, and complete the changeover within 15 calendar days. Climatic variations may require the KO to direct changeover of seasonal equipment at dates earlier or later than those specified. During changeover, the Contractor shall take corrective action via a SO or PWO if any operational deficiencies are identified.

5.10.3.3.5 Chiller. On an annual basis, the Contractor shall perform preventive maintenance on 2 chillers located at Camp Merrill TE 5.10-007. Maintenance tasks shall include, but are not limited to:

a. Checking oil levels, temperature controls, thermostats, pressure gauges, and electrical connections.

b. Checking for leaks and change oil and filters.

c. Cleaning condensers and rod tubes.

d. Lubricating and repacking bearings.

e. Checking alignment, operation, and amperage.

f. Cleaning and overhauling chiller pumps, and calibrating the system.

g. Check and change oil on vacuum pumps.

5.10.3.3.6 Circulating Water Pumps. As part of the changeover from heat to air conditioning and air conditioning to heat, the Contractor shall perform preventive maintenance on approximately three circulating water pumps on heating and air conditioning systems, to ensure their efficient and correct operation.

5.10.3.4 Grounds Maintenance

5.10.3.4.1 Airfields, Parachute Landing Platforms, and Mock Areas. Every two weeks during the growing season, the Contractor shall mow the airfield and the areas surrounding the parachute landing platforms, and mock areas.

5.10.3.4.2 Landfills. On a monthly basis, the Contractor shall mow the two landfills during the growing season and ensure that the ground cover for the landfills has not been compromised. Upon discovery of any bare spots within the landfill ground cover, the Contractor shall immediately re-establish cover in that area.

5.10.3.4.3 Landing Zones. The Contractor shall mow the landing zones on Yohna Mountain on a monthly basis during the growing season so as to avoid conflicts with mission requirements.

5.10.3.4.4 Landfill Operations and Maintenance. Camp Merrill has a separate service contract for the monitoring of methane gas and water quality at the two landfills that support the Installation; however, the Contractor shall support the service contractor’s quarterly inspections of these levels as required. In addition, on a semi-annual basis, the Environmental Protection Division (EPD) of the State of Georgia performs an inspection of the landfills. The Contractor shall support and facilitate this inspection as required and shall perform all required work to correct deficiencies identified during the inspection.

5.10.3.5 Surfaced Area Maintenance

Camp Merrill has a Letter of Understanding with the NFS regarding the use of NFS roads and training areas and their respective maintenance. This letter is contained within the Technical Library. The Contractor shall maintain the roads detailed in the Letter of Understanding in accordance with the agreement. The Contractor shall also meet with a representative from the NFS to coordinate and clarify road maintenance activities on Forest Service land. The contractor shall blade and grade approximately 40 miles of surfaced area as required. The contractor shall pull ditches, clean culvert heads and pipe a minimum of two times a year or more if needed. on and around approximately 40 miles of surfaced areas to prevent any blockage. In addition, on an annual basis, the Contractor shall inspect the NFS roads to identify necessary repairs. Upon identification of any deficiencies on the roads, the Contractor shall service the roads in accordance with the Letter of Understanding.

5.10.3.6 Water Plant and Sewage Processing

5.10.3.6.1 Water Treatment Plant. The Contractor shall operate, maintain, and repair the Micro Floc Water Treatment Plant located on Camp Merrill. All work shall be in accordance with EPD, State of Georgia regulations, TM 5-813-3 and TM 5-814-7, and the Plan of Operation Management for the Spray Irrigation System at Camp Merrill. A schematic of the water treatment plant at Camp Merrill can be found within the Technical Library.

5.10.3.6.2 Water Testing. The Contractor shall test the water at the water treatment plant for pH, turbidity, and chlorine every hour during operation. The Contractor shall record these test results in the daily operator's report and submit these results within the Water Treatment Report that is submitted on a monthly basis to the State of Georgia with a copy to Environmental Management Division (EMD) located at Fort Benning for approval (CD 51 0R003). For Violation of Regulation Report, submit CD 51 0R005.

5.10.3.6.3 State Sampling Requirements. On a monthly basis, the Contractor shall send a microbiological sample to the State of Georgia for analysis. Annually, the Contractor shall submit a nitrate and nitrite sample to the State for analysis.

5.10.3.6.4 Operational Maintenance Requirements. The Contractor performs all preventive maintenance of water treatment plant equipment in accordance with the timeframes as specified in EPD regulations.

5.10.3.6.5 Monitoring of Sewage Processing Operation.

5.10.3.6.5.1 The Contractor shall monitor each stage of the sewage process at Camp Merrill to ensure that the sewage output falls within all environmental requirements. The handling of all waste and sewage shall be in accordance with TM 5-813-3, TM 5-814-7, the Installation Operation Plan for Spraying of Fields, and all other applicable Federal, State, local, and Army regulations. If, through routine testing, the Contractor determines or is informed by the State of potential run-off from the spraying of sewage, the Contractor shall take corrective action to end the run-off. In addition, the Contractor shall continue to test the surrounding water systems for possible contamination for a period of time determined by the State of Georgia or the KO.

5.10.3.6.5.2 Low Meter on Retention Basin. On a daily basis, the Contractor shall read the water flow meter located on the north side of the lagoon or retention basis. The Contractor shall document the total number of gallons of water that came into the basin and the total number of gallons that was sprayed in the daily operators report.

5.10.3.6.5.3 Wastewater Report. The water test results and the daily flow meter results shall be incorporated into the monthly Wastewater Report which is submitted to the State of Georgia with a copy to EMD located at Fort Benning for approval (CD 510R004).

5.10.3.6.5.4 Sewage Spraying. When the sewage level in the retention basin reaches the designated level, the Contractor shall turn the pump house on to commence spraying. The Contractor shall inspect the pump house and the related sewage spraying equipment hourly during spraying operations. Requirements of the hourly inspections shall include, but are not limited to, visually checking the spray heads, checking the pump pressure gauges, and checking the line pressure. The Contractor shall perform all identified minor repairs or shall inform the KO of all major repairs exceeding the established thresholds. All work shall be in accordance with the Wastewater Treatment Facility SOP and all other applicable regulations.

5.10.3.6.5.5 Assistance to Other Organizations. Currently, a separate service contract is in place at Camp Merrill for the testing of phosphates, nitrogen, BOD, and pH at monitored wells (done on a quarterly basis); fluoride and electrical conductivity (done semi-annually); sodium, calcium, potassium, and soil samples (done annually); and lead, mercury, and organics at the monitored wells (done annually). However, the Contractor shall be required to support and facilitate each of these testing requirements on a scheduled basis.

5.10.4 FACILITIES ENGINEERING UNSCHEDULED TASKS

The Contractor shall perform the tasks described below when initiated through either a DMO or an approved PWO. All work shall be performed in accordance with the standards set forth in Section C-5, unless otherwise directed by the KO. In addition to the items listed, there are special event that are to be supported by the Contractor. These include but are not limited to Ranger Open House, Graduations, Change of Commands, and Family Day.

5.10.4.1 Buildings and Structures Maintenance

The Contractor shall perform all unscheduled maintenance on buildings and structures at Camp Merrill in accordance with the requirements as specified in Functional Area 5.2, Buildings and Structures Maintenance.

5.10.4.2 Utility Systems Operation and Maintenance

The Contractor shall perform all unscheduled maintenance of utility systems in accordance with the requirements as specified in Functional Area 5.4, Utility Systems Operation and Maintenance.

5.10.4.3 HVAC Systems Operation and Maintenance

The Contractor shall perform all unscheduled maintenance of HVAC systems in accordance with the requirements as specified in Functional Area 5.5, HVAC Systems Operation and Maintenance.

5.10.4.4 Grounds Maintenance

The Contractor shall perform all unscheduled grounds maintenance in accordance with the requirements as specified in Functional Area 5.6, Grounds Maintenance.

5.10.4.5 Surfaced Area Maintenance

The Contractor shall perform all unscheduled surfaced area maintenance in accordance with the requirements as specified in Functional Area 5.7, Surfaced Area Maintenance.

5.10.4.6 Water and Wastewater Plant Operations and Maintenance

The Contractor shall perform all unscheduled operations and maintenance of the water and wastewater plants in accordance with the requirements as specified in Functional Area 5.10, Water and Wastewater Plant Operation and Maintenance.

Functional Area 5.11
RANGE MAINTENANCE

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Functional Area 5.11

RANGE MAINTENANCE

5.11.1 INTRODUCTION

The Contractor shall perform replacement, installation, repair, and maintenance in the range facilities and areas at Fort Benning. The Contractor shall perform the tasks specified herein, based on requirements for training support of individuals, groups, or units in maneuver, equipment and weapons systems operation, and combined arms tactical operation. The Contractor effort shall be sufficient to provide functionally usable range facilities fully capable of supporting the Fort Benning training and support mission. The Contractor shall be responsible for providing timely service and maintenance to minimize any disruption of the training mission. The Contractor shall maintain ranges, associated range areas, range equipment, structures, and perform other special services. Qualified personnel in accordance with applicable laws, regulations, and documents in Section C-6 shall perform all work and Government developed annual and long range plans. Work must be coordinated with Range Division, Directorate of Operations and Training (DOT) at all times.

5.11.2 SCOPE OF SERVICES

5.11.2.1 Work Area and System Description

A list of ranges, including but not limited to, for which the Contractor shall be responsible is provided in Technical Exhibit 5.11-001. A map of range areas is provided in the Technical Library.

5.11.2.1.1 Types of Ranges. Types of ranges to be supported by the Contractor at Fort Benning include, but are not limited to, the following:

- Rifle ranges
- Pistol ranges
- Tank ranges

5.11.2.1.2 Areas Covered. Areas to be covered by the Contractor at Fort Benning include, but are not limited to, the following:

- Range facilities
- Training facilities such as obstacle courses
- Targets
- Berms
- Support structures
- Ammunition sheds
- General buildings
- Guard, watch, and control towers

- Platforms
- Latrines
- Water Crossing

5.11.2.2 Work Management and Control

5.11.2.2.1 Reporting Requirements. The Contractor shall prepare, submit, and maintain all records and reports as specified herein and in accordance with the Contract Data Requirements List (CDs) in Technical Exhibit 1-004.

5.11.2.2.2 Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.11.2.2.3 Coordination with Range Division Office. The Contractor shall coordinate with the Range Division Office to integrate maintenance operations with training support operations. Each and every Contractor activity (maintenance, grounds, repair, etc.) taking place in or adjacent to any range shall be cleared and approved by the Range Division Office to ensure the safety of Contractor personnel. Upon receipt of a Demand Maintenance Order (DMO) or project Work Order (PWO), the Contractor shall telephone the Range Division Office to schedule work during available timeframes. Upon arriving at impact areas (both high impact and low impact), the Contractor shall wait for Range Division to open all secured gates and provide access for equipment and Contractor employees. The Contractor shall use radio communications with Range Control when obtaining clearance for performing work forward of the firing line, only after obtaining clearance from the Range Facility Management Support System (RFMSS). The RFMSS is the software that is used to schedule all maintenance on range facilities.

5.11.2.2.3.1 Gate Clearance. The Contractor shall coordinate gate access to ranges through Range Division Operations. DOT may supply the Contractor with an updated list of contacts that could be called directly for gate access.

5.11.2.2.4 Organizational Maintenance. The units using the respective ranges perform certain preventive maintenance tasks and general replacement of components. This may include cleaning and oiling mechanisms and targets, as well as re-installing lifters and other mechanisms. Technical Exhibit 5.11- 002 provides a Maintenance Allocation chart to add further detail on this topic.

5.11.2.2.5 Response Time. Work shall be performed in accordance with the priority set by the Government. Due to the heavy use of ranges, ranges shall be repaired within the timeframes specified by the Government so as not to delay future use of the range. The Contractor shall be prepared to maintain and repair ranges in accordance with SO priorities listed in Section C-5.

5.11.2.2.6 EXODUS. The Contractor shall be aware that the flow of range work is often dictated by the period called EXODUS, the break in training which normally occurs between approximately December 15th and January 1st. During the EXODUS period, access to ranges will increase significantly thereby allowing easier repair and maintenance.

5.11.2.2.7 Working Inventory. The Contractor shall maintain a working inventory of essential supplies and materials readily available for use on the range to reduce down time. This may include, but not be limited to: skip plates, lumber, wood frames, security gates, and rails.

5.11.2.2.8 Safety Considerations and Staffing. It is suggested that Contractor assess risks inherent to some of the range and training sites. The Contractor shall determine appropriate staffing number.

5.11.2.2.9 Equipment Operators. All Contractor personnel operating mobile equipment and vehicles shall possess a valid state operator's license, to include Commercial Driver's Licenses (CDLs) with special permits to include air brakes and hazardous endorsements, if required for the equipment and vehicles operated. Contractor motor vehicle drivers and equipment operators shall be appropriately qualified, certified, and licensed.

5.11.3 SCHEDULED TASKS

5.11.3.1 Grading of Course and Range Roads

The Contractor shall grade course and range roads to include water crossing on a quarterly/semi-annual schedule IAW TE 5.11-003, allowing for weather conditions, (i.e. excessive moisture, cold weather where the ground is frozen). The Contractor shall execute this requirement. As part of this requirement, the contractor shall submit CD 511R001, one week prior to the beginning of each month describing the work need (i.e. severe roadside erosion, sinkholes or other problems) with locations, for the upcoming month, for KO approval.

Gravel/stone/rock will be utilized as required up to a maximum of 5,000 tons per quarter. Recycled material may be used where appropriate.

In addition to grading course/range roads, the Contractor shall also maintain approximately 100 miles of gravel and clay roads and repair erosion outside the maneuver boxes in the Good Hope Training area. Work will need to be coordinated with Range Control on an as-needed basis.

On an annual basis, contractor shall sweep debris off bridges and clear all drains associated with those bridges. All work covered under this requirement will be covered under a SOO.

A map of Ft. Benning identifying locations of the Range Roads and Access Roads listed in TE 5.11-003,003a & 003b is provided in the Technical Library.

5.11.3.2 Obstacle and Confidence Course

5.11.3.2.1 Sandhill Obstacle and Confidence Course: The Contractor shall replace all the ropes on obstacle number 2, at 6-month intervals, or as required by Directorate of Training (DOT), whichever comes first, at the Sand Hill Obstacle Course. The ropes shall be obtained from range maintenance, DOT. A record shall be kept on file of all rope replacements by position.

5.11.3.2.1.1 The Contractor shall provide a load of missile-free sand once quarterly (exact quantity needed to be determined at that time), for the low crawl obstacle at the Sand Hill Obstacle Course. The unit will spread the sand Self-Help. A running record shall be kept of the sand delivered.

5.11.3.2.1.2 The Contractor shall perform semi-annual inspections of all Obstacle and Confidence Courses identified in Technical Exhibit 5.11-004, to identify deficiencies. Deficiencies shall be submitted on CD 511R002. The installation Safety Office will schedule inspections with the contractor. The deficiencies shall be completed within 15 calendar days, and charged to an approved SOO. The inspections shall be conducted in the months of March through April, and October through November as outlined in Technical Exhibit 5.11-004

5.11.3.2.1.3 Static and drop testing on Obstacle and Confidence Courses: The Contractor shall conduct semi-annual stress and drop tests in conjunction with inspections described in the above para, 5.11.3.2.1.2 on Obstacle and Confidence Courses specified in Technical Exhibit 5.11-004, and IAW FM 21-20. The drop test and static test shall be conducted on all fall/climbing nets. A static load test will be conducted on all other anchor points at these courses. Examples of anchor points include eyebolts used for rappelling anchor points, eyebolts used for vertical climbing ropes, and anchor points on other climbing structures. The drop and static tests shall be conducted with a sand bag weighing at least 400-pounds.

5.11.4 UNSCHEDULED TASKS

The Contractor shall perform range maintenance and repair to include those tasks specified below when initiated through either a DMO, approved PWO, or OWO.

5.11.4.1 Equipment Maintenance and Repair

5.11.4.1.1 Targets. The Contractor shall maintain, repair, or replace components of the target mechanisms and target systems. Responsibilities for the maintenance of target mechanisms are further discussed in Functional Area 5.14, Materiel Maintenance. The Contractor shall be responsible for fabricating (but not installing) all targets required for the operation of the range. Fabrication must be in accordance with the requests of Range Division Operations, units, or other interested parties. Soldiers will be responsible for producing and installing silhouettes and paper targets on ranges. The target pop-up systems shall be maintained according to the manufacturer's instructions. The Contractor shall ensure that the target mechanisms operate freely, without binding, and in accordance with the FM 25-7, Range Training, which provides details of the construction of training aids.

5.11.4.1.2 Fabrication of Various Targets. The Contractor shall be responsible for fabricating and providing the hard targets (frames) for all of the ranges using such targets. The Contractor shall also fabricate T-72 and BMP Frontal, T-72 and BMP Flank, BRMD Front Targets, BBS Bradley Bore Sight Screening Panels, and various other wooded targets specified in TC 25-8. During the course of the year, an average of 600 targets are fabricated. More than 650 target frames are also constructed.

5.11.4.1.3 Skip Plates. The Contractor shall be responsible for repairing and replacing skip plates in front of targets, as required and as requested, to ensure maximum target protection.

5.11.4.1.4 Rails. The Contractor shall maintain rails used for targets in accordance with Installation specifications. Rails shall be bonded electrically and continually and shall be effectively grounded. The Contractor shall maintain the third rail and its protective covering by inspecting, repairing, and replacing components that have been damaged, separated, or in need of repair. The Contractor shall also inspect and maintain proper grounding, brackets, rail fasteners, and rail brackets.

5.11.4.1.5 Scoring and Tracking Systems. The Contractor shall be responsible for repairing and maintaining all scoring and tracking systems. The Contractor shall repair computer operated control systems to ensure accurate and timely recording of scores.

5.11.4.1.6 Lane Markers. The Contractor shall be responsible for fabricating, installing, and maintaining lane markers. This shall include, but not be limited to: ensuring that sight paths are clear down lanes; painting markers; installation of markers; and fabricating markers.

5.11.4.1.7 Communications-Electronics Equipment. The Contractor shall test, replace, and repair the control systems and components of all ranges so as to operate the targets and other apparatus without interfering with the range schedules or operations. The Contractor shall maintain electrically operated control systems to ensure accurate and timely control responses. The Contractor shall inspect the electrically operated control systems and their components and shall maintain, repair, and replace defective components, parts, and cards

5.11.4.1.8 Automated Tank Range Target System. The Contractor shall maintain and repair all components and radio controls of the automated tank range target system. The Contractor shall be responsible for the power from the tower to the targets themselves, as well as the targets.

5.11.4.1.9 Warning Lights. The Contractor shall repair and install warning lights including, but not limited to: down range right and left limits of the BFV/Tank ranges and heat thermal panels.

5.11.4.2 Grounds Maintenance and Repair (for Camp Merrill only)

5.11.4.2.1 Roads and Trails. The Contractor shall be responsible for maintaining all roads and trails to and from ranges and other training areas. The Contractor shall maintain access and entrance roads and trails of each range so that the wheeled and tracked vehicles are able to fully use trails in a safe and efficient manner at all times. Responsibilities shall include, but not be limited to, providing gravel in areas suffering from erosion, and grading gravel and dirt roads to and from the range. The Contractor shall maintain roads and trails to ensure that they are free from potholes, cracks, and obstacles and that the surfaces are well drained and in a high state of repair.

5.11.4.2.1.1 Sufficient and Proper Drainage. The Contractor shall maintain surfaces, ditches, culverts, and other water conveyances to adequately drain the ranges, roads, and other associated range areas to permit wheeled and tracked traffic to fully use the ranges safely. The Contractor shall repair surfaces, culverts, and other damaged water conveyances and remove debris from them to ensure proper drainage.

5.11.4.2.1.2 Erosion Control. The Contractor shall inspect and remediate erosion of all ranges and associated areas, in consultation with and under the direction of the Environmental Management Division (EMD). This includes, but is not limited to, ranges, contiguous areas, roads, trails, parking areas, turn pads, bivouac areas, target pits, and rail systems.

5.11.4.2.1.3 Terrain Restoration. The Contractor shall be responsible for restoring terrain to its original condition by providing soil, grading, and re-planting. The Contractor shall provide the EMD with any and all plans to restore terrain in and around range areas. The Contractor shall follow all applicable Federal, State, local, and Army environmental regulations.

5.11.4.2.2 Retaining Walls. The Contractor shall maintain and repair retaining walls used on ranges.

5.11.4.2.3 Turn Pads. The Contractor shall maintain concrete, gravel, and earth turn pads to ensure that surfaces and curbs are free from cracking, potholes, and other obstacles and that the surfaces are maintained with concrete or gravel to allow the smooth turning of wheeled and tracked vehicles.

5.11.4.2.4 Obstacle Courses, Ropes, and Other Training Devices. The Contractor shall maintain the grounds in and around all obstacle courses as well as the equipment on the courses.   The equipment includes, but is not limited to ropes, wooden obstacle courses, wooden structures, and rope ladders. The KO will provide rope and netting.

5.11.4.2.5 Tank Trails. The Contractor shall be responsible for the timely grading of tank trails and providing gravel to requested areas on a timely basis so not to detrimentally affect the training mission and general operation of all tank ranges and training grounds. This shall include, but not be limited to: repairing culverts; spreading gravel or clay; and cutting new trails.

5.11.4.2.6 Bridges, Fords, and Other Mobility Structures. The Contractor shall maintain bridges, bridge abutments, and spans so that they are able to safely and continuously carry tracked and wheeled vehicles in the weight class for which each bridge is designed. The Contractor shall inspect bridges upon request and record the results. Deficiencies will be reported within two hours to the KO. The Contractor shall fix and replace wood penetrated by ammunition.

5.11.4.2.7 Bivouac Areas. The Contractor shall maintain each bivouac area to ensure that utilities (water, electric, latrines, and storm systems) are in working condition.

5.11.4.2.8 Berms. The Contractor shall maintain berms by rebuilding and replacing fill material (i.e., clay, sand, or mulch) that has been displaced due to the impact of ordinance striking the berm during firing exercises. Berms shall be graded with a level area on top, shall have exactly the specified angle of attack, and shall be seeded and sodded to prevent erosion. Berms shall be constructed with maximum emphasis on protecting target mechanisms while simultaneously ensuring that targets are accessible for training purposes. Berms on tank ranges require a minimum width of 20 feet at their top.

5.11.4.2.9 Firing Positions. The Contractor shall be responsible for firing position structures. The Contractor shall be responsible for digging or re-digging holes on range facilities for foxhole installation; physically installing plastic, concrete, or wood coffins (as per specific requirement or DOT directions); providing appropriate drainage so to prevent pooling in firing positions; and to restore the terrain around foxholes and firing positions. Firing position cleaning will be the responsibility of soldiers. The Contractor shall be responsible for digging trenches per request to support the training function of the Installation.

5.11.4.2.10 Welding. Any repair associated with grounds maintenance equipment will be performed by the contractor. The personnel performing such work will have all applicable certifications.

5.11.4.3 Structures Maintenance and Repair

5.11.4.3.1 Painting. The Contractor shall only be responsible for the painting of exteriors of wooden structure buildings.

5.11.4.3.2 Electrical Systems. The Contractor shall maintain, repair, and install electrical system components including, but not limited to, wiring systems, conduit systems, cable systems, conductors, switches, receptacles, outlets, device plates, grounds, service equipment, motor control centers, control switchboards and consoles, lightning protection systems, fixtures, kitchen and building ventilating fans, sump pumps, and electrical heaters in all applicable buildings, structures, and facilities. Additional electrical maintenance requirements for motors, motor controllers (e.g., starters, contactors, or variable speed controllers), equipment status monitoring systems, pump controls, and similar electrical apparatus that is appurtenant to plant and mechanical equipment are included in the applicable section for the maintenance of that equipment. The Contractor shall also perform other electrical work such as: providing temporary electrical service as required; providing emergency lighting systems; determining estimates; testing circuits and load; providing standby electricians for special events.

5.11.4.3.3 Heating, Ventilation, and Air Conditioning (HVAC) Systems. The Contractor shall maintain, repair, and install HVAC system components located at range facilities upon the receipt of a valid work document. All work shall be performed in accordance with the standards specified in Functional Area 5.5, HVAC Systems Operation and Maintenance.

5.11.4.3.4 Welding. Any repair associated with structures maintenance will be performed by the contractor. The personnel performing such work will have all applicable certifications.

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Functional Area 5.12
CEMETERY SERVICES

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Functional Area 5.12

CEMETERY SERVICES

5.12.1 INTRODUCTION

The Contractor shall provide services necessary to operate and maintain the Main Post Cemetery located on the Main Post of Fort Benning, Georgia, and approximately 61 cemeteries throughout Georgia and Alabama maintained by the Installation. The Contractor shall perform administrative duties, arrange for and process interments and disinterment's, prepare for Special Ceremonies, open and close the Main Post Cemetery, perform maintenance, and perform all other tasks as specified in AR 210-190, and DA Pam 290-5. The Contractor shall perform all services required to assist and answer inquiries (either oral or written) from bereaved families and friends, funeral homes, Adjutant General (AG), hospitals, coroners, Department of the Army (DA), Veterans Administration (VA), and other official military channels as required by the KO.

5.12.2 SCOPE OF SERVICES

5.12.2.1 Work Area and System Description

The Main Post Cemetery consists of 8.26 acres of burial grounds (enclosed inside brick wall); Buildings 2679 (Office and Comfort Station) and 2680 (Break room and Storage Area); 3 unnumbered Storage Sheds; Columbarium; approximately 2 acres of grounds outside of the brick wall in the back of the Main Post Cemetery; and approximately 1 acre adjacent to Building 2679 (including the driveway and parking lot).

In addition to the Main Post Cemetery, the installation has 61 additional cemeteries that are located throughout the Installation's boundaries in Georgia and Alabama. These additional cemeteries shall only be maintained upon the receipt of a valid work document or as directed by the KO. The locations of the additional cemeteries maintained by Fort Benning can be found in Technical Exhibit 5.12-002.

5.12.2.2 Work Management and Control.

5.12.2.2.1 Reporting Requirements. The Contractor shall prepare, submit, and maintain all records and reports as specified herein and in accordance with the Contract Data Requirements List (CDs) in Technical Exhibit 1-004.

5.12.2.2.2 Publications and Forms. Specific publications and forms required for the accomplishment of work described in this Functional Area are listed in Section C-6.

5.12.2.2.3 Cemetery Control. The Contractor shall keep the Main Post Cemetery free of animals, unaccompanied children, bicycles, and any other unauthorized persons such as the news media that hasn't been cleared through appropriate Government channels.

5.12.2.2.4 Inspections. The Contractor shall be subject to inspections performed at the Main Post Cemetery. Inspections are performed in accordance with Par. 14, AR 210-190. The HQ, Casualty and/or Memorial Affairs, Washington, D.C. and the Commanding General inspect the Main Post Cemetery at least once each year. The Fire Marshal inspects the Main Post Cemetery for fire hazards as required or requested. The Occupational Health Office inspects and recommends protective equipment (eye protective items and protective clothing) and identifies safety hazards as required or requested. The DPW inspects at his discretion. The Real Estate Office, DPW inspects all buildings and their fixtures as required or requested. The Energy Conservation Office, DPW inspects at unannounced times. The Installation Safety Office will also perform unannounced inspections.

5.12.2.2.5 Sources of Work. The KO, Units, DS/GS personnel, other Government personnel, tenants or residents of the Installation, or the Contractor, can identify work. Contractor-generated work is the result of requests made by operators and maintenance personnel, who identified deficiencies during inspections, preventive maintenance, or other work performance. Sources of Work are provided in Technical Exhibit 5.15-003.

5.12.2.2.6 Workload. Workload for Cemetery Services is provided in Technical Exhibit 5.12-004.

5.12.2.2.7 Special Requirement, Dress Code at the Main Post Cemetery. The cemetery manager shall wear dress attire during operation hours at the cemetery. All other staff members shall wear an approved standardized uniform at all times. Contractor shall submit to the KO types of uniforms that shall be warned by contract personal, CD 0512002.

5.12.3 SCHEDULED TASKS

The Contractor shall perform the tasks described below on a recurring or scheduled basis and record the work under an approved Operation Work Order (OWO's). The Contractor shall be responsible for performing all work in accordance with the paragraphs that follow and with current Federal, State, local, and Army regulations. Technical Exhibit 5.12-001 provides a listing of required scheduled tasks for the Main Post Cemetery.

5.12.3.1Opening and Closing the Main Post Cemetery

The Contractor shall open the Main Post Cemetery gates and raise the Contractor-furnished 5' X 9.5' United States "Storm" flag to full staff on the cemetery pole no later than 0800 daily. The Contractor shall remove the flag, close, and secure the gates no earlier than 1700 daily, except Memorial Day that shall be closed no earlier than 1900. The Main Post Cemetery shall be open from 0800 to 1700 on a daily basis, and the Contractor shall display the cemetery operating hours in accordance with DA Pam 290-5, Par. 2-4a. The vehicle drive-through gates shall be closed and secured during the hours of darkness. The contractor shall comply with AR 840-10 in the flying of the United States flag.

5.12.3.2 Headstone Maintenance at the Main Post Cemetery

The Contractor shall align, clean, replace, and destroy headstones in accordance with Par. 4-18, 4-19, and 4-20, DA Pam 290-5. Contractor shall record headstone cleaning, alignment and maintenance using TE 5.12-005.

5.12.3.2.1The Contractor shall clean headstones and maintain an onsite record of cleaning, at the Main Post Cemetery twice per year in accordance with Par. 4-19, DA Pam 290-5. The contractor shall record cleaning of headstones using TE 5.12-005.

5.12.3.2.2 The Contractor shall realign headstones in accordance with Par. 4-18, DA Pam 290-5. The contractor shall record headstone alignment using TE 5.12-005.

5.12.3.3 Main Post Cemetery Grounds Maintenance

Fertilization. The Contractor shall apply fertilizer and lime so that distribution is uniform, without damaging flowers, shrubs, trees, or lawns. The Contractor shall fertilize twice per year, between 15 March and 31 March and between 15 July and 31 July, or as required by the KO. The Contractor shall lime twice per year, between 15 January and 31 January and between 15 October and 31 October, or as required by the KO. The Contractor shall make applications of fertilizer (8% nitrogen – 8% phosphorous – 8% potash) and lime in accordance with manufacturer's label. The Contractor shall also maintain a daily record of concentration per acre, approximate number of acres covered, trade name of product, manufacturer, and Environmental Protection Agency (EPA) Registration Number. The Contractor shall provide the KO a copy of the record after the end of each application. Flowers, shrubs, trees, and lawns damaged due to improper type, application, and distribution of fertilizer shall be replaced within ten working days.

5.12.3.3.1.The Contractor shall water and fertilize grass in accordance with Paragraph 5-3, DA PAM 290-5. Water shall be applied uniformly over the watered areas so that it does not run off into streets and drainage ditches or cause puddles of water. For single plants, such as trees, shrubs, and hedges, water shall be applied uniformly within the drip line of the plant. The Contractor shall correct all damage caused by the Contractor's failure to properly water within ten working days.

5.12.4 UNSCHEDULED TASKS

5.12.4.1 Interment/Inurnment at the Main Post Cemetery or Columbarium.

5.12.4.1.1 Interview Next of Kin. Upon verification of eligibility by the KO, the Contractor shall interview next of kin (NOK) of deceased person or persons for interment/inurnment in the Main Post Cemetery or Columbarium as required by the KO. The Contractor shall prepare Request for Interment/inurnment, and obtain certification (signature) of its accuracy from the NOK.

5.12.4.1.2 Grave Reservations. Single gravesite reservations made before the one-gravesite policy (where the spouse is interred in the same grave as the service member), will remain in effect as long as the reservee remains eligible for interment, as specified in AR 210-190, Chapter 6. Contractor shall maintain a log of these reservations. Additional references related to single gravesites are AR 210-190, paragraphs 7b, c, d, e, f and paragraph 10. Also refer to DA PAM 290-5, paragraphs 4-14b and c. No reservations will be permitted in the Columbarium. Assignments are without regard to military rank, race, color, creed, or gender of the qualifying service members. The next available grave or niche is assigned for the interment or inurnments.

5.12.4.1.2.1 The Contractor shall determine and assign the appropriate gravesite/niche as approved by the KO in accordance with AR 210-190, Chapter 7, and DA Pam 290-5, Par. 4-14.

5.12.4.1.3 Interment/Inurnment Arrangement. The Contractor shall coordinate with funeral directors for dates and times of interment/inurnment services. The Contractor shall schedule interment/inurnment services, normally about three hours from the beginning of one interment/inurnment service to the beginning of the next interment/inurnment service, so as not to interfere with each other. Interments/inurnments are not normally made on Saturdays, Sundays, or Holidays; however, the Contractor shall make interments/inurnments on Saturdays, Sundays, and Holidays as required by the KO.

5.12.1.4 Funeral Escorts. The Funeral Director will request Military Police funeral escorts to the Government designate entrance of the reservation.

5.12.4.1.5. Documentation. On a monthly basis, the Contractor shall prepare DA Form 2684 (CD 512R001), Cemetery Operations - Utilization of Gravesites, for signature of KO, and forward to HQ, DA in accordance with AR 210-190, Chapter 11 and DA Pam 290-5, Chapter 7 no later than the 5th of the following month. Grave Preparation and Funeral Services at the Main Post Cemetery. Contractor shall maintain a register of inurnments in the Columbarium in accordance with DA 290-5.

5.12.4.2 Grave /Niche Preparation and Funeral Services at the Main Post Cemetery.

5.12.4.2.1. The Contractor shall prepare the grave/niche to include, but not limited to, removing headstones/niche covers, flower vases, and markers to make entrance for equipment to gravesite, if necessary, to prevent damage. The Contractor shall prepare, close, and mark graves/niches in accordance with DA Pam 290-5, Par. 4-13, or as specified by the KO.

5.12.4.2.1.1. Upon completion of digging the grave, the Contractor shall cover dirt mounds with artificial grass. The Contractor shall border the gravesite with caution tape.

5.12.4.2.2. Committal Site Preparation. The Contractor shall maintain benches, sweep, and clean up around the committal service area prior to the arrival of the family. The Contractor shall ensure the metal awning is free of leaks, holes, stains, dirt, and dents. The casket bier shall be clean and in place not later than one hour prior to the interment service.

5.12.4.2.3 Flag Adjustments. The Contractor shall lower the flag to half-staff 30 minutes prior to interment services. Thirty minutes after the funeral services, the Contractor shall raise the flag back to full staff.

5.12.4.2.4. Escorts and Guards. The Contractor shall assist the Funeral Director in directing and parking vehicles that enter the Main Post Cemetery. The Funeral Director will request the honor guard for funeral services from AG Casualty when necessary.

5.12.4.2.5. Service Completion. After the funeral services and all people attending the interment/inurnment services have departed the Main Post Cemetery area, the Contractor shall assist the Funeral Director in placing the casket/urn in the vault/niche, shipping case, grave liner, or any other appropriate receptacle in the grave/niche. The Contractor shall close the grave/niche, fill in the dirt, level site with surrounding area casket bier, and replace headstones in their appropriate place. The Contractor shall also replace sod on gravesite as directed by the KO, leveling it to the same degree before removal as prescribed by the KO. The Contractor shall relocate flowers from the funeral site to the gravesite after the funeral services.

5.12.4.2.6. Suspension of Maintenance. The Contractor shall ensure maintenance is not performed inside the Main Post Cemetery boundaries from 30 minutes prior to expected arrival time of the family to the departure of all family and friends.

5.12.4.3. Headstones and Markers for the Main Post Cemetery and Columbarium.

The Contractor shall order, store, maintain, install, and replace headstones and markers in accordance with AR 210-190, Chapters 8 and 9 and DA Pam 290-5, Chapter 4.

5.12.4.3.1. Gravestone Order and Inscription

5.12.4.3.1.1. The Contractor shall prepare DA Form 2122, Record of Interment, in accordance with AR 210-190, Chapter 5 and DA Pam 290-5, Par. 7-3 no later than two working days after interment for KO signature. No later than one working day after receipt of the signed DA Form 2122, the Contractor shall send the original to the VA. The Contractor shall also send a copy of DA Form 2122 to the DA in accordance with AR 210-190, Par. 5e, within five working days after receipt of signed form from the KO and maintain a copy on file.

5.12.4.3.1.2. The VA will fax a copy of the proposed headstone inscription to the Contractor for verification of accuracy. This inscription is based on the DA Form 2122 forwarded to the VA by the Contractor. The Contractor shall verify accuracy and forward his acceptance of VA inscription or changes required to the VA in typed print by fax not later than five working days after receipt of the inscription from the VA.

5.12.4.3.21.3. Upon arrival of the headstones, the Contractor shall check the headstone inscription against DA Form 2122 in accordance with Par. 4-17, DA Pam 290-5. If the headstone is damaged, or if the inscription is incorrect, the Contractor shall reorder the headstone and destroy damaged and incorrect headstones within two working days after receipt. If the inscription is correct, the Contractor shall install the headstone within five working days after receipt in accordance with Par. 4-18, DA Pam 290-5. The contractor shall notify the COR electronically that a new headstone has arrived, with the gravesite location, any damages or inscription errors within one working day.

5.12.4.3.2. Temporary Markers for Gravesite/Columbarium. The Contractor shall provide, prepare, and place the temporary marker on the gravesite/Columbarium after the grave/niche is closed and leveled, but not later than the close of business on the date of the interment/inurnment, to remain there until the permanent headstone/niche marker arrives in accordance with AR 210-190, Chapter 8.

5.12.4.3.3. Headstone Replacement. The Contractor shall maintain the headstone replacement program in agreement with the DA and VA to replace damaged headstones, in accordance with Chapter 8c, AR 210-190, and Par. 4-20, DA Pam 290-5.

5.12.4.3.4. Documentation. The Contractor shall maintain and update DA Form 2123, Record of Interments/Inurnments and Reservations, interment log book, DA Form 2122 printout, reservation log, and Cemetery Layout Map, keeping them up to date and in order.

5.12.4.4. Flower Vase Assemblies for the Main Post Cemetery

The Contractor shall order, store, install, and maintain permanent bronze metal flower vase assemblies with plastic inserts. Only one flower vase may be purchased per gravesite. The Contractor shall not charge purchaser any more than the cost specified by the KO. This cost includes the vase assemblies, inserts, and shipping costs. The Contractor shall also install pre-approved similar vases purchased by individuals from other sources. No flowers are authorized in the Columbarium.

5.12.4.4.1. Vase Purchase. The Contractor shall maintain a minimum stock of four vases for purchase. The Contractor shall fill out the order for vase assemblies and provide the original copy to the KO, who will place the order with the Soldiers and Family Support Association. The Contractor shall not accept or install damaged vase assemblies and inserts.

5.12.4.4.1.1. The Contractor shall only accept payment for vases by money order or personal check No cash shall be accepted. The Contractor shall collect and secure money orders and checks from the sale of these vase assemblies, and provide the KO with a copy, (when requested), of the record of purchases. All vases purchased from the Contractor shall be placed on a grave in the Main Post Cemetery only.
The Contractor shall be held liable for all payments made to him. The Contractor shall follow the existing SOP for this requirement, i.e. purchase of vases, receiving payment, etc. The COR will provide the contractor with the SOP.

5.12.4.4.1.2. The Contractor shall require each purchaser to sign the receipt in duplicate for each vase assembly purchased, providing the purchaser a copy of that receipt and maintaining one for his record. The receipt shall contain, as a minimum, the name, address, and signature of the purchaser, amount paid by the purchaser, the name and signature of the Contractor's representative receiving payment, date of transaction, number of vases and inserts purchased, grave number where the vase is to be installed, and the name of the deceased.

5.12.4.4.1.3. The Contractor shall install the vase assembly no later than the following Monday after receipt of the money from the purchaser.

5.12.4.4.2. Vase Inscription. The Contractor shall permanently inscribe the vase sheath with the grave number and last name of the deceased in indelible ink. A record shall be maintained by the Contractor of the vase grave number, name of purchaser, and the date of purchase.

5.12.4.4.3. Vase Placement. The Contractor shall provide concrete vase rings to insert vase assemblies in, maintaining an adequate stock to take care of the need. The Contractor shall install the vase 18 inches from the front of the headstone. The ring shall be installed so it is flush with the ground and the vase shall be installed in the ring. The Contractor shall assure all vases are in an upright position at all times.

5.12.4.5. Disinterment's at the Main Post Cemetery and Columbarium.

The Contractor shall process all disinterments in accordance with Chapter 15, AR 210-190. The Contractor shall receive the initial request from the NOK requesting disinterment, explaining disinterment procedures to them. The Contractor shall provide FB FL 1 to the NOK. Upon receipt of notarized forms, the Contractor shall forward them to the KO for approval.

5.12.4.5.1. Disinterment. The Contractor shall coordinate the date and time of disinterment with the funeral home appointed by the NOK, a minimum of 48 hours prior to the requested service. Opening and closing the gravesite and removal of the remains of the deceased shall be at the expense of the NOK. The Contractor shall dispose of the headstone after the disinterment, as specified by the KO.

5.12.4.5.2. Reallocations and Re-interments. The Contractor shall refer all requests for re-interment in another grave within the cemetery to the KO.

5.12.4.5.3 Documentation. The Contractor shall indicate disinterment on DA Form 2684, Cemetery Operations-Utilization of Gravesites, delete the entry on the log book and printout, remove DA Form 2122 from the interment file and place in the disinterment file, and file the disinterment documents in the appropriate file folder.

5.12.4.6 Cemetery Grounds Maintenance

The Contractor shall perform the following tasks at the Main Post Cemetery and, Columbarium. The remaining 61 outlying cemeteries around Fort Benning, as identified in Technical Exhibit 5.12-002, upon the receipt of a valid work document or as directed by the KO.

5.12.4.6.1 Mowing. The Contractor shall mow in and around the cemeteries on an as needed basis. Prior to mowing, the Contractor shall pick up and remove all rubbish, debris, and trash (which includes, but is not limited to, leaves, rocks, paper, pine cones, limbs, and other portable objects) within the maintenance area and all trash, papers, leaves, and limbs lodged in shrubs, hedges, fences, and along foundation walls. All clipping rubbish, debris, and trash removed shall be properly disposed of as directed by the KO. In conjunction with mowing services, the Contractor shall inspect grounds for trash and debris buildup, possible fire hazards, and pest activity.

5.12.4.6.2 Trimming and Repair. Grass immediately around headstones, vases, and other obstacles shall be trimmed within twenty-four hours of each mowing operation, to the height of adjacent grass. The Contractor shall remove all vegetation growth in expansion joints and cracks in concrete or asphalt surfaces. The grass immediately around the headstones, building foundations, trees, shrubs, fences, poles, signs, structures, drainage ditches, roads, and curbs shall be trimmed each time the lawn is mowed. The Contractor shall replace with damaged or skinned trees, shrubs, or plants like trees, shrubs, or plants.

5.12.4.6.3 Sod Placement. The Contractor shall provide and place sod in all bare spots in turf as directed by the KO. Sod shall be capable of renewed growth after dormant periods. Centipede sod will be used for all new sod being placed.

5.12.4.6.4 Landscaping. The Contractor shall provide and plant shrubs, liriope, and flowers to beautify the cemetery grounds in accordance with Paragraph 5-3, DA PAM 290-5 and as required by the KO.

5.12.4.6.5 Funeral Flowers. Upon approval of the KO, the Contractor shall remove funeral flowers and receptacles when they become unsightly as required by Paragraph 5-3u, DA PAM 290-5. The Contractor shall place these unsightly flowers in an area designated by the KO. The Contractor shall restore eroded areas and take preventive measures to prevent recurrence.

5.12.4.6.6 Additional Grounds keeping Task. The Contractor shall continuously inspect grounds to ensure immediate removal of all visible debris (e.g. hoses, limbs, flowers, trash, etc.) during times gates are open and cemetery is available to the public.

5.12.4.6.7 Winterization of Water Fountains and Scattering Gardens. The contractor shall winterize all water fountains and scattering garden within the Cemetery to prevent damage to the equipment.

5.12.4.7 Special Ceremonies

The Contractor shall prepare for special ceremonies at the cemeteries listed in Technical Exhibit 5.12-002 and the Main Post Cemetery in accordance with Letters of Instruction; DA Pam 290-5, Chapter 2, and as required by the KO. These requirements shall include but not be limited to, providing set-up and removal of flags, chairs, and podium. The Contractor in accordance with AR 210-190, Par. 16, shall provide decorating flags for Memorial Day. The Contractor shall place decorating flags for the Italian and German Memorial Services.

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Functional Area 5.13

Chiller Maintenance and Repair

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List of Technical Exhibits

Exhibit Number	Title
5.13-001	Chiller Equipment
5.13-002	Chiller Annual Inspection Checklist
5.13-002a	Chiller Data
5.13-002b	Refrigerant Use Data
5.13-003	Building 100 Equipment
5.13-004	Deliverables Schedule
5.13-005	Operational Inspection Checklist
5.13-006	PRS
5.13-007	Government Furnished Property
5.13-008	Chiller Annual Maintenance Checklist

Functional Area 5.13

Chiller Maintenance and Repair

5.13.1 INTRODUCTION:

5.13.1.1 Provide all labor, material, and equipment to inspect, start-up, operate, and maintain the chillers as listed on Technical Exhibit 5.13-001 as well as all the associated auxiliary equipment and machinery required to support the chillers such as towers, motors, controls, tube cleaners, piping, expansion tanks, refrigerant monitors/sensors, exhaust fans, pumps, purge units, air compressors, drains, valves, unit heaters, lights, disconnects, variable speed drives, circuit panels, emergency lights, eye-lavages, switch gear, chart recorders. Functional Area work includes complete services required to repair all water-cooled chillers, cooling towers, pumps, valves, controls, and cold storage equipment (except buildings owned by MEDDAC) on Fort Benning, Georgia.

5.13.2 SCOPE OF SERVICE

5.13.2.1 Cooling Season: Chiller systems for buildings 9200, 9201, 9230, 5500 and the cooling tower for building 96 must be capable of operating on a year round basis. Chiller systems for buildings 4, 6 and 398 may be turned on or off year round as directed by the government. The remaining chillers will be turned on for the cooling season and off in the fall as directed by the Government. The Office in building 398 will be furnished to the Contractor as a base of operation.

5.13.3 Personnel:

5.13.3.1 Qualifications of Contractor: The contractor shall employ personnel regularly engaged in the operation and servicing of chillers and HVAC equipment of the general type indicated for operation, maintenance and repair service. The contractor shall have adequate stocks of new OEM parts to maintain equipment described herein and insure that all replacement parts are available. The contractor shall employ personnel knowledgeable with daily operation and maintenance of cooling plants and equipment indicated in the functional area.

5.13.4 SCHEDULED TASKS:

The contractor shall perform the tasks below on a recurring or scheduled basis and record the work under approved Operation Work Orders (OWO's). If a requirement for further work on any area beyond the scheduled requirements below is identified, the Contractor may be required to accomplish such work on an approved Project Work Order (PWO).

5.13.4.1 Perform an annual inspection on all chillers, towers, and pumps as shown on Technical Exhibit 5.13-001. Inspection shall be performed before 28 Feb with bound inspection or compact disc results provided to the government no later than 15 Mar. The initial annual inspection shall be performed within 30 days with bound inspection results provided to the Government no later than 30 days. Contractor shall make all repairs and correct all deficiencies as identified in annual inspection.

5.13.4.1.1 Detailed Schedule of Inspection: The contractor shall submit to the Contracting Officer or his authorized representative a detailed and complete schedule of the dates on which all weekly, monthly, and annual services are to be performed within 10 calendar days. The proposed schedule shall list the type of work to be performed, the areas to be worked and the estimated time to complete the work in each area. (See Technical Exhibit 5.13-004).

a. Provide a schedule within five (5) days of each annual contract period for chiller system outlining the periodic maintenance to be performed during each annual contract period.

b. Post this schedule at each equipment location.

c. Provide space blocks for each period and instruct chiller mechanic to initial when work has been completed.

5.13.4.1.2 The maintenance and inspection schedule shall provide a program to accomplish all maintenance and inspection as specified in TE 5.13-002.

5.13.4.1.3 Maintenance and Inspection Plan shall be updated every 60 calendar days thereafter, to indicate actual accomplishment for the year to date. The initial maintenance and inspection plan will be completed and submitted in writing to the Contracting Officer or his authorized representative for review to assure that the plan provides for all specified maintenance requirements in an organized and time phased manner. The Contracting Officer will approve or disapprove the schedule and, if disapproved, the contractor shall submit a corrected schedule within two (2) days after notification of disapproved schedule (see TE 5.13-004).

5.13.4.1.4 Provide the Contracting Officer or his authorized representative a bound copy or compact disc of each maintenance action taken on each piece of equipment. The report shall include the date and time maintenance was performed, equipment designation, building number, maintenance problem, action taken and replacement part information. In addition, maintain a log of all (planned and unplanned) outages, showing start and stop time of those outages. The log will be made available to the government as requested (usually quarterly) and included in the annual report. Report shall be delivered 2 weeks prior the end of contract year (see TE 5.13-004).

5.13.4.2 Start-up and place into operation all chillers, cooling towers, pumps and HVAC equipment when directed by the Contracting Officer. Contractor will be required to valve off steam/hot water heat in buildings listed in TE 5.13-001. Contractor will be provided five calendar days notice before start-ups are to begin and the contractor shall have all chillers operational within 3 calendar days. Seasonal start-up shall include, as a minimum, all of the items below:

a. Utilize manufacturer’s start-up recommendations.

b. Check chiller auxiliary equipment operation.

c. Check for proper voltages.

d. Check oil level, refrigerant level.

e. Check oil sump heater, gear case heater, and purge pump oil heaters.

f. Check vacuum.

g. Check relays, operating, and safety controls.

h. Start chilled water pump(s).

i. Start condenser water pump(s) and cooling tower.

j. Start water chiller.

k. Check purge unit operation.

l. Log operating conditions after unit stabilizes. Make sure unit maintains set point for 24 hours continuous.

m. Shutdown all chillers, towers, and pumps within 24 hours after receiving direction from the contracting officer, or his authorized representative.

5.13.4.2.1 Perform operational inspections on each chiller, tower, and pumps, throughout the air conditioning season. Operational inspections shall include as a minimum the inspections shown on Technical Exhibit 5.13-002. Operational inspections shall be performed at a frequency not less than that shown on TE 5.13-002. Contractor shall make all repairs and correct all deficiencies as identified in operational inspections. Contractor shall maintain records of inspections performed/repairs made and shall provide to contracting officer when requested.

5.13.4.2.2 Perform yearly maintenance on each chiller, tower, and pump as shown on TE 5.13-008.

5.13.4.2.3 Maintain the chillers to operate continuously throughout the air conditioning season. Make repairs as identified through operational inspections. Respond to chiller, tower, or pump failures, or shut down within one hour of contractor discovering or government notification, and restart chillers. Respond means, be on the site addressing why the chiller failed or shut down. Contractor may utilize a contractor provided telemeter system to provide notice of chiller failure. Notification may occur day or night (24 hours/day). The Contractor shall work continuously and diligently until equipment is operational.

5.13.4.4 The Contractor shall furnish the Contracting officer telephone numbers and names of qualified maintenance personnel who can be contacted to respond to emergency after normal working hours. (TE 5.13-004)

5.13.5 Maintain all HVAC equipment to operate continuously year round. Make repairs as identified through operational inspections. Respond to equipment failures, or shut down within one hour of contractor discovering or government notification, and restart HVAC equipment. Respond means be on the site addressing why the HVAC failed or shut down. Notification may occur day or night (24 hours/day). The Contractor shall work continuously and diligently until equipment is operational.

5.13.6 Operate and maintain the equipment at the cold storage facility located at building 100 (TE 5.13-003). Equipment shall be checked for proper operation on a daily basis**.**

a. Check compressors, pressure controls (high and low side), refrigerant leaks, oil leaks, oil level, and crank case heater.

b. Check evaporative condensers, spray water pumps, make-up float and valve assembly, discharge water eliminators, belts, fans, liquid receivers, discharge ducts, and roof penetrations of ductwork.

c. Check contacts, disconnects, starters, heaters, and electrical connections.

d. Check unit and product coolers; defrost cycle, motors, fans, drain pan and lines, heat tape, controls, thermostats, and calibrate controls.

e. Check doors and door frames and heaters in both, latches, and seals for securing and sealing rooms.

f. Check equipment base and supports, vibration isolators, frames of equipment, and fan guards.

g. Check air purifier, ductwork, and diffusers in ventilated storage area. Change air filters (new air filters will be dated and have technician's initials) annotated on the new air filters.

h. Check filter dryers, oil separators, gauges, strainers, and valves, including thermal expansion and solenoid valves.

i. Check and calibrate temperature charts.

j. Check drive couplings and alignment.

5.13.6.1 Correct all deficiencies found. If the government discovers failing or inoperative equipment, then the government will notify the contractor. The contractor shall respond within 1 hour of notification. Respond means that the contractor shall be on-site performing corrective action on the failed equipment. Contractor shall work continuously and diligently until all equipment is in proper operational condition.

5.13.7 Operate chillers and auxiliary equipment in the most energy efficient manner possible. Requirements include such items as:

a. When multiple chillers provide chilled water to a common distribution header, utilize the most energy efficient chiller(s) available.

b. Keeping strainers clean.

c. Keeping variable frequency drives in optimum operation.

d. Maintaining chilled water at highest temperature possible while still meeting space- conditioning requirements.

e. Keeping belts at proper tension.

f. Optimize condenser and evaporator flow and temperature.

g. Maintaining good water treatment program.

h. Replacing mechanical room light bulbs with compact fluorescent as they burn out.

i. Annually cleaning all tubes for optimum chiller operation.

j. Provide continuous cooling tower water treatment program.

5.13.8 The contractor shall be qualified and regularly engaged in the cooling tower water treatment program and shall submit a list of qualifications to the Government. Storage of chemicals shall be in an environmentally safe manner. Provide a list of chemicals to be used along with estimated quantities to the Government prior to air conditioning start-up. The treatment program must be approved by the Operations and Maintenance Mechanical Engineer. The chemical information will be provided to our Environmental Division for information only.

5.13.9 Maintain all existing ancillary items within each plant to include, but not limited to: Exhaust fans, relief valves, refrigerant monitors and sensors, piping, including insulation, water closets, sinks urinals, lighting, packing on valves, valves, strainers, pipe couplings, float valves, controls, sensors/gauges, air compressors, grounds around plant (cut grass with approved 4283), pumps (sump, circulating water, condenser), flooring (keep clean and safe), switchgear, eyelavages, unit heaters, strainers, variable speed drives, signage (hearing protection, confined spaces, etc.), air compressors, air dryers, fire extinguishers (wall mounted hand-held type), automatic tube cleaners emergency lighting, and chart recorders.

5.13.10 Operate and maintain 150-ton cooling tower at building 96 to include, pumps, motors, belts, chemical treatment, etc. The tower is the heat rejection source for water source heat pumps, which the contractor will not operate or maintain.

5.13.11 Spot and mark all underground chilled water and condenser water lines as required by the government and other contractors for repair (or new) work.

5.13.12 UNSCHEDULED TASKS

5.13.12.1 All parts replaced under the provisions of this Functional Area will be OEM components. The contractor shall replace or repair the component part rather than the entire unit in cases where the integrity of the unit will not be compromised or placed in jeopardy. If a part costs more than two thousand five hundred dollars ($2500.00), The contractor shall inform the Contract Officer Technical Representative verbally, and confirmed in writing to the Contracting Officer within one calendar day, of the nomenclature, description, and price of component part exceeding the limit that are required to restore the system to serviceable operation (See TE 5.13-004).

5.13.12.2 Clean-Up: The contractor shall be responsible for the general cleaning of spaces that house the equipment. General cleaning shall include removal of debris, dust, and other matter from any spaces utilized or maintained by the contractor.

5.13.12.3 Hazardous Conditions: The contractor shall secure the equipment and immediately notify the COR verbally and confirm the notification in writing within one calendar day.

5.13.12.4 Defective Parts: All defective parts replaced by the contractor/subcontractor shall remain the property of the Government. The contractor shall tag all parts with the following information, and deliver the parts to a disposal area designated.

a. Building and equipment number.

b. Description of malfunction.

c. Date of removal.

5.13.12.5   Equipment

5.13.12.6 The Contractor is expected to maintain and operate any new Government furnished chillers and associated equipment per the manufacturer’s instructions and as specified.

5.13.13 Rental Equipment

5.13.13.1 Chiller/cooling towers that require extensive repair during cooling cycles shall be required as directed by the Government to be backed up by a rental chiller.

5.13.14 Applicable Documents: Documents applicable to this Functional Area are listed below.

Government Publications:

Department of the Army

http://www.usace.army.mil/publications/

AR 11-27	Army Energy Conservation Program
AR 200-1	Environmental Quality, Environmental Protection and Enhancement
AR 420-53	Refrigeration
AR 420-54	Air-Conditioning and Refrigeration
AR 420-55	Food Service and Related Equipment
TM 5-670	Refrigeration, Air-Conditioning, Mechanical Ventilation, and Evaporative Cooling
TM 5-671	Preventive Maintenance for Refrigeration, Air- Conditioning, Mechanical Ventilation, and Evaporative Cooling
TM 5-785	Engineering Weather Data
TM 5-810-3	Mechanical Refrigeration and Ventilation in Cold Storage Facilities

Non-Governmental Publications

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

http://www.ndt.net/news/1999/cssinfo.htm#1

MSS SP-25	(1993) Standard Marking System for Valves, Fittings, Flanges and Unions
MSS SP-58	(1993) Pipe Hangers and Supports - Materials, Design and Manufacture

MSS SP-67	(1995) Butterfly Valves
MSS SP-69	(1996) Pipe Hangers and Supports - Selection and Application
MSS SP-70	(1990) Cast Iron Gate Valves, Flanged and Threaded Ends
MSS SP-71	(1990) Cast Iron Swing Check Valves, Flanges and Threaded Ends
MSS SP-72	(1992) Ball Valves with Flanged or Butt-welding Ends for General Service
MSS SP-73	(1991) Brazing Joints for Copper and Copper Alloy Pressure Fittings
MSS SP-78	(1987; R 1992) Cast Iron Plug Valves, Flanged and Threaded Ends
MSS SP-80	(1987) Bronze Gate, Globe, Angle and Check Valves
MSS SP-85	(1994) Cast Iron Globe & Angle Valves, Flanged and Threaded Ends
MSS SP-110	(1992) Ball Valves Threaded, Socket Welding, Soldered, Grooved /Flared Ends

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

http://www.nfpa.org/index.asp?cookie%5Ftest=1

NFPA 54	(1992) National Fuel Gas Code
NFPA 70	(1996) National Electrical Code
NFPA 37	(1994) Installation and Use of Stationary Combustion Engines and Gas Turbines
NFPA 90A	(1996) Installation of Air Conditioning and Ventilating Systems
NFPA 214	(1996) Water-Cooling Towers
NFPA 90B	Warm Air Heating and Air-Conditioning

AMERICAN WELDING SOCIETY (AWS)

http://www.aws.org/

AWS A5.8	(1992) Filler Metals for Brazing and Braze Welding
AWS B2.2	(1991) Brazing Procedure and Performance Qualification

COOLING TOWER INSTITUTE (CTI)

http://www.cti.org/

CTI ACT 105	(1990; Supple) Acceptance Test Code for Water Cooling Towers
CTI Std-103	(1986; Rev July 1994) The Design of Cooling Towers with Redwood Lumber
CTI Std-111	(1986) Gear Speed Reducers
CTI Std-114	(1986; Rev July 1994) Douglas Fir Lumber Specifications
CTI Std-134	(1985) Plywood for Use in Cooling Towers
CTI Std-137	(1988; Rev July 1994) Fiberglass Pultruded Structural Products for Use in Cooling Towers
CTI WMS-112	(1986) Pressure Preservative Treatment of Lumber for Industrial Water-Cooling Towers

AIR CONDITIONING AND REFRIGERATION INSTITUTE

http://ts.nist.gov/Standards/Global/upload/sp903-entries1.pdf

ARI 700-88	Specifications for Fluorocarbon Refrigerants
ARI 740-91	Performance of Refrigerant Recovery, Recycling and/or Reclaim Equipment

AMERICAN SOCIETY OF HEATING, REFRIGERATION AND AIR CONDITIONING

ASHRAE Guideline 3-1990, Reducing Emissions of Fully Halogenated Chlorofluorocarbon (CFC) Refrigerants in Refrigeration and Air-Conditioning Equipment and Application

ASHRAE 15-1994, Safety Code for Mechanical Refrigeration

HVAC Systems - Testing, Adjusting, and Balancing Installation Standards for Heating, Air Conditioning, and Solar Systems

AMERICAN NATIONAL STANDARDS INSTITUTE

http://publicaa.ansi.org/default.aspx

ANSI B16	Pipe, Flanges, and Fittings
ANSI B31.1	Power Piping
ANSI B36	Iron and Steel Pipe

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PUBLICATIONS AND FORMS

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PUBLICATIONS AND FORMS

6.1 GENERAL INFORMATION

Publications and forms applicable to this Contract are contained herein. The Contractor shall obtain all publications and forms required to perform the requirements of this Contract. The Contractor shall establish and update, as required, a file of all required publications listed in this Subsection as well as those cited in each Functional Area. Supplements or amendments to listed publications from any organizational level may be issued during the life of the Contract.

6.1.1 PROVISIONS

The Contractor shall ensure that all publications are posted and up-to-date. The Contractor shall accomplish the tasks set forth in this Contract in accordance with the references listed and in accordance with the following guidelines:

- If there is a conflict between Section C and the cited references, Section C shall control.
- If there is a conflict between or among two or more such references between or among those similarly coded, those issued by a higher authority shall control over those issued by a lower authority; and between or among those issued at the same level of authority, those with a later date of issue shall control over those with earlier dates of issue.
- Any task set forth in any such reference which calls for the exercise of discretionary Government authority that cannot be delegated shall be subject to the final approval of the Government official having such authority.
- All publications and forms will be current issue.
- Materiel Maintenance References are too numerous to list. The Contractor shall effect maintenance functions, repairs and services according to the appropriate level Technical Manual (TM), Technical Bulletin (TB), Lubrication Order (LO), or commercial equivalent.
- Text and copies of the publications, standards, and other referenced materials are available on various WWW sites. The following are provided as suggestions only and not intended as a complete list or the only acceptable sources:

1. www.benning2.army.mil\pubs
2. web7.whs.osd.mil

3. www.logsa.army.mil/pubs.htm

4. www.usapa.army.mil

5. www.tradoc.army.mil/publica.htm

6. www.cml.lib.oh.us/html/industry_standards.html

7. Department, Command, Agency, Organization Home Pages Army: http://www.army.mil

8. Army Forces Command (FORSCOM): http://www.forscom.army.mil

9. Army Materiel Command (AMC): http://www.amc.army.mil

10. Army Medical Department (AMEDD): http://www.armymedicine.army.mil/armymed/default2.htm

11. Army National Guard (ARNG): http://www-ngb5.ngb.army.mil

12. Army Reserve Personnel Command (AR-PERSCOM): http://www.army.mil/usar/ar-perscom/arpercom.htm

13. Defense Finance and Accounting Service (DFAS) Indianapolis: http://www.asafm.army.mil/DFAS

14. Department of the Army (DA): http://www.hqda.army.mil

15. Defense Information Systems Agency (DISA): http://www.disa.mil/disahomejs.html

16. Department of Defense (DOD): http://www.defenselink.mil

17. General Services Administration (GSA): http://www.gsa.gov

18. Logistics Support Activity (LOGSA) Redstone Arsenal, AL: http://www.logsa.army.mil/intro.htm

19. National Guard: http://www.ngb.dtic.mil

20. Occupational Safety and Health Administration (OSHA): http://www.osha.gov

21. Office of the Chief, Army Reserve (OCAR): http://www.army.mil/usar/ocar.htm

22. Office of Personnel Management (OPM): http://www.opm.gov

23. Reserve Affairs (Office of Assistant Secretary of Defense): http://raweb.osd.mil

24. Training and Doctrine Command (TRADOC): http://www-tradoc.monroe.army.mil

25. U.S. Army Publications Agency (USAPA): Http://www.usapa.army.mil

26. U.S. Army Reserve (USAR): http://www.army.mil/usar

27. U.S. Army Reserve Command (USARC) INTERNET web site: http://www.usarc.army.mil

28. U.S. Army Reserve Command (USARC) INTRANET web site: http://usarcintra (For authorized USAR users; no general public access.) Forms, Publications, Magazines, etc.

29. Army Corps of Engineers: http://www.usace.army.mil/usace-docs (Army engineering publications.)

30. Army Doctrine and Training Digital Library: http://www.adtdl.army.mil (Information on Army schools and Army documents.)

31. Army Reserve Magazine: http://www.army.mil/usar/armag/armag.htm

32. DOD electronic forms: http://web1.whs.osd.mil/icdhome/ddeforms.htm (Contains some forms not included on USAPA web site.)

33. FEDmanager: http://www.fedmanager.com

34. FORMDEPS (FORSCOM Regs 500-3-1 and 500-3-3: http://freddie.forscom.army.mil/mob

35. FORSCOM electronic pubs and forms: http://www.forscom.army.mil/pubs

36. GSA electronic forms: http://www.gsa.gov/forms

37. IRS forms and publications: http://www.irs.ustreas.gov/prod/forms_pubs/index.html (Includes link to State Tax forms.)

38. LOGSA pubs and forms: http://www.logsa.army.mil/pubs.htm(Supply catalogs, technical manuals, PS Magazine, and more.)

39. Military periodicals: http://www.dtic.mil/search97doc/aulimp/main.htm (Index to Military Periodicals.)

40. Optional Forms (OFs): http://web1.whs.osd.mil/icdhome/ofeforms.htm (Contains some forms not included on USAPA web site.)

41. Standard Forms (SFs): http://web1.whs.osd.mil/icdhome/sfeforms.htm (Contains some forms not included on USAPA web site.)

42. TRADOC pubs: http://www-tradoc.monroe.army.mil/publica.htm

43. USAPA electronic pubs and forms: http://www.usapa.army.mil (ARs, Pams, Cirs, OFs, SFs, DD, and DA forms; Pubs Ordering System)

44. USARC form files on INTERNET FTP server: ftp://www.usarc.army.mil (Access to USARC form files; download individual files or *.zip file from USARCForms" directory.)

45. USARC forms and pubs on INTRANET: http://usarcintra/hqs/im/ima/imap/pubsform/pubforms.htm (For authorized USAR users; no general public access.)

46. Pay and Finance Army Financial Operations: http://www.asafm.army.mil/financial.htm (Pay rates, drill pay, travel voucher information.)

47. DFAS: http://www.asafm.army.mil/DFAS (Defense Finance and Accounting Service Indianapolis.)

48. OCAR Pay Support Center: http://www.army.mil/usar/psc/ocarhp.htm (Links to important USAR pay information.)

49. Per diem rates: http://www.dtic.mil/perdiem/pdrates.html Education, Schools, and Training (other than Medical)

50. Army Doctrine and Training Digital Library: http://www.adtdl.army.mil (Information on Army schools and Army documents.)

51. Command and General Staff College: http://www-cgsc.army.mil

52. Combined Arms and Services Staff School (CAS3): http://www-cgsc.army.mil/cas3

53. TRADOC: http://www-tradoc.monroe.army.mil

6.2 PUBLICATIONS AND FORMS

Publication	Title	Date
	American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) Standards	
	American Wood Preservers Institute Standards	
	DFSC Contract Bulletin for Region VI, Vol. II	
	Fort Benning Hazardous Waste Management Plan	
	Fort Benning Integrated Natural Resources Management Plan	
	Fort Benning Severe Weather Emergency Action Plan (SWEAP)	
	Fort Benning Supply Division Micronic Filter, Differential Pressure - Fuel Servicing Equipment (Daily Log)	
	International Maritime Dangerous Goods (IMDG) Code	
	Joint Commission on Accreditation of Healthcare Organizations (JCAHO) Standards	
	Manual of Uniform Traffic Control Devices for Streets and Highways	
	National Arborist Association Standards for Pruning of Shades Trees, Guying of Shade Trees, Fertilizing Shade and Ornamental Trees, Lightning Protection, Installation Systems for Shade Trees, Standards for Hydraulic Sprayer Calibration	1988
	National Institute of Standards and Technology (NIST) codes	latest ed.
	Installation Design Guide, Fort Benning, GA	latest ed.
	Historic Building Maintenance and Repair Plan	latest ed.
	SAMS-I/TDA Supply Operations Manual	
	STAMIS Manuals	
	Fort Benning Historic Resources Interior Survey	latest ed.
	Historic District Management Plan, Fort Benning, GA	latest ed.
	Standard Property Book System Manuals	

	Uniform Mechanical Code, Chapter 20	
	United Nations (UN) "Orange Book" - Transport of Dangerous Goods	
29 CFR	Labor	Current ed.
29 CFR 100-198	Department of Transportation Regulations	Current ed.
29 CFR 1910	Occupational Safety and Health Standards	Current ed.
29 CFR 1910.1200	OSHA Hazard Communication Standard	Current ed.
29 CFR 1926	Construction Standards	Current ed.
29 CFR 1960	OSHA Basic Program Elements for Federal Employees	Current ed.
40 CFR	Protection of Environment	Current ed.
40 CFR 60	Clean Air Act (CAA)	1990
40 CFR 261	Identification and Listing of Hazardous Waste	Current ed.
40 CFR 279	Standards for the Management of Used Oil	01-Jul-98
49 CFR	Transportation Regulations	Current ed.
49 CFR 106-178	Department of Transportation Hazardous Materials Regulations	03-Jan-75
49 CFR 191-192	Transportation of Natural and Other Gas by Pipeline; Annual Reports, Incident Reports, and Safety Related Condition Reports / Minimum Federal Safety Standards	Current ed.
5 USC 6103(a)	Holidays	01-Jan-94
18 USC 1382	Entering Military, Naval, and Coast Guard Property	01-Jan-94
ACP 121	Communication Instructions-General (U)	current ed.
ACP 121 U.S. Suppl-1	Communication Instructions-General (U)	current ed.
ACP 131	Communication Instructions-Operating Signals	current ed.
ADSM 25-L-25/	SAMS-I/TDA User Manuals (T) Tech	10 Aug 96
1 and 2	SAMS-I/TDA User Manuals (F) End User	26 Feb 97
AFARS	Army FAR Supplement	Current ed.
AFR 75-25	Superseded By JFTR Vol. I, JTR Vol. II	

ANSI	American National Standards Institute Codes	
ANSI/IEEE Standard 142	Practice for Grounding of Industrial and Commercial Power Systems	
ANZI Z-60.1	American Standards for Nursery Stock	1986
AR 1-8	Smoking in DA Occupied Buildings and Facilities	
AR 5-9	Area Support Responsibilities	16-Oct-98
AR 11-2	Management Control	01-Aug-94
AR 11-9	The Army Radiation Safety Program	28-May-99
AR 11-27	Army Energy Program	03-Feb-97
AR 11-34	The Army Respiratory Protection Program	15-Feb-90
AR 15-6	Procedures for Investigating Officers and Boards of Officers	11-May-88
AR 25-1	The Army Information Resources Management Program	25-Mar-97
AR 25-30	Army Integrated Publications and Printing Program	28-Feb-89
AR 25-400-2	The Modern Army Recordkeeping System (MARKS)	26-Feb-93
AR 30-1	The Army Food Service Program	15-Aug-89
AR 37-1	Army Accounting and Fund Control	30-Apr-91
AR 37-100	Account/Code Structure	01-Jul-94
AR 40-3	Medical, Dental, and Veterinary Care	
AR 40-5	Preventive Medicine, Health, and Environment	15-Oct-90
AR 55-1	Conex/Milvan Equipment Control, Utilization, and Reporting	01-Apr-79
AR 55-4	CONUS Military Installation Materiel Outloading and Receiving Capability Report	15-Dec-84
AR 55-38	Reporting of Transportation Discrepancies in Shipments	31-Aug-92
AR 55-46	Travel Overseas	20-Jun-94
AR 55-71	Transportation of Personal Property and Related Services	01-Jun-83
AR 55-355	Defense Traffic Management Regulation	31-Jul-86
AR 58-1	Management, Acquisition, and Use of Administrative Use Motor Vehicles	15-Dec-79

AR 59-3	Air Transportation Movement of Cargo By Scheduled Military and Commercial Air Transportation-CONUS Outbound	01-Feb-81
AR 59-8	DOD Common User Airlift	20-Aug-82
AR 59-9	Special Assignment Airlift Mission Requirements	05-Mar-85
AR 71-13	The Department of the Army Equipment Authorization and Usage Program	03-Jun-88
AR 75-1	Malfunctions Involving Ammunition and Explosives	20-Aug-93
AR 105-10	Communications Economy and Discipline	01-Jul-77
AR 105-23	Administrative Policies and Procedures for Base Telecommunications Services	01-May-78
AR 105-24	Radio Frequency and Call Sign Assignments for U.S. Army Communications-Electronics Activities	28-May-77
AR 105-31	Record Communications	08-Aug-77
AR 105-32	Authorized Addresses for Electrically Transmitted Messages	15-Nov-84
AR 105-34	Reduction and Control of Telecommunications Traffic in an Emergency	16-Jul-73
AR 190-5	Motor Vehicle Traffic Supervision	08-Jul-88
AR 190-11	Physical Security of Arms, Ammunition, and Explosives	30-Sep-93
AR 190-13	The Army Physical Security Program	30-Sep-93
AR 190-22	Searches, Seizures, and Disposition of Property	07-Aug-87
AR 190-40	Serious Accident Report	30-Nov-93
AR 190-51	Security of Unclassified Army Property (Sensitive and Nonsensitive)	30-Sep-93
AR 200-1	Environmental Protection and Enhancement	21-Feb-97
AR 200-2	Environmental Effects of Army Actions	23-Dec-88
AR 200-3	Natural Resources: Land, Forest, and Wildlife Management	28-Feb-95
AR 210-12	Establishment of Rental Rates for Quarters Furnished Federal Employees	08-Feb-77
AR 210-20	Master Planning for Army Installations	30-Jul-93
AR 210-190	Post Cemeteries	15-Aug-82

AR 220-10	Preparation for Overseas Movement of Units (POM)	15-Jun-73
AR 210-190	Installation Post Cemeteries	
AR 310-25	Dictionary of U.S. Army Terms	
AR 310-34	The Department of the Army Equipment Authorization and Usage Program	12-Nov-86
AR 310-50	Authorized Abbreviations and Brevity Codes	15-Nov-85
AR 335-15	Management Control Information System	28 OCT 96
AR 340-3	Official Mail Cost Control Program	15-May-84
AR 380-5	Department of the Army Information Security Program	25-Feb-88
AR 380-19	Information Systems Security	01-Aug-90
AR 380-20	Restricted Areas	15-Mar-82
AR 380-40	(O) Policy for Safeguarding and Controlling Communications Security (COMSEC) Material (U)	01-Sep-94
AR 380-49	Industrial Security Program	15-Apr-82
AR 380-67	The Department of the Army Personnel Security Program	09-Sep-88
AR 385-10	The Army Safety Program	23-May-88
AR 385-11	Ionizing Radiation Protection (Licensing, Control, Transportation, Disposal, and Radiation Safety)	01-May-80
AR 385-40	Accident Reporting and Records	01-Nov-94
AR 385-55	Prevention of Motor Vehicle Accidents	12-Mar-87
AR 385-61	The Army Chemical Agents Safety Program	27-Feb-97
AR 385-64	Ammunition and Explosives Safety Standards	22-May-87
AR 415-28	Real Property Category Codes	10-Oct-96
AR 420-9	Energy Efficiency Program	25-Jun-75
AR 420-10	Management of Installation Directorates of Public Works	15-Apr-97
AR 420-17	Real Property and Resource Management	13-Dec-76
AR 420-18	Facilities Engineering Materials, Equipment, and Relocatable Building Management	03-Jan-92
AR 420-22	Preventive Maintenance and Self-Help Programs	06-Jul-76
AR 420-43	Electrical Services	27-Nov-87
AR 420-49	Utility Services	28-Apr-97
AR 420-54	Air-Conditioning and Refrigeration	05-Nov-90

AR 420-70	Building and Structures	01-Oct-97
AR 420-72	Surfaced Areas, Bridges, Railroad Track, and Associated Appurtenances	28-Mar-91
AR 420-90	Fire Protection	25-Sep-92
AR 420-90 TRADOC Suppl 1	Fire Prevention and Protection	Jun-77
AR 530-1	Operations Security (OPSEC)	03-Mar-95
AR 600-8-101	Personnel Processing (In and Out and Mobilization Processing)	12-Dec-89
AR 600-50	Standards of Conduct for Department of the Army Personnel	28-Jan-88
AR 600-55	The Army Driver and Operator Standardization Program (Selection, Training, Testing, and Licensing)	31-Dec-93
AR 611-5	Army Personnel Selection and Classification Testing	15-Nov-82
AR 700-43	Management of Defense-Owned Industrial Plant Equipment (IPE)	19-Nov-73
AR 700-53	Support of Railroad Equipment	15-Jul-81
AR 700-58	Packaging Improvement Report	31-Oct-74
AR 700-64	Radioactive Commodities in the DOD Supply Systems	19-Apr-85
AR 700-84	Issue and Sale of Personal Clothing	15-May-83
AR 700-88	Commercial Design Vehicles FSC Class 2300	22-Jun-72
AR 700-131	Loan and Lease of Army Materiel	01-Sep-96
AR 700-139	Army Warranty Program Concepts and Policies	10-Mar-86
AR 700-141	Hazardous Material Information System	20-Jan-87
AR 708-1	Cataloging of Supplies and Equipment Cataloging and Supply Management Data	20-Jul-94
AR 710-1	Centralized Inventory Management of the Army Supply System	01-Feb-88
AR 710-2	Inventory Management Supply Policy Below the Wholesale Level	31-Jan-92
AR 710-2-10-1	Using Unit Supply System	
AR 710-3	Asset and Transaction Reporting System	15-May-92

AR 725-50	Requisition, Receipt, and Issue System	15-Nov-95
AR 735-5	Policies and Procedures for Property Accountability	28-Feb-94
AR 730-58	Painting, Camouflage Painting and Marking of Army Material	
AR 735-11-2	Reporting of Item and Packaging Discrepancies	06-Dec-91
AR 740-1	Storage and Supply Activity Operations	23-Apr-71
AR 746-1	Packaging of Army Materiel for Shipment and Storage	08-Oct-85
AR 750-1	Army Materiel Maintenance Policy and Retail Maintenance Operations	01-Aug-94
AR 750-25	Army Test, Measurement, and Diagnostic Equipment (TMDE) Calibration and Repair Support Program	01-Sep-83
AR 750-43	Army Test, Measurement, and Diagnostic Equipment Program	29-Sep-89
AR 840-10	Flags, Guidons, Streamers, Tabards, and Automobile and Aircraft Plates	29-Oct-90
ASTM D-2513	American Society of Testing and Materials D-2513	
AWWA	American Water Works Association Standards	Current ed.
AWWA C 651-92	American Water Works Association Standards	Current ed.
AWWA C 652-92	American Water Works Association Standards	Current ed.
CTA 50-900	Clothing and Individual Equipment	01-Sep-94
CTA 50-909	Field and Garrison Furnishings and Equipment	01-Aug-93
DA Form 11-2	Management Control Evaluation Certification	Jul 94
DA Form 17	Requisition of Publications and Blank Forms	01-Oct-79
DA Form 17-1	Requisition of Publications and Blank Forms (continuation sheet)	01-Oct-79
DA Form 137	Installation Clearance Record	Dec-92
DA Form 348	Equipment Operator’s Qualification Record	10-Oct-64
DA Form 444	Inventory Adjustment Report (IAR)	Jan 82
DA Form 461-5	Vehicle Classification Inspection	Mar 89
DA Form 581	Request for Issue or Turn-In of Ammunition	Aug 89
DA Form 1687	Notice of Delegation of Authority - Receipt for Supplies	Jan 82
DA Form 1818	Individual Property Pass	Sep 82

DA Form 1854-R	Daily Transfer Summary	Aug 89
DA Form 1898	AVFUELS INTO-PLANE SALES SLIP	
DA Form 2028	Recommended Changes to Publications and Blank Forms	01-Feb-73
DA Form 2064	Document Register for Supply Actions	Jan 82
DA Form 2401	Organization Control Record for Equipment	01-Apr-62
DA Form 2404	Equipment Inspection and Maintenance Worksheet	01-Apr-79
DA Form 2405	Maintenance Request Register	01-Apr-62
DA Form 2406	Materiel Condition Status Report	Apr 93
DA Form 2407	Maintenance Request	Jul 94
DA Form 2407-1	Maintenance Request Continuation Sheet	Jul 94
DA Form 2408-14	Uncorrected Fault Record	Jun 94
DA Form 2408-9	Equipment Control Record	01-Oct-72
DA Form 2702	Bill of Materials	Jul 63
DA Form 2765	Request for Issue or Turn-in	Apr 76
DA Form 2765-1	Request for Issue or Turn-In	Apr 76
DA Form 2940-R	Unit Loading Inventory and Checklist (Worksheet)	Dec 75
DA Form 3020-R	Magazine Data Card	Aug 89
DA Form 3078	Personal Clothing Request	May 93
DA Form 3151-R	Ammunition Stores Slip	Apr 76
DA Form 3161	Request for Issue or Turn-in	May 83
DA Form 3329	Installation Property Record	Aug 78
DA Form 3590	Request for Disposition or Waiver	01-Jul-75
DA Form 3643	Daily Issues of Petroleum Products	Apr 85
DA Form 3644	Monthly Abstract of Issues of Petroleum Products and Operating Supplies	Apr 85
DA Form 3645	Organizational Clothing and Individual Equipment Record	Oct 91
DA Form 3645-1	Additional Organizational Clothing and Individual Equipment Record	Dec 83
DA Form 3782	Suspended Notice	Sep 71
DA Form 3853-1	Innage Gauge Sheet	
DA Form 3938	Local Service Request	Dec 77
DA Form 3941	Certificate of Proficiency	01-Oct-72
DA Form 3953	Purchase Request and Commitment	Mar 91

DA Form 3967	Facilities Engineering Operating Log (Boiler Plant)	Nov 72
DA Form 4013	Locator and Bin Card	01-Mar-73
DA Form 4283	Facilities Engineering Work Request - XFA, XFB, XFC	Aug 78
DA Form 4284	Facilities Engineering Work Order	Aug 78
DA Form 4287	Service Order	Jan 86
DA Form 4508	Ammunition Transfer Record	May 76
DA Form 4600	Travelope	May 94
DA Form 4702-R	Monthly Bulk Petroleum Accounting Summary	
DA Form 4949	Administrative Adjustment Report (AAR)	Jan 82
DA Form 5203	Dodic Master/Lot Locator Record	May 83
DA Form 5479-R	Contract Discrepancy Report	
DA Form 5504	Maintenance Request	Jun 93
DA Form 5504-1	Maintenance Request (Continuation Sheet)	Sep 88
DA Form 5811-R	Certificate-Lost or Damaged Class 5 Ammunition Items	Aug 89
DA Form 7531	Checklist and Reporting Document for Financial Liability of Property Loss	
DA Message 201502Z	Marking of Unserviceable Clothing and Individual Equipment	Mar 87
DA PAM 25-30	Consolidated Index of Army Publications and Blank Forms	01-Oct-96
DA PAM 25-32	Foreign Military Sales Publication Guide	14-Aug-87
DA PAM 25-40	Army Publishing, Action Officer's Guide	30 APR 02
DA PAM 25-380-2	(O) Security Procedures for Controlled Cryptographic Items	10-Jan-91
DA PAM 290-5	Administration, Operation, and Maintenance of Army Cemeteries	01-May-91
DA PAM 310-10	The Standard Army Publications System	1-Oct-82
DA PAM 600-8-101	Personnel Processing (In-, Out, Soldier Readiness, Mobilization And Deployment Processing	28-May-03
DA PAM 700-30	Logistic Control Activity (LCA) Information and Procedures	17-Jul-90
DA PAM 710-2-1	Using Unit Supply Systems (Manual Procedures)	01-Jan-82
DA PAM 710-2-2	Supply Support Activity Supply System: Manual Procedures	01-Mar-84
DA PAM 710-4	Management of Excess Materiel and Materiel Returns	13-Feb-87

DA PAM 738-750	Functional Users Manual for the Army Maintenance Management System (TAMMS)	01-Aug-94
DD Form 139	Pay Adjustment Authorization	May 53
DD Form 200	Financial Liability of Property Loss	
DD Form 250	Materiel Inspection and Receiving Report	Nov 92
DD Form 254	Department of Defense Contract Security Classification Specification	Dec 90
DD Form 314	Preventive Maintenance Schedule and Record	01-Dec-53
DD Form 362	Statement of Charges/Cash Collection Voucher	Jul 93
DD Form 441	DOD Security Agreement	01-Apr-97
DD Form 441-1	Appendage to DOD Security Agreement	01-Apr-97
DD Form 460	Provisional Pass	01-Mar-51
DD Form 577	Signature Card	01-May-88
DD Form 619	Statement of Accessorial Services Performed	01-Jul-74
DD Form 619-1	Statement of Accessorial Services Performed (SIT Delivery and Reweigh)	01-Jul-74
DD Form 805	Storage Management Report	Dec 93
DD Form 843	Requisition for Printing and Binding Service	01-Jul-55
DD Form 1070	Termite and Wood Decay Inspection	01-Feb-58
DD Form 1086	Export Traffic Release Requests - Part I (Basic Request)	01-May-76
DD Form 1131	Cash Collection Voucher	Apr 57
DD Form 1149	Requisition and Invoice Shipping Document	Dec 93
DD Form 1155	Order for Supplies or Services	01-Jun-94
DD Form 1164	Service Order for Personal Property	Dec 85
DD Form 1299	Application for Shipment and Storage of Personal Property	01-Dec-85
DD Form 1341	Report of Commercial Air Carrier Passenger Service	01-Feb-90
DD Form 1348-1	DOD Single Line Item Release/Receipt Document	Jul 91
DD Form 1348-2	DOD Issue Release/Receipt Document With Address Label	Feb 89
DD Form 1348-6	DOD Single Line Item Requisition System Document (Manual Long-Form)	Feb 85
DD Form 1348M	DOD Single Line Item Requisition System Document (Mechanical)	Mar 79
DD Form 1351-2	Travel Voucher or Subvoucher	Oct 91

DD Form 1556	Request, Authorization, Agreement, Certification of Training and Reimbursement	Mar 87
DD Form 1577-2	Unserviceable (Repairable) Tag - Materiel	01-Oct-66
DD Form 1662	DOD Property in the Custody of Contractors	01-Jan-97
DD Form 1701	Inventory of Household Goods	Jun 74
DD Form 1797	Personal Property Counseling Checklist	01-Jul-82
DD Form 1800	Mobile Home Inspection Record	Feb 83
DD Form 1814	Carrier-Warning-Suspension-Reinstatement-Cancellation of Warning	01-Sep-74
DD Form 1857	Temporary Commercial Storage at Government Expense	Mar 89
DA Form 1898	AVFUELS Into Plane Sales Slip	
DD Form 2278	Application for Do-It-Yourself Move (DITY) and Counseling Checklist	01-Dec-87
DFARS	Defense FAR Supplement	Current ed.
DFSCH 4280.1	Defense Fuel Supply Center Handbook	Sep-97
DIC TK4	In Transit Data Card	
DOD 4140.25-M	DOD Management of Bulk Petroleum Products, Gas and Coal Volumes I through IV	Jun 94
DOD 4145.19-R	Storage and Warehousing Facilities and Services	15-Jun-78
DOD 4160.19-M	Defense ADPE Reutilization Manual	
DOD 4160.21-M	Defense Reutilization and Marketing Manual	Mar 90
DOD 4500.32-R Vol. I and II	Military Standard Transportation and Movement Procedures (MILSTAMP)	15-Mar-87
DOD 4500.34-R	Personal Property Traffic Management Regulation (PPTMR)	01-Oct-91
DOD 4500.9-R (DTR) Part I	Department of Defense Transportation Regulation, Part I, Passenger Movement	04-Aug-95
DOD 4500.9-R (DTR) Part II	Department of Defense Transportation Regulation, Part II, Cargo Movement	04-Aug-95
DOD 4500.9-R (DTR) Part III	Department of Defense Transportation Regulation, Part III, Mobility	04-Aug-95
DOD 4500.9R (DTR) Part V	Department of Defense Transportation Regulation, Part V, Movement of Personal Property	04-Aug-95
DOD 4525-8-M	DOD Official Mail Manual	Jul 87

DOD 5100.76-M	Physical Security of Sensitive Conventional Arms, Ammunition, and Explosives	03-Feb-83
DOD 5200.1-R	DOD Information Security Program Regulation	14-Jan-97
DOD 5220.22-M	National Industrial Security Program for Operating Manual	Jan 95
DOD 5220.22-R	Industrial Security Regulation	04-Dec-85
DOD 5500.7-R	Standards of Conduct	Aug 93
EM-385-1-1	U.S. Army Corps of Engineers Safety and Health Requirements	03-Sep-96
EO 11643	Environmental Safeguards	08-Feb-72
EO 12088	Federal Compliance with Pollution Control Standards	13-Oct-78
FAR	Federal Acquisition Regulation	Current ed.
FM 1-55	Guide for Operation of Army Airfields	
FM 5-446	Military Nonstandard Fixed Bridging	03-Jun-91
FM 5-742	Concrete and Masonry	14-Mar-85
FM 9-13	Ammunition Handbook: A Guide for Ammunition Specialists	04-Nov-86
FM 9-38	Conventional Ammunition Unit Operations	02-Jul-93
FM 10-18	Petroleum Terminal and Pipeline Operations	03-Dec-86
FM 10-20	Organizational Maintenance - Military Petroleum, Pipelines, Tanks, and Related Equipment	
FM 10-68	Aircraft Refueling	
FM 10-69	Petroleum Supply Point Equipment and Operations	22-Oct-86
FM 10-70	Inspecting and Testing Petroleum Products	
FM 10-71	Petroleum Tank Vehicle Operations	12-May-78
FM 19-30	Physical Security	01-Mar-79
FM 43-5	Unit Maintenance Operations	25-Sep-88
FM 55-30	Army Motor Transport Units and Operations	27-Jun-97
FORSCOM Reg. 525-2	Emergency Deployment Readiness Exercise (EDRE)	01-Nov-89
FORSCOM Reg. 700-3	Ammunition Basic Load	31-Mar-93
FORSCOM Reg. 700-4	Ammunition	01-Oct-90

GSA Form 494	Monthly Motor Vehicle Usage Report	Sep 96
GSA Form 3478	Motor Vehicle Service Authorization	
ISO 9002	Quality Systems-Model for Quality Control in Production, Installation, and Servicing	
Local Form 11	Request for Shipment	
MIL-A-8243	Anti-Icing and De-Icing - Defrosting Fluid	Nov 80
MIL-HDBK-129	DOD Handbook Military Marking	15-May-97
MIL-STD-129N	DOD Standard Practice for Military Marking	15-May-97
MTMC Form 363-R	DOD Standard Tender of Freight Services	
NAVSUP Pub 490	Transportation of Personal Property	30-Aug-90
NEC	National Electrical Code	1998
NFPA Codes	National Fire Protection Association Codes	Current ed.
NFPA 780	Lightning Protection Installation Systems	1992
NFPA 70	National Electrical Code	Current ed.
NFPA 72	National Fire Alarm Code	1996 ed.
NFPA 96	Installation of Equipment for the Removal of Smoke and Grease Laden Vapors from Commercial Cooking Equipment.	1991
NFPA 101	Life Safety Code	1997 ed.
OF 346	U.S. Government Motor Vehicle Operator's Identification Card	Nov 85
PL 91-190	National Environmental Policy Act (NEPA)	13-Sep-82
PL 93-205	Endangered Species Act	1973
PL 94-469	Toxic Substances Control Act (TSCA)	01-Jan-77
PL 94-580	Resource Conservation and Recovery Act (RCRA)	1976
PPP-B-580	Federal Specification	
PTWB 420-43-1	Uninteruptible Power Supply (UPS) System Design and Operation/Maintenance	01-May-93
QMFL Form 307	Equipment Justification and Approval Form	
QMFL Form 580	Request for Transportation	
QQ-S-781	Federal Specification for Steel, Strapping, Flat	

RCS-CSGLD-1732	CCI and Small Arms Serial Number Registration and Reporting	
SARA Title III	Emergency Planning and Community Right-To-Know Act (EPCRA)	1986
SB 700-20	Army Adopted/Other Items Selected for Authorization/List of Reportable Items	01-Mar-95
SB 708-3	Department of Defense Ammunition Code	01-Jan-88
SB 708-3400-2 through 708-7440-1	Industrial Plant Equipment Handbooks	Various
SB 742-1	Ammunition Surveillance Procedures	12-Nov-90
SF 91	Motor Vehicle Accident Report	Feb 93
SF 215	Deposit Ticket	
SF 361	Transportation Discrepancy Report	Mar 84
SF 364	Report of Discrepancy (ROD)	Feb 80
SF 368	Product Quality Deficiency Report	Oct 85
SF 700	Security Container Information	Aug 85
SF 1034 & 1034A	Public Vouchers for Purchases and Services Other Than Personal	Oct 87
SF 1103	U.S. Government Bill of Lading-Original	Apr 85
SF 1109	U.S. Government Bill of Lading-Original Continuation Sheet	Apr 85
SF 1200	Government Bill of Lading (GBL) Correction Notice	Aug 82
ST9-170	Ammunition Handler	
TB 9-1300-385	Munitions Restricted or Suspended	01-Apr-97
TB 9-2300-247-40	Tactical Wheeled Vehicles: Repair of Frames	04-Dec-90
TB 9-2300-422-20	Security of Tactical Wheeled Vehicles	27-Aug-88
TB 43-0116	Identification of Radioactive Items in the Army	01-Aug-93
TB 43-0142	Safety Inspection and Testing of Lifting Devices	28-Feb-97
TB 43-0151	Inspection and Test of Air and Other Gas Compressors	17-Mar-89
TB 43-0180	Calibration and Repair Requirements for the Maintenance of Army Materiel	30-Apr-96
TB 43-0210	Nonaeronautical Equipment Army Oil Analysis Program (AOAP)	10-Dec-84

TB 750-25	Maintenance Of Supplies And Equipment: Army Test, Measurement, And Diagnostic Equipment (TMDE) Calibration And Repair Support (C&RS) Program	01-Mar-97
TB 750-97-74	Maintenance Expenditure Limits for FSC Group 74; FSC Classes 7420, 7430; and 7490	13-Jan-72
TB 750-103	Maintenance and Care of Common Type Machine Tools and Shop Equipment Before, During, and After Operation	19-Oct-65
TB ENG 405	Repairs and Utilities Standards	10-Aug-65
TIM 11	Hydrogen Phosphide Fumigation of Subsistence With Aluminum Phosphide	01-Feb-87
TM 3-6665-312-30	Maintenance Manual for M8A1 Automatic Chemical Agent Alarm	17-Mar-85
TM 5-551K	Plumbing and Pipefitting	29-Jul-71
TM 5-609	Military Custodial Services Manual	25-Sep-69
TM 5-610	Preventive Maintenance for Facilities Engineering, Buildings and Structures	01-Nov-79
TM 5-617	Facilities Engineering: Maintenance and Repair of Roofs	30-Jan-74
TM 5-624	Maintenance and Repair of Surface Areas	27-Nov-95
TM 5-626	Unsurfaced Road Maintenance Management	16-Jan-95
TM 5-627	Maintenance of Trackage	01-Jan-80
TM 5-628	Railroad Track Standards	08-Apr-91
TM 5-630	Natural Resources - Land Management	01-Jul-82
TM 5-650	Repairs and Utilities: Central Boiler Plants	13-Oct-89
TM 5-652	Steam, Hot Water, and Gas Distribution Systems: Repairs and Utilities	11-Apr-47
TM 5-653	Steam, Hot Water, and Gas Distribution Systems: Inspection and Preventive Maintenance Service	01-Jun-46
TM 5-654	Maintenance and Operation of Gas Systems	03-Nov-70
TM 5-665	Operation and Maintenance of Domestic and Industrial Waste Water Systems	01-Jan-82

TM 5-682	Facilities Engineering - Electrical Facilities Safety	01-Jun-83
TM 5-683	Facilities Engineering - Electrical Interior Facilities	15-Dec-95
TM 5-684	Facilities Engineering - Electrical Exterior Facilities	20-Nov-96
TM 5-685	Operation, Maintenance, and Repair of Auxiliary Generators	26-Aug-96
TM 5-805-14	Roofing and Waterproofing	01-May-93
TM 5-811-3	Electrical Design: Lightning and Static Electricity Protection	29-Mar-85
TM 5-818-4	Backfill for Subsurface Structures	01-Jun-83
TM 5-822-7	Standard Practice for Concrete Pavements	16-Aug-87
TM 5-823-4	Marking of Army Airfield - Heliport Operational and Maintenance Facilities	07-Jul-87
TM 9-1300-200	Ammunition, General	03-Oct-69
TM 9-1300-206	Ammunition and Explosives Standards	30-Aug-73
TM 9-6140-200-14	Operator's, Unit, Direct Support, and General Support Maintenance Manual for Lead-Acid Storage Batteries	13-Jul-89
TM 10/20 series		
TM 10-1101	Petroleum Handling Equipment and Operation	03-May-72
TM 38-230-1	Packaging of Materiel: Preservation (Volume I)	01-Aug-82
TM 38-230-2	Packaging of Materiel: Packing (Volume II)	15-Jun-77
TM 38-250	Preparing Hazardous Materials for Military Air Shipments	25-Nov-94
TM 38-410	Storage and Handling of Hazardous Materials	29-May-92
TM 38-600	Management of Administrative Use Motor Vehicles	13-Sep-79
TM 38-750	The Army Maintenance Management System (TAMMS)	31-May-81
TM 38-750-1	The Army Maintenance Management System (TAMMS) Field Command Procedures	29-Dec-78
TM 38-L03-19	Standard Army Intermediate Level Supply System (SAILS)	01-Jan-85
TM 38-L03-20	Standard Army Intermediate Level Supply (SAILS) Subsystem: Codes	01-Jul-86

TM 743-200-1	Storage and Materials Handling	15-Jan-58
TM 743-200-2	Storage Modernization	24-Nov-67
TM 743-200-3	Storage and Materials Handling	01-Jul-71
TN 420-43-2	How to Evaluate Power Quality for Automated Data Processing (ADP) System	30-Jun-91
TN 420-43-3	How to Protect Automated Data Processing (ADP) and Communication System Against Lightning Strikes	30-Aug-91
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TN 420-43-5	How to Provide a Proper Ground for Electronic Equipment	02-Jan-92
TN 420-43-6	Recommended Ground for Automated Data Processing (ADP) Installation	22-Dec-92
TN 420-43-7	Facilities Engineering - Key Installation and Utility Maps	26-Feb-96
TN 420-43-9	Power Factor Correction	03-May-93
TN 420-43-11	Application Problems with Transformer Differential Relaying Systems	20-Aug-95
TRADOC Form 406-R	TRADOC Activity Report	
TRADOC Reg. 420-05	Preventive Maintenance and Self-Help	
TRADOC Reg. 420-11	Energy and Utility Systems Management	23-Aug-85
TRADOC Reg. 700-2	Ammunition (Logistics)	10-Mar-89
UPC	Uniform Plumbing Code	current ed.
USAIC 190-5	Motor Vehicle Traffic Regulation	
USAIC 210-5	Garrison Regulations	
USAIC 385-2	USAIC Accident Prevention Program	
USAIC 420-22	Self Help Program	
USAIC 420-90	Fire Prevention and Protection	
USAIC 703-1	Waste Oil Recovery	
USAIC 703-2	Petroleum Management	

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